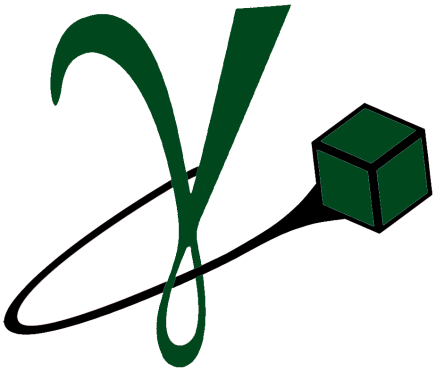

***Universidade de Brasília* — UnB**

Gama CubeDesign

Critical Design Review – CDR

Cubetschek Mission

CubeDesign 2026

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Critical Design Review – CDR

Cubetschek Mission

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Abstract

This document presents the Critical Design Review (CDR) of Cubetschek, a 2U Cube Satellite (CubeSat) developed by the Gama CubeDesign team from *Universidade de Brasília* for the CubeDesign 2026 competition. The satellite's primary mission is to receive and process Automatic Dependent Surveillance–Broadcast (ADS-B) signals from aircraft. All subsystems undergo rigorous testing and verification, with engineering decisions grounded in theoretical basis, numerical modeling, environmental analyses and design standards.

Keywords: CubeSat, ADS-B signal processing, 2U nanosatellite, critical design review (CDR), space systems engineering, CubeDesign 2026

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List of Acronyms

1U One-unit CubeSat (10 × 10 × 10 cm).
2U Two-unit CubeSat (10 × 10 × 20 cm).
ACK Acknowledgement.
ADCS Attitude Determination and Control System.
ADR Architecture Decision Record.
ADS-B Automatic Dependent Surveillance–Broadcast.
Al 6061 Aluminum Alloy 6061.
AOCS Attitude and Orbit Control System.
ASD Acceleration Spectral Density.
back-EMF Back Electromotive Force.
BLDC Brushless Direct Current.
CAD Computer Aided Design.
CDR Critical Design Review.
CDS CubeSat Design Specification.
ConOps Concept of Operations.
CRC Cyclic Redundancy Check.
CubeSat Cube Satellite.
DC Direct Current.
EPDM Ethylene-Propylene-Diene-Monomer.
EPS Electrical Power Subsystem.
FOC Field-Oriented Control.
FR-4 Flame Retardant 4.
FSPL Free-Space Path Loss.
GPIO General-Purpose Input/Output.
GSFC Goddard Space Flight Center.
HK Housekeeping.
HLR High-Level Requirement.
I²C Inter-Integrated Circuit.
ICD Interface Control Document.
IDE Integrated Development Environment.
IMU Inertial Measurement Unit.
IPC Inter-Process Communication.
ISM Industrial, Scientific and Medical.
LBT Listen Before Talk.
LoRa Long Range.
MEMS Micro-Electro-Mechanical Systems.
MLI Multi-Layer Insulation.
NASA-GEVS NASA's General Environmental Verification Standard.
NDJSON Newline-Delimited JSON.
NiCr Nickel-Chromium.
OBC On-Board Computer.
PBS Product Breakdown Structure.
PCB Printed Circuit Board.
PER Packet Error Rate.
PETG Polyethylene Terephthalate Glycol.
PLA Polylactic Acid.

PSD Power Spectral Density.
PWM Pulse-Width Modulation.
RSSI Received Signal Strength Indicator.
RTC Real-Time Clock.
SDR Software-Defined Radio.
SF Spreading Factor.
SI International System of Units.
SNR Signal-to-Noise Ratio.
SPI Serial Peripheral Interface.
TBC To Be Confirmed.
TC Telecommands.
TM Telemetry.
ToA Time on Air.
TRL Technology Readiness Level.
TT&C Telemetry, Tracking and Control.
TVAC Thermal Vacuum Chamber.
UART Universal Asynchronous Receiver-Transmitter.
V&V Verification and Validation.

List of Symbols

Latin letters

Symbol	Description and unit of measurement
A	State matrix
A	Heat transfer area (perpendicular to the heat flow or surface area) [m^2]
A_{cond}	Conduction area of each face of the satellite [m^2]
A_e	Effective area between the electrodes of the accelerometer [m^2]
A_{rad}	External surface area for radiation and convection [m^2]
$A(\vec{e}, \vartheta)$	Rotation matrix about axis $\vec{e}$ by angle ϑ
A_{ijk}, A_{321}	Attitude matrix in Euler angles
$\vec{a}$	Linear acceleration [m/s^2]
a, b	Satellite edges perpendicular to the axis of rotation [m]
a_x, a_y, a_z	Components of linear acceleration [m/s^2]
B	Channel bandwidth [Hz]
B	Input matrix
$\vec{B}$	Magnetic field [T]
b	Triad of body-frame unit vectors
$\vec{b}_1$, $\vec{b}_2$, $\vec{b}_3$	Unit vectors of the body frame
C	Capacitance of the branches of a symmetric balun [F]
C_e	Capacitance between the electrodes of the accelerometer [F]
C	Output matrix
C_1, C_2	Series and shunt capacitances of the balun [F]
$\mathcal{C}$	Channel capacity [bit/s]
$\mathcal{C}_o$	Controllability matrix
c	Speed of light in vacuum [m/s]
c_p	Specific heat [J/(kg K)]
D	Antenna directivity [dimensionless]
D	Feedthrough matrix
D_0	Maximum antenna directivity [dimensionless]
d	Distance between the electrodes of the accelerometer [m]
E_{in}, E_{out}	Energy entering and leaving the system [J]
$E_{\text{generated}}$	Thermal energy generated by the system [J]
$\vec{e}$, $\vec{e}_i$	Rotation axis; orthonormal basis vectors
e_b	Back electromotive force of the motor [V]

Symbol	Description and unit of measurement
$\vec{F}$	Force [N]
$\vec{F}_B$	Lorentz magnetic force [N]
$\vec{F}_C$	Coriolis force [N]
$\vec{F}_{\text{res}}$	Resultant force [N]
F, G, H, I	Final, intermediate and initial reference frames
f	Operating frequency [Hz]
f	Stall torque per volt of the reaction wheel motor (K_t/R) [N m/V]
f_0	Design center frequency [Hz]
G_0	Maximum antenna gain [dimensionless]
h	Reaction wheel thickness [m]
h	Convection heat transfer coefficient [W/(m^2 K)]
h_c	Depth of the motor coupling recess [m]
I	Satellite inertia matrix [kg m^2]
I	Electric current flowing through the resistor [A]
I_0	Motor no-load current [A]
I_3	Identity matrix of order 3
I_n	Identity matrix of order n
I_{sat}	Moment of inertia of the satellite about the z-axis, without the reaction wheel [kg m^2]
I_{wheel}	Moment of inertia of the reaction wheel about its axis [kg m^2]
i	Current generated by the solar panel [A]
i_1, i_2, i_3	Currents measured on the illuminated face of each pair [A]
$i_{\max}$, $i_{k,\max}$	Maximum current (normal incidence) of panel k [A]
$i_{\pm k}$	Current generated by face $\pm k$ [A]
j	Imaginary unit ($j^2 = -1$)
K	State-feedback gain matrix
K_1	Attitude angle feedback gain [N m/rad]
K_2	Angular velocity feedback gain [N m s/rad]
K_e	Back electromotive force constant [V s/rad]
K_t	Motor torque constant [N m/A]
k	Dipole shortening factor [dimensionless]
k	Thermal conductivity [W/(m K)]

Symbol	Description and unit of measurement
L	Inductance of the branches of a symmetric balun [H]
L	Length of the conduction path (wall thickness) [m]
L_1, L_2	Series and shunt inductances of the balun [H]
$\vec{L}_{\text{sat}}$, $\vec{L}_{\text{wheel}}$	Angular momentum of the satellite and of the reaction wheel [N m s]
$\vec{L}_{\text{total}}$	Total angular momentum of the system [N m s]
M_{sat}	Satellite mass, without the reaction wheel [kg]
M_{wheel}	Reaction wheel mass [kg]
m	Mass of the proof mass or vibrating element [kg]
m	Satellite mass in the lumped thermal model [kg]
N	Average noise power over the bandwidth [W]
n	System order
n_{faces}	Number of faces considered in the conduction path [dimensionless]
P	Power dissipated by the resistor [W]
P_{rad}	Total radiated power [W]
p	Closed-loop pole parameter ($s = -p$) [s^{-1}]
$\vec{p}$	Linear momentum [kg m s^{-1}]
Q	Component quality factor [dimensionless]
Q	Heat transferred [J]
$\dot{Q}$	Heat transfer rate [W]
$\dot{Q}_{\text{cond}}$, $\dot{Q}_{\text{conv}}$, $\dot{Q}_{\text{rad}}$	Heat transfer rate by conduction, convection and radiation [W]
Q_{in}, Q_{out}	Heat entering and leaving the system [J]
$Q_{\text{generated}}$	Internal heat generated by the satellite components and heaters [W]
Q_{stored}	Heat stored in the satellite [W]
Q_{rad}	Heat dissipated by radiation to deep space [W]
Q_{total}	Net heat flux between the satellite and the oven [W]
$Q_{\text{cond+conv+rad}}$	Combined heat flux between the satellite and the oven [W]
q	Electric charge [C]
q_{solar}, q_{albedo}, $q_{\text{planetshine}}$	Heating by direct solar radiation, albedo and planetary infrared radiation [W]

Symbol	Description and unit of measurement
R	Motor armature resistance [Ω]
R	Electrical resistance of the heater resistor [Ω]
R_{cond}, R_{conv}, R_{rad}	Thermal resistances by conduction, convection and radiation [K/W]
R_{total}	Total thermal resistance between the satellite and the oven [K/W]
R_{in}	Antenna input resistance [Ω]
r	Reaction wheel radius [m]
r_c	Radius of the motor coupling recess [m]
r_x, r_y	Radii of the cones with axes x and y [dimensionless]
S	Average received signal power over the bandwidth [W]
s	Laplace variable [s^{-1}]
$\vec{s}_b$	Sun vector in the body frame
$\vec{T}$	Torque applied to the satellite [N m]
T	Temperature of the satellite [K]
T, T_z	Component of the torque applied to the satellite about the z-axis [N m]
$\vec{T}_m$, T_m	Motor torque applied to the reaction wheel and its z component [N m]
$T_{\max}$	Maximum motor torque [N m]
$T_{\max}$, $T_{\min}$	Maximum and minimum satellite temperatures in the simulation [°C]
T_{amb}	Temperature of the environment or of neighboring bodies [K]
T_g	Glass transition temperature [°C]
T_{oven}	Temperature of the thermal cycling oven [K]
T_s	Surface temperature [K]
T_∞	Temperature of the fluid far from the surface [K]
t	Time [s]
U_{avg}	Average radiation intensity [W/sr]
$U_{\max}$	Maximum radiation intensity [W/sr]
$\vec{u}$	Control signal (general state-space form)
$\vec{u}_c$, u_c	Desired torque on the satellite and its z component [N m]
V	Voltage applied to the motor [V]
V	Voltage applied to the heater resistor [V]
V_2, V_3	Voltages at the balanced outputs of the balun (ports 2 and 3) [V]
V_c	Commanded motor voltage [V]
$V_{\max}$	Maximum voltage applied by the motor driver [V]
V_{nom}	Nominal motor supply voltage [V]
$\vec{v}$	Velocity [m/s]
W	Work transferred [J]
X, Y, Z	Sign (± 1) of the illuminated face of each pair
X_{in}	Antenna input reactance [Ω]
$\vec{x}$	State vector (general state-space form)
x	State of the single-axis model ($x = \omega$) [rad/s]
x	Coordinate along the direction of heat flow [m]
x, y, z	Cartesian coordinates of the body frame
$\hat{x}$, $\hat{y}$, $\hat{z}$	Unit vectors along the x, y and z axes of the body frame
$\vec{y}$	Output vector
Z_{in}	Antenna input impedance [Ω]

Greek letters

Symbol	Description and unit of measurement
α	Angle of incidence of light relative to the panel normal [rad]
α_x, α_y	Angles of incidence on faces +1 and +2 (half-aperture of the cones) [rad]
$\alpha_1, \ldots, \alpha_n$	Coefficients of the desired characteristic polynomial
β, β_x, β_y	Rotation of the generatrix about the cone axis [rad]
ΔE_{system}	Change in the energy of the system [J]
ΔT	Temperature difference [K]
Δt	Time interval; simulation timestep [s]
ΔU	Change in internal energy [J]
ε	Permittivity of the medium [F/m]
ε	Surface emissivity [dimensionless]
η	Radiation efficiency [dimensionless]
θ, $\vec{\theta}$	Angular position of the rotor or of the satellite [rad]
ϑ	Angle of rotation about an axis [rad]
λ	Wavelength [m]
λ_0	Free-space wavelength at the design frequency [m]
$\mu_1, \ldots, \mu_n$	Desired closed-loop poles [s^{-1}]
ρ	Density of the reaction wheel material [kg/m^3]
σ	Stefan–Boltzmann constant [$\text{W}/(\text{m}^2\,\text{K}^4)$]
τ	Mechanical time constant of the motor and wheel assembly [s]
ϕ, θ, ψ	Euler angles (3-2-1 sequence) [rad]
ϕ, θ, ψ	Angles defining the direction of the sun vector [rad]
$\vec{\omega}$, ω	Angular velocity of the satellite and its z component [rad/s]
$\vec{\omega}_0$	Initial angular velocity of the satellite [rad/s]
ω_0, ω_f	Magnitudes of the initial and final angular velocities of the satellite [rad/s]
ω_{nl}	Motor no-load speed [rad/s]
$\vec{\omega}_r$, ω_r	Angular velocity of the reaction wheel relative to the satellite and its z component [rad/s]
$\vec{\omega}_{\text{sat}}$	Angular velocity of the satellite [rad/s]
$\vec{\omega}_{\text{wheel}}$, ω_{wheel}	Absolute angular velocity of the reaction wheel [rad/s]
ω_x, ω_y, ω_z	Components of angular velocity [rad/s]
$\vec{\Omega}_{\text{wheel}}$, Ω_{wheel}	Operating angular velocity of the reaction wheel relative to the satellite [rad/s]

Operators

Operator	Description
$[\vec{e}\times]$	Skew-symmetric matrix of the cross product with $\vec{e}$
$\dot{(\,)}$	Time derivative
$\text{rank}(\cdot)$	Rank of a matrix
$\text{sat}(\cdot,\ \pm L)$	Symmetric saturation function with limit L

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1 Introduction

1.1 Project Introdution

The Cubetschek Mission is a project developed by the Gama CubeDesign team for the 5th CubeDesign, in the CubeSat category, hosted by the Universidad Nacional de San Martín in Buenos Aires, Argentina. The mission involves the development and testing of subsystems that make up a CubeSat capable of executing an ADS-B signal reception objective, fulfilling the required technical and environmental tests. Thus, the nanosatellite will consist of the Payload (ADS-B Payload) and the Antenna Deployment Mechanism, Attitude and Orbit Control System (AOCS), Electrical Power Subsystem (EPS), OBC, Structural Design, Thermal Control, and TT&C subsystems.

Furthermore, there is an extra Payload as a secondary mission, consisting of an ambipolar thruster developed at LPF-UnB. The theoretical scientific development and theoretical integration are cited in Section 13. With the addition of the extra payload, the cubesat would become a 3U, however, the integration is only theoretical and the physical satellite remains 2U.

The purpose of this CDR is to provide a critical analysis of the project scope, which involves the maturity of the product in the current phase, the engineering decisions made, and the necessary validations.

Below, Figure 1 presents the official mission patch, developed as a visual identity element for the project. The emblem represents the main technical objectives of the satellite and combines the team's colors with elements that refer to the goals of the mission.

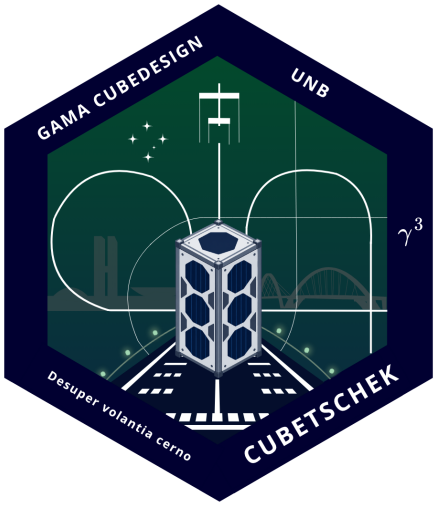

Figure 1: Official Mission Patch Cubetschek
Source: Authors (2026).

The mission patch pays tribute to *Brasília*, the capital of Brazil and the city where our university is located. The choice reflects the shape of the city's original urban plan, the *Plano Piloto*, which resembles an airplane when seen from above. This image links the city to the theme of the mission.

Each element holds a memory of the capital. The roads follow the pattern of the *tesourinhas*, the cloverleaf-shaped interchanges that stitch together Brasília's urban design. Above them, the Southern Cross guides the eye toward the southern sky the satellite will travel. At the center, a runway welcomes Cubetschek, whose name pays tribute to Juscelino and Sarah Kubitschek, the founder of the city that inspired this work and Brazil's First Lady at his side.

The Latin motto *Desuper volantia cerno*, "from above, I behold what flies", captures the essence of the mission: observing, from orbit, the aircraft that cross the skies. On the right,

gamma cubed (γ^3) signs the work on behalf of the Gama CubeDesign team, which carries into space a piece of the city where it all began.

1.2 Document Objectives

The purpose of this document is to record the design stages of the project, focusing on verifying and validating the CubeSat against the specified mission requirements. This report provides a clear rationale for the design decisions made throughout mission conceptualization, as well as the technical contributions of each engineering subsystem. It documents the engineering analyses, in-house development of the satellite components, and the internal evaluation framework, including tests already performed and those planned in accordance with the Cubedesign 2026 requirements (CubeDesign, 2026).

Additionally, this document details the system architecture, engineering budgets (mass, power, link), and satellite integration strategy, ensuring full functional performance both at the subsystem and system levels. All technological decisions are supported by technical references and justified within the mission's scope and constraints.

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2 Mission Overview

2.1 Mission Objectives

The mission objectives for the Cubetschek nanosatellite are derived from the official rules provided by the 5th CubeDesign organizing committee. The primary payload mission focuses on detecting, decoding, and analyzing flight data from aircraft ADS-B signals. Beyond meeting the competition requirements, the Cubetschek mission serves as an integrated systems engineering initiative across multiple disciplines. The specific primary and secondary mission objectives are established as follows (CubeDesign, 2026):

Primary Mission Objectives

- Detecting ADS-B signals from aircraft.
- Decoding position, velocity, heading, and altitude.
- Reconstructing trajectories and estimating origin and destination.
- Transmitting processed data as telemetry.

Secondary Mission Objectives

- Evaluate system architecture decisions.
- Analyze trade-offs between coverage, latency, and power.
- Strengthen technical documentation practices.
- Each team must propose a secondary mission compatible with the main misson. Implementation is not mandatory.

3 Mission Requirements

A CubeSat must comply with size, assembly, and operation standards, aiming for the standardization of the model and the fulfillment of the established objectives. The general mission requirements specified by the 5th CubeDesign rules (CubeDesign, 2026) comprise:

General Requirements

Table 1: High-Level General Requirements

Code	Requirement Information	Verification
HLR-GEN-01	The system shall conform to a mechanical envelope compatible with the CubeSat Design Specification (CDS v14.1), within a volume equivalent to 1U, 2U, or 3U.	Inspection (Fit-Check)
HLR-GEN-02	The system shall include, at minimum, the following subsystems: • Structural platform • Electrical Power System (EPS) including battery • On-Board Computer (OBC) • Communication subsystem • Attitude Determination and Control System (ADCS, minimum functional capability) • ADS-B payload • Flight and mission software	Inspection + Documentation review
HLR-GEN-03	The system shall operate autonomously during mission execution, supporting remote interaction via Telecommands (TC) and generating Telemetry (TM) without manual intervention.	Documentation review
HLR-GEN-04	Each team shall define, document, and maintain a consistent set of subsystem-level requirements derived from system-level requirements.	Documentation review

Source: (CubeDesign, 2026).

Communications (TM/TC)

Table 2: Communications Requirements

Code	Requirement Information	Verification
HLR-COMM-01	The system shall receive telecommands (TC) from the ground segment and execute corresponding actions within a bounded response time defined by the team.	Test
HLR-COMM-02	The system shall generate and transmit structured telemetry (TM), including at minimum: • System status • Power parameters • Attitude information • Mission data	Test + Data inspection
HLR-COMM-03	The communication architecture (including physical layer, protocols, and data rates) shall be defined, justified, and consistent with mission requirements.	Documentation review + Test

Source: (CubeDesign, 2026).

Power and Energy Requirements (EPS)

Table 3: Power and Energy Requirements

Code	Requirement Information	Verification
HLR-EPS-01	The system shall include a rechargeable energy storage element enabling autonomous operation for the full mission duration.	Test
HLR-EPS-02	The system shall demonstrate the capability to recharge its battery using a light-based energy source representative of solar input.	Test
HLR-EPS-03	The system shall define and justify a power budget covering all operational modes, including nominal and peak consumption.	Documentation review
HLR-EPS-04	The system shall maintain safe battery operating conditions (voltage, current, temperature) throughout all test phases.	Test + Telemetry analysis

Continued on the next page.

Table 3 — continued

Code	Requirement Information	Verification
HLR-EPS-05	The system shall support operation from an external power source during testing phases, including thermal vacuum (Thermal Vacuum Chamber (TVAC)), without requiring modification of core subsystems. If an internal battery is present, the system shall allow it to be removed during such tests while maintaining full system functionality. The external power interface shall provide electrical characteristics equivalent to nominal operating conditions and shall not interfere with system performance.	Test + Inspection

Source: (CubeDesign, 2026).

Attitude Requirements (AOCS)

Table 4: Attitude Requirements

Code	Requirement Information	Verification
HLR-ADCS-01	The system shall estimate its attitude along at least one axis using onboard sensors.	Test
HLR-ADCS-02	The system shall reduce an initial rotational condition (detumbling) to below 5 rpm (TBC) within a finite and measurable time.	Test
HLR-ADCS-03	The system shall quantify and report attitude determination performance, including error metrics and limitations.	Analysis + Documentation review

Source: (CubeDesign, 2026).

ADS-B Mission Requirements

Table 5: ADS-B Mission Requirements

Code	Requirement Information	Verification
HLR-ADS-01	The system shall receive ADS-B signals at 1090 MHz and process them during mission execution.	Test

Continued on the next page.

Table 5 — continued

Code	Requirement Information	Verification
HLR-ADS-02	The system shall decode standard ADS-B messages and extract, at minimum: • ICAO identifier • Position (latitude, longitude) • Altitude • Velocity	Test + Data validation
HLR-ADS-03	The system shall detect and process multiple aircraft simultaneously within a single mission scenario.	Test
HLR-ADS-04	The system shall implement and justify a processing architecture (onboard, ground, or hybrid).	Documentation review
HLR-ADS-05	The system shall execute a continuous mission of 10 minutes, processing up to 20 aircraft per test.	Test + Data validation
HLR-ADS-06	Each team shall define a secondary mission concept compatible with the primary mission. Note: This mission does not necessarily have to be implemented in hardware or software.	Documentation review
HLR-ADS-07	The system shall associate a timestamp with each received or processed ADS-B message, with a time resolution sufficient to enable trajectory reconstruction and temporal ordering of events within the mission duration.	Test + Data inspection
HLR-ADS-08	Mission Success Criteria The system shall meet defined mission success criteria during the 10-minute operational window, including at minimum: • Successful reception of ADS-B signals • Decoding of valid ADS-B messages • Extraction of required parameters (ICAO, position, altitude, velocity) • Transmission of processed data as telemetry Each team shall define quantitative success metrics, including at least: • Minimum number or percentage of correctly decoded messages • Maximum acceptable data loss rate • Maximum acceptable latency between reception and telemetry transmission	Test + Data inspection

Source: (CubeDesign, 2026).

Software and Data Requirements

Table 6: Software and Data Requirements

Code	Requirement Information	Verification
HLR-SW-01	The system shall implement operational modes consistent with the defined Concept of Operations (ConOps).	Test + Documentation review
HLR-SW-02	Mission and telemetry data shall be: • Structured • Documented • Reproducible	Data inspection

Source: (CubeDesign, 2026).

Verification and Validation Requirements

Table 7: Verification and Validation Requirements

Code	Requirement Information	Verification
HLR-VV-01	Each team shall define a Verification and Validation (V&V) strategy demonstrating compliance with all system requirements.	Documentation review
HLR-VV-02	Verification methods (test, analysis, inspection, demonstration) shall be explicitly defined and justified.	Documentation review

Source: (CubeDesign, 2026).

Environmental Requirements

Table 8: Environmental Requirements

Code	Requirement Information	Verification
HLR-ENV-01	The system shall successfully execute the mission prior to environmental testing.	Test
HLR-ENV-02	The system shall demonstrate functional survival after thermal and vibration testing, or provide justified analysis of observed limitations.	Test + Analysis

Source: (CubeDesign, 2026).

Cost and Fair Play Requirements

Table 9: Cost and Fair Play Requirements

Code	Requirement Information	Verification
HLR-COST-01	Each team shall provide a cost estimation for the complete system, including all subsystems and major components.	Documentation review
HLR-COST-02	Design decisions shall be justified based on performance, cost, and complexity trade-offs.	Documentation review

Source: (CubeDesign, 2026).

System Coherence Requirement

Table 10: System Coherence Requirements

Code	Requirement Information	Verification
HLR-SYS-01	The mission definition, system architecture, requirements, verification plan, and results shall be consistent and defensible from a systems engineering perspective.	Integrated review

Source: (CubeDesign, 2026).

3.1 Technical Tests

The on-site stage of the competition will be held in person at the host university, featuring evaluative tests for each subsystem to verify compliance with mission requirements. In addition to physical testing, this report will also be evaluated. The various tests and their respective requirements are outlined below (CubeDesign, 2026):

Basic Level

- Communication test: The CubeSat must receive telecommands (TC) and send telemetry (TM).

Intermediate Level

- Mechanism test: The CubeSat must deploy an antenna or similar device via remote command. The deployed element must be at least 10 cm. Teams must bring their own ground station.
- Battery charging using a light source: current and voltage will be evaluated.

Advanced Level

- Detumbling: reduce initial rotation (60 rpm TBC) to below 5 rpm (TBC). Target times: $<$10 s, $<$25 s, $<$60 s.
- Attitude determination (azimuth): measurement based on a light source. [$\leq$5°, $\leq$15°, $\leq$30°]

3.2 Environmental Testing

Environmental testing simulates the harsh conditions of the space environment, from launch to in-orbit operation. Thus, evaluating the vibrations to which the satellite is subjected

in the launch vehicle and the thermal variation conditions experienced in each orbit around the Earth.

Thermal Tests

- The system will undergo thermal cycling testing to evaluate its functional performance and the response of critical subsystems to temperature variations. Initially, two (2) thermal cycles are planned, with the following reference parameters:
  - Range: –10 °C to +50 °C (TBC)
  - Ramp rate: 1.5 °C/min
  - Stabilization: 30 minutes
  - Battery must remain above 0 °C

Vibration Test

- Random vibration test in accordance with NASA (Goddard Space Flight Center, 2021) requirements: 2 minutes per axis at 1 g.

4 System Architecture

4.1 System

The subsystems comprising the project span multiple engineering disciplines, each executing specific satellite functions essential to fulfilling the primary mission: the detection of aircraft ADS-B signals. The Product Breakdown Structure (PBS) and primary functional decomposition of the Cubetschek project are illustrated in Figure 2.

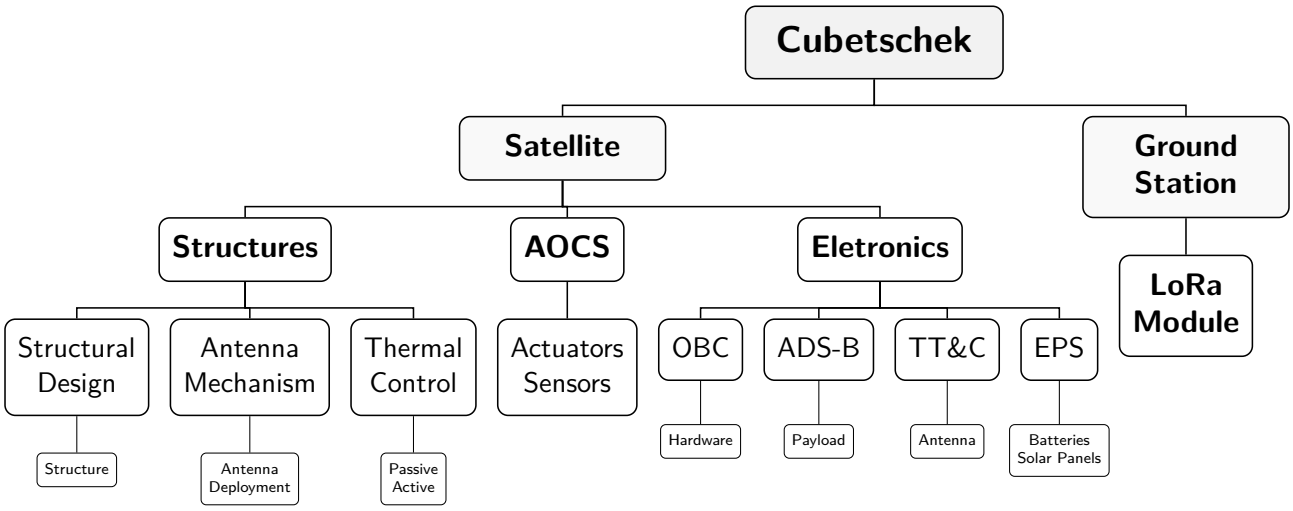


Figure 2: Product Breakdown Structure
Source: Authors (2026).

4.2 Master Schedule

Table 11 details the master schedule for the execution of the Cubetschek mission, listing the primary project activities, their respective start and end dates, and total duration in working days. A visual representation of this schedule, formatted as a Gantt chart, is provided in Figure 69, in Appendix A.

Table 11: Master Schedule for the Cubetschek Mission

ID	Activity	Start	End	Duration (Days)
1	Project Outline	08/04/26	29/04/26	16
2	Subsystem Definition	08/04/26	29/04/26	16
3	Literature Review	09/04/26	16/05/26	28
4	CAD Modeling and Design	01/05/26	14/09/26	93
5	Engineering Simulations	01/06/26	09/10/26	93
6	Component Procurement	05/06/26	21/09/26	75
7	Manufacturing and Assembly	01/08/26	10/11/26	72
8	Testing and Verification	08/09/26	20/11/26	53
9	CDR Documentation and Reporting	15/09/26	09/10/26	19

Source: Authors (2026).

4.3 Engineering Budgets

This subsection presents the primary engineering budgets for the mission, including mass, power, data, link and financial budgets. These budgets are established to ensure full project oversight, enabling the team to monitor the CubeSat's progress and margins.

4.3.1 Mass Budget

The mass budget is an essential aspect of a CubeSat, maintaining a maximum mass limit of 4.0 kg for a 2U satellite (The CubeSat Program, Cal Poly SLO, 2022a). This restriction satisfies requirement HLR-GEN-01, accounting for all components in the total mass calculation.

Table 12: Mass Budget

Component	Total Mass (g)
External Structure	640
Nuts and Screws Assembly	10.7
Rods	68
Spacers	3
Top-PCB	22.12
Bottom-PCB	22.12
Side-PCBs	192.96
Mechanism-PCB	50
TT&C-PCB	52.12
OBC-PCB	72.12
EPS-PCB	30
Batteries-PCB	105
Batteries	187.64
AOCS-PCB	142.36
Reaction Wheel	430

Continued on the next page.

Table 12 — continued

Component	Total Mass (g)
Solar Panels	37.36
EPDM lining	288
Total	**2353.50**

Source: Authors (2026), Section 5.2.

4.3.2 Power Budget

The power budget outlines the electrical consumption of each CubeSat subsystem during normal operations. Serving as a key design instrument, it validates the proposed Electrical Power System architecture by confirming that the solar panels, battery pack, and DC–DC converters can adequately support all mission demands.

Table 13: Power Budget

Subsystem	Component	Model	Qty.	Voltage [V]	Current [A]	Power [W]
AOCS	Accelerometer / Gyroscope	BMI088	1	3.3	0.006	0.0198
AOCS	Microcontroller	ESP32	1	3.3	0.080	0.264
AOCS	Encoder	AS5048A	1	3.3	0.002	0.0066
Electronic	Voltage and Current Sensor	INA219	1	3.3	0.001	0.0033
Electronic	Temperature Sensor	DS18B20	3	3.3	0.002	0.020
Electronic	LoRa	RA-02 SX1278	1	3.3	0.110	0.350
Total Current Draw (3.3 V Loads)						**0.20 A**
Electronic	OBC	Pi Zero 2W	1	5.0	0.700	3.500
Electronic	SDR	NESDR Nano 3	1	5.0	0.300	1.500
Mechanisms	SDR	NiCr Wire	1	5.0	1.000	5.000
Total Current Draw (5 V Loads)						**2.00 A**
AOCS	Driver	DRV8313	1	12.0	0.001	0.012
AOCS	Motor	GM2804	1	12.0	0.800	9.600
Total Current Draw (12 V Loads)						**0.80 A**
Total Power Consumption: 20.27 W						
Total Power Consumption with 20% Design Margin: 24.33 W						
Estimated Operating Time – 2S2P Battery Pack: 2.14 h						

Source: Authors (2026), Section 10.6.

4.3.3 Financial Budget

This section presents the financial budget of the Cubetscheck CubeSat project, including the estimated costs of components, materials, and resources required for the satellite development. The budget was structured according to the main subsystems and considers the current design proposed for the mission. The values are presented in Brazilian Real (BRL) and provide an overview of the financial requirements necessary for the manufacturing, integration, and testing phases of the CubeSat.

Table 14: Financial Budget of the Cubetscheck CubeSat project. Values are expressed in Brazilian Real (BRL)

No.	Subsystem	Component	Quantity	Unit Cost	Cost
1	Structural Design	Al 6061-T6 Structure	1	386.00	386.00
2	Structural Design	Carbon Stainless Steel Rods	1	36.00	36.00
3	Structural Design	M3x16 mm Countersunk Screws	8	0.35	2,80
4	Structural Design	M3x5 mm Countersunk Screws	10	0.85	8.50
5	Structural Design	M3 Self- locking Threads	8	0.45	3.60
6	Structural Design	PLA Filament	1	89.99	89.99
7	Thermal Control	1 Ω, 5 W Resistors	8	1.20	9.60
8	Thermal Control	1 kΩ, 1/4 W Resistors	2	3.60	7.20
9	Thermal Control	DS18B20 Temperature Sensors	3	5.98	17.94
10	Thermal Control	TIP120 NPN Transistors	2	2.45	4.90
11	Thermal Control	EPDM Tape	1	28.06	28.06
12	Antenna Mechanism	NiCr Wire	1	20.00	20.00
13	Antenna Mechanism	TIP120 NPN Transistors	2	2.45	4.90
14	Antenna Mechanism	Nylon Thread	1	5.40	5.40
15	Antenna Mechanism	M3x8mm Countersunk Screws	6	0.35	2.10
16	Antenna Mechanism	M3 Nuts	10	0.15	1.50
17	AOCS	ESP32 WROOM-32	1	32.88	32.88
18	AOCS	AS5048A Encoder	1	34.90	34.90
19	AOCS	SimpleFOC Mini DRV8313	1	25.49	25.49
20	AOCS	IMU BMI088	1	62.69	62.69
21	AOCS	GM2804 Brushless Gimbal Motor	1	203.59	203.59
22	AOCS	2 mm HSS Drill Bit Set	1	19.95	19.95
23	AOCS	3 mm Carbide End Mill	1	43.57	43.57
24	AOCS	SAE 1045 Steel Bar	1	64.99	64.99
25	AOCS	JST Connectors	5	4.40	22.00
26	EPS	2V 270 mAh Solar Panels	12	7.87	94.44
27	EPS	NCR18650 Batteries	4	30.00	120.00
28	EPS	18650 Battery Case	2	19.90	39.80
30	EPS	JST Connectors	10	1.49	14.90
31	EPS	BMS hx-2s-jh20	2	7.00	14.00
32	EPS	MP1584 Conversor Step-Down	2	8.50	17.00
33	EPS	Conversor Step-Up TPS61088	1	9.00	9.00
34	TT&C	GNS5851	1	486.70	486.70
35	TT&C	LoRa SX1278 RA-02 Module	2	70.49	140.98
36	TT&C	Antenna	2	7.15	14.30
37	TT&C	Pigtail Connector	2	3.90	7.80

Continued on next page

Table 14 — continued

No.	Subsystem	Component	Quantity	Unit Cost	Cost
38	TT&C	Pigtail Cable	1	5.00	5.00
39	TT&C	Pigtail SMA	1	20.00	20.00
40	TT&C	Coaxial Cable	2	15.41	30.82
41	TT&C	Ceramic disc Capacitor, +5 V	1	0.19	0.19
42	TT&C	Ceramic disc Capacitor, +3.3 V	1	0.90	0.90
43	TT&C	USB-A Connector	1	15.00	15.00
44	OBC	Raspberry Pi Zero 2W	1	228.79	228.79
45	OBC	SD Card	1	35.31	35.31
46	OBC	INA219 Sensor	1	24.41	24.41
47	OBC	DS18B20 Sensor	1	11.30	11.30
48	OBC	PCB	12	74.40	892.80
49	OBC	Solder/Thermal Paste	1	100.00	100.00
50	OBC	2×20 Pin Header	6	10.00	60.00
51	OBC	Resistor 4.7 kΩ	3	1.12	3.36
52	ADS-B	SDR NESDR Nano 3	1	374.00	374.00
				Total Budget BRL	**3,731.21**

Source: Authors (2026).

4.3.4 Data Budget

The data budget compares what the ADS-B payload produces during the mission with what the telemetry link can carry in the same time (HLR-ADS-05, HLR-ADS-08, HLR-COMM-03). Table 15 summarises it; the derivation of each figure is given where it is used, and every figure is reproduced by a script that the project test suite runs on every build (Gama CubeDesign, 2026). The source is 129 times the link capacity, which is why the payload decodes and aggregates on board and transmits a compact state of each aircraft (Section 11.3.2).

Table 15: Data budget over the 10-minute mission

Item	Value	Derivation
ADS-B messages decoded on board (20 aircraft, 4 msg/s, 600 s)	48 000; 7.34 MiB	Section 12
Link capacity (nominal profile, transmitting half of the time)	58.3 KiB	Section 11.3.2
Source over capacity	129:1	
Transmitted: one snapshot of every aircraft each 5 s	48.5 KiB (reduction 155:1)	Section 11.3.2
Transmit time used (snapshots, housekeeping, statistics, acknowledgements)	274 s of 300 s; 46 % of the mission	Section 11.3.2
Onboard record	12.5 KiB/s; capped at 1 GiB	Section 11.3.9

Source: Authors (2026), computed by `lora_budget.py` (Gama CubeDesign, 2026), Section 11.3.2.

4.3.5 Link Budget

The link budget checks that the telemetry and telecommand link closes with margin (HLR-COMM-03). Table 16 gives the parameters and the margin of the nominal profile; the

derivation, the other profiles and the degradation allowance are in Section 11.3.4. At 100 m the margin survives 70 dB of accumulated pessimistic allowances with 9.8 dB to spare.

Table 16: Link budget of the nominal profile

Parameter	Value
Carrier frequency	433.92 MHz
Transmit power, both ends	14 dBm
Antenna gain, each end	2.15 dBi
Losses, both ends (baluns, cables, connectors)	2.3 dB
Profile	SF9, 125 kHz, CR 4/5
Receiver sensitivity	−129 dBm
Margin at 10 m / 100 m / 1 km (free space)	99.8 / 79.8 / 59.8 dB
Margin at 100 m after the 70 dB allowance	9.8 dB

Source: Authors (2026), Section 11.3.4.

5 Structural Design

CubeSats are small, standardized satellites in a cubic format of 10 cm per unit (U) (NASA, 2023a). This architecture reduces development costs and accelerates manufacturing, democratizing access to space through multiple launches (The CubeSat Program, Cal Poly SLO, 2022a). The rigorous standardization of interfaces with deployers allows for the use of highly reliable commercial-off-the-shelf components (Science Learning Hub Pokapu Akoranga Putaiao, 2022).

5.1 Theoretical Background

The material choice for the Cubetschek project was made based on structural, thermal, and manufacturing requirements. The structural design must reconcile high stiffness and low mass against the severe conditions of launch, including intense vibrations, shocks, and axial accelerations (Oliveira, J., 2022). In space, environmental conditions are extreme due to such factors as large thermal ranges, high radiation levels, and microgravity. Therefore, each material must be carefully defined so that the satellite can operate within its ideal range and prevent damage to internal components. Table 17 consolidates the mechanical properties of the materials used.

Table 17: Physical and mechanical properties of structural materials

Material	**Density (g/cm^3)**	**Elastic Modulus (GPa)**	**Poisson's Ratio**	**Yield Strength (MPa)**
Aluminum 6061-T6	2.70	69.0	0.33	275
Carbon Steel	7.87	205	0.29	370
FR-4	1.85	24.1	0.136	345
PETG	1.27	2.09	0.38 – 0.40	48

Source: Adapted from (MatWeb, 2024), (MatWeb, s.d.), (Mercado-Colmenero *et al.*, 2020), (Accurate Plastics, Inc., 2016), (Laminated Plastics, s.d.).

The Al 6061-T6 alloy was selected for the primary structure due to its excellent extrudability, strength, and strength-to-weight ratio, supporting the loads from the deployer. For the internal spacers, Polyethylene Terephthalate Glycol (PETG) was used due to its easy additive manufacturing and $T_g \approx 80$ °C (MatWeb, s.d.), while the PCB were designed in Flame Retardant 4 (FR-4) (Yogendrappa, 2021).

5.2 Modeling

The geometric design of Cubetschek was developed in SolidWorks 2026, structured into eight Al 6061-T6 aluminum components: two main side walls with embossed hexagonal cutouts, top and bottom covers, and closure frames also featuring a hexagonal embossed pattern. This modular division concentrates the loads on the walls directly connected to the covers and thereby preventing misalignments and tortions caused by tightening torque.

The top and bottom covers connect the walls and rods, which act as structural supports for stacking PCB inside the satellite, forming a unified box that evenly distributes the compression from the deployer. Conceptual validation was initiated through polymer prototyping prior to final machining in Al 6061-T6 aluminum, Figure 3.

The hexagonal geometry was inspired by the honeycomb structural principle, which is widely supported in nanosatellite design literature as an effective strategy for structural optimization, excellent resistance against vibration and bending, and mass reduction (Li; Xing, 2025; Reda *et al.*, 2024). Due to machining limitations, the original honeycomb concept was adapted. This geometric choice was structured to optimize manufacturing and prevent structural twisting or bending during the satellite integration and assembly stage. Table 18 contains a list of the structural components and their respective materials.

Figure 3: Final machining in Al 6061-T6 aluminum
Source: Authors (2026).

Table 18: Structural components of the satellite

Component	Material	Quantity
M3×16 mm Countersunk Screws	Stainless Steel	8
M3×8 mm Countersunk Screws	Stainless Steel	10
M3×5 mm Countersunk Screws	Stainless Steel	16
M3 Nuts	Stainless Steel	12
M3 Self-Locking Threads	Stainless Steel	8

Source: Authors (2026).

Another critical aspect in CubeSat design is the mass budget, which must not exceed 4.0 kg, in accordance with the 2U (The CubeSat Program, Cal Poly SLO, 2022a). This limit is part of the compliance with the mechanical envelope required by HLR-GEN-01, which demands conformity with the CubeSat Design Specification within a volume equivalent to 12U, 22U, or 32U (CubeDesign, 2026). At this stage, every component, even screws and nuts, counts toward the satellite's total mass balance. Table 19 shows the detailed mass budget analysis of Cubetschek, ensuring it remains within the established limit.

Table 19: Mass Budget

Component	Total Mass (g)
External Structure	640
Nuts and Screws Assembly	10.7
Rods	68
Spacers	3
Top-PCB	22.12
Bottom-PCB	22.12
Side-PCBs	192.96
Mechanism-PCB	50
TT&C-PCB	52.12
OBC-PCB	72.12
EPS-PCB	30
Batteries-PCB	105
Batteries	187.64
AOCS-PCB	142.36
Reaction Wheel	430
Solar Panels	37.36
EPDM lining	288
Total	2353.50

Source: Authors (2026).

The 3D Computer Aided Design (CAD) modeling was performed using the SolidWorks 2026 software. This made it possible to test several pilot versions of the design prior to machining the final one. In the initial attempts, aspects such as the required space were analyzed for each subcore, ease of machining due to constraints of the available machinery, material economy and optimization, and potential points of fragility and stress concentration. In this way, it was possible to design a structure that met all the requirements, both of the mission and those specific to each subcore. Moreover, modeling ensured that all PCB had the holes and connections in the correct locations before placing the final order.

The complete final model can be seen in Figure 4, and all technical drawings can be found in Appendix B .

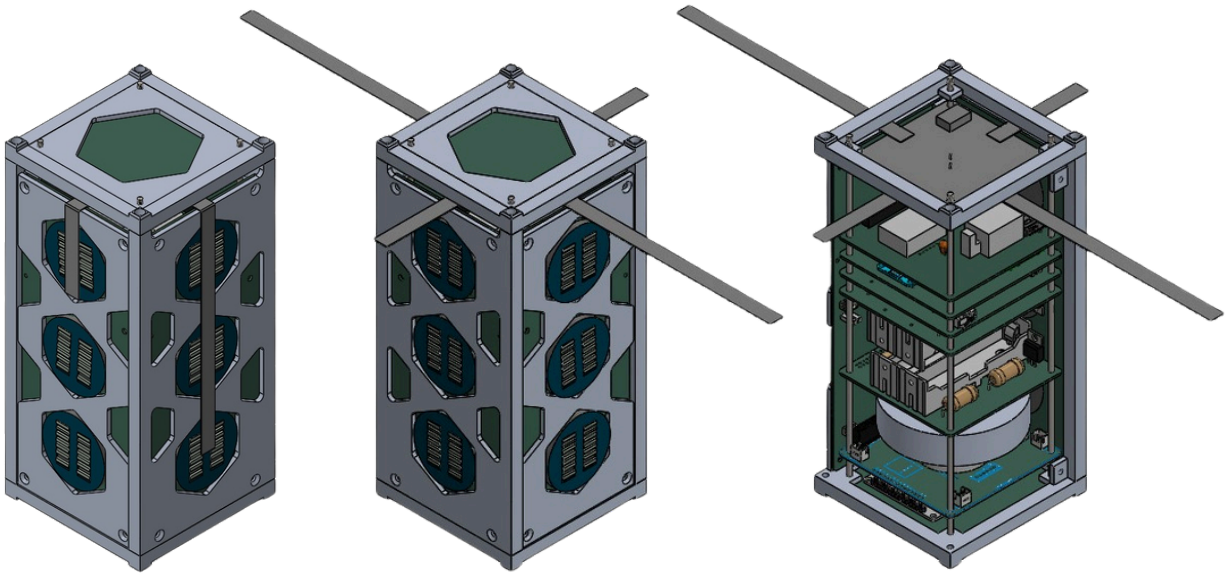


Figure 4: Full CAD of Cubetschek
Source: Authors (2026).

5.3 Simulation

For the satellite model validation, the random vibration test was simulated according to NASA-GEVS, a Goddard Space Flight Center (GSFC) document that defines how to test and verify whether space hardware can withstand the environmental conditions of launch and space (Goddard Space Flight Center, 2021).

The random vibration test is a non-sinusoidal vibration test, meaning the satellite is subjected to a random distribution of frequencies and amplitudes within a specific band. Thus, the shaker applies an Acceleration Spectral Density (ASD) to the satellite, which allows for a more faithful simulation of the actual conditions of a launch, unlike a sinusoidal test in which the object is subjected to only a specific frequency.

The parameters used that describe how the ASD varies with frequency throughout the test can be observed in Table 20 and represented graphically in Figure 5. The test is carried out in the frequency range of 20 to 2000 Hz, where the central region presents different test profiles depending on the total mass of the equipment or payload to be tested, and represents the area of highest vibrational energy.

Table 20: Random Vibration Test Specifications, NASA-GEVS standard

Frequency (Hz)	**ASD Level** (g^2/Hz)
20	0.026
20-50	+6 dB/oct
50-800	0.16
800-2000	-6 dB/oct
2000	0.026
Overall	14.1 G_{rms}

Source: (Goddard Space Flight Center, 2021).

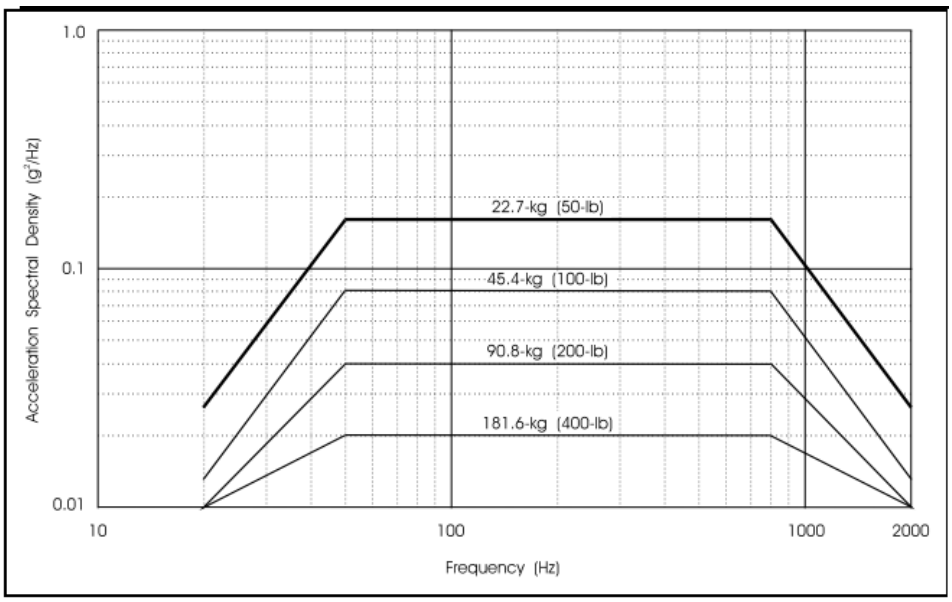


Figure 5: Acceleration Spectral Density
Source: (Goddard Space Flight Center, 2021).

The simulation was performed using Ansys Workbench 2026 R1 software, specifically within the Modal Analysis and Random Vibration tools. Thus, a simplified version of the Cubetschek structure was modeled, considering the PCB as flat plates without holes or connections.

5.3.1 Modal Analysis

For the modal analysis, the simplified geometry was imported from SolidWorks 2026 in .step format into the Geometry tab in the Modal window, as shown in the Figure 6.

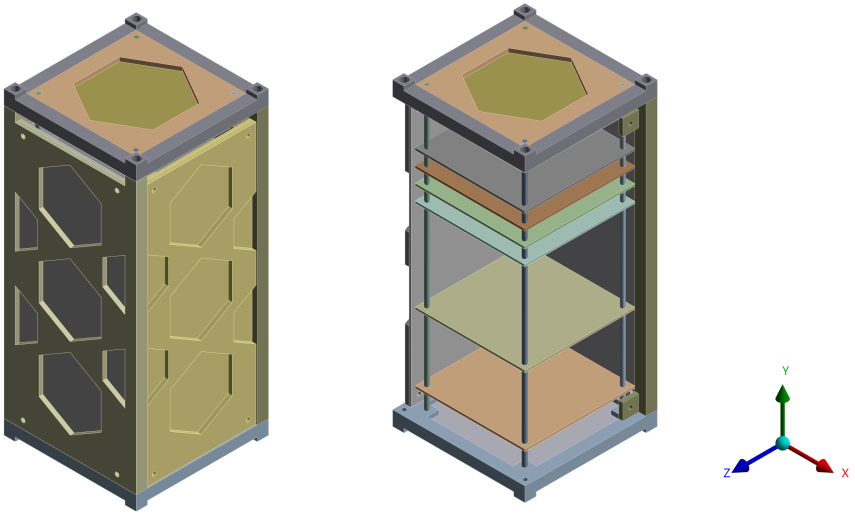


Figure 6: Cubetschek geometry
Source: Authors (2026).

Still in the Modal window, boundary conditions were defined in Model. The entire lateral surface was fixed using the Fixed Support command, Figure 7, which represents the mounting of the satellite to the shaker. Additionally, all contact regions were defined as bonded.

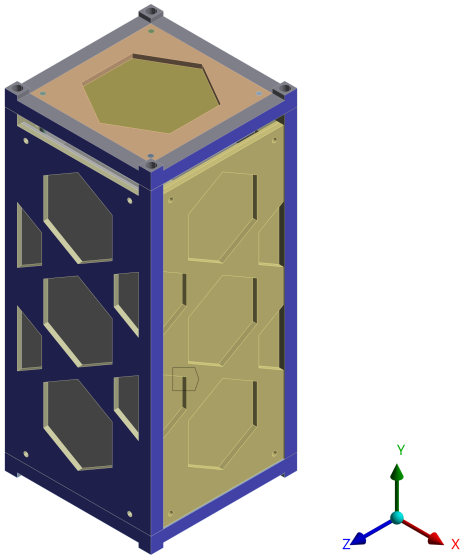


Figure 7: Boundary conditions
Source: Authors (2026).

Next, the materials for each component were assigned. Aluminum 6061-T6 was defined for the sides, cover, side frame, and cover frame; carbon steel for the support rods; and FR-4 for the internal PCB, as well as the side and cover PCB.

For this study, a 3 mm element size was used for the model's mesh. The mesh representation can be seen in Figure 8, and the statistics were 38623 finite elements and 184534 nodes.

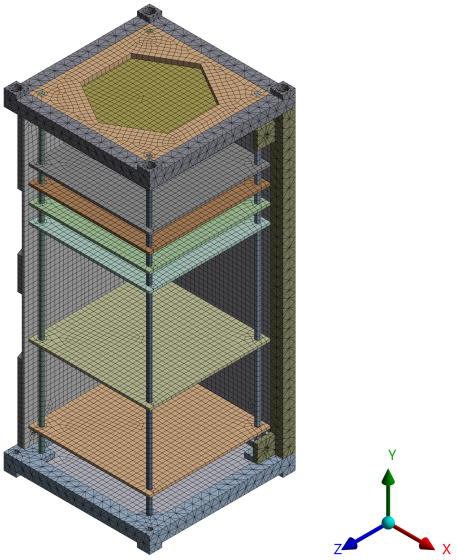
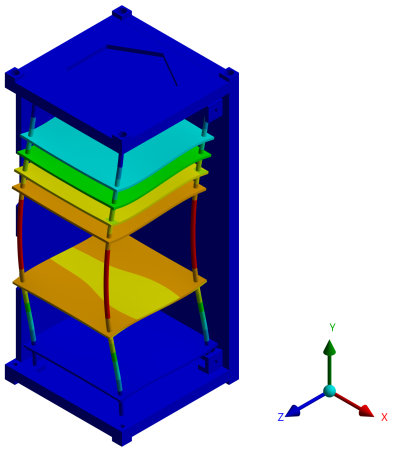
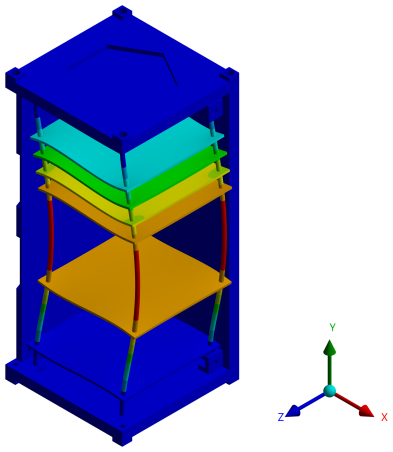


Figure 8: Model's mesh
Source: Authors (2026).

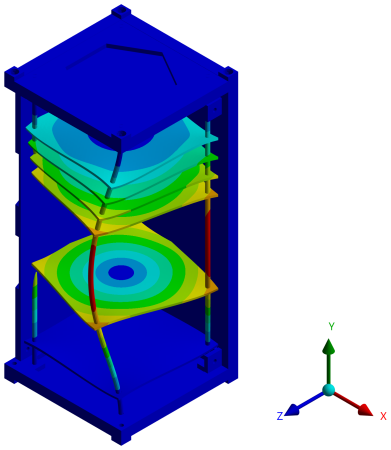
In the Solution section, the first 50 vibration modes of the satellite and their respective natural frequencies were analyzed, within the range of 0 to 2500 Hz. These values are presented in Tables 21 to 22 and will be used in the random vibration analysis. Each vibration mode behaves differently in the structure and can be characterized as bending, when there are bendings or curvatures, or torsion, when there is rotation around its own axis. There is also the occurrence of mixed modes, in which the vibration mode is not restricted to a single axis, affecting both the internal PCB and the mounting rods. Figures 9 exemplify each type of mode mentioned.



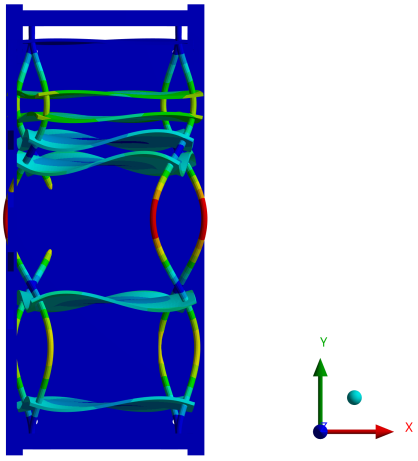
(a) Example of a bending mode about the Z-axis



(b) Example of a bending mode about the X-axis

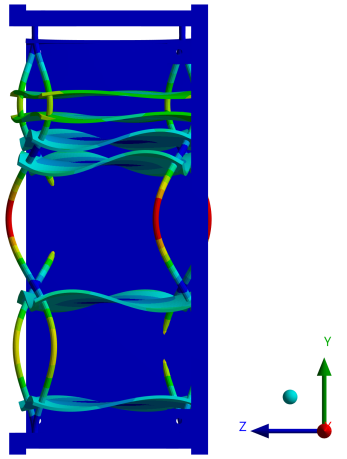


(c) Example of a torsional mode about the Y-axis



(d) Example of mixed modes in the YX plane

Figure 9: Vibration modes



(e) Example of mixed modes in the YZ plane

Figure 9: Vibration modes (continuação)
Source: Authors (2026).

Table 21: Rod vibration mode and respective natural frequencies of Cubetschek

a) Bending modes in the Z-axis

Mode	Frequency (Hz)
1	244.32
2	485.49
3	1257

c) Torsional modes in the Y-axis

Mode	Frequency (Hz)
1	339.15
2	682.57
3	1557.3

b) Bending modes in the X-axis

Mode	Frequency (Hz)
1	252.84
2	490.5
3	1275.5
4	1914.7

d) Mixed modes

Mode	Frequency (Hz)
1	1880.3
2	1988.7
3	2114.9
4	2117.1

Source: Authors (2026).

Table 22: Internal PCB vibration mode and respective natural frequencies of Cubetschek

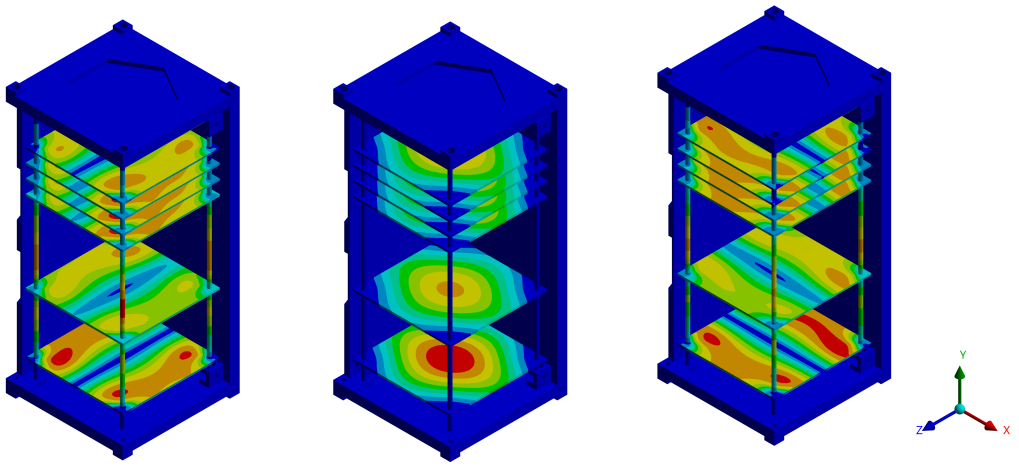
Mode	Frequency (Hz)	Mode	Frequency (Hz)	Mode	Frequency (Hz)
1	522.78	13	949.7	25	1146.9
2	533.57	14	990.94	26	1148.7
3	535.21	15	1013.6	27	1154.1
4	535.68	16	1017.9	28	1287.4
5	537.33	17	1019.2	29	1761.1
6	539.01	18	1020.8	30	1781.6
7	568.4	19	1023.1	31	1784.9
8	925.89	20	1037.7	32	1785.8
9	942.51	21	1107.8	33	1787.1
10	945.6	22	1116.1	34	1787.5
11	946.63	23	1141.7	35	1884.3
12	947.86	24	1145.9	36	2127.2

Source: Authors (2026).

5.3.2 Random Vibration Analysis

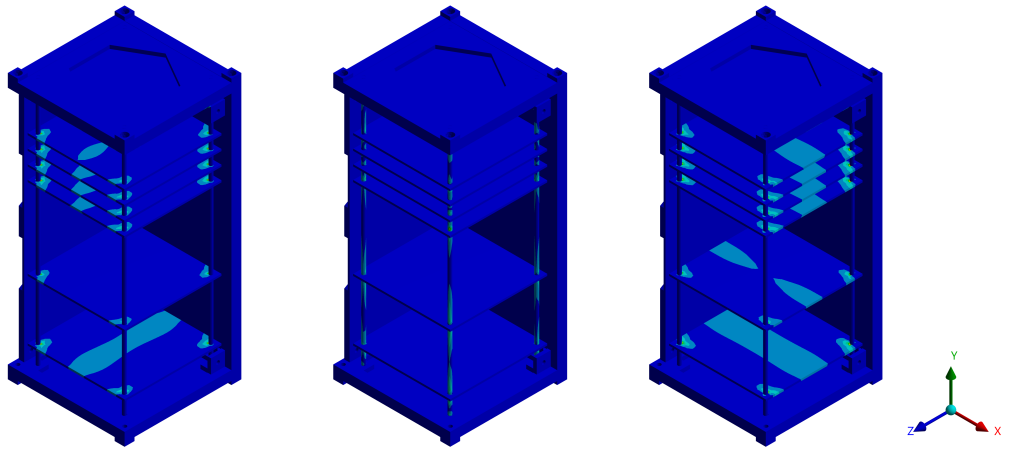
After the modal analysis, it was possible to carry out the second step in the Random Vibration window. The simulation was based on the NASA-GEVS standards mentioned previously in Table 20 and Figure 5. To do this, the Power Spectral Density (PSD) G Acceleration command was initially applied to the fixed supports in the direction of the Y-axis.

For the test, it is desired to obtain the values of directional (τ), normal stress (σ), and shear stress (τ) for each of the three axes of the CubeSat. In the study of random vibrations, the response is statistical, given that the excitation is random noise, determined by a PSD. Thus, the choice of the 2-sigma configuration, which includes twice the standard deviation of the vibrational response, generates a 95.45% probability of occurrence for the obtained values. The results can be viewed in Figures 10 to 12, and the extreme values are listed in Table 23.



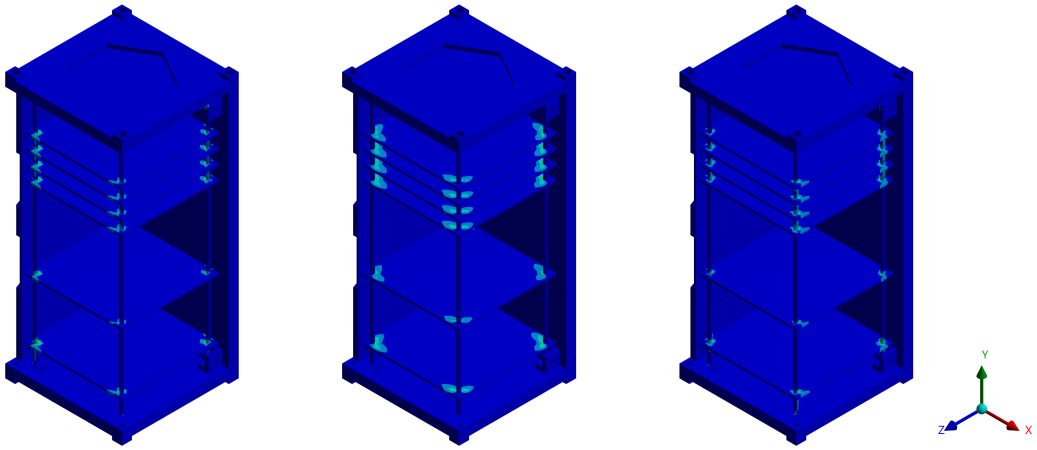
(a) (b) (c)

Figure 10: a) Directional deformation about the X-axis, b) Directional deformation about the Y-axis and c) Directional deformation about the Z-axis
Source: Authors (2026).



(a) (b) (c)

Figure 11: a) Normal stress about the X-axis, b) Normal stress about the Y-axis and c) Normal stress about the Z-axis
Source: Authors (2026).



(a) (b) (c)

Figure 12: a) Shear stress in the XY plane, b) Shear stress in the XZ plane and c) Shear stress in the YZ plane
Source: Authors (2026).

Table 23: Critical directional deformation, normal stress, and shear stress results for Cubetschek

Directional deformation	δ_x	δ_y	δ_z
Maximum (mm)	3.8219e003	0.23422	4.9644e003
Minimum (mm)	0	0	0
Normal stress	σ_x	σ_y	σ_z
Maximum (MPa)	33.058	32.099	28.993
Minimum (MPa)	9.8027e-016	1.9473e-015	9.9197e-016
Shear stress	τ_{xy}	τ_{yz}	τ_{xz}
Maximum (MPa)	7.2534	7.7215	16.139
Minimum (MPa)	8.4461e-018	3.2866e-016	1.7157e-017

Source: Authors (2026).

From the images, it is noticeable that the critical behavior occurred mainly in the plates, at stress concentration points, such as around the holes at the extremities. It was also observed a higher stress along the rods and, mostly, at the contact points with the plates. The values obtained for strain and stress are considered acceptable and indicate good chances of withstanding the vibration test.

5.4 Testing and Validation

To ensure the structural integrity, safety margins, and reliability of the satellite during the operational and launch phases, a verification and validation (V&V) strategy was implemented in accordance with aerospace standards required for the CubeDesign competition (CubeDesign, 2026).

5.4.1 Vibration Environment and FEA Numerical Verification

Due to local laboratory infrastructure constraints preventing physical vibration table testing under full launch profile conditions, numerical simulation via Finite Element Analysis (FEA) using ANSYS was adopted as the primary verification method, as detailed in the structural simulation (Section 5.3). This approach addresses HLR-ENV-02, which allows functional survival after vibration testing to be demonstrated by test or by a justified analysis of the observed limitations, and HLR-VV-02, which requires the verification methods to be explicitly defined and justified (CubeDesign, 2026).

The structural model was subjected to modal and harmonic analyses to determine fundamental eigenfrequencies and verify that structural resonance avoids launch vehicle characteristic frequencies. In addition, the random vibration simulation was performed using the PSD profile specified by the NASA-GEVS(GSFC-STD-7000) standard, yielding an integrated overall response of approximately 14.1 g_{rms} (qualification level) / 10.0 g_{rms} (acceptance level). Therefore, the computational analysis adopted a conservative approach by subjecting the model to NASA's full flight dynamic envelope. The compliance demonstrated under the GEVS spectrum validates the structural design for the 1 g_{rms} requirement (Section 3.2). Furthermore, quasi-static load verifications demonstrated that stress concentrations around the hexagonal relief pockets and fastener holes remain well below the yield strength of the Al 6061-T6 aluminum alloy, ensuring conservative safety factors.

5.4.2 Dimensional and Mass Verification

Physical validation of the external structure was conducted through precision dimensional inspection and mass measurement. The manufactured 2U structure strictly complies with the envelope constraints defined by the CubeSat Design Specification (CDS) (The CubeSat Program, 2022), confirming that its dimensions precisely match the standard 2U requirements ($100 \times 100 \times 227\,\text{mm}$) to ensure proper fit within the deployer interfaces (HLR-GEN-01). As detailed in the system mass budget (Table 19), the external structure achieved a total measured mass of 690.00 g, successfully fulfilling the targeted mass allocation for the mission architecture.

Figure 13: Physical mass measurement of the 2U aluminum external structure
Source: Authors (2026).

This combined numerical simulation compensating for facility limitations, paired with strict dimensional compliance and mass verification ensures the structural flight readiness of the satellite subsystem.

6 Antenna Deployment Mechanisms

The deployment mechanism in CubeSats is crucial for achieving the objectives of small satellites, enabling the controlled deployment of components post-launch. By extending the antennas, this mechanism adapts the satellite to the specific space requirements of each launch vehicle, ensuring that the satellite occupies the minimum possible volume during launch to optimize costs and, upon reaching its orbit, can deploy its mechanisms to fulfill the mission (Vilán; López Estévez; Aguado Agelet, 2012).

6.1 Theoretical Background

One of the main challenges of the antenna deployment mechanism is space constraint, as the compact design of CubeSats requires them to be engineered to occupy minimal volume. Furthermore, controlled antenna deployment is crucial, requiring a precise and reliable actuation mechanism (Rivera; Stewart, 2021). Maintenance and ease of component replacement present additional challenges, as the system must be capable of being reset and reactivated multiple consecutive times during testing and for troubleshooting potential issues. CubeSat antennas are essential components that enable communication between the satellite and Earth, as well as scientific data collection. The operating principle of these antennas relies on their deployable structure, which allows them to unfold post-launch. In their stowed state, the antennas occupy minimal space, but once deployed, they extend relative to the CubeSat, enabling the transmission and reception of radio signals. There are various types of deployable antennas, including dipole/monopole, Patch or Meshed Patch, and helical antennas (Liu *et al.*, 2022). Each type possesses its own specific characteristics and applications, but all share the same basic operating principle.

The system features two pairs of distinct antennas, aligned at 90 degrees relative to the satellite, which are folded downwards and secured at the base of the CubeSat. They are tied through the holes at their tips, so that upon actuation of the mechanism, the antennas deploy back to a 90-degree angle relative to the satellite. The antenna behaves like a spring due to its elastic nature, storing elastic potential energy while in its stowed state. The properties that influence the antenna's ability to return to its original shape include its material, stiffness, length, and thickness, as well as its concave geometry (Encinas Plaza *et al.*, 2010). This spring force is held restrained by a tensioned line

Mechanism actuation occurs when a resistor heats the line until it severs. Upon release, the antenna instantly returns to its original shape. The material of this line must be sufficiently strong to withstand the antenna's deployment force while remaining susceptible to thermal cutting for actuation.

6.2 Modeling

6.2.1 Design Concept and Objectives

The antenna deployment mechanism system was developed to enable the controlled and reliable deployment of the CubeSat's dipole antennas post-launch, ensuring that the satellite achieves its full in-orbit communication capabilities (Bhagavathula; Gupta, 2015). The system design prioritized simplicity, low mass, modularity, and reliability, adhering to the spatial and volume constraints imposed by the CubeSat standard.

The mechanism incorporates two antennas: a smaller dipole antenna for the ADS-B module and a larger dipole antenna for ground station communication using the Long Range (LoRa) module. This configuration was selected because it facilitates satellite integration, enabling testing and maintenance to be performed individually.

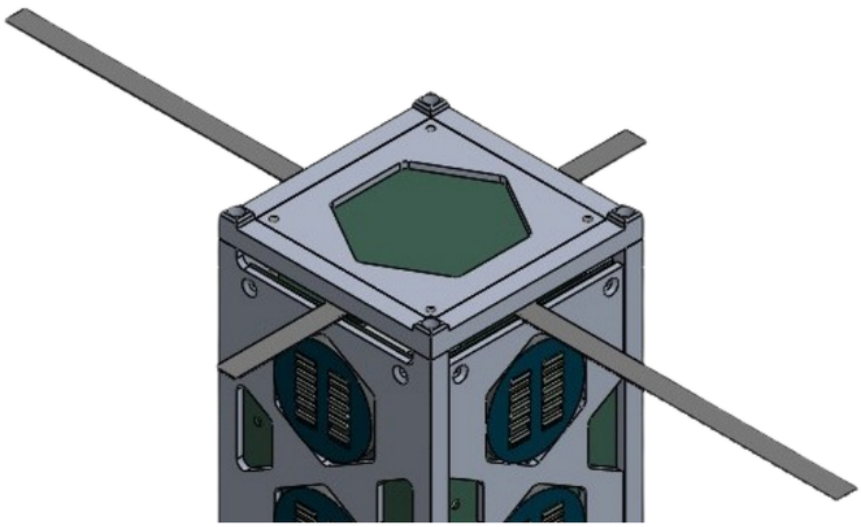


Figure 14: CAD of Cubetschek, focus on the antennas
Source: Authors (2026).

6.2.2 Materials and Construction

The antennas are fabricated from metallic measuring tape blades, which exhibit elastic properties and shape memory. When folded, these tape blades store elastic potential energy and, upon release, spontaneously return to their extended state. This characteristic makes the

behavior of the antenna analogous to that of a tape spring, allowing the strain energy stored in the fold to drive automatic deployment once the restraint is released.

6.2.3 Principle of Operation

The operation of the deployment mechanism is based on a thermo-mechanical controlled release system, in which the tension holding the antennas stowed is severed through resistive heating. During launch and the initial orbital phase, the antennas remain restrained by a nylon line acting as a retention element. This line is sufficiently robust to withstand launch vibrations and accelerations, but is engineered to sever when subjected to elevated temperatures.

Heating is generated by a Nickel-Chromium (NiCr) wire positioned in contact with the nylon line (Gardiner, 2015). Upon receiving a command from the On-Board Computer (OBC), the drive circuit—comprising the batteries, a TIP120 transistor, and the NiCr resistive wire—allows electrical current to flow through the heating element. Electrical power dissipation via Joule heating elevates the temperature of the NiCr wire, which conductively heats the nylon to its melting point.

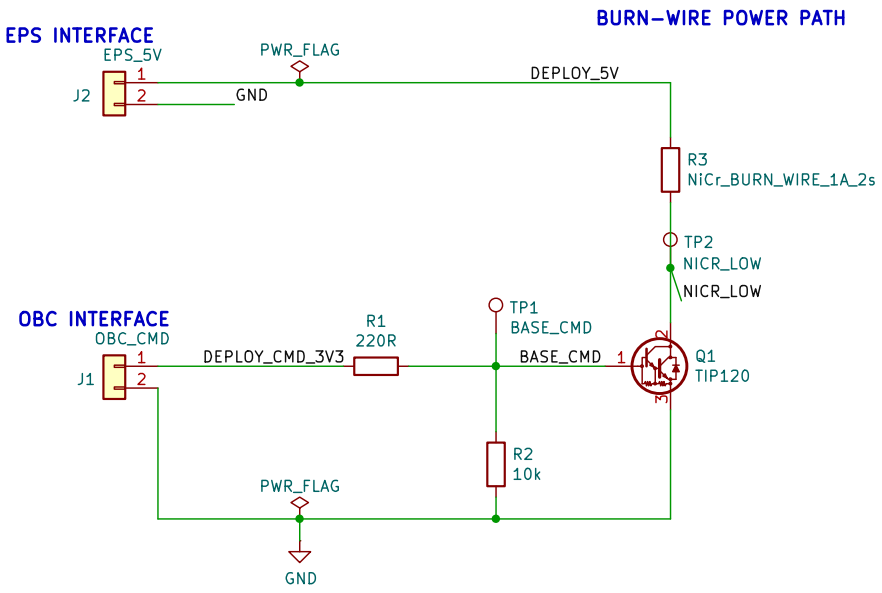


Figure 15: Burn-wire circuit schematic for antenna deployment actuation
Source: Authors (2026).

The severance of the nylon line releases the tension applied at the tips of the antennas, allowing the stored elastic strain energy to be released and driving full antenna deployment. This operating principle exhibits high reliability and low power consumption, making it widely adopted in CubeSats due to its simplicity, low mass, and absence of complex moving parts (Oh; Lee, 2015).

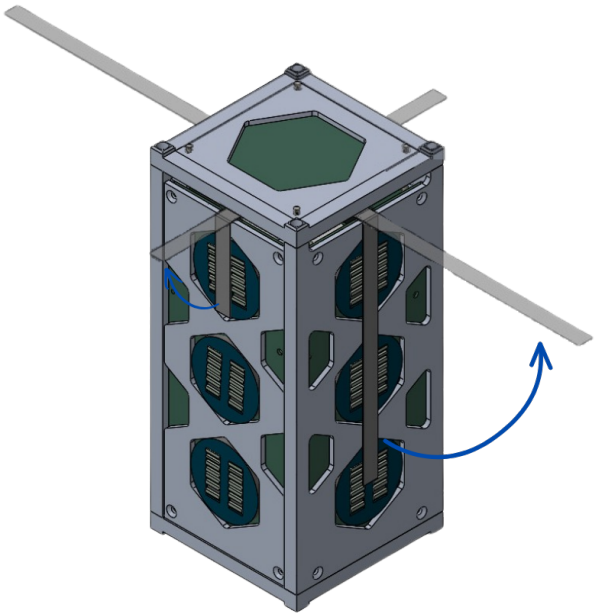


Figure 16: Basic demonstration of antenna deployment
Source: Authors (2026).

6.3 Integration and Assembly

The assembly process for the antenna deployment mechanism was designed focusing on simplicity, safety, and the ease of multiple assemblies and activations. The antennas are fixed directly onto the PCB, which also contains the activation circuit. By following this antenna mounting pattern, the aim was to minimize interfaces that could cause signal loss. The antennas are tied to the nylon line through holes at their tips; these nylon lines then loop around the satellite, tensioning the antennas and bending them against the body of the CubeSat. After looping around, the nylon line is secured with screws, providing greater assembly convenience and allowing finer control over tension. As for the NiCr wire, it is carefully placed in a strategic location where the nylon line passes directly against it, ensuring good thermal contact and preventing overheating points.

During integration, each deployment was tested individually to validate the correct positioning of the antennas and the proper operation of the actuation system. The activation circuit was configured to operate upon a signal from the OBC, with electronic control via a transistor, ensuring that activation occurs only under authorized command and preventing accidental deployments.

6.4 Testing and Validation

To validate system performance, benchtop and integration tests were conducted. In initial trials, the mechanisms were actuated individually by cutting the nylon line with scissors to evaluate whether the spring force of the metallic tape would overcome the inertia associated with its stowed position. The tests were successful, confirming the full deployment of the antennas when released from their stowed configuration, as illustrated in Figure 16, which demonstrates the operating principle of the antenna deployment mechanism.

Additionally, the mechanical integration and arrangement of the antenna deployment system can be verified through the CAD of the mechanisms PCB, presented in Figure 78. This technical drawing provides a reference for the physical configuration of the PCB used to support and integrate the antenna deployment mechanism.

Subsequently, testing of the activation circuit was carried out. These preliminary tests were conducted on a benchtop setup using breadboards. The circuit comprised one TIP120 transistor, one NiCr heating wire for thermal cutting (burn-wire), one 10Ω, 1 resistor, one 220Ω, resistor, an ESP32 microcontroller to send the command, an external power supply, and nylon lines to verify line severance, as shown in Figure 17. Initially, the TIP120 was kept continuously on, allowing current to flow through the NiCr wire, which was in contact with the nylon. The external power supply was set to 0V; upon gradually increasing the voltage, it was observed that at 5V, the circuit drew 1 A, an ideal level for heating the wire sufficiently to snap the nylon. The parameters that ensured the best efficiency for reaching the nylon's breaking temperature were a supply voltage of 5V and a current of 1 A.

Figure 17: Benchtop Testing of the Antenna Deployment Mechanism Actuation Circuit
Source: Authors (2026).

According to this result, tests were conducted varying the duration for which the ESP32 kept the TIP120 supplying current to the NiCr wire. These tests confirmed that the safest duration for maintaining current through the resistive wire is 2 seconds, and this duration was adopted for the final version of the activation code.

Based on the results obtained, the Cubetschek antenna deployment mechanism proved to be an efficient, lightweight, and functional solution, compatible with the mechanism test requirement (Section 3.1). The use of accessible materials, combined with a controlled thermal cutting actuation principle, resulted in a compact, low-cost system that is fully compatible with the requirements and constraints of the CubeSat standard.

7 Thermal Control

The Thermal Control subcore has as its main objective to ensure that all components of the Cubetschek nanosatellite operate within the temperature ranges established in the design during all phases of the mission, ensuring the proper functioning of the onboard systems. The complexity of the thermal design is determined by the mission requirements (Section 3), which define the thermal environments to be faced and the operating modes of the onboard components.

In low orbits, the space environment imposes extreme thermal conditions, since direct incidence of solar radiation, added to the albedo effect, which is the radiation reflected by the Earth's surface, can significantly raise the temperature of the satellite. On the other hand, during eclipse periods, in which there is no exposure to solar radiation, the CubeSat is subjected to negative temperatures that can compromise the performance of its systems. In the vacuum of space, radiation is the main form of heat exchange, which makes thermal control an essential challenge for the operational integrity of the mission (Gilmore, 2002).

In Cubetschek, both thermal control strategies will be implemented. Passive thermal control is based on an insulation method similar to Multi-Layer Insulation (MLI) and on the

use of thermal tapes, which consists of applying two layers of thermal insulation that act to minimize radiative transfer between the space environment and the satellite structure (Gilmore, 2002). The system will rely on a passive thermal control, composed of four walls of the satellite internally lined with Ethylene-Propylene-Diene-Monomer (EPDM) rubber, a material selected for its properties of low thermal conductivity, dimensional stability, and resistance to temperature variations (NETZSCH Polymers, 2026). The covers will also be lined with EPDM foam sealing tape, ensuring continuity of the thermal insulation across the entire structure. In addition, the solar panel and cover PCBs, made of FR-4, will also serve as the main thermal insulating material of the system, contributing to the reduction of heat exchange between the external environment and the interior of the satellite (Accurate Plastics, Inc., 2016). Components considered critical, such as the batteries, will be insulated with Kapton tape, a material widely used in the space sector due to its thermal stability and resistance to extreme environments (DuPont, 2026). Kapton also acts as an electrical insulating element, helping to mitigate surface electrostatic charges and contributing to the preservation of the functional integrity of the onboard circuits (DuPont, 2026).

Active thermal control will be based on an electrical resistance heating system, a widely established methodology validated in space applications, with a high with a high Technology Readiness Level (TRL) (NASA, 2023b). The choice of this type of system is based on its high reliability and good synergy with the chosen passive control, resulting in efficient and stable thermal regulation throughout the operation of the CubeSat.

7.1 Theoretical Background

The transfer of energy between systems can occur through two main mechanisms: heat transfer (Q) and work transfer (W). When energy is transferred due to a temperature difference, it is considered a *heat transfer*, a quantity commonly represented by the letter Q. On the other hand, when the transfer occurs due to any other type of interaction, it is characterized as work (Çengel; Ghajar, 2012).

The total amount of heat transferred during a process can be determined by integrating the heat transfer rate over time, as expressed in Equation 1:

$$Q = \int_0^{\Delta t} \dot{Q}\, dt \quad [J] \tag{1}$$

where $\dot{Q}$ represents the heat transfer rate as a function of time, and Δt corresponds to the time interval considered.

In cases where the heat transfer rate remains constant over time ($\dot{Q} = \text{constant}$), Equation 1 simplifies to:

$$Q = \dot{Q}\, \Delta t \tag{2}$$

This simplification facilitates the mathematical development of the problem.

Based on the first law of thermodynamics, also called the principle of energy conservation, we have:

$$E_{\text{in}} - E_{\text{out}} = \Delta E_{\text{system}}$$

In a steady state, that is, when the state of the system does not change, $\Delta E_{\text{system}} = 0$. Thus, it follows that:

$$E_{\text{in}} = E_{\text{out}}$$

Maintaining the considerations of a closed and steady system and assuming that the other forms of energy are negligible, the energy balance can be expressed as $\Delta E = \Delta U$:

$$\Delta E_{\text{therm. system}} = Q_{in} - Q_{out} + E_{\text{generated}} \tag{3}$$

The term $E_{\text{generated}}$ refers to the thermal energy generated by the system during operation, which may come from electrical resistances, mechanical resistances, or other sources (Çengel; Ghajar, 2012).

Having seen how the energy balance of a system in steady state is carried out, there is a need to understand how the heat entering and leaving the system is quantified. Heat transfer can occur through three fundamental mechanisms: **conduction**, **convection**, and **radiation**. Each mechanism has distinct characteristics and specific laws that describe the flow of thermal energy between bodies or within materials (Borgnakke; Sonntag, 2015). In addition, since the active thermal control relies on electrical heaters, the generation of heat by Joule heating in resistors is also addressed.(Borgnakke; Sonntag, 2015)

7.1.1 Conduction

Conduction is the process of heat transfer through a solid or stationary material, without macroscopic movement of the fluid. This phenomenon occurs due to the temperature difference between regions of the material. This behavior can be evaluated through Fourier's Law (Çengel; Ghajar, 2012):

$$\dot{Q}_{\text{cond}} = -kA\frac{dT}{dx} \quad [W] \tag{4}$$

Where:

- $\dot{Q}_{\text{cond}}$ is the heat transfer rate by conduction;
- k is the thermal conductivity of the material $(W/m \cdot K)$;
- A is the area perpendicular to the direction of heat flow (m^2);
- $\frac{dT}{dx}$ is the temperature gradient in the direction of flow (K/m).

For one-dimensional cases and steady state, with length L and temperature difference ΔT, the equation simplifies to:

$$\dot{Q}_{\text{cond}} = \frac{kA\Delta T}{L} \tag{5}$$

7.1.2 Convection

Convection involves heat transfer between a surface and a moving fluid, or between regions of the fluid itself. The transfer rate is given by Newton's Law of Cooling (Borgnakke; Sonntag, 2015):

$$\dot{Q}_{\text{conv}} = hA(T_s - T_\infty) \quad [W] \tag{6}$$

where:

- h is the convection coefficient $(W/m^2 \cdot K)$,
- A is the surface area (m^2),
- T_s is the surface temperature (K),
- T_∞ is the temperature of the fluid far from the surface (K).

The value of h depends on the type of fluid, the velocity, the flow regime, and the physical properties of the fluid, and is normally obtained through experimental correlations or state-of-the-art tables.

7.1.3 Radiation

Radiation is the transfer of thermal energy through electromagnetic waves, and can occur in a vacuum. The amount of heat radiated by a surface is described by the Stefan-Boltzmann Law (Çengel; Ghajar, 2012):

$$\dot{Q}_{\text{rad}} = \varepsilon \sigma A \left(T_s^4 - T_{\text{amb}}^4 \right) \quad [W] \tag{7}$$

where:

- ε is the emissivity of the surface, ranging from 0 to 1,
- $\sigma = 5.670 \times 10^{-8}\ W/m^2 \cdot K^4$ is the **Stefan-Boltzmann constant**,
- A is the surface area (m^2),
- T_s is the surface temperature (K),
- T_{amb} is the temperature of the environment or of neighboring bodies (K).

Radiation is especially important in space environments, where convection is negligible due to the absence of air.

7.1.4 Heat Dissipation in Resistors

One of the main characteristics of resistors is the conversion of electrical energy into thermal energy. This phenomenon is described by *Joule's First Law* (Perea Martins, 2018), which establishes that the amount of heat generated in a conductor is proportional to the square of the electric current flowing through it, to the electrical resistance of the material, and to the conduction time.

Mathematically, this relationship is expressed by:

$$Q = I^2 R t \tag{8}$$

where:

- **Q** is the heat generated, expressed in joules [J];
- **I** is the electric current flowing through the resistor, in amperes [A];
- **R** is the electrical resistance of the conductor, in ohms [Ω];
- **t** is the time interval during which the current flows through the resistor, in seconds [s].

The power dissipated by the resistor is equal to the rate of conversion of electrical energy into heat, and is given by:

$$P = I^2 R \tag{9}$$

Thus, the total amount of heat generated during time t is:

$$Q = Pt \tag{10}$$

Considering the relationship between voltage, current, and resistance ($V = IR$), it is possible to also express the power as a function of voltage:

$$P = \frac{V^2}{R} \tag{11}$$

Therefore, the power, and consequently the dissipated heat, is proportional to the square of the voltage applied to the system (Perea Martins, 2018).

Finally, it is important to point out that the maximum thermal dissipation capacity of a resistor depends on factors such as mass, dimensions, shape, and environmental conditions, which limit the heat transfer rate to the external environment (Çengel; Ghajar, 2012).

7.2 Space Environment and Thermal Systems

The space environment presents significant thermal challenges due to the absence of atmosphere, direct exposure to solar radiation, and large temperature variations. The thermal conditions of a satellite are determined by the balance between the energy absorbed, stored, internally generated, and dissipated to space:

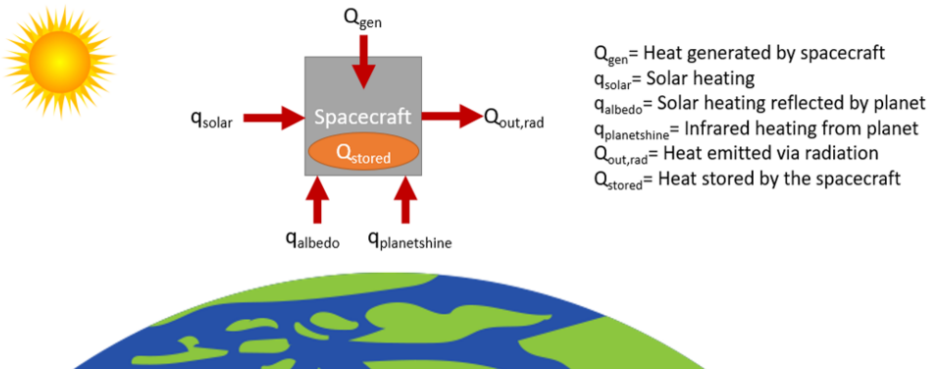


Figure 18: Simplified representation of the thermal boundary conditions of a space artifact
Source: (NASA, 2023b).

$$q_{\text{solar}} + q_{\text{albedo}} + q_{\text{planetshine}} + Q_{\text{generated}} = Q_{\text{stored}} + Q_{\text{rad}} \tag{12}$$

where:

- q_{solar}: heating due to direct solar radiation;
- q_{albedo}: radiation reflected by the planet;
- $q_{\text{planetshine}}$: infrared radiation emitted by the planet;
- $Q_{\text{generated}}$: internal heat generated by the satellite components;
- Q_{stored}: stored heat, determined by the thermal capacitance of the satellite;
- Q_{rad}: heat dissipated by radiation to deep space.

In space, it is possible to disregard external conduction and convection, since these forms of heat transfer require a medium to occur. Consequently, radiation becomes the main heat dissipation mechanism. Internally, conduction is the dominant mode of heat transfer between components (NASA, 2023b; Çengel; Ghajar, 2012).

Aiming to reach the thermal equilibrium that is optimal for the operation of satellites, there is a wide range of active and passive systems. Tables 24 and 25 present some categories of thermal controllers.

Table 24: Main passive thermal control technologies for CubeSats

Passive Systems
Multilayer Insulation (MLI)
High-Emissivity Surfaces
Thermal Tapes
Sun Shades
Louvers

Source: Adapted from (NASA, 2023b; Muraoka, 2000; Hengeveld *et al.*, 2010).

The passive technologies, presented in Table 24, are generally employed for their simplicity and low energy consumption. However, in missions that require more precise thermal control, active systems are also used, shown in Table 25.

Table 25: Main active thermal control technologies for CubeSats

Active Systems
Electric Heaters
Thermoelectric Coolers (Peltier)
Pumped Fluid Loops
Deployable Radiators
Microvascular Composites

Source: Adapted from (NASA, 2023b; Muraoka, 2000; Hengeveld *et al.*, 2010).

Thus, Cubetschek combines MLI-like insulation with thermal tapes (passive) and electric heaters (active), solutions of high technological maturity and compatible with the mass, volume, and power consumption constraints of a CubeSat.

7.3 Modeling

7.3.1 Active Thermal Control

The active thermal control designed for Cubetschek aims to monitor and maintain the satellite's temperature always positive, ensuring all components operate within their optimal operational temperature range. The heat is generated by Joule heating in the resistors, as described in Section 7.1.4 (Perea Martins, 2018).

For this purpose, a closed-loop circuit was developed, which is activated when the satellite temperature reaches 10°C and heats the interior up to 12°C, at which point it is turned off. This temperature range was selected and sized to meet the requirement that the battery remain above 0°C and operate under safe temperature conditions (HLR-EPS-04) (CubeDesign, 2026). The 10°C activation threshold keeps a safety margin relative to the 0°C limit, yet is close enough to it, so that the heating has time to stabilize the temperature and make it rise again before the limit is reached. The 12°C shutdown threshold, in turn, provides a hysteresis band that allows the interior to heat up without the circuit switching on and off at every small temperature variation. The power dissipated by the circuit, combined with the losses generated by the internal electronic components during nominal operation, guarantees the optimal operating temperature for the system.

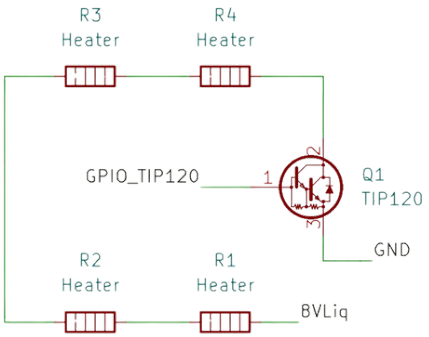


Figure 19: Active Thermal Control circuit schematic
Source: Authors (2026).

The circuit consists of four resistors, one specifically for current limiting, and a TIP120 (NPN) transistor acting as an electronic switch connected to the on-board computer. It receives the command that closes the circuit as soon as the DS18B20 sensor, attached to the battery,

detects a temperature drop down to the activation threshold. The battery directly powers the circuit at a voltage of 8 V and a current of approximately 1.7 A.

In this way, each battery PCB will feature an independent circuit for higher precision in temperature measurement across each cell set. The schematic is shown in Figure 19, and its implementation on the PCB can be seen in Figures 82 and Figures 92, 93, 94 and 95 in the Appendix D.2.

7.3.2 Passive Thermal Control

The passive thermal control of the satellite relies on materials whose thermal properties promote the conduction, insulation, and rejection of heat, while also being low cost and readily available. The selected materials and their properties are listed in Table 26, where t is the thickness, k the thermal conductivity, ε the emissivity, and c_p the specific heat.

Table 26: Thermal Properties of the Passive Thermal Control Materials

Material	**Thickness [mm]**	**Thermal Conductivity [W/m·K]**	**Emissivity**	**Specific Heat [J/kg·K]**
Al 6061-T6	TBD	167	0.10	896
EPDM	2.0	0.26	0.86	1900
FR-4	1.6	0.25	0.80	1100
Kapton	0.05	0.12	0.74	1090

Source: (MatWeb, 2026), (NETZSCH Polymers, 2026), (DuPont, 2026).

The Al 6061-T6 alloy was selected for the primary structure due to its excellent extrudability, strength, and strength-to-weight ratio, which allow it to withstand the loads imposed by the deployer (MatWeb, 2026). It is also one of the alloys recommended by the CubeSat Design Specification for the structure and rails (The CubeSat Program, Cal Poly SLO, 2022b). Its high thermal conductivity makes the frame the main conductive path of the satellite: the heat dissipated by the onboard electronics and the absorbed solar radiation are distributed across the structure and carried to the external faces, avoiding localized hot spots. The low emissivity of the bare alloy limits radiative losses and can be increased through anodizing or surface coatings wherever greater heat rejection is required (Gilmore, 2002).

The EPDM was adopted as an insulating layer due to its low thermal conductivity and high emissivity. Placed between the structure and thermally sensitive components, it reduces the conductive heat transfer between them while favoring radiative balance (NETZSCH Polymers, 2026).

The PCBs were designed in FR-4, the standard substrate for printed circuit boards (Yogendrappa, 2021). Besides supporting the electronic circuits, the boards fixed to the side walls carry the solar cells and close the faces of the structure, acting as an additional barrier between the internal modules and the space environment. For the internal spacers, PETG was used due to its ease of additive manufacturing and its glass transition temperature of $T_g \approx 80\,^{\circ}\mathrm{C}$ (MatWeb, s.d.), which also limits heat conduction between stacked boards.

Finally, Kapton tape is applied as a complementary thermal protection element, mainly around the batteries and at the interfaces between sensitive components. This polyimide film combines high thermal stability, chemical inertness, and low degradation under ultraviolet radiation (DuPont, 2026), and presents low outgassing in vacuum (NASA Goddard Space

Flight Center, 2026). Besides insulating these regions, it helps keep the batteries within their operating temperature range and mitigates the buildup of surface electrostatic charges.

Together, these materials provide a conductive path through the aluminum structure, local insulation through the EPDM, the PETG spacers, and the Kapton tape, and radiative coupling through the FR-4 boards, ensuring a stable and predictable thermal behavior of the satellite throughout its operating conditions (Gilmore, 2002).

7.4 Testing and Validation

The verification and validation of the thermal control were based on the design previously established in the modeling of the active (Section 7.3.1) and passive (Section 7.3.2) systems and on the following requirements imposed by CubeDesign 2026 (CubeDesign, 2026): HLR-EPS-04, which requires safe battery operating conditions (voltage, current, and temperature) throughout all test phases; HLR-ENV-02, which requires functional survival after thermal and vibration testing, or a justified analysis of the observed limitations; and HLR-VV-02, which requires the verification methods to be explicitly defined and justified.

For verification, we used the mathematical computational simulation presented in Section 7.4.1, which corresponds to verification by analysis. For the validation of the active control system, we carried out a physical test, presented in Section 7.4.2, which corresponds to validation by test.

7.4.1 Simulation

To evaluate the thermal control performance of the CubeSat, a numerical model was developed in Python, using the *NumPy* and Matplotlib libraries for data processing and visualization, and a **lumped parameter model** approach to represent the energy balance of the system.

The CubeSat was modeled as a homogeneous body with a mass of $m = 2.0\,\text{kg}$ and a specific heat capacity of $c_p = 1700\,\text{J/kg}\cdot\text{K}$, subjected to thermal exchange with the environment through conduction, convection, and radiation. The external environment was represented by a thermal oven with a time-varying temperature, simulating the thermal cycling test to which the CubeSat will be submitted during testing.

The thermal profile of the oven was built numerically, considering the thermal cycling parameters released by the organizers in Section 3.2 (CubeDesign, 2026). Furthermore, the heating and cooling ramps were assumed to be linear at 1.5 °C/min, with 30 min plateaus at the temperature extremes (+50 °C and −10 °C). Five complete cycles were simulated with a continuous and smooth temperature variation obtained by linear interpolation.

The thermal balance was determined by the ordinary differential equation:

$$\frac{dT}{dt} = \frac{Q_{\text{total}}}{m\,c_p} \tag{13}$$

where Q_{total} represents the net heat flux between the CubeSat and the oven, including the internal power generated by the active control system.

The total thermal resistance of the system (R_{total}) was obtained by combining the resistances associated with each heat transfer mechanism:

$$R_{\text{cond}} = \frac{L}{n_{\text{faces}}\,k\,A_{\text{cond}}}, \quad R_{\text{conv}} = \frac{1}{h\,A_{\text{rad}}}, \quad R_{\text{rad}} = \frac{1}{\varepsilon\,\sigma\,A_{\text{rad}}\,(T_{\text{oven}}^2 + T^2)(T_{\text{oven}} + T)} \tag{14}$$

The resistances were combined to represent the different faces of the satellite (top, bottom, and sides), resulting in a total thermal resistance R_{total}. Thus, the heat flux between the

satellite and the oven is expressed by:

$$Q_{\text{cond+conv+rad}} = \frac{T_{\text{oven}} - T}{R_{\text{total}}} \tag{15}$$

The model was developed considering a dissipated power value originating from the active control and operation of the satellite. For this simulation, an activation temperature of 15 °C was used to achieve better graphical results, given that the satellite naturally remains above 10 °C through passive thermal control alone. As a reminder, whenever the CubeSat temperature fell below 15 °C, there was an increase in heat dissipation from the heaters. During these intervals, the term $Q_{\text{generated}}$ was added to Q_{total}, simulating the action of internal resistors or heaters.

Time integration was performed explicitly with a 1 s timestep, applying the **Forward Euler method**:

$$T(t + \Delta t) = T(t) + \frac{dT}{dt}\,\Delta t \tag{16}$$

This approach allowed updating the temperature at every second of simulation, while also recording the moments when the heating system was activated.

Finally, the results were plotted as a function of time (in minutes), comparing the oven temperature and the satellite temperature, with visual highlights for the thermal control activation periods. The resulting plot allowed observing the dynamic thermal behavior of the CubeSat and the efficiency of the thermal control in keeping the temperature within the desired operational range. In Figure 20, the behavior of the satellite throughout the thermal cycling test can be observed. The code for this simulation can be checked at the appendix B.2.

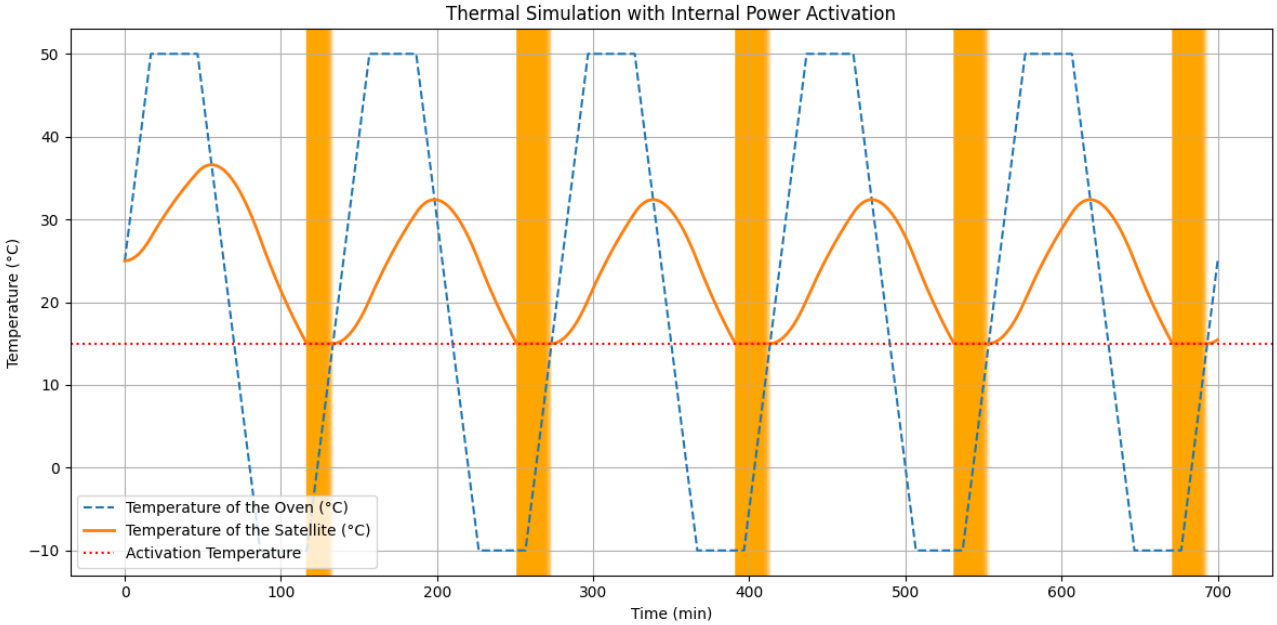


Figure 20: Thermal Simulation with Internal Power Activation
Source: Authors (2026).

The maximum temperature recorded was $T_{\text{max}} = 36.60$ °C, while the minimum was $T_{\text{min}} = 14.99$ °C.

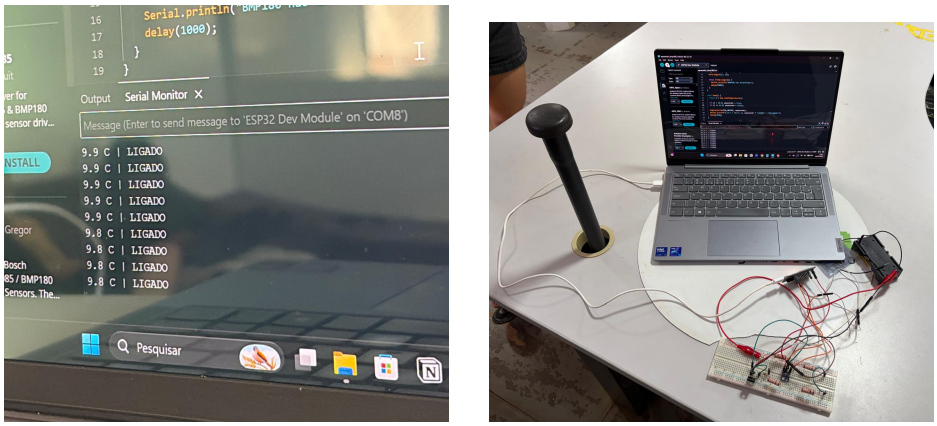
7.4.2 Tangible Validation

The active thermal control test was conducted using the battery pack, an ESP32 microcontroller connected to a laptop, and the heater resistor array. Once assembled, the setup

was placed inside an insulated container and put into a freezer, with a temperature sensor transmitting real-time data to a display. The test was monitored from room temperature down to the activation threshold, confirming the successful triggering of the active control system at 10 °C. The temperature dropped to approximately 6.5 °C before stabilizing and halting any further decrease, successfully fulfilling the requirement cited in Section 3.2.

For future testing, plans are underway to use the fully integrated satellite with all subsystems combined. Additionally, a high-temperature test will be conducted, making it possible to cover all operational temperature ranges and ensure that the satellite is thermally balanced.

In Figure 21, it is possible to see the temperature sensor reading at the moment of activation, at 10°C, and the setup used for the complete test.



(a) Temperature sensor data (b) Testing setup

Figure 21: Active Thermal Control
Source: Authors (2026).

8 AOCS

The AOCS is the subsystem responsible for maintaining the satellite's pointing with respect to a previously established reference frame. To this end, attitude and orbit control relies on sensors that determine the satellite's position and orientation, along with other necessary parameters, which are processed by a controller that commands the actuators accordingly (Cappelletti; Battistini; Malphrus, 2020).

This section presents the development of the AOCS of a 2U CubeSat, focusing on single-axis attitude control and on performing the detumbling mode by means of a reaction wheel. It covers sun vector determination, the electronic architecture associated with the system, the modeling and design of the control system, as well as its evaluation through simulations and bench tests.

8.1 Theoretical Background

This section presents the theoretical foundations underlying the modeling of the attitude control system, covering the concepts necessary for attitude representation, sun vector determination, satellite dynamics, and the development of the controller and the actuator.

8.1.1 Coordinate Systems

To describe the translation and rotation of a body in space, coordinate systems are employed, which simplifies the dynamic analysis for each specific situation. For this CubeSat,

the *Spacecraft Body Frame* is adopted.

This system is defined by an origin and three Cartesian axes fixed at a specific point on the spacecraft's body, normally used to align components during assembly. It is common to establish the body coordinate system based on the orientation of a rigid navigation base, which houses the attitude sensors and whose origin generally coincides with the satellite's center of mass. The goal is to determine and control the orientation of this base with respect to an external reference frame (Markley; Crassidis, 2014).

The system is defined by the triad of orthogonal unit vectors $\vec{b}_1$, $\vec{b}_2$, and $\vec{b}_3$, aligned with the body axes:

$$b = \{\vec{b}_1, \vec{b}_2, \vec{b}_3\}^T \tag{17}$$

The unit vectors along the x, y and z axes of the body frame are also denoted by $\hat{x} = \vec{b}_1$, $\hat{y} = \vec{b}_2$ and $\hat{z} = \vec{b}_3$.

For the development of the AOCS presented in this document, the z-axis is adopted as the reaction wheel's actuation axis and, consequently, as the attitude control axis. This axis corresponds to the longitudinal axis of the 2U structure, which is labeled as the y-axis in the structural simulations (Section 5). The difference is only a naming convention between the reference frames of the subsystems: the physical axis is the same, so it does not affect the moment of inertia, the dynamic model or the performance of the attitude control system. Thus, even though the satellite's orientation is presented in three-dimensional form, the dynamic modeling and controller design are subsequently reduced to rotational motion about this axis.

8.1.2 Euler Angles

The Euler angle representation describes the change from an initial reference frame I to a final reference frame F by means of three successive rotations. The first takes I to an intermediate frame H; the second, from H to a second intermediate frame G; and the third, from G to the final frame F (Markley; Crassidis, 2014).

Thus, a rotation expressed in Euler angles is composed of three elementary rotations, as given by Equation (18):

$$A_{ijk}(\phi, \theta, \psi) = A(\vec{e}_k, \psi)\, A(\vec{e}_j, \theta)\, A(\vec{e}_i, \phi) \tag{18}$$

where $\vec{e}_i$, $\vec{e}_j$, and $\vec{e}_k$ are orthonormal basis vectors defining the rotation axes:

$$\vec{e}_1 = \begin{bmatrix}1\\0\\0\end{bmatrix}, \quad \vec{e}_2 = \begin{bmatrix}0\\1\\0\end{bmatrix}, \quad \vec{e}_3 = \begin{bmatrix}0\\0\\1\end{bmatrix} \tag{19}$$

The rotation matrix $A(\vec{e}, \vartheta)$ about an axis $\vec{e}$, by an angle ϑ, is given by:

$$A(\vec{e}, \vartheta) = I_3 - \sin\vartheta\, [\vec{e}\times] + (1 - \cos\vartheta)\, [\vec{e}\times]^2 \tag{20}$$

where I_3 is the 3×3 identity matrix and $[\vec{e}\times]$ is the skew-symmetric matrix associated with the cross product by $\vec{e}$.

Considering the 3-2-1 rotation sequence, Equation (18) becomes:

$$A_{321}(\phi, \theta, \psi) = A(\vec{e}_1, \psi)\, A(\vec{e}_2, \theta)\, A(\vec{e}_3, \phi) \tag{21}$$

whose explicit matrix form is:

$$A_{321}(\phi,\theta,\psi) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\psi & \sin\psi \\ 0 & -\sin\psi & \cos\psi \end{bmatrix} \begin{bmatrix} \cos\theta & 0 & -\sin\theta \\ 0 & 1 & 0 \\ \sin\theta & 0 & \cos\theta \end{bmatrix} \begin{bmatrix} \cos\phi & \sin\phi & 0 \\ -\sin\phi & \cos\phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{22}$$

In the context of this project, the Euler angle representation is used to describe the satellite's orientation in a general sense. Since the developed system acts on a single axis, the system modeling is subsequently simplified to rotation about the z-axis, corresponding to the reaction wheel's actuation axis.

8.1.3 Attitude Dynamics

A satellite can be modeled as a rigid body whose motion involves, in addition to translation, a rotational component that is essential for controlling its orientation. Attitude dynamics is directly related to the torques applied to the satellite and to the mass distribution about its principal axes of inertia. Thus, the spacecraft's rotational behavior can be described from the rigid-body equations of motion (Markley; Crassidis, 2014).

In a real system, in addition to the torques associated with the actuators, the spacecraft's attitude can be affected by external disturbances and by the interaction between its different components. In the development presented in this document, however, a simplified model is considered for the study of detumbling, in which the rotational motion is analyzed only about the z-axis, corresponding to the reaction wheel's actuation axis.

The rotational dynamics of a rigid body can be expressed by Euler's equation:

$$I\dot{\vec{\omega}} + \vec{\omega} \times (I\vec{\omega}) = \vec{T}, \tag{23}$$

where I is the satellite's inertia matrix, $\vec{\omega}$ is the angular velocity vector, and $\vec{T}$ represents the vector of torques applied to the body.

For the model adopted in this project, only rotation about the z-axis is considered. Under this condition, the dynamics can be reduced to a single-degree-of-freedom model, given by:

$$I_{\text{sat}}\,\dot{\vec{\omega}} = \vec{T}, \tag{24}$$

where I_{sat} represents the satellite's moment of inertia about the z-axis, $\vec{\omega} = \omega_z\,\hat{z}$ is its angular velocity, and $\vec{T} = T_z\,\hat{z}$ is the torque applied about that axis.

In the system studied, the control torque is produced by the reaction wheel. By conservation of angular momentum, a change in the wheel's angular velocity results in a torque of opposite sense on the satellite, making it possible to change its angular velocity about the control axis.

8.1.4 Sun Vector

Most satellite missions depend on some subsystem whose orientation is defined relative to the Sun. This orientation is represented by the sun vector, a unit vector that points from the satellite to the Sun and is expressed in the axes of the body coordinate system.

The electromagnetic radiation emitted by the Sun can be detected by different sensors, depending on the mission's needs. Among the components and systems that interact with the Sun and, consequently, depend on the sun vector, the following stand out: the solar sail, the sun sensor, photovoltaic cells, and, indirectly, components sensitive to solar radiation (Silva, 2023).

The panels' own photovoltaic cells can be used as sun sensors, since the current they generate depends on the angle of incidence of the light. Considering that sunlight strikes a panel at an angle α with respect to its normal, indicated as γ in Figure 22, the generated current is given by (Silva, 2023):

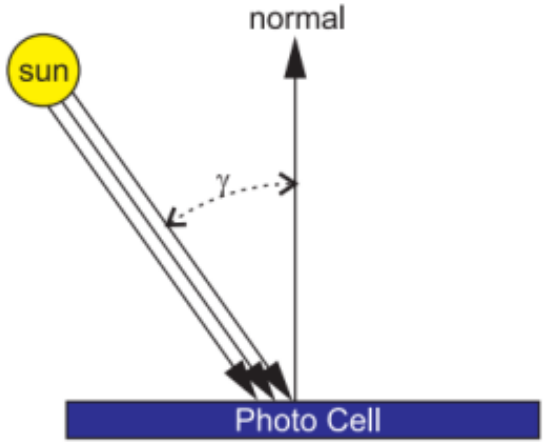


Figure 22: Incidence of sunlight on a photovoltaic cell
Source: (Silva, 2023).

$$i = i_{\max} \cos \alpha \tag{25}$$

where $i_{\max}$ is the current generated when the light strikes the panel perpendicularly ($\alpha = 0$). The model is valid for $0 \leq \alpha \leq \frac{\pi}{2}$; for larger angles, the face is not illuminated and the current is zero.

The computation of the sun vector depends on several factors, most notably the geometric arrangement of the panels, which directly affects the interpretation of the signals generated by the photovoltaic cells. For this reason, a trigonometric analysis is required. A two-dimensional arrangement of sun sensors is initially considered, as shown in Figure 23.

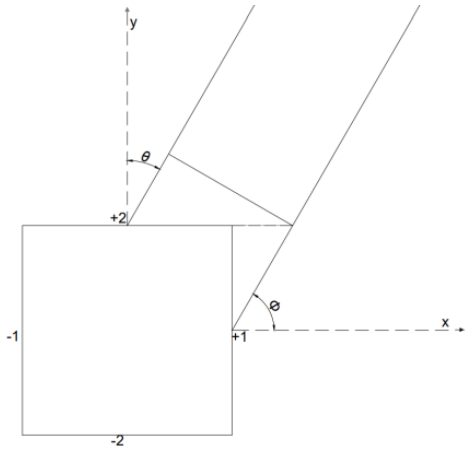


Figure 23: Sunlight incident on a CubeSat in the xy plane
Source: (Silva, 2023).

The angles θ and ϕ are complementary, which allows the following relations to be established:

$$\theta + \phi = \frac{\pi}{2} \tag{26}$$

$$\cos\theta = \cos\left(\frac{\pi}{2} - \phi\right) = \sin\phi \tag{27}$$

Since the behavior of each panel is described by Equation (25), the currents produced are given by:

$$i_{+1} = i_{1,\max}\cos\phi \tag{28}$$

$$i_{+2} = i_{2,\max}\cos\theta = i_{2,\max}\sin\phi \tag{29}$$

For the three-dimensional model, the incidence of sunlight shown in Figure 24 is considered.

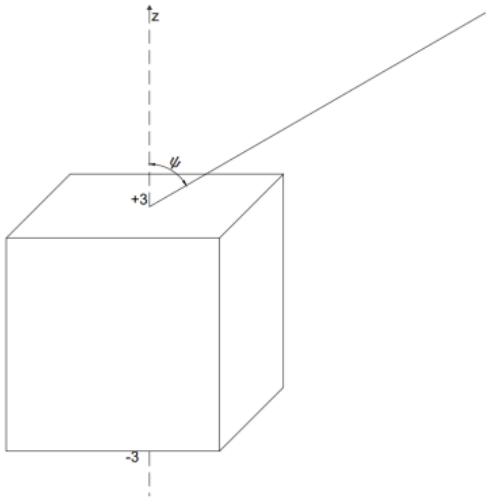


Figure 24: Sunlight incident on a CubeSat in the three-dimensional model
Source: (Silva, 2023).

From the previous relations, the three-dimensional model is obtained by incorporating the effect of the third angle, ψ, on the sunlight incident on faces ±1 and ±2 (Silva, 2023):

$$i_{+1} = i_{1,\max}\sin\psi\cos\phi \tag{30}$$

$$i_{+2} = i_{2,\max}\sin\psi\sin\phi \tag{31}$$

$$i_{+3} = i_{3,\max}\cos\psi \tag{32}$$

Similarly, the sun vector in the body frame can be expressed as a function of the angles:

$$\vec{s_b} = \begin{bmatrix}\sin\psi\cos\phi \\ \sin\psi\sin\phi \\ \cos\psi\end{bmatrix} \tag{33}$$

Assuming that the maximum current generated by the photovoltaic panels is the same for all faces, the sun vector can be rewritten as a function of the measured currents:

$$\vec{s_b} = \frac{1}{i_{\max}}\begin{bmatrix}X & 0 & 0 \\ 0 & Y & 0 \\ 0 & 0 & Z\end{bmatrix}\begin{bmatrix}i_1 \\ i_2 \\ i_3\end{bmatrix} \tag{34}$$

where i_1, i_2, and i_3 are the currents measured on the illuminated face of each pair, and X, Y, and Z take the value $+1$ or -1, indicating which face of each pair (± 1, ± 2, and ± 3, respectively) is illuminated and, therefore, producing current at that instant.

Due to imperfections in the manufacturing of the semiconductor elements, the maximum current may vary between panels. To reduce the error caused by this variation, the simplest strategy is to normalize the current of each face by its respective maximum value (Silva, 2023):

$$\vec{s}_b = \begin{bmatrix} X & 0 & 0 \\ 0 & Y & 0 \\ 0 & 0 & Z \end{bmatrix} \begin{bmatrix} i_1/i_{1,\max} \\ i_2/i_{2,\max} \\ i_3/i_{3,\max} \end{bmatrix} \tag{35}$$

8.1.5 Cone Intersection Method

Equation (25) shows that the current generated by a panel depends solely on the angle of incidence of the light on the plane where it is located. Once this angle is known, an infinite conical shell can be drawn containing all possible positions of the sun vector.

The cone intersection method, introduced by (Grubin, 1977) and extended by (Nygren, 2012), is based on this principle. From the angles measured on two panels, two cones are constructed whose intersection determines the vector, or set of vectors, compatible with the direction of the Sun.

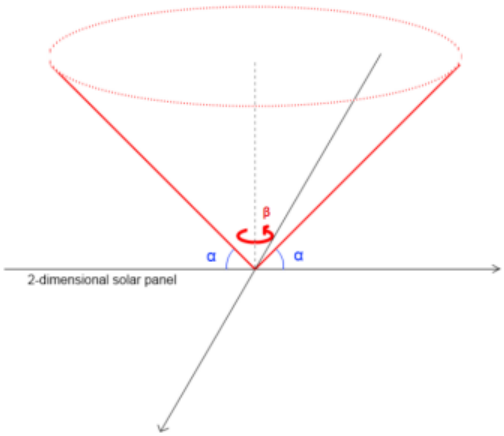


Figure 25: Projection of the cone onto the angle of incidence on the panel
Source: (Silva, 2023).

Figure 25 shows the relationship between the angles, taken from the plane in which the solar panel is positioned. The angle α corresponds to the angle formed between the sun vector, represented by the cone's generatrix, and the cone's axis, normal to the panel. The angle β describes the rotation of this generatrix about the cone's central axis.

Considering faces $+1$ and $+2$ of the CubeSat, the cone associated with face $+1$, with its axis along x, is described by:

$$y^2 + z^2 = r_x^2 = x^2 \tan^2 \alpha_x \tag{36}$$

$$y = x \sin \beta_x \tan \alpha_x \tag{37}$$

$$z = x \cos \beta_x \tan \alpha_x \tag{38}$$

Similarly, the cone associated with face $+2$, with its axis along y, is described by:

$$x^2 + z^2 = r_y^2 = y^2 \tan^2 \alpha_y \tag{39}$$

$$x = y \sin \beta_y \tan \alpha_y \tag{40}$$

$$z = y \cos \beta_y \tan \alpha_y \tag{41}$$

Equations (36) and (39) share the term z^2. Isolating it in both and equating the results gives:

$$y^2 \tan^2 \alpha_y - x^2 = x^2 \tan^2 \alpha_x - y^2 \tag{42}$$

$$y^2 \left(\tan^2 \alpha_y + 1\right) = x^2 \left(\tan^2 \alpha_x + 1\right) \tag{43}$$

$$\left(\frac{y}{x}\right)^2 = \frac{\tan^2 \alpha_x + 1}{\tan^2 \alpha_y + 1} \tag{44}$$

Substituting y in Equation (44) with Equation (37) yields:

$$\left(\frac{x \sin \beta_x \tan \alpha_x}{x}\right)^2 = \frac{\tan^2 \alpha_x + 1}{\tan^2 \alpha_y + 1} \tag{45}$$

$$\sin^2 \beta_x \tan^2 \alpha_x = \frac{\tan^2 \alpha_x + 1}{\tan^2 \alpha_y + 1} \tag{46}$$

$$\sin^2 \beta_x = \frac{\tan^2 \alpha_x + 1}{(\tan^2 \alpha_y + 1) \tan^2 \alpha_x} = \frac{\cos^2 \alpha_y}{\sin^2 \alpha_x} \tag{47}$$

$$\beta_x = \arcsin\left(\pm \frac{\cos \alpha_y}{\sin \alpha_x}\right) \tag{48}$$

The angle β_x represents the rotation about the cone's central axis to the sun vector's direction, with origin (0°) at the xz plane. Equation (48) has a real solution only when $\cos \alpha_y \leq \sin \alpha_x$, that is, when $\alpha_x + \alpha_y \geq \frac{\pi}{2}$. For $\alpha_x + \alpha_y < \frac{\pi}{2}$, the cones do not intersect. For $\alpha_x + \alpha_y = \frac{\pi}{2}$, they are tangent. For $\alpha_x + \alpha_y > \frac{\pi}{2}$, they intersect in two distinct solutions, symmetric with respect to the xy plane (Silva, 2023).

8.1.6 Pole Placement Controller

The pole placement method is a controller design technique formulated in state space, in which the system's state variables are used in the feedback. Unlike root-locus-based methods, pole placement allows the closed-loop pole locations to be set directly, provided the system satisfies the necessary controllability conditions (Ogata, 2010).

Consider a linear, time-invariant system described, in state space, by:

$$\dot{\vec{x}} = A\vec{x} + B\vec{u}, \qquad \vec{y} = C\vec{x} + D\vec{u}, \tag{49}$$

where $\vec{x}$ is the state vector of dimension n, $\vec{u}$ is the control signal, $\vec{y}$ is the output vector, A is the state matrix, B is the input matrix, C is the output matrix, and D is the feedthrough matrix.

In the regulator problem, the goal is to drive the system's state to the origin from a nonzero initial condition, caused, for example, by a disturbance, without the need for a reference

signal to be tracked (Ogata, 2010). To this end, a state-feedback control law is adopted, given by:

$$\vec{u} = -K\vec{x}, \tag{50}$$

where K is the state-feedback gain matrix. Substituting Equation (50) into Equation (49) yields the closed-loop dynamics:

$$\dot{\vec{x}} = (A - BK)\vec{x}. \tag{51}$$

The solution of Equation (51) can be written as:

$$\vec{x}(t) = e^{(A-BK)t}\vec{x}(0). \tag{52}$$

Equation (52) shows that the system's stability and transient response characteristics are determined by the eigenvalues of the matrix $A - BK$, referred to as the closed-loop poles. For a linear, time-invariant system, asymptotic stability is achieved when all closed-loop poles have negative real part (Ogata, 2010).

The necessary and sufficient condition for the closed-loop poles to be arbitrarily assignable is that the system be completely controllable. This property can be verified by means of the controllability matrix:

$$\mathcal{C}_o = \begin{bmatrix} B & AB & A^2B & \cdots & A^{n-1}B \end{bmatrix}. \tag{53}$$

The system is completely controllable if, and only if, the controllability matrix has rank equal to n, that is:

$$\text{rank}(\mathcal{C}_o) = n. \tag{54}$$

Once this condition is satisfied, the desired poles $\mu_1, \mu_2, \ldots, \mu_n$ can be chosen, and the gain matrix K is determined such that the characteristic equation of the closed-loop matrix matches the desired characteristic equation:

$$|sI_n - A + BK| = (s - \mu_1)(s - \mu_2)\cdots(s - \mu_n), \tag{55}$$

which can be expressed in the form:

$$(s - \mu_1)(s - \mu_2)\cdots(s - \mu_n) = s^n + \alpha_1 s^{n-1} + \cdots + \alpha_{n-1}s + \alpha_n. \tag{56}$$

Thus, the feedback gains are determined so that the eigenvalues of the matrix $A - BK$ coincide with the poles specified in the design. For low-order systems, the gains can be obtained by direct substitution, equating the coefficients of equal powers of s in the two characteristic polynomials.

8.1.7 Actuator

Actuators are devices responsible for converting a form of energy or a command signal into a physical action on a system, such as force, torque, or motion. In a control system, they are the elements that physically carry out the actions determined by the controller, making it possible to modify the system's behavior according to the established objectives (Markley; Crassidis, 2014).

In the context of space systems, actuators play a fundamental role in attitude control, as they generate the torques needed to change or stabilize the spacecraft's orientation (Fortescue; Swinerd; Stark, 2011). Thus, they can be used to perform maneuvers, reduce unwanted rotations, and keep the satellite oriented in a specific direction.

Reaction Wheel

Reaction wheels are electromechanical actuators used for the attitude control of spacecraft. In general, they consist of a rotating mass coupled to an electric motor, whose rotational speed can be controlled. Changing the wheel speed alters its angular momentum and, by conservation of the system's angular momentum, causes the satellite body to react in the direction opposite to the wheel's acceleration (Markley; Crassidis, 2014).

Thus, reaction wheels make it possible to generate torque on the satellite without expelling mass to the external environment, which allows maneuvers to be performed and attitude to be maintained with high precision. In addition to providing torque, these actuators are also capable of storing angular momentum (NASA, 1997).

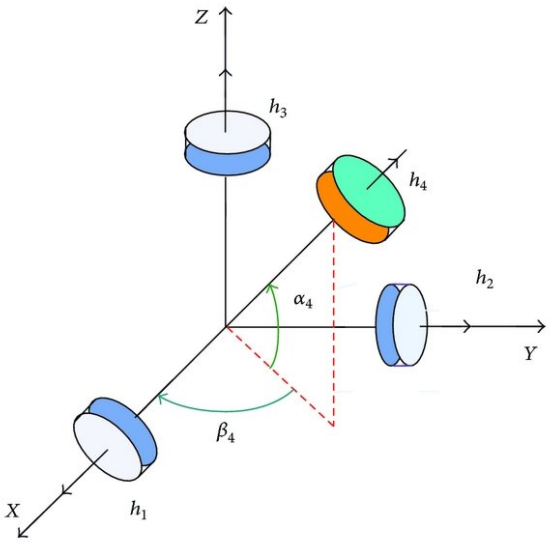


Figure 26: Reaction wheel configuration along the x, y and z axes
Source: (Zhang; Hu; Huo, 2014).

Their operation can be described mathematically from the principle of conservation of angular momentum. Considering the system formed by the satellite and the reaction wheel as an isolated system, that is, neglecting external torques, the total angular momentum $\vec{L}_{\text{total}}$ remains constant (Nussenzveig, 2013):

$$\vec{L}_{\text{total}} = \vec{L}_{\text{sat}} + \vec{L}_{\text{wheel}} = \text{constant} \tag{57}$$

In an analysis restricted to a single axis of rotation, the angular momentum of each component can be written as the product of its moment of inertia and its angular velocity. During detumbling, the satellite starts with an initial angular velocity $\vec{\omega}_0$, with the wheel at rest relative to it, so that the wheel initially rotates together with the satellite. The total angular momentum is therefore $\vec{L}_{\text{total}} = (I_{\text{sat}} + I_{\text{wheel}})\,\vec{\omega}_0$ and, by conservation of angular momentum:

$$(I_{\text{sat}} + I_{\text{wheel}})\,\vec{\omega}_0 = I_{\text{sat}}\,\vec{\omega}_{\text{sat}} + I_{\text{wheel}}\,\vec{\omega}_{\text{wheel}} \tag{58}$$

where $\vec{\omega}_{\text{sat}}$ is the angular velocity of the satellite and $\vec{\omega}_{\text{wheel}}$ is the absolute angular velocity of the wheel. At the end of detumbling, with $\vec{\omega}_{\text{sat}} = \vec{0}$, all of the initial angular momentum of the system is stored in the wheel:

$$(I_{\text{sat}} + I_{\text{wheel}})\,|\vec{\omega}_0| = I_{\text{wheel}}\,|\vec{\omega}_{\text{wheel}}| \tag{59}$$

The moment of inertia of a rectangular cuboid, adopted as the geometric representation of the satellite, about an axis passing through its center of mass is given by (Nussenzveig, 2013):

$$I_{\text{sat}} = \frac{M_{\text{sat}}\,(a^2 + b^2)}{12} \tag{60}$$

where a and b are the edges perpendicular to the axis of rotation and M_{sat} is the satellite mass, without the reaction wheel. In this CubeSat, the axis of rotation is the z-axis and the edges perpendicular to it are equal, that is, $a = b$. Therefore:

$$I_{\text{sat}} = \frac{M_{\text{sat}}\,(a^2 + a^2)}{12} = \frac{M_{\text{sat}}\,a^2}{6} \tag{61}$$

In turn, the reaction wheel is geometrically represented as a solid cylinder, whose moment of inertia about its axis of symmetry is given by (Nussenzveig, 2013):

$$I_{\text{wheel}} = \frac{M_{\text{wheel}}\,r^2}{2} \tag{62}$$

where M_{wheel} is the mass of the reaction wheel and r is its radius. The wheel adopted in this project also has a coaxial recess for coupling to the motor, whose effect on the moment of inertia is considered in Section 8.2.2.

8.1.8 Sensors

Sensors are devices capable of detecting physical quantities and converting them into signals that can be interpreted by an electronic system. In control systems, the information obtained by the sensors is used to determine the state of the system and to provide the feedback required for the controller to compute the appropriate control actions.

In the context of attitude determination and control, it is necessary to obtain information related both to the motion of the satellite body and to the motion of certain internal components, such as the reaction wheel. For this purpose, inertial sensors, which are responsible for measuring accelerations and angular velocities, can be employed, as well as sensors intended for determining angular positions.

Inertial Measurement Unit

An Inertial Measurement Unit (IMU) consists of sensors capable of providing information about the motion of a body. A widely used configuration combines a triaxial accelerometer and a triaxial gyroscope, making it possible to obtain linear accelerations and angular velocities along the three spatial axes.

Thus, a six-degree-of-freedom IMU provides the quantities a_x, a_y and a_z, the components of linear acceleration, and ω_x, ω_y and ω_z, the components of angular velocity.

Most modern inertial sensors are manufactured using Micro-Electro-Mechanical Systems (MEMS) technology, which enables the integration of mechanical structures and electronic circuits in small-sized devices (Yazdi; Ayazi; Najafi, 1998). The mechanical structures respond to the motion applied to the sensor, while the electronic circuits convert this response into electrical signals that can later be processed.

Accelerometer

The accelerometer is a sensor designed to measure linear acceleration. Its operating principle can be understood, initially, through Newton's Second Law, which states that the resultant force acting on a body equals the time rate of change of its linear momentum:

$$\vec{F}_{\text{res}} = \frac{d\vec{p}}{dt} \tag{63}$$

where $\vec{F}_{\text{res}}$ is the resultant force and $\vec{p} = m\,\vec{v}$ is the linear momentum of the body, given by the product of its mass m and its velocity $\vec{v}$. For a body of constant mass, which is the case of the proof mass of an accelerometer, the time derivative acts only on the velocity. Since the time derivative of the velocity is the acceleration $\vec{a}$, Equation (63) reduces to its most familiar form, $\vec{F}_{\text{res}} = m\,\vec{a}$, which relates the force applied to a body directly to the acceleration it experiences.

This relation is the basic idea behind the accelerometer: if the resultant force acting on a known mass, called the proof mass, can be measured, the acceleration of the sensor can be determined directly as:

$$\vec{a} = \frac{\vec{F}_{\text{res}}}{m} \tag{64}$$

In a MEMS accelerometer, a small mass, called the proof mass, is suspended by flexible structures. When the sensor package is subjected to an acceleration, the inertia of the mass causes a small relative displacement between it and the surrounding structure. In operation, the elastic force of the suspensions balances the inertial force, so that this displacement is proportional to the applied acceleration.

One of the techniques used to detect this displacement is capacitive sensing. The capacitance between two conductive surfaces can be represented, in a simplified manner, by:

$$C_e = \frac{\varepsilon A_e}{d} \tag{65}$$

where ε is the permittivity of the medium, A_e is the effective area between the electrodes and d is the distance between them. When the proof mass moves, the distance between the electrodes changes and, consequently, the measured capacitance changes. This variation is then converted into an electrical signal proportional to the acceleration experienced by the sensor.

Gyroscope

The gyroscope is used to determine the angular velocity of a body. In a three-dimensional system, the angular velocity can be written as:

$$\vec{\omega} = \omega_x\,\hat{x} + \omega_y\,\hat{y} + \omega_z\,\hat{z} \tag{66}$$

In vibratory MEMS gyroscopes, a microscopic mass is kept in oscillatory motion along a given direction, called the drive mode. When the sensor is subjected to a rotation, this mass experiences the Coriolis force, which transfers part of the motion to a direction orthogonal to the original vibration, called the sense mode (Passaro *et al.*, 2017). The Coriolis force is expressed by:

$$\vec{F}_C = -2m\,(\vec{\omega} \times \vec{v}) \tag{67}$$

where m is the mass of the vibrating element, $\vec{v}$ is its velocity and $\vec{\omega}$ is the angular velocity of the system.

In the absence of rotation, therefore, the mass remains predominantly in its original vibration mode. When a rotation occurs, the Coriolis force causes an additional motion in the perpendicular direction, whose amplitude is proportional to the angular velocity and can be measured, for example, by capacitive sensing.

The Coriolis force is a phenomenon associated with rotating systems and can be observed in different physical situations, such as the motion of masses relative to the Earth's surface. Figure 27 presents a representation of this phenomenon on Earth.

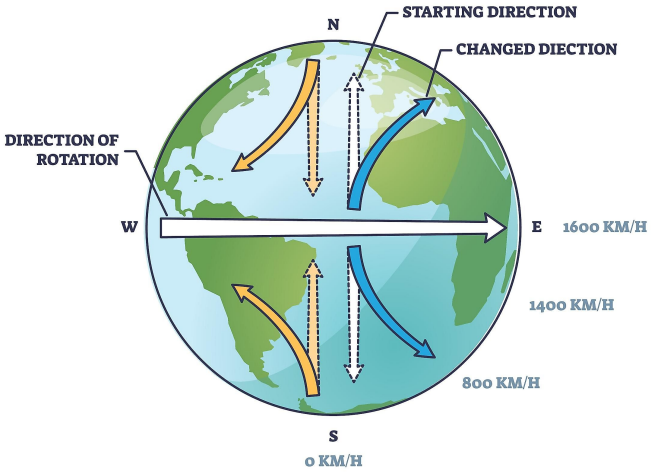


Figure 27: Coriolis effect on Earth
Source: (Shawki, 2023).

Encoder

Encoders are sensors used to determine the angular position of rotating elements. The angular position can be represented by the variable θ and, when observed over time, it also allows the direction of rotation and the angular velocity of the monitored element to be obtained.

In the encoder adopted in this project, the position is determined using the Hall effect. This phenomenon occurs when moving electric charges are subjected to a magnetic field with a component perpendicular to the direction of their motion. Under these conditions, the charges are subjected to the Lorentz magnetic force (Nussenzveig, 2013):

$$\vec{F}_B = q\left(\vec{v} \times \vec{B}\right) \tag{68}$$

where q is the electric charge, $\vec{v}$ is its velocity and $\vec{B}$ is the applied magnetic field. As a consequence of this force, a redistribution of charges occurs within the material, giving rise to a transverse potential difference called the Hall voltage, as illustrated in Figure 28. In this way, Hall elements make it possible to convert information about the magnetic field into measurable electrical signals (Popovic, 2004).

To determine the angular position, a magnet is attached to the rotating shaft and different Hall elements detect the variations of the magnetic field as the shaft rotates. The combination of the obtained signals makes it possible to relate the orientation of the detected field to the angular position of the shaft. In the encoder used, an array of Hall elements provides signals corresponding to the detected magnetic flux, which are digitized and processed to compute the angular position (Ams AG, 2018).

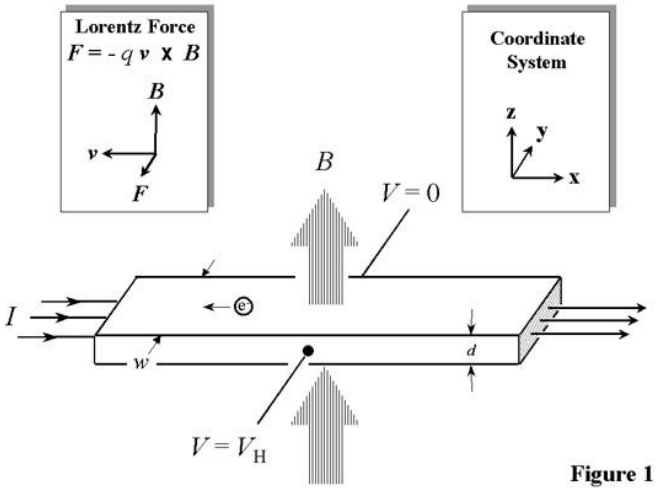


Figure 28: Representation of the Hall effect
Source: (NIST, 2022).

8.2 Modeling

This section aims to represent the main elements involved in the operation of the AOCS and to establish the relationships among them. Based on the electronic architecture, the characteristics of the reaction wheel, and the mathematical representation of the system dynamics, it is possible to analyze the system's behavior and evaluate the control performance.

The developed model is divided into three main parts: the electronic modeling, which presents the sensors, the processing, and the actuator command; the structural modeling, related to the geometry, mass, and moment of inertia of the reaction wheel; and the control system modeling, which brings together the satellite dynamics, the behavior of the wheel, and the feedback of the measured variables.

8.2.1 Electronics

The electronics of the attitude control system consist of the components listed in Table 27. The selection criteria for each component and the integrated operation of the system are discussed below.

Table 27: Electronic components of the attitude control system

Component	Function
ESP32-WROOM-32	Processing and control
BMI088	Inertial measurement unit
AS5048A	Magnetic encoder for the motor
iPower GM2804	Reaction wheel motor
DRV8313	Motor driver
XTJ Prime solar panels	Sun vector estimation

Source: Authors (2026).

ESP32-WROOM-32

To give the subsystem a control architecture independent of the other satellite subsystems, a dedicated microcontroller was chosen: the ESP32-WROOM-32 (Espressif Systems,

2025), shown in Figure 29. The module features a dual-core processor running at up to 240 MHz, which is sufficient to run the attitude control loop and the motor control simultaneously. It also provides Serial Peripheral Interface (SPI) and Inter-Integrated Circuit (I^2C) interfaces for communication with the sensors, Pulse-Width Modulation (PWM) peripherals for driving the motor, and Wi-Fi and Bluetooth connectivity, which is useful for telemetry during testing. In addition, the module is widely available on the market and has a low cost.

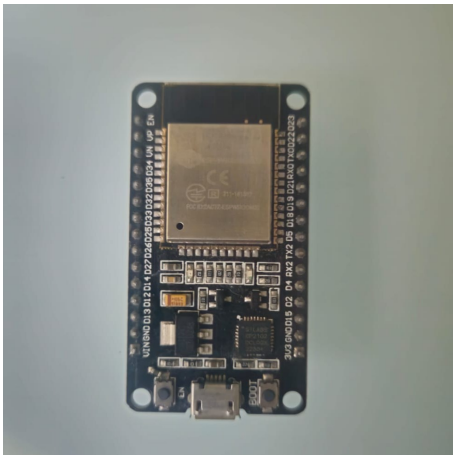


Figure 29: ESP32-WROOM-32 module
Source: Authors (2026).

BMI088

The closed-loop control relies on two sensors. The first is the BMI088 IMU (Bosch Sensortec, 2018), shown in Figure 30, which combines a triaxial accelerometer and a triaxial gyroscope to provide measurements of linear acceleration and angular velocity along all three axes. This sensor was selected for its robustness to vibration, an important feature in a system with a spinning reaction wheel, and for its angular velocity measurement range, which is suitable for monitoring the rotation of the satellite. A magnetometer was not used, since its measurements could be corrupted by magnetic interference from the permanent magnets and coil currents of the motor, as well as from the encoder magnet, all of which are mounted close to the sensor.

Figure 30: BMI088 inertial measurement unit
Source: Authors (2026).

AS5048A

The second sensor is the AS5048A (Ams AG, 2018), shown in Figure 31. It is a Hall-effect magnetic encoder with 14-bit resolution and an SPI interface, which provides feedback on the angular position of the rotor. This information is used by the Field-Oriented Control (FOC) method to determine the rotor position and drive the motor accordingly.

Figure 31: AS5048A magnetic encoder
Source: Authors (2026).

GM2804

The reaction wheel actuator is the iPower GM2804 Brushless Direct Current (BLDC) motor (IPower Motor, s.d.), shown in Figure 32, which is designed for three-axis camera gimbals. Motors of this type are optimized for precise operation at low speeds and have torque characteristics suitable for this application, as shown later in the control system modeling. The motor is compatible with the SimpleFOC library (Skuric *et al.*, 2022), which is used in the ESP32 firmware to implement FOC.

Figure 32: iPower GM2804 brushless motor
Source: Authors (2026).

DRV8313

The DRV8313 driver (Texas Instruments, 2016), shown in Figure 33, was selected to drive the motor. It acts as the power stage between the microcontroller and the motor windings. The

circuit integrates three half-bridges that, based on three PWM signals generated by the ESP32, switch the supply voltage to the three motor phases at high frequency, enabling controlled operation of the motor.

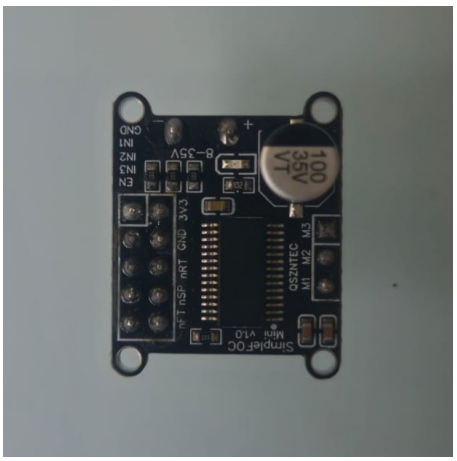


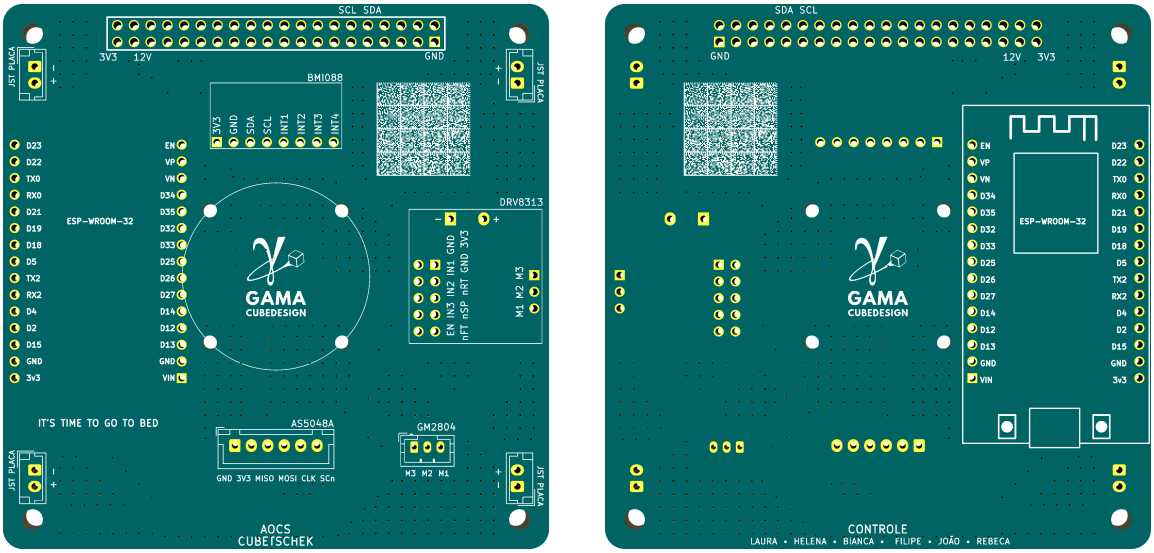
Figure 33: DRV8313 motor driver
Source: Authors (2026).

Sun Sensor

The sun vector is estimated using the satellite's own solar panels, whose selection is justified in the section on the EPS. This approach eliminates the need for dedicated sensors. As described in Section 8.1.4, the current generated by each panel depends on the angle of incidence of the sunlight, which allows the panels to be used as sun sensors. The panels use XTJ Prime triple-junction cells (Spectrolab, 2022), made of layers of different semiconductors, each optimized for a specific range of the solar spectrum.

Printed Circuit Board

During operation, the system works as follows: the BMI088 measures the angular velocity of the satellite, and the solar panels provide the information used to estimate the sun vector. To meet requirement HLR-ADCS-01, which requires the system to estimate its attitude about at least one axis using onboard sensors, the attitude about the z-axis is obtained by integrating the angular velocity measured by the gyroscope and is complemented by the sun vector estimated from the solar panels. Based on the angular velocity measured by the BMI088, the ESP32 computes the control action of the detumbling mode; the sun vector estimate is not used by this control law. This control command is then passed to the motor controller, which uses the rotor position provided by the AS5048A to generate the PWM signals sent to the DRV8313. The driver, in turn, energizes the motor phases, changing the speed of the reaction wheel. By conservation of angular momentum, the change in the angular momentum of the wheel produces a reaction torque on the satellite, which makes it possible to change its angular velocity about the control axis. The printed circuit board is shown in Figure 34, and its electrical schematic is presented in Appendix C.1.



(a) Front (b) Back

Figure 34: AOCS PCB
Source: Authors (2026).

8.2.2 Structures

The reaction wheel consists of a steel disk with radius $r = 30$ mm and a coaxial cylindrical recess with diameter 36 mm and depth 5 mm, used to couple the wheel to the iPower GM2804 motor, as detailed in the technical drawing presented in Appendix C.3. Its dimensions were defined based on the moment of inertia required to absorb the initial angular momentum of the satellite during the detumbling process.

Since the wheel initially rotates together with the satellite, its own angular momentum must also be absorbed, so the satellite mass without the wheel (1.9235 kg) was used as input and the wheel inertia was obtained from the conservation of angular momentum of the whole system. The sizing was performed using the code presented in Appendix C.4, considering an operating speed of 1000 rpm for the wheel at the end of the detumbling.

Assuming the wheel initially at rest relative to the satellite and the satellite at rest at the end of the detumbling, the conservation of angular momentum gives:

$$(I_{\text{sat}} + I_{\text{wheel}})\,\vec{\omega}_0 = I_{\text{wheel}}\,\vec{\Omega}_{\text{wheel}} \tag{69}$$

where I_{sat} is the moment of inertia of the satellite without the wheel, $\vec{\omega}_0$ is the initial angular velocity of the satellite and $\vec{\Omega}_{\text{wheel}}$ is the wheel operating speed relative to the satellite. This corresponds to Equation (58) with $\vec{\omega}_{\text{sat}} = \vec{0}$, in which case the absolute angular velocity of the wheel equals its velocity relative to the satellite, $\vec{\omega}_{\text{wheel}} = \vec{\Omega}_{\text{wheel}}$. Since both vectors are aligned with the z-axis, projecting Equation (69) onto this axis and solving for the wheel inertia gives:

$$I_{\text{wheel}} = \frac{I_{\text{sat}}\,\omega_0}{\Omega_{\text{wheel}} - \omega_0} \tag{70}$$

where $\omega_0 = |\vec{\omega}_0|$ and $\Omega_{\text{wheel}} = |\vec{\Omega}_{\text{wheel}}|$.

The wheel is modeled as a full cylinder of radius r and thickness h minus the coaxial recess of radius r_c and depth h_c, so its moment of inertia is:

$$I_{\text{wheel}} = \frac{1}{2}\,\rho\,\pi\left(r^4\,h - r_c^4\,h_c\right) \tag{71}$$

where ρ is the density of the material. Solving for the thickness:

$$h = \frac{1}{r^4}\left(\frac{2\,I_{\text{wheel}}}{\rho\,\pi} + r_c^4\,h_c\right) \tag{72}$$

and the mass of the wheel is:

$$M_{\text{wheel}} = \rho\,\pi\left(r^2\,h - r_c^2\,h_c\right) \tag{73}$$

Based on the parameters defined for the satellite and the wheel, the code determined a required moment of inertia of 2.046×10^{-4} kg m^2 and a wheel mass of 0.4292 kg. The main parameters obtained from the sizing are presented in Table 28.

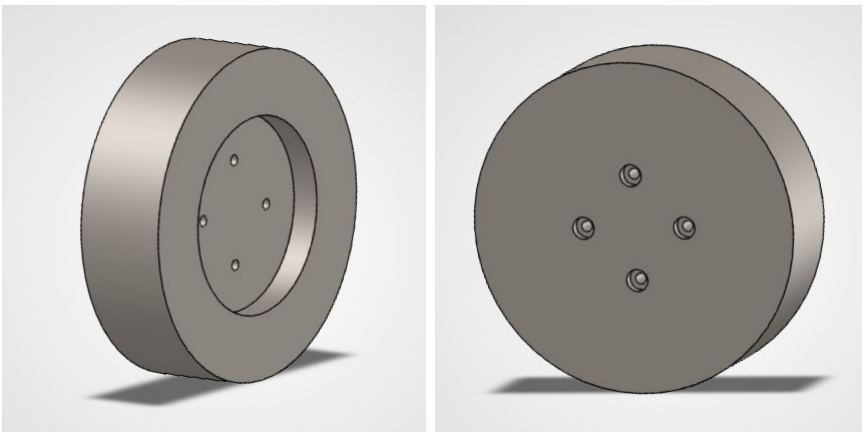
Table 28: Physical parameters determined for the reaction wheel

Parameter	**Value**
Radius	30 mm
Recess diameter	36 mm
Recess depth	5 mm
Operating speed	1000 rpm
Angular velocity	104.72 rad/s
Required moment of inertia	2.046×10^{-4} kg m^2
Mass	0.4292 kg
Material	Steel
Density	7850 kg/m^3
Thickness	21.136 mm

Source: Authors (2026).

Steel ($\rho = 7850$ kg/m^3) was chosen for its high density, which allows the required moment of inertia to be achieved with a smaller thickness. For comparison, keeping the same radius, recess and required moment of inertia, using aluminum with a density of 2700 kg/m^3 would result in a thickness of 60.214 mm. For steel, the calculated thickness is 21.136 mm, resulting in a more compact wheel.

The CAD model of the wheel is shown in Figure 35, and a printed model of the wheel coupled to the motor, used to verify the fit between the components, is shown in Figure 36.



(a) Rear view (b) Front view

Figure 35: Reaction wheel 3D model

Source: Authors (2026).

The motor rotor fits into the recess of the wheel, and the wheel is bolted directly to it, ensuring torque transmission between the components without relative slip. The motor and wheel assembly is mounted directly on the satellite's printed circuit board, with the rotation axis aligned with the z-axis of the body reference frame. This alignment is necessary so that the torque produced by the wheel is applied predominantly about the control axis. Any angular misalignment between the wheel's rotation axis and the z-axis introduces torque components about the other axes of the satellite, which are not controlled by the developed system.

Figure 36: Printed model of the reaction wheel coupled to the motor
Source: Authors (2026).

8.2.3 Control System

The control system was developed to meet the detumbling High-Level Requirement (HLR) of the competition (HLR-ADCS-02): reducing the angular velocity of the satellite from 60 rpm to less than 5 rpm, evaluated in the advanced-level detumbling test described in Section 3.1, with target times of less than 10, 25 and 60 s, the first of which corresponds to the maximum score. A single reaction wheel is used, acting about the z-axis of the 2U CubeSat, with $M_{\text{sat}} = 1.9235$ kg, edges perpendicular to the rotation axis of 10 cm and, from Equation (61), $I_{\text{sat}} = 3.206 \times 10^{-3}\ \text{kg}\,\text{m}^2$. The mass and inertia of the satellite do not include the reaction wheel, whose axial inertia I_{wheel} is considered separately in the wheel dynamics (Equation (82)). The adopted controller is a state-feedback controller with pole placement.

Satellite Dynamics

For the controller design, the dynamics of a rigid body about a single axis are considered, with the torque $\vec{T}$ applied by the wheel as the input and the angular velocity $\vec{\omega}$ as the controlled variable:

$$I_{\text{sat}}\,\dot{\vec{\omega}} = \vec{T} \tag{74}$$

Since the rotation occurs only about the z-axis, $\vec{\omega} = \omega\,\hat{z}$ and $\vec{T} = T\,\hat{z}$, where $\omega = \omega_z$ and $T = T_z$ are the components along this axis, and the same applies to the other vector quantities in this section ($\vec{\theta}$, $\vec{u}_c$, $\vec{\omega}_r$ and $\vec{T}_m$). In the relations that involve scalar quantities, such as voltages, the components along z are used. Defining the state $x = \omega$ and the input $u_c = T$, the state-space model is:

$$\dot{x} = A\,x + B\,u_c, \qquad A = 0, \qquad B = \frac{1}{I_{\text{sat}}} \tag{75}$$

The controllability matrix is $\mathcal{C}_o = B = 1/I_{\text{sat}} \neq 0$; therefore, the system is controllable and the pole can be placed arbitrarily. The actuator dynamics, including the wheel speed, are incorporated in the actuator model presented below, where it is shown that this design model is recovered by compensating for the Back Electromotive Force (back-EMF).

Control Law

The general control law feeds back the angle $\vec{\theta}$ and the angular velocity $\vec{\omega}$ of the satellite, with gains K_1 and K_2, respectively:

$$\vec{u}_c = -\left(K_1\,\vec{\theta} + K_2\,\vec{\omega}\right) \tag{76}$$

where $\vec{u}_c$ is the desired torque on the satellite. Since the detumbling requirement does not specify a final angle, only a reduction in angular velocity, $K_1 = 0$ is adopted and only $\vec{\omega}$ is fed back, as measured by the gyroscope of the BMI088 IMU:

$$\vec{u}_c = -K_2\,\vec{\omega} \tag{77}$$

In closed loop, $I_{\text{sat}}\,\dot{\vec{\omega}} = -K_2\,\vec{\omega}$, with a single pole at $s = -K_2/I_{\text{sat}}$. Choosing the desired pole at $s = -p$, the gain is given by:

$$K_2 = p\,I_{\text{sat}} \tag{78}$$

and the system response is:

$$\vec{\omega}(t) = \vec{\omega}_0\,e^{-p\,t} \qquad \Rightarrow \qquad t = \frac{1}{p}\ln\left(\frac{\omega_0}{\omega_f}\right) \tag{79}$$

where ω_0 and ω_f are the magnitudes of the initial and final angular velocities. A pole of $p = 0.5\ \text{s}^{-1}$ was adopted, which results in $K_2 = 1.603 \times 10^{-3}\ \text{N m s/rad}$. For an ideal actuator, Equation (79) gives $t \approx 4.97$ s for the reduction from 60 to 5 rpm.

Actuator Model

The motor was modeled as an equivalent Direct Current (DC) motor, with armature resistance $R = 9\ \Omega$ and nominal supply voltage $V_{\text{nom}} = 12$ V (IPower Motor, s.d.). Assuming $K_t = K_e$ in SI units, the torque constant and the stall torque per volt are obtained from the no-load speed, taken as the mean of the range given in the datasheet ($\omega_{\text{nl}} = 1686.5$ rpm, between 1602 and 1771 rpm), and from the no-load current ($I_0 = 0.09$ A):

$$K_t = \frac{V_{\text{nom}} - R\,I_0}{\omega_{\text{nl}}} \approx 0.0634\ \text{N m/A}, \qquad f = \frac{K_t}{R} \approx 7.04 \times 10^{-3}\ \text{N m/V} \tag{80}$$

where f is the stall torque per volt. Neglecting the inductance, the torque applied by the motor to the wheel depends on the applied voltage V and on the back-EMF $e_b = K_t\,\omega_r$, which is proportional to the component ω_r of the wheel angular velocity relative to the satellite, $\vec{\omega}_r$:

$$T_m = \frac{K_t}{R}\left(V - K_t\,\omega_r\right) = f\left(V - e_b\right) \tag{81}$$

By conservation of angular momentum, the motor torque $\vec{T}_m = T_m\,\hat{z}$ accelerates the wheel and produces on the satellite a torque of equal magnitude and opposite direction:

$$I_{\text{sat}}\,\dot{\vec{\omega}} = -\vec{T}_m, \qquad I_{\text{wheel}}\left(\dot{\vec{\omega}}_r + \dot{\vec{\omega}}\right) = \vec{T}_m \tag{82}$$

Since the back-EMF increases with wheel speed, a constant voltage does not produce a constant torque: the wheel accelerates until e_b approaches V, and the torque tends to zero. The associated mechanical time constant is $\tau = I_{\text{wheel}} R / K_t^2 \approx 0.459$ s. For the delivered torque to match the controller command, the voltage command V_c includes back-EMF compensation, based on the wheel speed measured by the AS5048A encoder, and the voltage actually applied, V, is limited by the driver:

$$V_c = -\frac{u_c}{f} + K_t\,\omega_r, \qquad V = \text{sat}\left(V_c,\ \pm V_{\max}\right), \qquad V_{\max} = 12 \text{ V} \tag{83}$$

Substituting Equation (83) into Equation (81) yields $T_m = -u_c$ and, therefore, $I_{\text{sat}}\,\dot{\vec{\omega}} = \vec{u}_c$, which recovers the design dynamics as long as there is no saturation. In addition to voltage saturation, the motor torque is limited to $\pm T_{\max}$, where $T_{\max}$ is the mean of the 250 to 350 gf cm range given in the datasheet, that is, $T_{\max} = 300$ gf cm $= 29.43$ mN m.

With the wheel included, the linearized model (without saturation) has the states $(\omega,\ \omega_r)$. With compensation, its poles are $s = -p = -0.5$ s^{-1} and $s = 0$; the latter corresponds to the transfer of angular momentum to the wheel, whose speed accumulates, and does not prevent $\vec{\omega}$ from converging to zero. Without compensation, that is, with $V_c = -u_c/f$, the poles become $s = 0$ and $s \approx -2.82$ s^{-1}, and the motor torque vanishes whenever $K_2\,\omega = f\,K_t\,\omega_r$. In this case, the back-EMF of the wheel cancels the command, and the satellite stops decelerating at $|\vec{\omega}| \approx 49$ rpm, approximately 1 s after the start of the maneuver, failing to meet the requirement. This result justifies the use of Equation (83), implemented in practice by a driver that controls the motor torque or current.

Simulink Model

The simulated loop, shown in Figure 87, is organized into six subsystems: *Attitude Feedback*, with gains K_1 and K_2; *Attitude Control*, which applies the sign inversion of Equation (76) and converts the desired torque u_c into voltage through the factor $-1/f$; *Voltage Conversion*, which adds the back-EMF compensation and applies the ± 12 V saturation (Equation (83)); *Motor Model*, which computes the torque from Equation (81), with saturation at $\pm T_{\max}$; *Satellite Dynamics*; and *Reaction Wheel Dynamics & Back-EMF*. The last two integrate Equations (82), providing the components ω and ω_r of the satellite and wheel angular velocities, from which the back-EMF e_b is obtained. Since the model is single-axis, all signals in the diagram are components along z. The correspondence between the notation in the text and in the diagram is given in Table 29, and the block parameters are given in Table 30.

Table 29: Correspondence between the notation in the text and in the Simulink diagram

Text	Diagram	Description
I_{sat}	`J`	Satellite inertia (without the wheel)
I_{wheel}	`J_w`	Axial inertia of the wheel
ω, θ	ω, θ	Satellite angular velocity and angle (z components)
ω_r	ω`_w`	Wheel speed relative to the satellite (z component)
u_c	`u_c`	Desired torque on the satellite (z component)
V_c, V	`V_c`, `V`	Commanded voltage and applied voltage
$e_b = K_t\,\omega_r$	`e_b`	Back electromotive force
$V - e_b$	`V_net`	Net armature voltage
T_m	`T_m`	Motor torque on the wheel (z component)

Source: Authors (2026).

Table 30: Parameters used in the Simulink model

Parameter	Value
$1/I_{\text{sat}}$ ($\text{kg}^{-1}\,\text{m}^{-2}$)	311.93
$1/I_{\text{wheel}}$ ($\text{kg}^{-1}\,\text{m}^{-2}$)	4886.81
K_1 (N m/rad)	0
K_2 (N m s/rad)	1.603×10^{-3}
K_t (N m/A)	0.06336
f (N m/V)	7.04×10^{-3}
$V_{\max}$ (V)	12
$T_{\max}$ (mN m)	29.43
ω_0 (rad/s)	6.283
$\omega_{r,0}$ (rad/s)	0

Source: Authors (2026).

The initial condition is $\omega_0 = 6.283$ rad/s (60 rpm), with the wheel at rest relative to the satellite. The angle θ is obtained from an additional integrator; since $K_1 = 0$, this signal is used for visualization only. The simulation was run for 30 s with an output step of 0.005 s, and the results were verified by the MATLAB script presented in Appendix C.5, which integrates the same equations with the `ode45` solver (relative tolerance of 10^{-8}).

Results

The wheel was sized to absorb the total initial angular momentum of the system, $|\vec{L}_{\text{total}}| = (I_{\text{sat}} + I_{\text{wheel}})\,\omega_0 = 2.143 \times 10^{-2}$ N m s, which includes the wheel itself since it initially rotates together with the satellite, at the operating speed of 1000 rpm, considering a steel disk with radius $r = 30$ mm and a coaxial recess for the motor coupling (Section 8.2.2). Table 31 and Figures 37 and 38 present the results obtained for this configuration.

Table 31: Detumbling simulation results

Parameter	Value
Wheel operating speed (rpm)	1000
I_{wheel} ($\text{kg}\,\text{m}^2$)	2.046×10^{-4}
Wheel mass (g)	429.2
Wheel thickness (mm)	21.1
Maximum applied voltage (V)	6.63
Maximum current (A)	0.16
Maximum delivered torque (mN m)	10.07
Maximum wheel speed (rpm)	1000
Time to $\lvert\vec{\omega}\rvert < 5$ rpm (s)	4.97
Maximum wheel angular momentum (N m s)	2.14×10^{-2}

Source: Authors (2026).

The satellite reaches $|\vec{\omega}| < 5$ rpm in 4.97 s, within the maximum-score range, defined for times below 10 s. Since the back-EMF compensation makes the motor deliver the commanded torque and no saturation is reached, the satellite angular velocity follows the exponential response of Equation (79), with a pole at $s = -0.5\ \text{s}^{-1}$. The response shows no overshoot: the satellite angular velocity converges to zero without reversing its direction of rotation.

The maximum delivered torque of 10.07 mN m occurs at the initial instant, when the satellite speed is highest, and then decays exponentially. This value remains below the motor

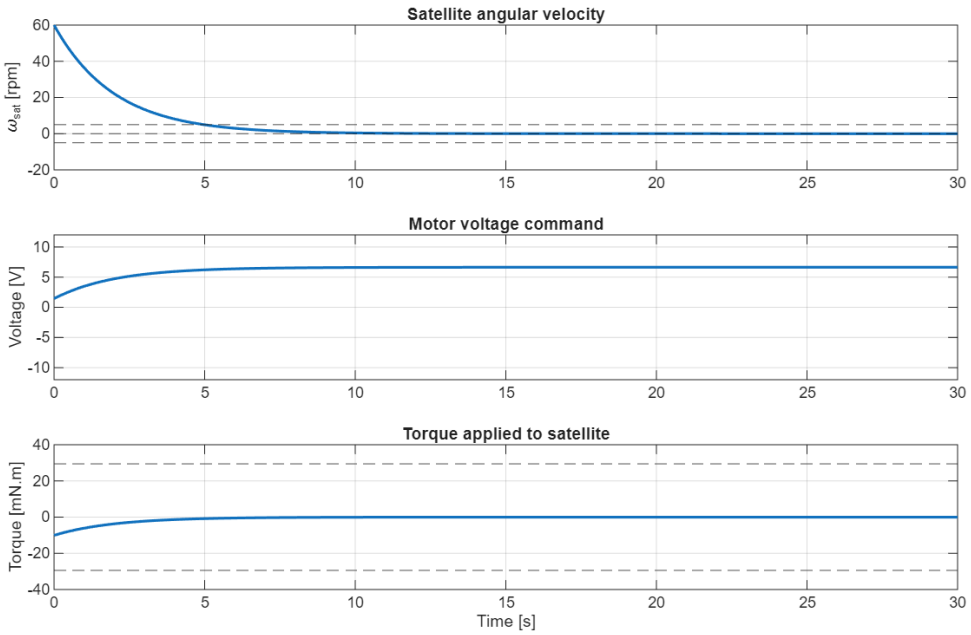


Figure 37: Satellite angular velocity, commanded motor voltage, and torque applied to the satellite during detumbling
Source: Authors (2026).

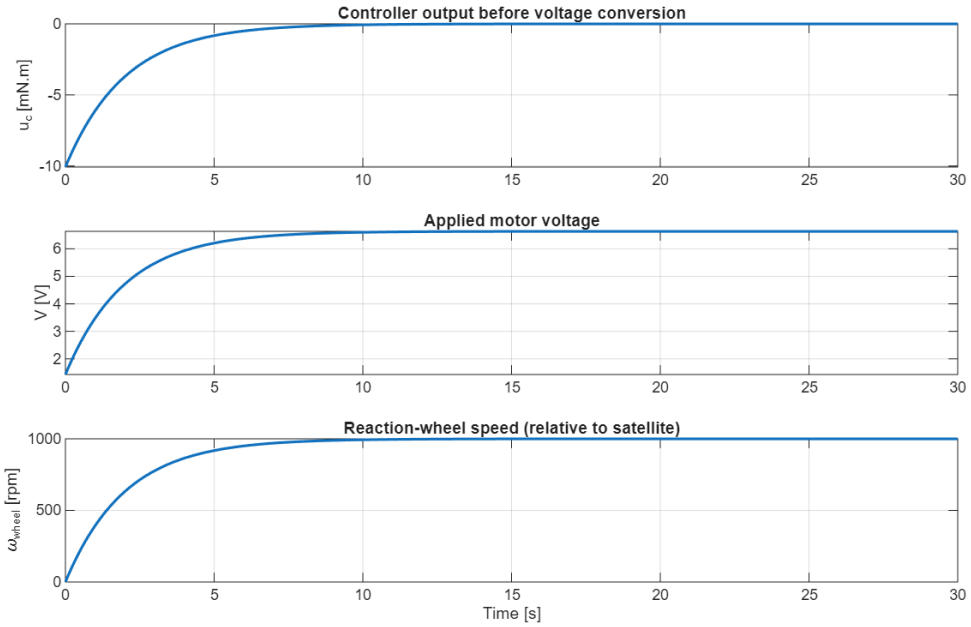


Figure 38: Control signal, voltage applied to the motor, and reaction wheel speed relative to the satellite during detumbling
Source: Authors (2026).

limit of 29.43 mN m by a margin of almost three times and corresponds to a maximum current of 0.16 A. This value represents only the torque-producing component of the current (T_m/K_t); since the model neglects motor friction and losses, represented by the no-load current $I_0 = 0.09$ A, the actual current should be on the order of 0.25 A. Unlike the torque, the applied voltage starts at approximately 1.43 V and increases throughout the maneuver, following the back-EMF of the wheel, until it reaches 6.63 V at the end, below the 12 V saturation. The system therefore operates in the linear regime throughout the maneuver. In the plots, the control signal and the torque are negative because they represent the torque applied to the satellite, whereas the voltage is positive because it is applied to the motor, which accelerates

the wheel in the opposite direction.

The wheel speed shown in Figure 38 is measured relative to the satellite. At the end of the maneuver, the wheel reaches its operating speed of 1000 rpm, as defined in the sizing, which already accounts for the angular momentum of the wheel itself at the initial instant, $I_{\text{wheel}}\,\omega_0 \approx 1.29 \times 10^{-3}$ N m s, when it rotates together with the satellite. The final angular momentum of the wheel, 2.14×10^{-2} N m s, therefore corresponds to the total angular momentum of the system. Thus, the sized wheel is capable of absorbing all the angular momentum associated with the initial condition of 60 rpm at its nominal speed, well below the no-load speed of the motor.

8.3 Testing and Validation

Two tests were carried out to validate the Attitude and Orbit Control System: the determination of the sun vector from the solar panel voltages, and the characterization of the motor that drives the reaction wheel used in the detumbling maneuver.

8.3.1 Sun Vector Determination Test

The first test aimed to validate the estimation of the sun vector direction from the voltages generated by the CubeSat's solar panels, in preparation for the advanced-level attitude determination (azimuth) test described in Section 3.1, in which the azimuth measured from a light source is evaluated in the accuracy ranges of $\leq 5°$, $\leq 15°$ and $\leq 30°$. The estimation relies on the coarse sun sensor principle: the current of a photovoltaic cell, and consequently its voltage, varies approximately with the cosine of the angle of incidence of light relative to the normal of the cell surface.

A 12U structural mock-up was used, serving only as a mechanical support. A solar cell was attached to the center of each of the four lateral faces ($X+$, $X-$, $Y+$ and $Y-$, corresponding to faces $+1$, -1, $+2$ and -2 of Section 8.1.4), with the cell normal aligned with the corresponding body axis. The cells were connected through twisted pairs to the analog inputs of an ESP32 microcontroller mounted on top of the structure. Data were transmitted over serial communication at 115 200 baud and monitored in real time through the serial monitor of the Arduino Integrated Development Environment (IDE). The experimental setup is shown in Figure 39.

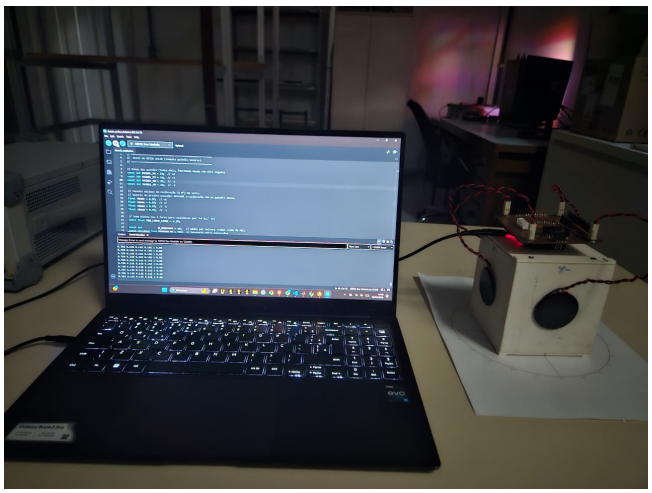


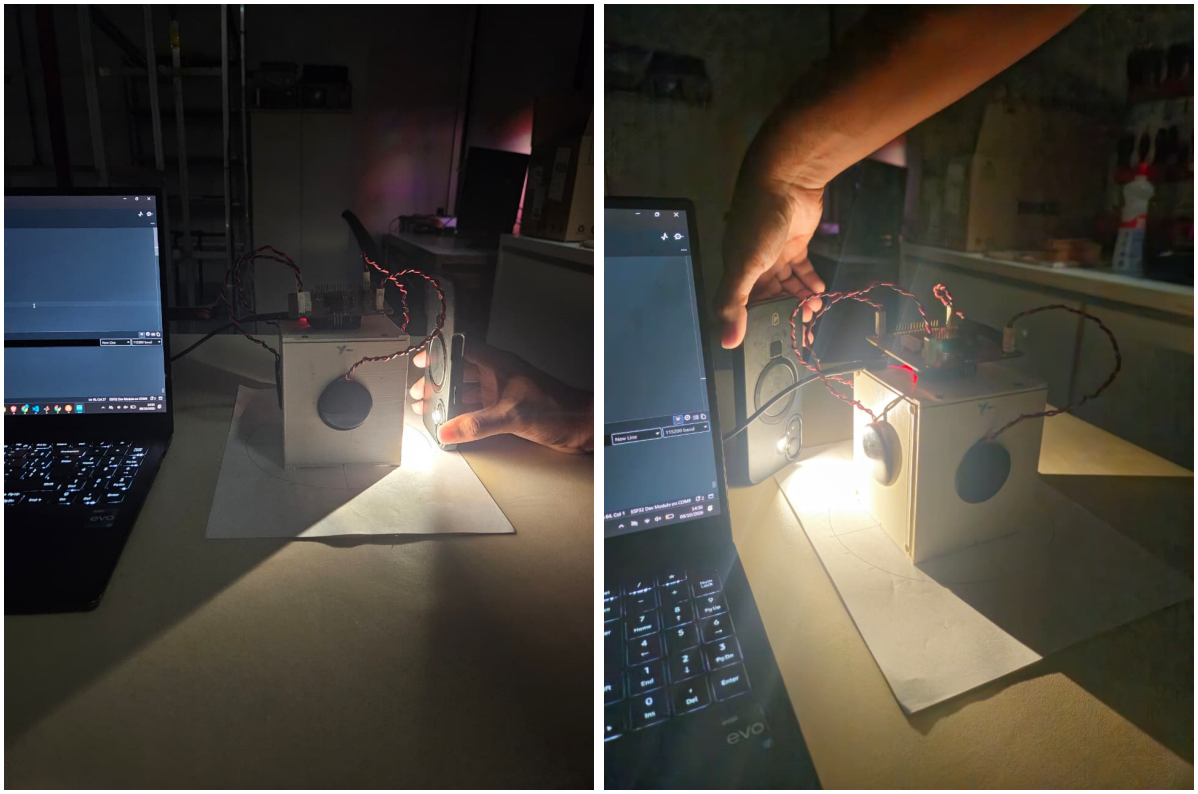
Figure 39: Experimental setup for the sun vector determination test: 12U mock-up with solar cells on the lateral faces, placed on the polar chart and connected to the ESP32, with the serial monitor displaying the cell voltages and the estimated angle

Source: Authors (2026).

Since the final printed circuit boards of the subsystem had not yet been delivered when the tests were carried out, a provisional double-sided PCB was manufactured using the toner thermal transfer method. The top and bottom layer layouts were printed on paper with a laser printer and heat-transferred onto both sides of a copper-clad phenolic board, aligned with the aid of reference holes. The exposed copper was then removed by chemical etching in a ferric chloride solution. The electrical connection between layers was made manually, with wires soldered through the via holes. The provisional board reproduces the connections essential to the test, namely the connection of the solar cells to the analog inputs and the interface with the ESP32 module. This made it possible to validate the acquisition hardware and the sun vector determination algorithm without depending on the final boards.

This solution has limitations compared to an industrially manufactured PCB. The traces have lower dimensional accuracy, there is no solder mask or plated vias, and some misalignment between the two layers may occur. Nevertheless, the board meets the requirements of this test, whose focus is to verify the operation of the system.

The mock-up was placed on a printed polar chart, with the geometric center of the cube coinciding with the origin of the chart. The adopted convention was $X+ = 0°$, $Y+ = 90°$, $X- = 180°$ and $Y- = 270°$. The light source used was a smartphone flashlight, moved manually around the mock-up to known angular positions. The test was performed in a dark environment to reduce the influence of external lighting, as shown in Figure 40.



(a) Overview of the setup during illumination (b) Detail of the illuminated face and solar cell

Figure 40: Sun vector determination test: the light source is moved around the mock-up on the polar chart while the ESP32 reads the voltages of the solar cells and estimates the sun vector angle
Source: Authors (2026).

The results showed that the system correctly identifies the quadrant and follows the displacement of the source. This validates the operation of the complete chain, from voltage acquisition to processing on the ESP32 and computation of the sun vector angle.

In accordance with requirement HLR-ADCS-03, which requires the attitude determina-

tion performance to be quantified and reported, including error metrics and limitations, some sources of error must be considered when interpreting these results. Because the flashlight is close to the mock-up, it emits divergent light, unlike the practically parallel rays of the Sun, which causes the angle of incidence to vary across the surface of each cell. The white paper and nearby surfaces reflect light onto mock-up faces that are not directly illuminated, in a manner analogous to Earth's albedo in orbit. Manufacturing and assembly variations cause identical cells to generate different voltages under the same illumination. The manual positioning of the source introduces uncertainty in the reference angle and in the distance.

8.3.2 Reaction Wheel Motor Parameters for Detumbling

The second test aimed to evaluate the actuator responsible for the detumbling maneuver, which consists of reducing the angular velocity of the CubeSat. For this purpose, the operation of the BLDC motor that drives the reaction wheel was verified, focusing on its rotational speed, current consumption and speed control.

It was not possible to measure the motor speed under load, that is, with the reaction wheel mass coupled to its shaft, since the steel wheel had not yet been manufactured. The test was therefore carried out with the motor in the no-load condition. Under a supply voltage of 12 V, the motor exceeded 1300 rpm with a current below 400 mA. The speed was measured with a digital tachometer, as shown in Figure 41. Since no load was applied, these values represent an upper bound for the speed and a lower bound for the current consumption, and the inertia of the wheel is expected to reduce the acceleration and increase the current, mainly during the transients.

The test also demonstrated control of the motor acceleration through a 5 kΩ potentiometer. The potentiometer was connected as a voltage divider to an analog input of the ESP32, whose reading was converted into a PWM signal applied to the motor through the motor driver. This made it possible to vary the speed and acceleration of the motor continuously. In the final system, the manual reference provided by the potentiometer will be replaced by the command generated by the attitude control algorithm.

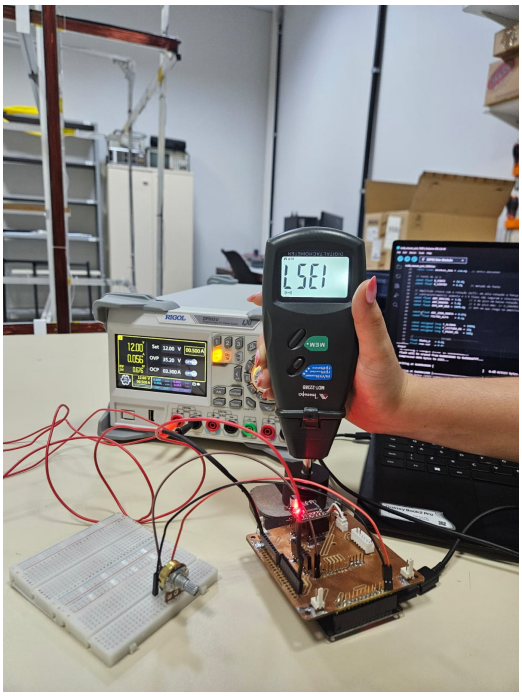


Figure 41: No-load speed measurement of the reaction wheel motor using a digital tachometer
Source: Authors (2026).

Although incomplete, the results confirm that the motor and its drive circuit operate as expected and that its speed can be controlled. The test is to be repeated with the reaction wheel installed, in order to characterize the speed, current consumption and response time of the motor under load, which are necessary to evaluate the angular momentum available for the detumbling maneuver.

9 OBC (On-Board Computer)

The On-Board Computer, or OBC, is the unit that runs the satellite's embedded software, that is, the programs that manage the system's main functions. It is commonly referred to as the brain of the satellite and has several duties, such as coordinating all actions, generating commands for the different modules, receiving and storing CubeSat data, hosting the processing of the primary ADS-B reception mission, and sending this information back to Earth through the Telemetry and Telecommand (TT&C) System, including error handling within the CubeSat. In this document, OBC names both the computer, a Raspberry Pi Zero 2 W shared by all onboard processes, and the OBC software, the process that runs the mission modes; the TT&C and payload software are separate processes on the same computer (Section 11.3.3).

9.1 Theoretical Background

Since the On-Board Computer plays a central role in the CubeSat, the choice of OBC was grounded in technical criteria that ensure the satellite operates reliably and efficiently. To accomplish the CubeDesign 2026 mission, the OBC was designed to meet specific *High Level Requirements* (HLRs).

In accordance with requirements **HLR-GEN-02** and **HLR-GEN-03**, the system must operate autonomously during the mission, supporting interaction via TC and generating TM. The acceptance criterion states that the OBC must boot the operating system, start the onboard processes under the system supervisor and enter `PRE_TEST_MODE` (defined in the ConOps, Section 9.2.3), with the telecom daemon listening, without any physical manual intervention.

For TM/TC communications (**HLR-COMM-01**, **HLR-COMM-02** and **HLR-COMM-03**), the responsibility is shared with the TT&C software: the telecom daemon receives each telecommand over the LoRa link, acknowledges it and delivers mission events to the OBC, and transmits the telemetry, to which the OBC contributes the power, temperature and attitude data (Section 11.3.3). As an acceptance criterion, a command executed by the OBC takes effect within the TT&C response bound plus one cycle of the OBC main loop, 1 s (Table 54) and the housekeeping telemetry must contain, at minimum: system status, power parameters (battery voltage/current), attitude data (when applicable to the mode), and mission data. The physical layer, protocol, and data rate used for communication must be defined and justified in the documentation (HLR-COMM-03).

For payload execution (**HLR-ADS-01**, **HLR-ADS-02** and **HLR-ADS-04**), the SDR receiver is powered whenever the satellite is on and is operated by the payload software, a separate process, which runs the 1090 MHz decoder only in `MISSION_ADSB_MODE`, processes the data onboard according to the hybrid architecture and extracts the ICAO address, Position, Altitude, and Velocity. In accordance with **HLR-ADS-03**, the system must be able to detect and process multiple aircraft simultaneously within the same mission scenario, and in accordance with **HLR-ADS-07**, each received ADS-B message must be associated with a *timestamp* with sufficient temporal resolution for trajectory reconstruction. The OBC's share of these requirements is the mission window: as an acceptance criterion, `MISSION_ADSB_MODE` reaches the payload software through the telecom daemon and ends after 600 s or earlier by telecommand. The acceptance criteria of the reception chain itself, up to 20 simultaneous aircraft with a

timestamp for every message, are in Table 73.

In compliance with **HLR-ADS-08**, the quantitative mission success metrics adopted by the team — correct decoding, data loss and latency — are given in Table 74; the OBC contributes the mode transitions that open and close the 600 s window over which they are measured.

Additionally, to cover the software and data requirements (**HLR-SW-01** and **HLR-SW-02**), the system implements operational modes strictly consistent with the ConOps. As acceptance criteria, every transition of Figure 44 is executed and recorded in the OBC log, and the generated data are saved in structured log files (JSON lines) in the onboard memory, stamped with the monotonic clock and the wall-clock anchor sent by the ground (Section 11.3.6); during the mission a compact state of each aircraft is transmitted via RF every 5 s, and the complete record is retrieved over USB after the test (Section 11.3.2).

Finally, in accordance with **HLR-EPS-04**, the battery must remain within safe operating conditions (voltage, current, and temperature) throughout all test phases. The protection itself is done in hardware, by the battery management module of the EPS power board (Figure 90) and by the active thermal control (Section 7); as an acceptance criterion, the OBC measures the battery voltage, current and temperature in every mode and delivers them to the telecom daemon, so that they reach the ground in every housekeeping frame, every 10 s.

9.2 Modeling

The subsystem uses the Raspberry Pi Zero 2W as the OBC, running the 64-bit Raspberry Pi OS Lite operating system. The Raspberry Pi Zero 2W is a small-form-factor computer board, particularly well suited for compact projects, with enough processing power to manage the CubeSat thanks to its 1 GHz *quad-core* CPU and 512 MB of RAM. Its multiple cores are crucial for simultaneously supporting the satellite's state machine and the ADS-B payload's SDR (*Software Defined Radio*) receiver.

The board features a 40-pin HAT-compatible (*Hardware Attached on Top*) header, which allows expansion boards and additional modules to be connected; in this project, through the OBC board and the 40-pin bus between the boards (Figure 99), it connects the temperature and power sensors, the LoRa radio module, the actuation of the antenna deployment and of the heaters, and the ESP32 of the attitude control.

The CubeSat's sensors are fundamental, as they provide the information needed to monitor the environment and the satellite's own health status, allowing the OBC to read and process the information and coordinate actions. For the satellite's power management, the INA219 sensor was chosen, responsible for measuring the battery voltage and the current drawn from it in real time. This sensor makes it possible to track battery health, preventing issues that could compromise the mission.

For measuring the CubeSat's internal and external temperature, the DS18B20 digital sensor was chosen. This sensor offers good accuracy and can operate over a wide temperature range, suitable for the thermal variation of the competition. Integrating this sensor with the Raspberry Pi is straightforward, using the 1-Wire communication protocol, which operates over a single wire and allows multiple sensors to be connected on the same bus.

9.2.1 Hardware Selection Rationale

In accordance with **HLR-COST-02**, the OBC hardware decisions were based on performance, cost, and complexity criteria. The Raspberry Pi Zero 2W was chosen over alternatives such as the Raspberry Pi 4 or pure microcontroller-based boards (e.g., a standalone ESP32) because it offers the best balance between processing capability (a quad-core CPU, sufficient to run the state machine and the SDR receiver simultaneously) and power consumption, in

(a) Raspberry Pi Zero 2W (b) INA219 sensor (c) DS18B20 sensor

Figure 42: OBC hardware components
Authors (2026).

addition to its compact form factor, compatible with the dimensional constraints. A Raspberry Pi 4 would offer more processing power, but at higher power consumption and cost, which is not justified given that the ADS-B and state machine processing load is already handled by the Zero 2W.

For power monitoring, the INA219 sensor was chosen over alternatives such as the ACS712 or the INA226 because it combines a native I2C interface (simplifying integration with the bus already used by the other peripherals) and adequate accuracy for the CubeSat battery's voltage/current range, without requiring additional signal conditioning circuitry. Likewise, the DS18B20 was selected for its 1-Wire protocol, which allows multiple sensors on the same bus while occupying the fewest GPIO pins, reducing wiring complexity compared to analog sensors that would require dedicated A/D conversion.

9.2.2 Software Architecture

To ensure continuous operation without blocking the satellite's vital processes, the onboard software runs as separate processes on Linux: the OBC software, described here, and the TT&C and payload processes, described in Section 11.3.3, which own the LoRa radio and the SDR receiver. Inside the OBC software, the work is divided into threads:

- **Main Thread:** Runs the ConOps state machine in a 1 s cycle: it reads, without blocking, the mission events and the time anchor that the telecom daemon delivers from the ground telecommands, applies them, and reports the current mode and the platform telemetry to it over a local socket (HLR-COMM-01, HLR-COMM-02). It is a persistent thread, active in every mission phase.
- **Mission window:** On entering `MISSION_ADSB_MODE`, starts the 600 s timer that ends the mission (`EV_ADSB_TIMEOUT`). The SDR receiver stays powered in every mode; the payload process, informed of the mode by the telecom daemon, starts and stops the decoder (HLR-ADS-01 to HLR-ADS-03, HLR-ADS-05).
- **Sensors and EPS Thread:** Performs continuous reading (*polling*) of the INA219 and DS18B20 and keeps the latest battery voltage, current and temperature, which the main thread delivers to the telecom daemon (HLR-EPS-04). It does not change the mode: `ENV_SURVIVAL_MODE` is entered by telecommand. It is a persistent thread, active in every mission phase.
- **AOCS Communication Thread (UART/I2C):** Responsible for offloading the heavy attitude determination and control task directly to the ESP32 microcontroller, which stays powered in every mode. It is instantiated on demand, only during `ADVANCED_AOCS_MODE`.

Below, the Per-Thread Processing Flow diagram illustrates the logical processing flow, the decision points, and the division of these tasks between the Raspberry Pi (OBC) and

the ESP32 (AOCS), and the messages it exchanges with the TT&C and payload processes. This diagram documents the internal software architecture — the formal ConOps definition (operational modes and their transitions) is given in Section 9.2.3.

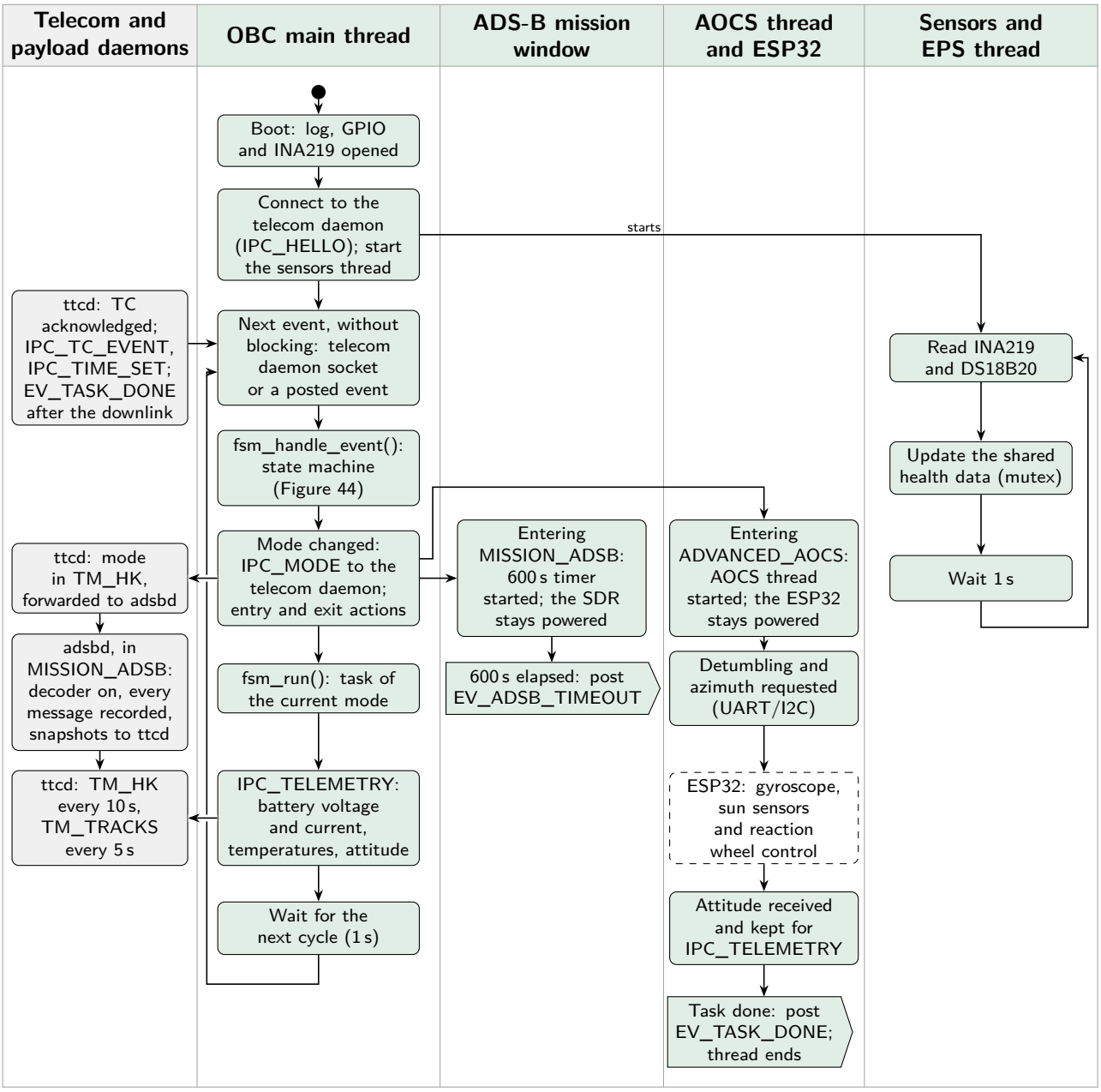


Figure 43: OBC per-thread processing flow and its interfaces with the TT&C and payload processes. Green: OBC software; grey: TT&C and payload processes (Section 11.3.3); dashed: ESP32 firmware; pentagons: events posted to the main thread

Source: Authors (2026).

9.2.3 Concept of Operations (ConOps)

The OBC software's operating cycle is managed by a State Machine, whose transitions occur mostly via Telecommand (TC), delivered to the OBC as events by the telecom daemon; Figure 44 shows each transition with the event that triggers it. Operation is mapped according to the following phases:

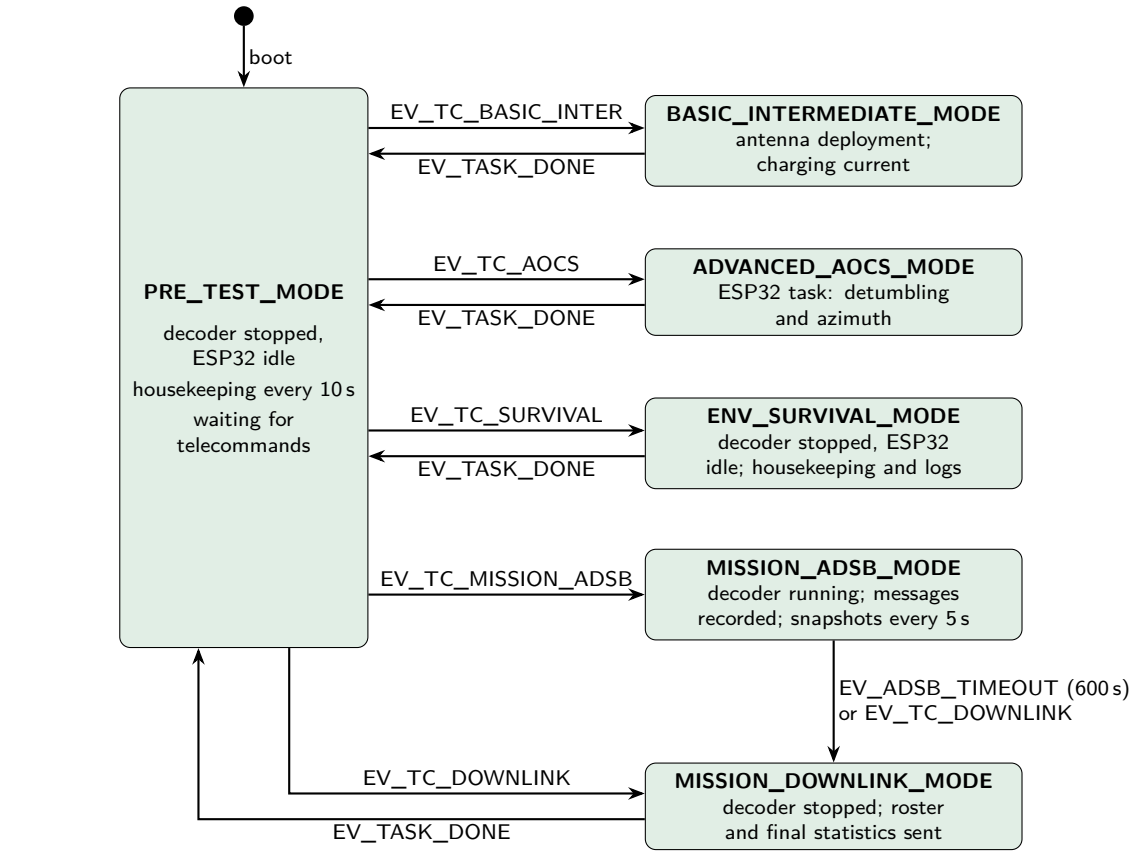
In **PRE_TEST_MODE** (Standby / *Fit-Check*), the satellite starts the phase completely de-energized. After the initial system *boot*, the Raspberry Pi checks the battery volt-

age through the EPS and enters standby mode awaiting commands, with the ADS-B decoder stopped and the ESP32 idle. Upon receiving the appropriate command, the system advances to **BASIC_INTERMEDIATE_MODE**. In this phase, the Raspberry Pi triggers the antenna release mechanism and reads the battery voltage to follow the recharging from the light source, and the OBC returns to `PRE_TEST_MODE` when the task is done.

When **ADVANCED_AOCS_MODE** (Attitude Control) is activated, the Raspberry Pi fully delegates the *detumbling* task — reducing the rotation from 60 rpm to less than 5 rpm — and azimuth determination to the ESP32, which stays powered in every mode but receives this task only here. The ESP32 computes the pointing error, uses the sun sensors, and controls the reaction wheel, then sends the processed data back to the Raspberry Pi, which forwards the attitude to the telecom daemon for transmission (Figure 43); the OBC returns to `PRE_TEST_MODE` when the task is done.

The primary phase is **MISSION_ADSB_MODE** (Main Mission), in which the payload software starts the ADS-B decoder and the ESP32 stays idle. During this mode, the Raspberry Pi dedicates its resources to capturing ADS-B signals, extracting the aeronautical information (ICAO, position, velocity, altitude), and saving it to local memory for 10 minutes. Once the capture is complete, after 600 s or earlier by telecommand, the board transitions to **MISSION_DOWNLINK_MODE**, in which the decoder is stopped, the TT&C software sends the roster of the aircraft observed and the final statistics, and signals the end of the task, after which the OBC returns to `PRE_TEST_MODE`. The complete record, 7.34 MiB for 20 aircraft, would take about 3.4 hours of transmission even at the fastest profile, so it is retrieved over USB after the test (Table 55).

Finally, when the satellite is subjected to vibration and temperature extremes, **ENV_SURVIVAL_MODE** (Environmental Testing) is activated, meeting **HLR-ENV-01** and **HLR-ENV-02**. The Raspberry Pi enters survival mode, stopping the non-essential processing (the ADS-B decoder and the attitude task) and monitoring temperature and battery voltage, which keep being reported in the housekeeping telemetry; the mode is entered and left by telecommand. As an acceptance criterion, the system must successfully execute the mission before the environmental tests (HLR-ENV-01) and, after the thermal and vibration tests, demonstrate functional survival — that is, the ability to resume telemetry capture and report its status — or, if this does not occur, present a justified analysis of the observed limitations based on the telemetry data collected before, during, and after the tests (HLR-ENV-02).



EV_TC_*: a SET_MODE telecommand, acknowledged by the telecom daemon and delivered to the OBC as IPC_TC_EVENT.
EV_TASK_DONE: the task of the mode ended: antenna deployed, detumbling done, or roster and final statistics sent (posted by the telecom daemon). ENV_SURVIVAL_MODE is entered and left by telecommand.
EV_ADSB_TIMEOUT: OBC timer, 600 s after entering MISSION_ADSB_MODE. Every transition is reported to the telecom daemon (IPC_MODE) and in the housekeeping telemetry.

Figure 44: OBC ConOps state machine diagram
Source: Authors (2026), after the state machine of the OBC software.

9.3 Testing and Validation

Verification and validation follow the definitions of Section 11.4.1: verification is evidence obtained by analysis, inspection or software tests, and validation is evidence obtained on the flight hardware. The OBC computer is the platform on which the TT&C and payload software were tested, so its validation as a platform already exists; the OBC software has not yet been tested on the target, and its tests are planned. Table 32 traces each acceptance criterion of this section to its evidence, with the status codes of Table 66: **A** verified by analysis, **D** verified by documentation review, **S** verified in software, **H** validated on hardware, **P** planned.

Platform

On 7 and 9 October 2026, the Raspberry Pi Zero 2 W ran the 64-bit operating system (Debian 13), compiled the onboard software without warnings and passed its 23 test suites (Section 11.4.3). With the SDR sampling at 2.4 MS/s, the ADS-B decoder used 21.9 % of one core while the whole computer was 5 % busy, at 46 °C, without throttling and with 283 MB of memory free (Table 63). This supports the hardware selection of Section 9.2.1 at low traffic; the load with about 20 aircraft is part of the remaining validation.

Interface with the telecom daemon

The interface is verified in software from the TT&C side, against a stand-in OBC client. The integration tests run the real telecom daemon: it receives the client's handshake, mode and telemetry, reports them in the housekeeping, and delivers to the client, as an event, a `SET_MODE` telecommand received over the simulated radio; the core tests check that the housekeeping stops reporting the platform values when the client disconnects. In the channel simulation, 100 `SET_MODE` telecommands with one acknowledgement lost delivered exactly 100 events, the worst case from submission to delivery being 2.42 s (Table 61), within the one-frame-lost bound of Table 54.

Table 32: Acceptance criteria of the OBC, with their verification and validation

ID	Acceptance criterion	Verification	Validation	Status
OBC-1	Boots, starts the onboard processes and enters `PRE_TEST_MODE` without manual intervention (HLR-GEN-02, HLR-GEN-03)	—	power-on test of the flight image	P
OBC-2	The computer runs the decoder and the onboard processes with margin (HLR-COST-02)	—	bench: decoder 21.9 % of one core, computer 5 % busy, 283 MB free, with 0 to 3 aircraft; about 20 aircraft	H, P
OBC-3	A command executed by the OBC takes effect within the TT&C bound plus 1 s (HLR-COMM-01)	channel simulation: 100 events for 100 `SET_MODE`, worst case 2.42 s	integration: time from submission to the transition record of the OBC log	S, P
OBC-4	The housekeeping carries the mode and the power, temperature and attitude data of the OBC (HLR-COMM-02)	integration test with a stand-in client; reference vectors of the messages	integration with the reference client; on the bench, without it, these fields read as unknown	S, P
OBC-5	Every transition of Figure 44 is executed and recorded in the OBC log (HLR-SW-01)	—	state machine on the target with the keyboard telecommand simulator, then with `SET_MODE` over the link	P
OBC-6	`MISSION_ADSB_MODE` reaches the payload software and ends after 600 s or by telecommand (share of HLR-ADS-01 to HLR-ADS-05)	integration test: `SET_MODE` delivered as the mission event	10-minute rehearsal	S, P
OBC-7	Battery voltage, current and temperature measured in every mode and reported every 10 s (HLR-EPS-04)	codec tests: voltage and current; temperature in protocol version 2	readings against a multimeter and a reference thermometer, during the rehearsal	S, P
OBC-8	Mission executed before the environmental tests; status reported after them (HLR-ENV-01, HLR-ENV-02)	—	telemetry before, during and after the environmental tests	P

Continued on the next page.

Table 32 — continued

ID	Acceptance criterion	Verification	Validation	Status
OBC-9	Logs in JSON lines with the monotonic time and the wall-clock anchor (HLR-SW-02)	log format documented by the OBC team; anchor delivery verified on the TT&C side	log inspected after the rehearsal	D, S, P

Source: Authors (2026).

The planned tests run in the order in which they can change the design: the state machine on the target first, then the integration with the telecom daemon through the reference client, the sensor readings and the actuation of the antenna deployment and of the heaters, the boot of the flight image, and finally the rehearsal and the environmental tests shared with the TT&C and payload software (Table 67).

10 EPS (Electrical Power System)

The Electrical Power System (EPS) is responsible for supplying and distributing electrical power to all satellite subsystems, ensuring that the CubeSat remains operational even under critical operating conditions. Throughout this document, the acronym EPS is used to refer to the CubeSat Electrical Power System (NASA, 2023b).

Solar irradiance is converted into electrical energy by the photovoltaic panels. Part of this energy is stored in the battery pack, while the remaining portion is delivered to the satellite subsystems. Before distribution, however, the electrical power is conditioned and converted to the voltage levels required by each subsystem (Shakoor; Alayedi; Elsayed, 2025).

10.1 Theoretical Background

In CubeSats, solar panels constitute the primary source of electrical power, being responsible for the direct conversion of solar radiation into electricity through the photovoltaic effect. These panels are typically composed of monocrystalline silicon, gallium arsenide (GaAs), or multi-junction solar cells, which provide higher conversion efficiencies under the limited surface area and extreme thermal variations encountered in the space environment (Yamada; Kim; Torres, 2024; Wang; Kumar; Chen, 2024). The efficiency of the solar panels is a critical factor affecting satellite performance, as the available area for energy harvesting is severely constrained, requiring devices with high power density and excellent thermal stability. Figure **??** illustrates the structure of a photovoltaic cell.

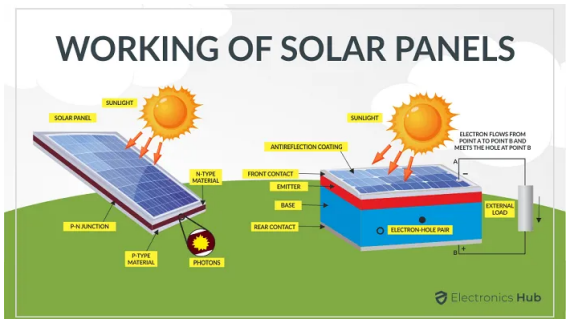


Figure 45: Solar cell

Source: Medium, How do solar panels work?

The operating principle of photovoltaic cells is based on the interaction between incident light and a semiconductor material, resulting in the generation of free charge carriers. Solar cells are primarily composed of semiconductor materials characterized by a completely filled valence band and an initially empty conduction band. When the semiconductor is doped, a p–n junction is formed through the controlled introduction of acceptor impurities (such as boron, a p-type dopant) and donor impurities (such as phosphorus, an n-type dopant), thereby modifying the electrical conductivity of the material (Abdullah; Singh; Patel, 2023).

The p–n junction establishes a permanent internal electric field due to the diffusion of electrons from the n-region to the p-region and holes in the opposite direction. This depletion region inhibits the continuous flow of charge carriers but allows incident photons with energy greater than the semiconductor bandgap (E_g) to excite electrons from the valence band to the conduction band under illumination, generating electron–hole pairs. The separation of these charge carriers by the internal electric field produces a potential difference across the p- and n-regions, resulting in the generation of photocurrent (Abdullah; Singh; Patel, 2023). The electrical behavior of a photovoltaic cell can be represented by an equivalent circuit consisting of a current source connected in parallel with a diode and parasitic resistances, following the widely adopted single-diode model (Wang; Kumar; Chen, 2024).

The characteristic equation describing the electrical current of the photovoltaic cell is expressed as

$$I = I_{ph} - I_0 \left[e^{\frac{(V+IR_s)}{nV_T}} - 1 \right] - \frac{(V + IR_s)}{R_{sh}} \tag{84}$$

where I_{ph} is the photocurrent proportional to the incident irradiance, I_0 is the diode saturation current, R_s and R_{sh} are the series and shunt resistances, respectively, n is the diode ideality factor, and V_T is the thermal voltage. This relationship demonstrates the direct dependence between the generated current and voltage, both of which are influenced by operating temperature, solar irradiance, and the semiconductor material employed.

The characteristic I–V and P–V curves define the Maximum Power Point (MPP), whose location depends on the operating conditions. Maximum Power Point Tracking (MPPT) techniques are employed to ensure that the system continuously operates at this point, thereby maximizing the conversion efficiency of solar energy into electrical power in CubeSat applications (Yamada; Kim; Torres, 2024).

10.2 Buck and Boost DC–DC Converters

The electrical power generated by the solar panels must be conditioned to satisfy the voltage and current requirements of the CubeSat subsystems. For this purpose, DC–DC converters are employed. These power electronic devices are responsible for converting DC voltage levels while maintaining high conversion efficiency and low structural weight, both of which are essential characteristics for space applications (Santos; Huang; Rodríguez, 2025).

The *Buck* converter, also referred to as a step-down converter, is used to reduce a higher input voltage (V_{in}) to a lower output voltage (V_{out}). The output voltage is regulated by the switching duty cycle (D) according to

$$V_{out} = D \cdot V_{in}. \tag{85}$$

During the interval in which the semiconductor switch (typically a MOSFET) is turned on, energy is stored in the inductor. When the switch is turned off, the diode becomes forward biased, transferring the stored energy to the load while maintaining current continuity (Rahman; Silva; Novak, 2023). This operating principle makes the Buck converter particularly suitable for supplying sensitive electronic circuits requiring stable voltage levels, such as

onboard microcontrollers and sensors.

Conversely, the *Boost* converter, also known as a step-up converter, performs the opposite function by increasing the output voltage from a lower input voltage source. Its voltage conversion ratio is given by

$$V_{out} = \frac{V_{in}}{1 - D}. \tag{86}$$

In this configuration, energy is stored in the inductor while the semiconductor switch remains on. Once the switch is turned off, the stored energy is transferred to both the capacitor and the load, adding to the input voltage and producing an output voltage greater than the input voltage. The *Boost* converter is widely employed in CubeSat electrical power systems to supply subsystems requiring higher voltage levels, such as radio transmitters and power electronics (Lee; Costa; Ahmad, 2025). Together, the *Buck* and *Boost* converters ensure that the electrical power generated by the solar panels is efficiently conditioned and distributed to the various satellite subsystems, thereby providing a stable and reliable electrical power architecture.

The selection of Buck converters was motivated by the need to supply subsystems operating at voltage levels lower than those provided by the battery pack. In addition to reducing the input voltage, these converters exhibit high conversion efficiency, minimizing thermal losses and maximizing the utilization of the energy stored in the batteries.

The Boost converter was incorporated into the proposed architecture following the same design philosophy, supplying subsystems that require voltage levels higher than those directly available from the battery pack. Employing a Boost converter eliminates the need for a higher-voltage battery pack, thereby reducing the overall mass, volume, and complexity of the CubeSat Electrical Power System.

10.3 System Modeling

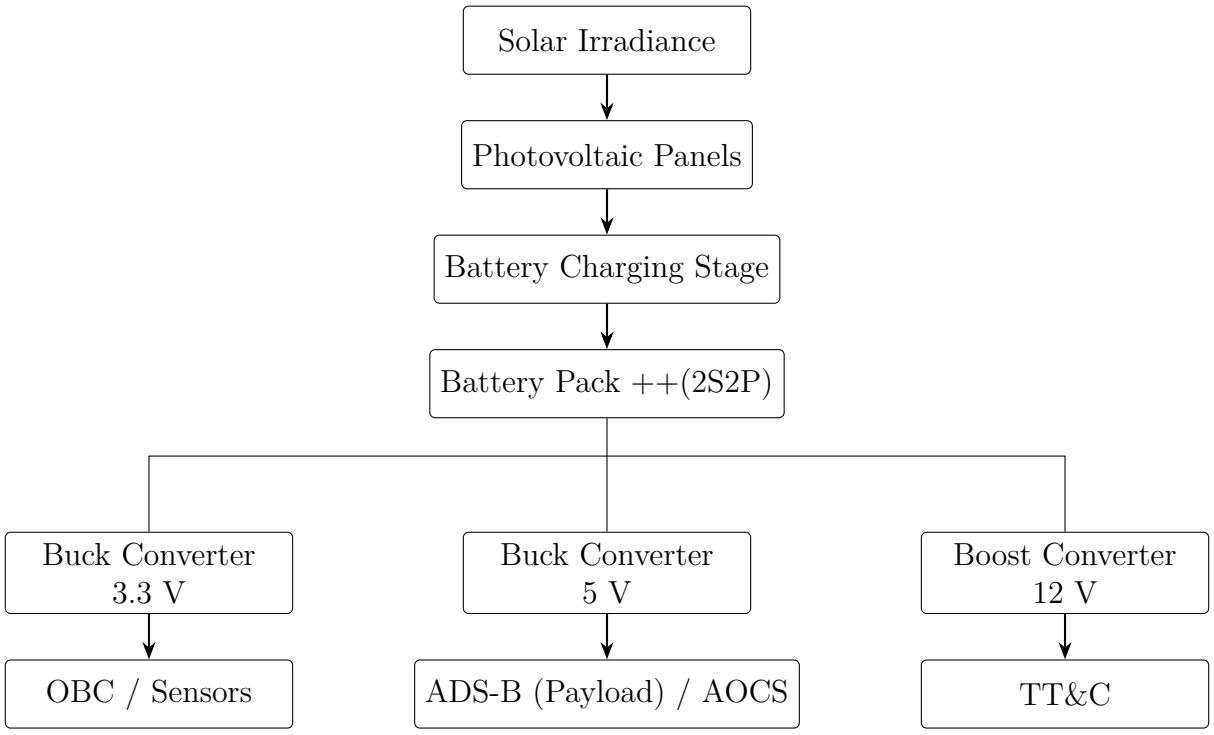


Figure 46: CubeSat eletrical power system diagram.
Source: Authors (2026).

The EPS architecture was defined to simultaneously satisfy the CubeSat requirements for power supply, implementation simplicity, and operational reliability. The proposed centralized architecture, consisting of photovoltaic panels, a battery pack, and DC–DC converters, enables efficient management of both the generated and stored electrical energy while ensuring that all satellite subsystems receive the voltage levels required under the different operating conditions expected throughout the mission.

The EPS was designed to satisfy the CubeSat's power requirements throughout all planned mission phases. To provide a systematic overview of the proposed Electrical Power System (EPS) architecture, an electrical block diagram was initially developed, as shown in Figure 46. For clarity, the modeling section is divided into three (3) subsections, namely:

1. Solar panels;
2. Battery pack;
3. Switching converters

10.3.1 Solar Panels

Miniature photovoltaic panels were selected to convert solar radiation into electrical energy (NASA, 2023b). The EPS design was driven by two primary requirements: maximizing the charging current supplied to the battery pack and ensuring mechanical compatibility with the dimensions of the CubeSat side printed circuit board (PCB), measuring 36 mm in diameter. Consequently, miniature photovoltaic panels with a nominal voltage of 2 V and a nominal current of 35 mA were selected, as illustrated in Figure 47. Each panel has overall dimensions of 36 mm diameter.

To increase the available charging current and reduce the battery charging time, 3 photovoltaic panels were installed on each CubeSat face using a paralel and serie configuration, as illustrated in Figure 46 (Yusof; Baharuddin, 2020). Under this configuration, each face provides a nominal voltage of 6 V and a nominal current of 270 mA. Furthermore, the lateral faces are interconnected in parallel. As a result, the absence of electrical power generation on one satellite face does not affect the power generation capability of the remaining illuminated faces.

Figure 47: Solar photovoltaic panel
Source: Mercado Libre.

The parallel interconnection between the CubeSat faces also increases the robustness of the power generation subsystem. Since the satellite attitude continuously changes throughout the mission, different faces are exposed to varying levels of solar irradiance. Consequently, if one face is not illuminated, the remaining illuminated faces continue generating electrical power independently, thereby improving the reliability of the Electrical Power System.

The selection of the photovoltaic panels was based on electrical, mechanical, and structural design criteria. From a mechanical perspective, the panel dimensions had to be compatible with the available area on the CubeSat side PCBs while ensuring seamless integration with the remaining onboard subsystems. Additionally, the satellite structure adopts a design in which

hexagonal regions are formed on the side PCBs. To accommodate this geometry, miniature circular photovoltaic panels were selected instead of the conventional square or rectangular modules commonly found in commercial applications.

From an electrical perspective, priority was given to maximizing the charging current delivered to the battery pack, thereby reducing the time required to replenish the energy consumed during satellite operation.

Another factor considered during the component selection process was the commercial availability of the photovoltaic panels. This facilitates component replacement throughout the prototyping, assembly, and experimental validation stages of the EPS development.

10.3.2 Battery Pack

For energy storage, Panasonic NCR-18650B lithium-ion cells, shown in Figure 48, were selected. Each cell provides a nominal capacity of 3,350 mAh (**panas**). The proposed CubeSat design employs four (4) cells arranged in a 2S2P configuration (two cells connected in series and two series strings connected in parallel), providing an output voltage greater than 8 V, which is well suited for the selected DC–DC converters (the rationale for this choice is discussed in Subsection Switching Converters), while providing a total battery pack capacity of 6,700 mAh.

Figure 48: Panasonic NCR-18650B battery
Source: Panasonic.

The selection of the Panasonic NCR-18650B cells was based on their high energy density, wide commercial availability, widespread adoption in aerospace applications (reference required), and electrical characteristics compatible with the requirements of the proposed EPS. Furthermore, the nominal capacity of the battery pack provides sufficient stored energy to supply the CubeSat subsystems during periods without photovoltaic power generation, such as orbital eclipse phases and ground-based testing.

The 2S2P configuration was selected as a compromise between output voltage, energy storage capacity, and simplicity of the power conversion architecture. Connecting the cells in series increases the battery pack voltage, allowing the Buck converters to operate within the manufacturer's recommended input voltage range. The parallel connection, in turn, increases the overall storage capacity, thereby extending the operational autonomy of the CubeSat without significantly increasing the complexity of the Electrical Power System.

The parallel arrangement was also adopted to increase the current supply capability while maintaining the nominal output voltage of the battery pack. This configuration contributes to shorter battery charging times and improves current availability during periods of high power demand from the satellite subsystems.

10.3.3 Switching Converters

To convert the battery pack voltage into the voltage levels required by each CubeSat subsystem, switching DC–DC converters were adopted. The proposed architecture employs

two Buck converters, providing regulated output voltages of 3.3 V and 5 V, respectively (Figure 49(a)), and one Boost converter providing a regulated output voltage of 12 V (Figure 49(b)).

Switching converters were selected instead of linear voltage regulators because, in linear regulators, the transistor operates as a variable resistance, resulting in significantly lower conversion efficiency and greater power dissipation through the Joule effect within the satellite (Hart, 2016).

Since the selected Buck converter requires a minimum voltage difference of 1.5 V between its input and output terminals to ensure proper operation (MPS, 2016), the 2S2P battery configuration was adopted. A configuration with all battery cells connected exclusively in parallel would not provide the minimum input-to-output voltage differential required by the converter, preventing its correct operation.

(a) Buck DC–DC Converter – MP1584EN

(b) Boost DC–DC Converter

Figure 49: Switching Converters
Source: Mercado Libre, FLYROBO.

The decision to employ switching converters instead of linear regulators was primarily motivated by their high power conversion efficiency. In space applications, the available electrical power is inherently limited by the energy generated by the solar panels and the storage capacity of the battery pack, making it essential to minimize losses during power conditioning. Furthermore, reducing the power dissipated as heat alleviates the thermal management challenges associated with CubeSat operation, thereby improving the overall efficiency and reliability of the Electrical Power System.

10.4 Experimental Validation

Several experimental tests were conducted to validate the proposed Electrical Power System (EPS) architecture. The first experiment evaluated the battery charging process using the photovoltaic panels as the energy source, while the second assessed the performance of the motor supplied by the Boost DC–DC converter presented in Subsection 10.3.3.

10.5 Battery charging validation

Under the proposed parallel configuration, the photovoltaic panels are capable of supplying up to 270 mA per satellite face to the battery pack. However, the electrical power generated by the panels depends on several environmental factors, including the operating temperature of the photovoltaic cells (Dubey; Sarvaiya; Seshadri, 2013) and the angle of incidence of the incoming solar radiation.

During the charging experiments, the battery pack initially exhibited a voltage of 1.660 V. After approximately six minutes of charging using the four-panel array shown in Figure **??**, the

battery voltage increased to 1.725 V, demonstrating the capability of the proposed photovoltaic arrangement to recharge the battery pack under the adopted test conditions.

10.6 Power Budget

The power budget provides a quantitative assessment of the electrical power required by each CubeSat subsystem under nominal operating conditions. It is an essential design tool for validating the proposed Electrical Power System architecture, ensuring that the photovoltaic generation subsystem, battery pack, and DC–DC converters are capable of meeting the mission power requirements.

The analysis presented in this section summarizes the voltage, current, and power consumption of each subsystem, allowing the verification of the overall energy balance throughout the mission, including eclipse periods and peak power demand conditions.

Table 33: Cubetscheck power budget

Subsystem	Component	Model	Qty.	Voltage [V]	Current [A]	Power [W]
AOCS	Accelerometer / Gyroscope	BMI088	1	3.3	0.006	0.0198
AOCS	Microcontroller	ESP32	1	3.3	0.080	0.264
AOCS	Encoder	AS5048A	1	3.3	0.002	0.0066
Electronic	Voltage and Current Sensor	INA219	1	3.3	0.001	0.0033
Electronic	Temperature Sensor	DS18B20	3	3.3	0.002	0.020
Electronic	LoRa	RA-02 SX1278	1	3.3	0.110	0.350
Total Current Draw (3.3 V Loads)						**0.20 A**
Electronic	OBC	Pi Zero 2W	1	5.0	0.700	3.500
Electronic	SDR	NESDR Nano 3	1	5.0	0.300	1.500
Mechanisms	SDR	NiCr Wire	1	5.0	1.000	5.000
Total Current Draw (5 V Loads)						**2.00 A**
AOCS	Driver	DRV8313	1	12.0	0.001	0.012
AOCS	Motor	GM2804	1	12.0	0.800	9.600
Total Current Draw (12 V Loads)						**0.80 A**
Total Power Consumption: 20.27 W						
Total Power Consumption with 20% Design Margin: 24.33 W						
Estimated Operating Time – 2S2P Battery Pack: 2.14 h						

Source: Authors (2026).

10.7 External Power Interface

The implementation of an external power interface ensures that environmental qualification tests can be performed safely without the presence of lithium-ion batteries inside the thermal vacuum chamber, while maintaining full functionality of the CubeSat throughout the test campaign.

In addition to its nominal operating configuration, the Electrical Power System (EPS) was designed to support CubeSat operation without the internal battery pack, complying with requirement HLR-EPS-05. This requirement specifies that the satellite shall remain fully operational during Thermal Vacuum (TVAC) testing while being powered exclusively by an external power source located outside the test chamber.

The external power interface is connected to the EPS board through connector J2. The external supply is injected into the 8 V power rail, located downstream of diode D1 and upstream of the Remove Before Flight (RBF) switch. Diodes D1 and D2 provide electrical isolation between the battery pack and the external power source, preventing reverse current flow between the two power paths, as defined by the EPS topology.

Connector J2 consists of two terminals. Pin 1 is connected to the common ground plane (GND), while Pin 2 is connected to the anode of diode D2. The cathode of D2 is connected directly to the 8 V power rail, which supplies the common terminal of the RBF switch through a dedicated PCB trace. This configuration allows the external power source to be seamlessly integrated into the CubeSat power distribution architecture, enabling normal satellite operation without the internal battery pack installed.

During TVAC testing, the lithium-ion battery pack will be completely removed in accordance with the competition safety requirements. Under these conditions, the CubeSat will be powered exclusively through the external power interface using a regulated DC power supply compatible with the electrical specifications of the EPS. The connection will be established through an appropriate power cable and, when required, an adapter compatible with the electrical feedthrough interface of the thermal vacuum chamber.

11 TT&C

The telemetry and telecommand (TT&C) subsystem is responsible for the communication link between the satellite and the ground station: it transmits the telemetry (TM) generated on board and receives the telecommands (TC) sent from the ground, interfacing with the on-board computer, which aggregates the data from the other subsystems (European Space Agency, 2025). Information is transmitted and received by a radio module, and link performance is assessed through the link budget, which accounts for the gains and losses between transmitter and receiver and determines the available margin. In the uplink, the CubeSat receives information from the ground station; in the downlink, it transmits data to it, as illustrated in Figure 50.

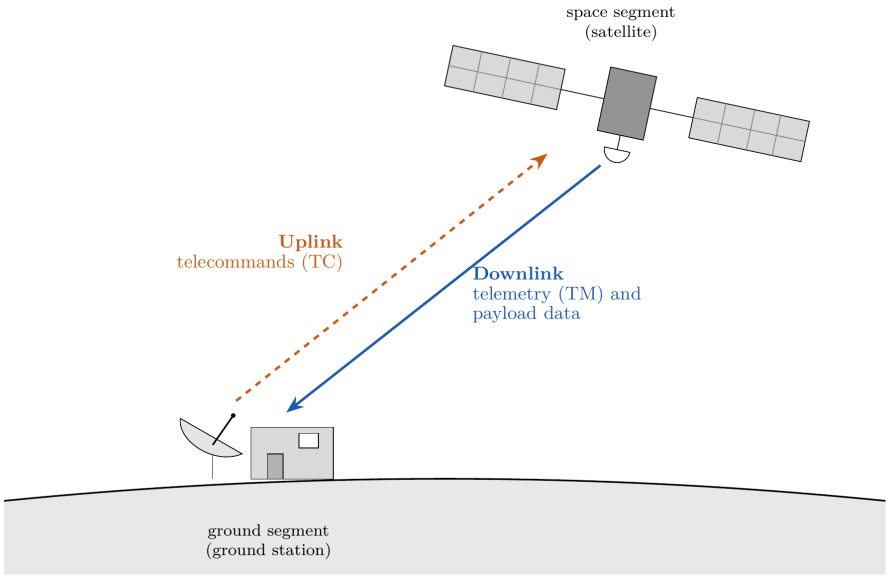


Figure 50: Uplink and downlink between the space segment and the ground station
Source: author (2026).

11.1 Theoretical Background

Radio-frequency Systems

Radio-frequency systems play an essential role in the exchange of data between satellites and ground stations, using electromagnetic waves that propagate through free space and the atmosphere (NASA, 2025). Architecturally, a radio communication link comprises a transmitter, a propagation channel, and a receiver (NASA, 2025). In the transmitting segment, the digital information passes through a data interface and is impressed onto a carrier wave by a modulator, and is then radiated by the antenna (NASA, 2025). At the opposite end, the receiving station captures these electromagnetic waves through its antenna and uses a demodulator circuit to recover the original digital message; these functions are often combined in a single transceiver device (NASA, 2025).

Because the space environment imposes severe challenges to signal integrity, channel coding techniques are employed to provide error detection and correction capabilities, ensuring reliable communication under the conditions imposed by the satellite transmission path (NASA, 2025). The theoretical limit on the transmission rate over a noisy channel is given by the Shannon–Hartley theorem:

$$\mathcal{C} = B \log_2 \left(1 + \frac{S}{N}\right) \tag{87}$$

In this equation, $\mathcal{C}$ is the channel capacity in bits per second, B is the channel bandwidth in hertz, S is the average received signal power over the bandwidth, and N is the average noise or interference power over the same band, with the ratio S/N expressed on a linear scale rather than in decibels (Rohini College of Engineering and Technology, 2025). As this relation shows, the channel capacity can be increased either by widening the bandwidth or by improving the signal-to-noise ratio (SNR) or the carrier-to-noise ratio (CNR), fundamental principles exploited by various modulation and coding schemes (NASA, 2025).

Satellite communications are carried out over a wide range of frequency bands (NASA, 2025). The bands typically considered for small satellites include UHF, S, X, and Ka, with VHF and UHF being the most established and mature frequencies for CubeSat communications (NASA, 2025), as detailed in Table 34.

Table 34: Radio-frequency bands used in satellite systems

Band	Frequency range
VHF	30 MHz to 300 MHz
UHF	300 MHz to 1000 MHz
L	1 GHz to 2 GHz
S	2 GHz to 4 GHz
C	4 GHz to 8 GHz
X	8 GHz to 12 GHz
Ku	12 GHz to 18 GHz
K	18 GHz to 27 GHz
Ka	27 GHz to 40 GHz
V	40 GHz to 75 GHz

Source: Adapted from (Institute of Electrical and Electronics Engineers, 2020).

Antennas

Antennas act as fundamental transducers in radio-frequency links, enabling both the transmission (Tx) and the reception (Rx) of electromagnetic waves between the transmission lines and the propagation medium (Ribeiro, 2019). When transmitting, they convert electrical quantities into radiating electromagnetic fields, whereas when receiving they perform the inverse process to feed the electronic circuits. This behavior is directly governed by the wavelength (λ), which is related to the operating frequency (f) and the speed of light in vacuum (c) by the expression:

$$\lambda = \frac{c}{f} \tag{88}$$

Directivity (D) expresses the ability of the device to concentrate the radiated energy in preferred directions compared with an ideal isotropic source; its maximum value (D_0) is the ratio of the maximum radiation intensity (U_{max}) to the average radiation intensity (U_{avg}) (Oliveira, A. L. A., 2020):

$$D_0 = \frac{U_{\text{max}}}{U_{\text{avg}}} = \frac{4\pi U_{\text{max}}}{P_{\text{rad}}} \tag{89}$$

where P_{rad} is the total power radiated by the structure. Combining this directivity with the radiation efficiency (η) yields the antenna gain (G_0), which indicates how much the antenna concentrates the input power in the direction of maximum radiation relative to a lossless isotropic antenna — the antenna is passive, so there is no power amplification (Castro; Franco, 2015):

$$G_0 = \eta D_0 \tag{90}$$

At the circuit level, the input impedance (Z_{in}) at the antenna terminals plays a critical role; it is represented as a complex quantity comprising a resistive component (R_{in}) — associated with the power actually dissipated and radiated — and a reactive component (X_{in}) — associated with the electromagnetic energy stored in the near-field region (Castro; Franco, 2015; Oliveira, A. L. A., 2020):

$$Z_{\text{in}} = R_{\text{in}} + jX_{\text{in}} \tag{91}$$

Proper matching of this impedance to the transmission line is imperative to maximize power transfer and mitigate reflection losses. Among the most traditional structural topologies,

the dipole antenna stands out for its simplicity and effectiveness, consisting of two collinear conducting elements fed at a central point (Hangar, 2025). Since each arm corresponds to approximately one quarter of a wavelength ($\lambda/4$), the structure totals about half a wavelength ($\lambda/2$) (Hangar, 2025). A dipole of exactly $\lambda/2$ has an input impedance of approximately $73\,\Omega + j42.5\,\Omega$; by shortening it slightly, the reactive part vanishes and the dipole becomes resonant, with a resistance close to $73\,\Omega$ (Balanis, 2016).

Radio Module

Radio-frequency modules are electronic devices that enable the transmission and reception of radio signals between two points without physical connections. They are essential components of wireless communication systems and may be transmitters, receivers, or transceivers, each serving a specific function.

The Ra-02, based on the SX1278 integrated circuit, is a radio module that both receives and transmits signals, i.e., it acts as a transceiver, capable of modulating and demodulating them. LoRa (Long Range) is a radio-frequency technology developed by Semtech Corporation and promoted by the LoRa Alliance, an open, non-profit association of companies seeking to spread the use of LoRa technology in the Internet of Things (IoT).

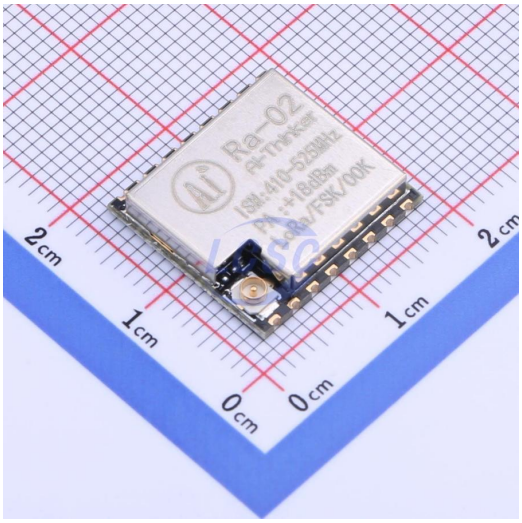


Figure 51: Ra-02 LoRa module (Ai-Thinker), based on the SX1278 transceiver
Source: (LCSC Electronics, 2023).

LoRa modulation is a spread-spectrum technique known as CSS (Chirp Spread Spectrum), originally employed in military and radar applications. In this technique, each symbol is a chirp — a pulse whose frequency varies linearly across the entire channel bandwidth — and the information is encoded in the offset of the chirp's starting frequency. Transmission rate is thus traded for processing gain: the data rate is low relative to the occupied bandwidth, but the receiver can detect very weak signals, which results in long range.

The Ra-02 module operates in the 410 MHz to 525 MHz range, which includes 433 MHz, with a maximum output power of (18 ± 1) dBm, and communicates with the microcontroller through SPI (Serial Peripheral Interface) (Shenzhen Ai-Thinker Technology, 2017). For the link-budget calculations, a receiver sensitivity of -137 dBm is adopted for spreading factor 12 and a bandwidth of 125 kHz, consistent with Table 48. The resulting link margin depends on the selected transmission parameters, antenna gains, losses, and propagation conditions. The output power of the module must be defined in the link budget, considering the 2.15 dBi gain of the dipole, the 0.15 dB loss of the balun, and the limits set by local spectrum regulations.

With LoRa™ spread-spectrum technology, the SX1278 achieves transmission distances considerably greater than those of traditional FSK- or OOK-based systems. At the highest data rates, its sensitivity is 8 dB better than that of FSK; using low-cost components, such as a 20 ppm crystal, this improvement can exceed 20 dB (Semtech Corporation, 2020). In addition, LoRa™ modulation increases communication reliability, with better selectivity and greater blocking immunity. For maximum flexibility, modulation parameters such as the bandwidth (BW), the spreading factor (SF), and the error-correction coding rate (CR) can be configured.

An additional benefit of this modulation is that the spreading factors are mutually orthogonal, allowing multiple signals to share the same channel with minimal interference, which also facilitates coexistence with already deployed FSK systems (Semtech Corporation, 2020). To ensure interoperability with existing standards and systems, such as Wireless M-Bus and IEEE 802.15.4g, the device also supports GFSK, FSK, OOK, and GMSK modulation. The SX1278 synthesizer operates in the 137 MHz to 175 MHz and 410 MHz to 525 MHz bands, with a configurable LoRa bandwidth between 7.8 kHz and 500 kHz (Semtech Corporation, 2020).

Figure 52 shows the block diagram of the SX1278 (Semtech Corporation, 2020).

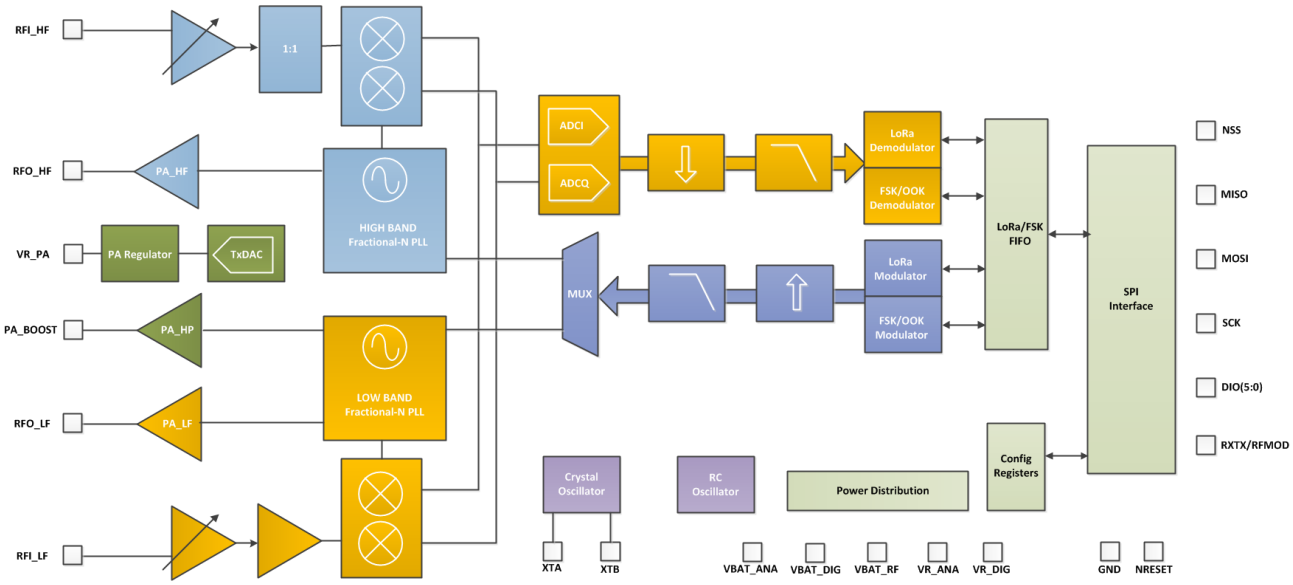


Figure 52: Block diagram of the SX1276/77/78/79 transceiver
Source: (Semtech Corporation, 2020, p. 9).

11.2 Modeling – LoRa and ADS-B Antennas

The hardware architecture of the telecommunications subsystem was developed to meet the applicable high-level requirements summarized in Table 35. Two independent radio links share a single antenna board: a LoRa link at 433.92 MHz for telemetry and telecommand (TM/TC) and a receive-only ADS-B link at 1090 MHz for the mission payload. Each antenna is a dipole fed by an LC lattice balun. The design was verified by electromagnetic co-simulation of the complete board: with commercial components, the worst in-band reflection is −18.4 dB (LoRa) and −19.0 dB (ADS-B), against a −10 dB requirement, and the −23.2 dB isolation between channels keeps the SDR input at about −5 dBm during a 18 dBm LoRa transmission.

Table 35: High-level requirements applicable to the RF subsystem

Requirement	Statement (summary)	Implication for balun and antenna
HLR-GEN-01	Mechanical envelope compatible with the CDS v14.1, within a volume equivalent to 1U, 2U or 3U	LoRa tape-springs stowed within the envelope until deployment; ADS-B dipole printed on the board (no deployable part)
HLR-GEN-02	Minimum set of subsystems, including the communication subsystem and the ADS-B payload	Two RF chains on one board: TM/TC at 433.92 MHz and ADS-B reception at 1090 MHz
HLR-GEN-04	Subsystem-level requirements derived from system-level requirements, documented and consistent	Derived RF requirements and acceptance criteria in Table 36
HLR-COMM-01	Receive TC from the ground segment and execute the corresponding actions within a defined response time	Bidirectional link: the LoRa balun operates in both transmission and reception
HLR-COMM-02	Structured TM including system status, power parameters, attitude information and mission data	The TM link must accommodate the processed ADS-B data
HLR-COMM-03	Physical layer, protocols and data rates defined, justified and consistent with the mission	The choice of 433.92 MHz requires explicit justification
HLR-ADS-01	Receive ADS-B signals at 1090 MHz and process them during the mission	Antenna and balun dedicated to reception; receiver protected against the LoRa transmission by inter-channel isolation
HLR-ADS-08	Success criteria over the 10 min window, with metrics for decoded messages, data loss and latency	ADS-B reception degraded during LoRa transmissions; transmit time logged and reported with the decoding metric
HLR-ENV-02	Functional survival after thermal and vibration testing, or justified limitations	Tape-spring attachments, connectors and components must survive the tests; matching re-measured afterwards
HLR-VV-02	Verification methods (test, analysis, inspection, demonstration) defined and justified	Method assigned to each derived requirement (Table 36); status in Table 41

Source: Adapted from (CubeDesign, 2026).

The frequency of 433.92 MHz, at the centre of the 433.05 MHz to 434.79 MHz ISM band, was adopted for the TM/TC link. A band below 1 GHz offers lower propagation loss than the 2.4 GHz bands, low-cost transceivers are widely available, and the LoRa data rate can be adjusted to the telemetry required by the ADS-B processing architecture (HLR-COMM-03). The link uses an Ra-02 module (SX1278, up to 18 dBm). The 1090 MHz link feeds the payload receiver, an RTL-SDR (NESDR Nano 3, R820T2 tuner), which captures the ADS-B messages from aircraft (HLR-ADS-01); since this band is reserved for aeronautical surveillance, the link is used exclusively for reception.

The CubeSat Design Specification (CDS) v14.1 envelope (The CubeSat Program, 2022) (HLR-GEN-01) rules out a rigid 433.92 MHz dipole. The LoRa antenna is therefore a dipole made of two flexible steel tape-springs, stowed during launch and deployed by a release mechanism. The same mechanism serves the intermediate-level test of the competition, which requires the telecommanded deployment of an antenna at least 10 cm long; each tape-spring is 125 mm long. A deployable dipole was also studied for ADS-B, but the simulations showed an anti-resonance at the usable arm lengths, and the length that would match ($\leq$ 35 mm per arm) is

too short to justify a mechanism. The ADS-B antenna is therefore a fixed dipole printed on the bottom layer of the board. Table 36 lists the RF requirements derived from the high-level requirements, with their acceptance criteria and verification methods (HLR-GEN-04, HLR-VV-02). The verification performed and its results are summarized in Table 41.

Table 36: Derived requirements, acceptance criteria and verification methods

ID	Requirement	Acceptance criterion	Traces to	Method
R1	LoRa matching	$\lvert S_{11}\rvert \leq -10\,\text{dB}$ at the LoRa connector over 433.05 MHz to 434.79 MHz, antenna deployed	HLR-COMM-01, HLR-COMM-03	A, T
R2	ADS-B matching	$\lvert S_{11}\rvert \leq -10\,\text{dB}$ at the ADS-B connector over 1085 MHz to 1095 MHz	HLR-ADS-01	A, T
R3	Receiver protection	Power at the SDR input $\leq 0\,\text{dBm}$ during LoRa transmission at flight power (goal: $\leq -10\,\text{dBm}$)	HLR-ADS-01, HLR-ADS-08	A, T
R4	Deployable LoRa antenna	Stowed within the envelope; deployed by telecommand; length $\geq 10\,\text{cm}$	HLR-GEN-01, intermediate test	I, D
R5	Fixed ADS-B antenna	Printed on the board, no deployable part	HLR-GEN-01	I
R6	Board constraints	Single two-layer FR4 board, 1.6 mm, no plated vias; no filters or layout changes for isolation purposes only	Fabrication constraint	I
M1	Internal design margin	$\lvert S_{11}\rvert \leq -15\,\text{dB}$ in both bands; $\lvert S_{21}\rvert \leq -20\,\text{dB}$ in the LoRa band	R1–R3	A
DC	Design criterion	Matching to $50\,\Omega$ has priority; output phase/amplitude balance is not enforced	Faculty advisor	—

A: analysis; T: test; I: inspection; D: demonstration.

Source: Authors (2026).

Criterion R3 follows from the power budget at the receiver. With the LoRa radio transmitting P_{TX}, the SDR receives

$$P_{\text{SDR}} = P_{\text{TX}} + |S_{21}|_{\text{dB}}, \tag{92}$$

where $|S_{21}|$ is the transmission between the two connectors. At 18 dBm, $|S_{21}| \leq -18\,\text{dB}$ is needed for 0 dBm and $-28\,\text{dB}$ for $-10\,\text{dBm}$. Linear operation of the tuner (below about $-30\,\text{dBm}$) would require about 48 dB, which is not achievable without a filter (R6). The ADS-B receiver is therefore expected to be saturated while the LoRa radio transmits. Messages corrupted during a transmission fail the decoder's parity check and are not recorded, and the telecom daemon logs every transmit interval, so the reception time lost is known and is reported with the HLR-ADS-08 metrics (Section 11.3.4).

11.2.1 Balun Topology

The radio and the coaxial cable are unbalanced (signal referenced to the cable shield), whereas the dipole is a balanced load that requires currents of equal amplitude and opposite phase in its arms (Balanis, 2016). Fed directly, part of the current returns along the outside of the shield as a common-mode current: the cable radiates, the pattern is distorted, and the

matching becomes dependent on the position of cables and structure, which in a CubeSat are only a few centimetres from the antenna. The balun is intended to provide a balanced feed and, in the same circuit, to transform the 50 Ω of the radio to the feed-point impedance of the antenna. The adopted topology, a lumped-element LC lattice balun (Bakalski *et al.*, 2002), is shown in Figure 53.

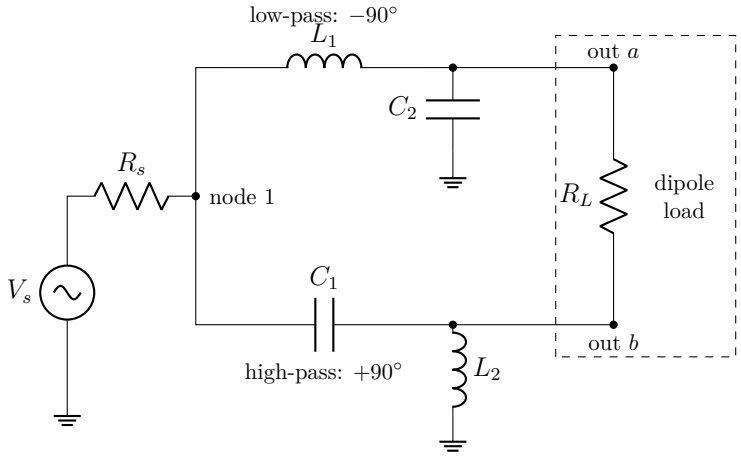


Figure 53: Adopted LC lattice balun topology. In the final LoRa balun, C_2 is not populated (Table 39)

Source: Authors (2026).

The low-pass branch (series L_1, shunt C_2) delays the signal by 90° and the high-pass branch (series C_1, shunt L_2) advances it by 90°. With equal L and C in both branches,

$$\frac{V_b}{V_a} = -\left(\frac{f}{f_0}\right)^2, \qquad f_0 = \frac{1}{2\pi\sqrt{LC}}, \tag{93}$$

so the 180° difference is structural, while the amplitude imbalance, $40\log_{10}(f/f_0)$ dB, is only 0.03 dB at the edges of the 433 MHz band. For a source resistance R_S and a resistive load R_L, the classical values are

$$L = \frac{\sqrt{R_S R_L}}{\omega_0}, \qquad C = \frac{1}{\omega_0\sqrt{R_S R_L}}, \tag{94}$$

which, for 50 Ω and the 73 Ω of an ideal half-wave dipole, give 22 nH and 6.0 pF at 433.92 MHz, and 8.8 nH and 2.4 pF at 1090 MHz.

These values were only the starting point. The simulation of the complete board showed that the load seen by each balun (antenna, feed traces and ground planes) is neither 73 Ω nor resistive, and differs between the two outputs; with the theoretical values the match was only about −3 dB. With four degrees of freedom, perfect matching and perfect balance cannot be imposed together, so, following criterion DC, the final values were optimized for matching and isolation, which makes both baluns asymmetric. The structural 180° of Eq. (93) no longer applies; the balance actually obtained is discussed in Section 11.2.4.

11.2.2 Dimensions and Board Definition

The initial dipole length was estimated as $0.95\,\lambda_0/2$ (Balanis, 2016). The final antennas are shorter (Table 37) because the feed traces between balun and antenna have no ground plane underneath and are collinear with the arms, so they radiate as the first segment of each arm, and because the 19 mm LoRa tape-spring is wide, which shortens the resonant length. The board is also not electrically small ($0.12\lambda_0$ at 433 MHz, $0.30\lambda_0$ at 1090 MHz), so its ground

planes take part in radiation; the electrically short ADS-B dipole has a capacitive reactance that the balun compensates.

Table 37: Dipole dimensions: initial estimate and final design

	LoRa (433.92 MHz)	ADS-B (1090 MHz)
Free-space wavelength λ_0	690.9 mm	275.0 mm
Initial estimate ($0.95\,\lambda_0/2$)		
Total length	328.2 mm	130.6 mm
Per arm	164.1 mm	65.3 mm
Final design		
Total length (tip to tip)	304.2 mm ($0.44\,\lambda_0$)	$\approx$ 83 mm ($0.30\,\lambda_0$)
Radiating element per arm	125 mm tape-spring	14.4 mm printed arm

Source: Authors (2026).

Both circuits share one two-layer board (Figure 54, Table 38): LoRa on the top layer and ADS-B, with its printed dipole, on the bottom layer. Each channel has its connector, balun and feed on the same layer, so there is no via in the RF path; at 1090 MHz, a via through 1.6 mm of FR4 ($\approx$ 0.5 nH) would represent about 20 % of the 2.4 nH series inductor of the final balun.

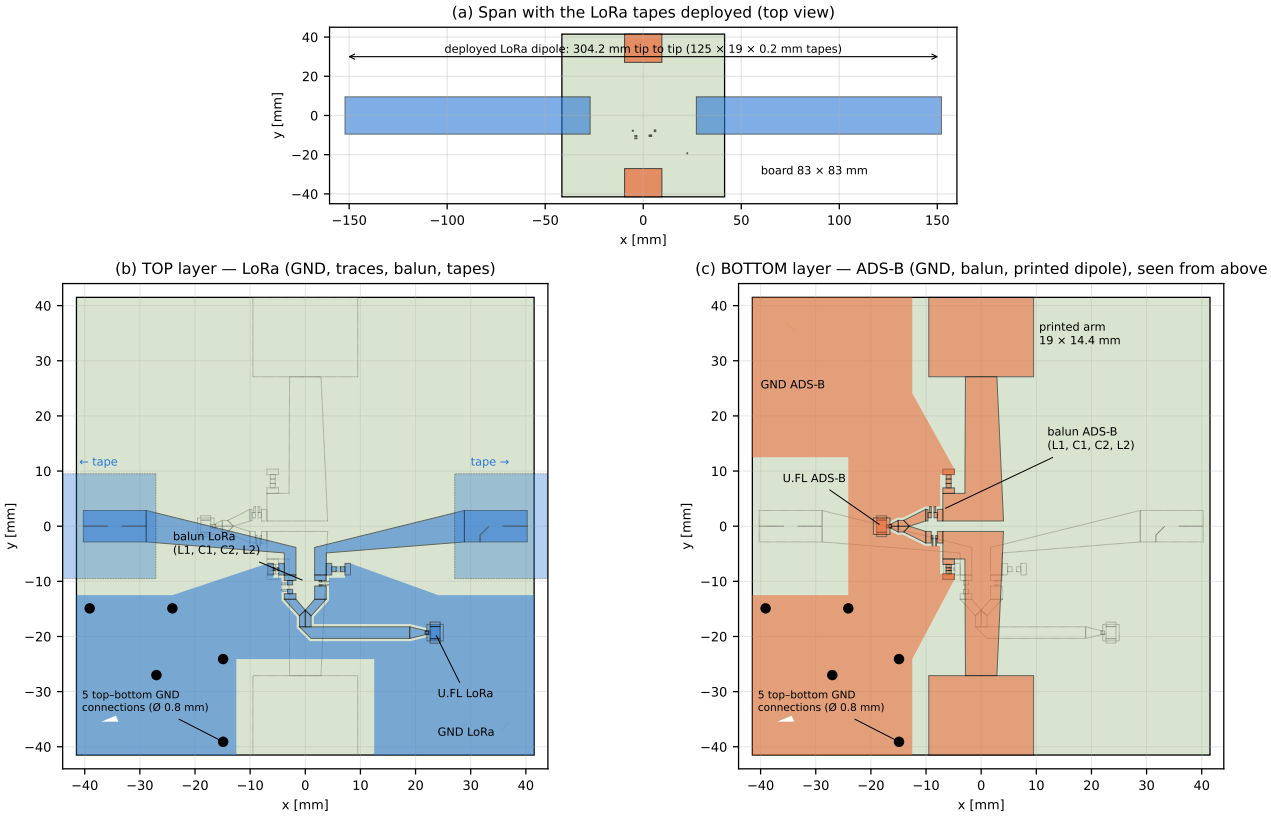


Figure 54: Final geometry. (a) Span with the LoRa tape-springs deployed. (b) Top layer: LoRa ground plane, tapered feed traces, balun, U.FL connector and tape-spring mounting area (dashed). (c) Bottom layer, seen from above: ADS-B ground plane, balun, U.FL connector and printed dipole. Dotted outlines show the copper on the opposite layer; black circles mark the five ground-plane interconnections

Source: Authors (2026), from the final HFSS model.

Table 38: Board, antenna and component definitions

Common to both channels		
Substrate	FR4, 83 mm × 83 mm, 1.6 mm, 2 layers; $\varepsilon_r = 4.4$, $\tan\delta = 0.02$; 1 oz copper	
Components	C0G/NP0 capacitors (±0.1 pF); high-Q RF inductors (±2 % or ±0.1 nH); package 0603	
Connectors	U.FL, one per channel, on the layer of its circuit	
Ground links	5 links, 0.8 mm in diameter, where the ground planes overlap	
	*LoRa (*433.92 MHz*)*	*ADS-B (*1090 MHz*)*
Layer	Top	Bottom
Antenna	2 steel tape-springs, 125 mm × 19 mm × 0.2 mm	2 printed arms, 19 mm × 14.4 mm
Position	27.1 mm ≤ $\lvert x\rvert$ ≤ 152.1 mm (14.4 mm overlap with the board)	27.1 mm ≤ $\lvert y\rvert$ ≤ 41.5 mm (up to the board edge)
Feed traces	Tapered, collinear with the arms	5 mm wide, 6 mm lateral offset
Component pads	L: 0.9 mm wide, 0.8 mm gap C: 0.5 mm wide, 0.5 mm gap	L1, C1, C2: 0.5 mm gap L2: 0.8 mm gap

Source: Authors (2026).

Two features of this geometry govern its electrical behaviour. First, the feed traces have no ground plane underneath and are collinear with the antennas, so they are not transmission lines but the first segment of each dipole arm. Second, the board ($0.12\lambda_0$ at 433 MHz and $0.30\lambda_0$ at 1090 MHz) is not electrically small and its two ground planes are connected, so the ground planes take part in radiation and carry common-mode currents. Consequently, the ADS-B dipole, electrically short at $0.30\lambda_0$, has a capacitive input reactance that the balun compensates, and part of its radiation comes from the board itself.

11.2.3 Simulation Methodology

Electromagnetic model

The board was solved in Ansys HFSS 2024 R1 (ANSYS. . . , 2024) (Driven Modal, finite elements with adaptive mesh). Copper and steel were modelled as PEC: the skin depth (3.2 µm in copper at 433 MHz) is far below the conductor thickness, and the steel loss is below 0.01 dB, so the dominant loss is the FR4. The air region has a radiation boundary 190 mm from the structure in every direction (more than $\lambda/4$ at 433 MHz). Fourteen 50 Ω lumped ports were placed: one per connector and one per component terminal, with the pads left open, so that no component exists in the field model. The mesh was refined to 0.2 mm on the ports and 0.5 mm on the baluns, and the multi-frequency adaptation at 433 MHz and 1090 MHz converged in six passes (636 913 tetrahedra) to $\max|\Delta S| = 0.0034$ and 0.0048, below the 0.01 criterion. An interpolating sweep from 350 MHz to 1300 MHz produced the 14×14 S-matrix.

Circuit co-simulation and optimization

The S-matrix was imported into Ansys Circuit (Nexxim) by dynamic link, where each component is connected between its ports and both connectors are terminated in 50 Ω. Since the components exist only in the circuit, each new set of values is evaluated in seconds without a new field solution. High-impedance probes at the balun outputs give the phase difference and voltage imbalance between the two nodes. The components were optimized with the Optimetrics quasi-Newton algorithm, first the ADS-B balun ($|S_{11}| \leq -15$ dB in its band and $|S_{21}| \leq -20$ dB in the LoRa band) and then the LoRa balun with the ADS-B one fixed. Each optimum was followed by a sweep of the neighbouring commercial values (27 combinations for ADS-B, 54 for LoRa, including C2 not populated), and the combination with the largest

worst-case margin over M1 was selected. Both baluns were always simulated together, because the channels are strongly coupled: changing only the ADS-B balun moved the LoRa $|S_{11}|$ from $-13.8\,\mathrm{dB}$ to $-10.7\,\mathrm{dB}$.

Currents and patterns

Push Excitations returns the circuit's port voltages and currents to HFSS, so currents and patterns include the components. One connector was excited at a time, with the other terminated in $50\,\Omega$. The reported gain is *SystemGainTotal*, referenced to the available source power, so it includes mismatch, dielectric loss and coupling into the other channel, as required by the link budget. Coverage metrics were computed in Python from the exported patterns, weighted by solid angle.

11.2.4 Results

Table 39 lists the selected components and Figure 55 the resulting S-parameters. The worst in-band reflection is $-18.4\,\mathrm{dB}$ for LoRa ($-17.1\,\mathrm{dB}$ over a wider 430 MHz to 438 MHz, giving robustness to detuning) and $-19.0\,\mathrm{dB}$ for ADS-B; the impedances at the centre frequencies are $(50.2 + \mathrm{j}11.9)\,\Omega$ and $(46.7 - \mathrm{j}5.6)\,\Omega$. The isolation in the LoRa band is $-23.2\,\mathrm{dB}$, which by Eq. (92) gives $-5.2\,\mathrm{dBm}$ at the SDR for 18 dBm and $-9.2\,\mathrm{dBm}$ for 14 dBm. It is helped by the ADS-B balun itself, which reflects almost all the power at 433 MHz ($|S_{11}^{\text{ADS-B}}| \approx -0.06\,\mathrm{dB}$) and acts as a filter without additional components. At 1090 MHz the coupling is $-12.3\,\mathrm{dB}$, which costs about $0.3\,\mathrm{dB}$ of the ADS-B received power.

Table 39: Final balun components (commercial values)

Balun	L1 (series)	C1 (series)	C2 (shunt)	L2 (shunt)
LoRa	12 nH	5.1 pF	not populated	22 nH
ADS-B	2.4 nH	10 pF	4.7 pF	3.0 nH

Source: Authors (2026).

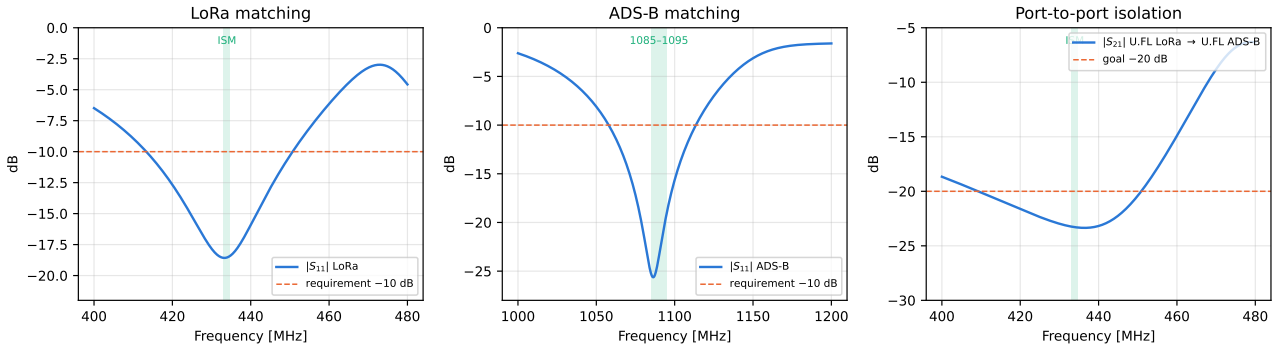
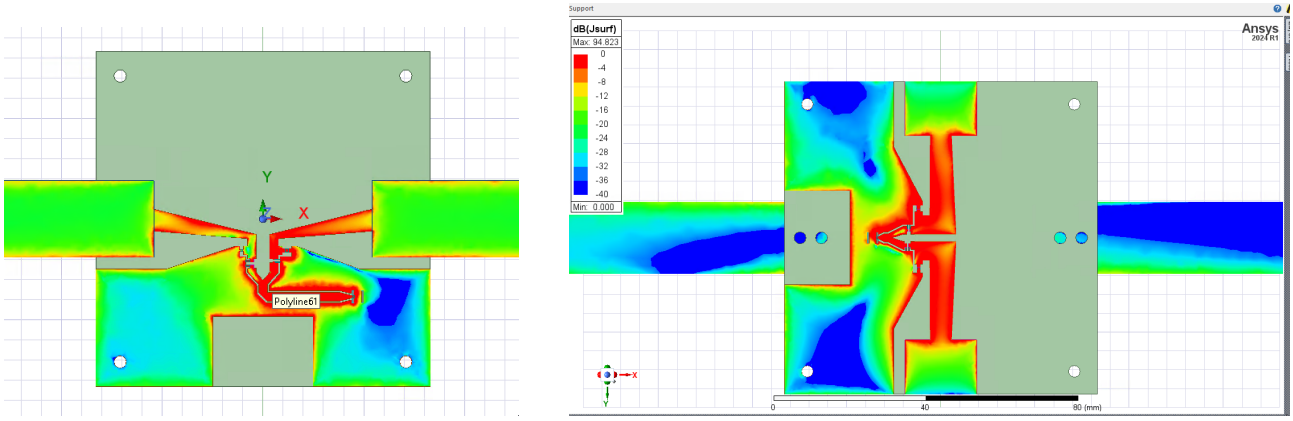


Figure 55: S-parameters with the commercial components: LoRa matching (left), ADS-B matching (centre) and isolation between the connectors (right). Shaded bands: operating bands

Source: Authors (2026), Ansys Circuit.

Replacing one component by the adjacent commercial value shows that the LoRa C1 is the most sensitive ($5.1 \to 5.6\,\mathrm{pF}$ degrades $|S_{11}|$ to $-14.0\,\mathrm{dB}$), followed by the LoRa L1 (about 2 dB per nH); L2 and C2 have moderate or negligible effect, and the isolation varies by less than 0.5 dB. With $\pm 0.1\,\mathrm{pF}$ capacitors the expected worst-case degradation is about 1 dB, well within the 8 dB margin. The LoRa L1 (12/11 nH) and the ADS-B L2 (3.0/2.7 nH) are the designated bench-tuning components.

The probes indicate 170.7° and −15.0 dB for LoRa and −53° and −2.3 dB for ADS-B, far from the ideal 180°, as expected from asymmetric baluns. These are, however, node voltages relative to a ground plane that is itself radiating, and they can differ widely while the differential current in the arms remains balanced. The current distribution confirms this: both LoRa tape-springs carry similar currents (Figure 56(a)), and with ADS-B excited the current concentrates on the printed arms (Figure 56(b)), while the current induced on the tape-springs is 20 dB to 40 dB lower.



(a) LoRa excited (434 MHz), top layer (b) ADS-B excited (1090 MHz), bottom layer

Figure 56: Surface current density $|\mathbf{J}_s|$ (−40 dB(A/m) to 0 dB(A/m))
Source: Authors (2026), Ansys HFSS.

Both antennas radiate as dipoles (Figures 57 and 58, Table 40). The LoRa pattern is a toroid along x with deep nulls at the tape-spring tips and nearly constant gain in the YZ plane. The ADS-B pattern, along y, has shallow and asymmetric nulls filled by radiation from the board, but practically the same coverage above −3 dBi. Of the ADS-B power, 5.9 % couples into the LoRa connector and about 12 % is dissipated, mostly in the FR4; for LoRa, losses are below the numerical resolution (about 4 %).

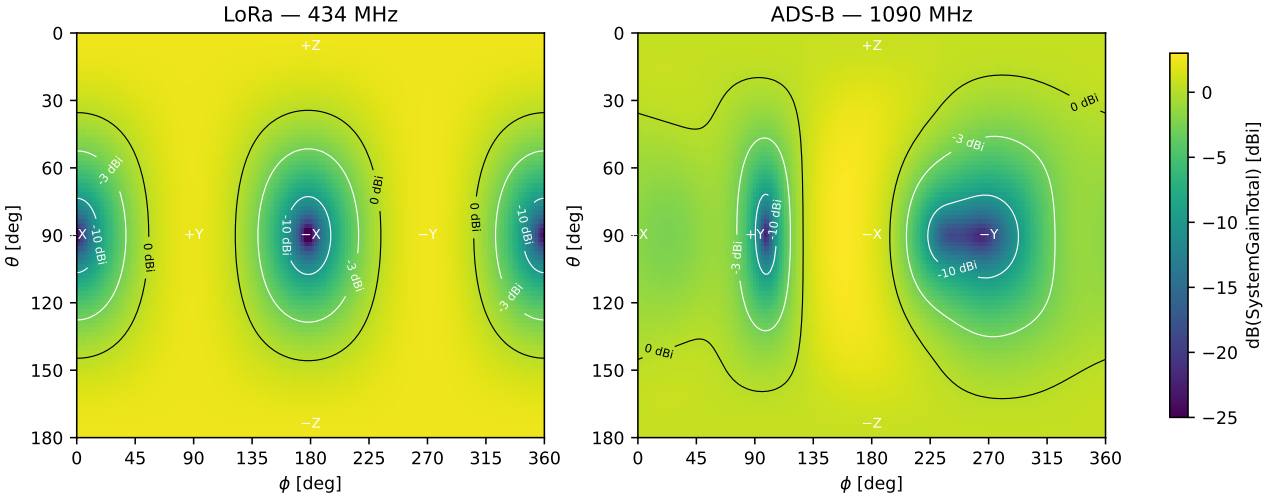


Figure 57: Gain maps (*SystemGainTotal*) with contours at 0, −3 and −10 dBi. Board axes: x along the LoRa tape-springs, y along the ADS-B dipole, z normal to the board
Source: Authors (2026).

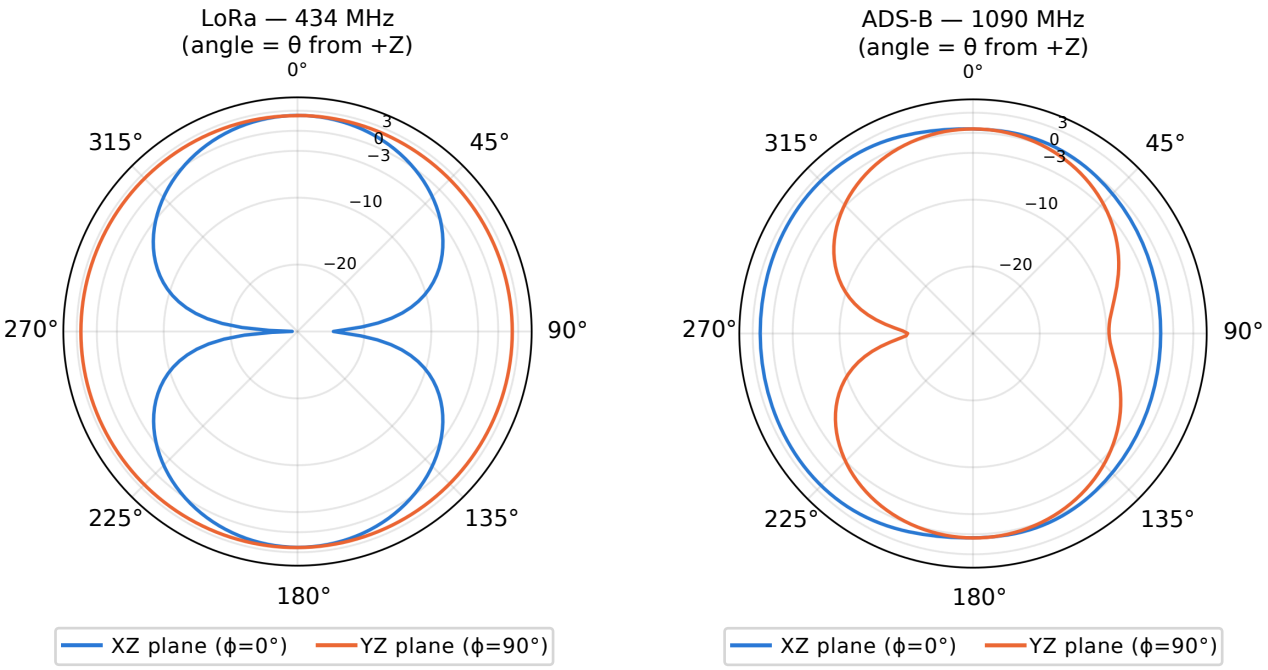


Figure 58: Principal cuts in the XZ ($\phi = 0°$) and YZ ($\phi = 90°$) planes; θ measured from $+z$
Source: Authors (2026).

Table 40: Radiation metrics

Metric	LoRa (434 MHz)	ADS-B (1090 MHz)
Peak gain	2.4 dBi	2.9 dBi
Gain along $\pm z$ (board normal)	2.3 dBi	0.6 dBi
Nulls (direction, depth)	$\pm x$, -25 dBi to -41 dBi	$\pm y$, -10 dBi to -23 dBi
Sphere with $G \geq 0$ dBi	56.9 %	31.5 %
Sphere with $G \geq -3$ dBi	78.6 %	75.1 %
Sphere with $G \geq -10$ dBi	95.6 %	94.1 %
Total efficiency	≈ 100 %	81.5 %

Source: Authors (2026).

11.2.5 Verification Status and Remaining Risks

Table 41 closes the loop with Table 36, stating for each derived requirement the internal verification performed, its result and the remaining test. Bench tests use a vector network analyzer (VNA) with ferrite chokes on the cable, and are repeated after the environmental tests (HLR-ENV-02).

Table 41: Verification matrix of the derived requirements

ID	Internal verification (tool)	Result	Remaining test	Status
R1	HFSS + Circuit co-simulation, tape-springs deployed	-18.4 dB (8.4 dB margin)	VNA S_{11}, deployed and stowed	Met by analysis; test pending
R2	HFSS + Circuit co-simulation	-19.0 dB (9.0 dB margin)	VNA S_{11}	Met by analysis; test pending
R3	Co-simulated $\lvert S_{21}\rvert$ and Eq. (92)	-5.2 dBm at 18 dBm	Calibrated attenuator or power meter in place of the SDR	Met; goal only at ≤ 14 dBm
R4	Inspection of the mechanical design	125 mm per tape-spring	Deployment by TC (intermediate test)	Met by design; demo pending
R5, R6	Inspection of the final layout	Printed dipole; 2 layers, no vias, no filters	—	Met
M1	Optimetrics goals and commercial-value sweep	$\lvert S_{11}\rvert \leq -18.4$ dB; $\lvert S_{21}\rvert = -23.2$ dB	Two-port VNA isolation	Met by analysis

Source: Authors (2026).

11.3 Software

This subsection describes the software of the TT&C subsystem and the onboard part of the ADS-B payload software, as designed for the on-site test: the facts that drive the design and the decisions taken from them, the onboard architecture, the radio link, the operation in each mission mode, the ground segment, the data the software produces and the practices used to implement it. The design is described as a whole. The verification and validation subsection that follows states, criterion by criterion, which parts are verified in software, which were validated on hardware and which are still planned. The source code, the decision records, the budgets and the test logs cited here are in the project repository (Gama CubeDesign, 2026).

11.3.1 Design Drivers and Decision Records

Four facts shape the software:

1. **The payload produces 129 times more data than the radio can carry.** Twenty aircraft over the 10-minute mission generate 7.34 MiB of decoded messages, while the LoRa link carries 58.3 KiB in the same time (Table 70). Everything that moves bytes over the radio is designed against this ratio.
2. **The radio is half-duplex.** While the satellite transmits it cannot hear a telecommand, and if both ends transmit at once both frames are lost. The rules nevertheless require a bounded telecommand response time defined by the team (HLR-COMM-01) and measured data loss and latency (HLR-ADS-08).
3. **The onboard computer has no real-time clock**, and there is no network at the venue, yet every message needs a timestamp good enough to reconstruct trajectories (HLR-ADS-07).
4. **Three groups write the onboard software in two repositories**: the OBC team, the

ADS-B sub-team and the LoRa sub-team. The interfaces between them must be explicit and testable by each side alone.

Every architectural decision is recorded as an Architecture Decision Record (Architecture Decision Record (ADR)): a short numbered document, written in English when the code materialises the decision, stating the context, the decision, the alternatives rejected and the consequences accepted (Nygard, 2011). An accepted record is immutable. A decision that changes receives a new record, and a wrong figure receives an erratum at the end of the record, so the reasoning that was once correct remains as evidence. A figure that comes from a calculation is produced by a script in the repository, and the test suite fails if a record quotes a value the script no longer reproduces. The records answer the requirement that design decisions be justified against performance, cost and complexity (HLR-COST-02) and that the communication and processing architectures be justified (HLR-COMM-03, HLR-ADS-04). Table 42 summarises the records that define the architecture; two further records fix the process itself and the documentation language.

Table 42: Architecture decision records of the TT&C and payload software

ADR	Decision	Main alternative rejected	Deciding fact	HLR
0002	C11 on board and on the ground station; one codec compiled from the same source files at both ends of the link	C++ on board and C on the ground station	the wire format must be compiled once, not written twice, so that the two ends cannot diverge	SW-02, SYS-01
0003	Hybrid processing, split at the track-state boundary (Section 11.3.2)	everything on the ground; store and forward after the mission	the source is 129 times the link capacity; store and forward makes the latency equal to the mission duration	ADS-04, ADS-05, ADS-08
0004	Three fixed LoRa profiles at 125 kHz, selected by telecommand, with automatic revert (Section 11.3.4)	spreading factor 12 at 62.5 kHz; automatic rate adaptation	spreading factor 12 at 62.5 kHz buys 8 dB of sensitivity on a link with more than 70 dB of margin, at 22 times less capacity	COMM-03, ADS-08
0005	Separate processes under the system supervisor, with the telecom daemon as hub; the OBC as sole owner of the mission modes; decision logic in a pure core	one multi-threaded program	the radio is the only way to diagnose and recover the satellite, so no payload fault may stop it	GEN-03, COMM-01
0006	Local sockets that preserve message boundaries, each message one radio frame	files, pipes, TCP, message bus	a frame arrives whole or not at all, and a dead peer is detected at once; 10 000 frames each way without loss	COMM-01, COMM-02
0007	The link protocol: reply gap and listen-before-talk, acknowledgement of every telecommand, execution at most once, profile change with revert, fallback to the safe profile, track age, response bound (Section 11.3.5)	transmit at any time and rely on retransmission; scheduled uplink slots	the first channel simulation had 170 collisions in 401 telecommands; with the rules, 69, and the 95th percentile fell from 2.44 s to 1.78 s	COMM-01, COMM-03, ADS-07, ADS-08

Continued on the next page.

Table 42 — continued

ADR	Decision	Main alternative rejected	Deciding fact	HLR
0008	The payload daemon owns the SDR, through the dump1090-fa decoder	the OBC opens the SDR; a hardware decoder module	the procured receiver delivers raw samples and decodes nothing; dump1090-fa provides parity check, single-bit correction and position decoding	ADS-01 to ADS-04
0009	Monotonic clocks in every process; wall-clock time from an anchor sent by the ground (Section 11.3.6)	the system wall clock; setting the clock; network time; a GPS receiver	no real-time clock and no network on site; the crystal drifts 30 ms over a 10-minute mission	ADS-07, SW-02
0012	Radio access through the kernel's SPI and GPIO character-device interfaces, without root	a GPIO library with a root daemon and a callback thread	the radio interrupt becomes one more file descriptor in the event loop: no thread, no privileges	COMM-01, COMM-03

Source: Authors (2026), from the decision records in the project repository (Gama CubeDesign, 2026).

Two parameters are set in this review and will be recorded in a new decision record: the carrier frequency of 433.92 MHz, at the centre of the band the antennas were designed for, and the flight transmit power of 14 dBm, chosen to protect the ADS-B receiver (Section 4.3.5).

11.3.2 Processing Split

Link capacity and allocation

At the nominal radio profile (Section 11.3.4), a frame of $L_f = 208$ B takes $T_{\text{air}} = 1.046$ s on air, a goodput of $R_g = 8L_f/T_{\text{air}} = 1592$ bit/s. The radio is half-duplex: while it transmits it cannot hear a telecommand, so at most half of the mission is allocated to transmission ($\delta = 0.5$), keeping the other half for listening (HLR-COMM-01). The capacity of the link over the mission is

$$C = T\,\delta\,\frac{R_g}{8} = 600 \times 0.5 \times \frac{1592}{8} \approx 59.7\,\text{kB} = 58.3\,\text{KiB}, \tag{95}$$

129 times less than the 7.34 MiB of decoded messages the payload produces (Table 70). The link carries one snapshot of every tracked aircraft per period: 20 aircraft at 20 B per track record fill two frames (247 B and 167 B), 2.09 s on air. Table 43 allocates the 300 s of transmit time. A snapshot every 4.2 s is the fastest cadence the air time allows, but it would use 299 s of the 300 s and leave no room for housekeeping; a 5 s period gives 120 trajectory points per aircraft and keeps 26 s (9 %) in reserve. The radio therefore transmits 46 % of the mission time, a figure that the power budget and the coexistence with the ADS-B receiver both depend on.

Table 43: Allocation of the transmit time over the 600 s mission (nominal profile)

Item	Frame size	Cadence	Count	Air time	Volume
Track snapshot, 20 aircraft	2 frames, 414 B	5 s	120	251 s	48.5 KiB
Housekeeping (`TM_HK`)	32 B	10 s	60	15 s	1.9 KiB
Statistics (`TM_STAT`)	33 B	30 s	20	5 s	0.6 KiB
Telecommand acknowledgements	11 B	on demand	20 (budgeted)	3 s	0.2 KiB
Total				**274 s of 300 s**	

Source: Authors (2026), computed by `lora_budget.py` (Gama CubeDesign, 2026).

Where to split

HLR-ADS-04 offers three ways to process the mission data: on board, on the ground, or hybrid. The arithmetic of the data budget shows that they are not equivalent. Processing on the ground does not relieve the downlink bottleneck, because the data must cross the link first: only the *analysis* of the tracks can move to the ground, not the *decoding and aggregation* of the messages. Each candidate boundary was measured against the 58.3 KiB the link carries (Table 44).

Table 44: Candidate processing boundaries against the link capacity over the mission

Downlink the…	Volume in 600 s	Fits in 58.3 KiB?
raw IQ samples (2.4 MS/s, 2 bytes each)	≈ 2.9 GB	no, about 50 000 times over
raw 1090ES frames (14 B each)	0.64 MiB	no, 11 times over
decoded messages (the onboard record)	7.34 MiB	no, 129 times over
the state of each aircraft, every 5 s	**48.5 KiB**	**yes, 83 % of the capacity**
one summary per aircraft at the end	≈ 0.4 KiB	yes, but without trajectory

Source: Authors (2026), computed by `lora_budget.py` (Gama CubeDesign, 2026).

The processing is therefore hybrid, with the split at the *track state*: the first boundary that fits in the link and the last one that still carries a trajectory. A single summary per aircraft would satisfy the letter of "transmitting processed data as telemetry" while making trajectory reconstruction, a primary objective, impossible.

- **On board**, the decoder demodulates and decodes the 1090ES messages from the SDR samples; the payload daemon timestamps each message on arrival, appends every message to the onboard record and keeps the current state of each aircraft, identified by its ICAO address.
- **Over the radio**, one snapshot of every tracked aircraft is transmitted every 5 s, 20 B per aircraft, together with housekeeping, statistics and acknowledgements (Table 43).
- **On the ground**, the received snapshots are turned into trajectories, and the origin and destination of each aircraft are estimated (Section 12).
- **After the test**, the complete onboard record is retrieved over USB. HLR-SW-02 requires mission data to be structured, documented and reproducible, and the rules add "available for post-analysis"; none of these requires the data to arrive by radio.

Three alternatives were rejected. Processing everything on board, down to the origin and destination, would run the analysis on a computer already shared with the decoder, for no gain, and would make the result unauditable: a reconstructed trajectory can be checked, a conclusion computed out of sight cannot. Storing everything and downlinking it after the mission would make the latency between reception and transmission equal to the mission duration, which HLR-ADS-08 asks to bound; at the fastest profile the complete record would take about 3.4 hours of continuous transmission. Sending only the aircraft whose state changed would save air time but make loss self-concealing, because a missing aircraft would be indistinguishable from one that did not move, while HLR-ADS-08 requires the loss rate to be measured.

The split has accepted costs. Positions are quantised to about 1.2 m in latitude and 2.4 m in longitude, and time to 100 ms, below the accuracy of ADS-B itself. A lost frame loses up to 12 aircraft for one snapshot; snapshots are not retransmitted, because the next one is 5 s away and sequence numbers make the loss countable. The onboard processor becomes mission-critical: the decoder load was the largest unquantified risk of the architecture until it was measured on the target at 21.9 % of one core (Section 11.4.3). The choice of boundary was checked in software by replaying complete captures through an emulation of the downlink: the ground estimator reached the same conclusions from the snapshots, one fifth of the observations,

as from the complete stream, also with 10 % of the frames lost, in a synthetic scenario (Gama CubeDesign, 2026).

11.3.3 Onboard Software Architecture

All onboard software runs on the Raspberry Pi Zero 2 W of the OBC (four cores at 1 GHz, 512 MB of memory) under the 64-bit Raspberry Pi OS (Raspberry Pi Ltd, 2026). It is split into four processes, supervised by systemd and run without administrator privileges (Figure 59 and Table 45). The criterion for the split is what must keep working when something else fails. The radio is the only way to diagnose and recover the satellite, so no fault of the payload or of the OBC software may stop it: a crash in the payload parser terminates only the payload daemon, which systemd restarts (Systemd project, 2026), while the telecom daemon keeps transmitting housekeeping and answering telecommands (ADR-0005).

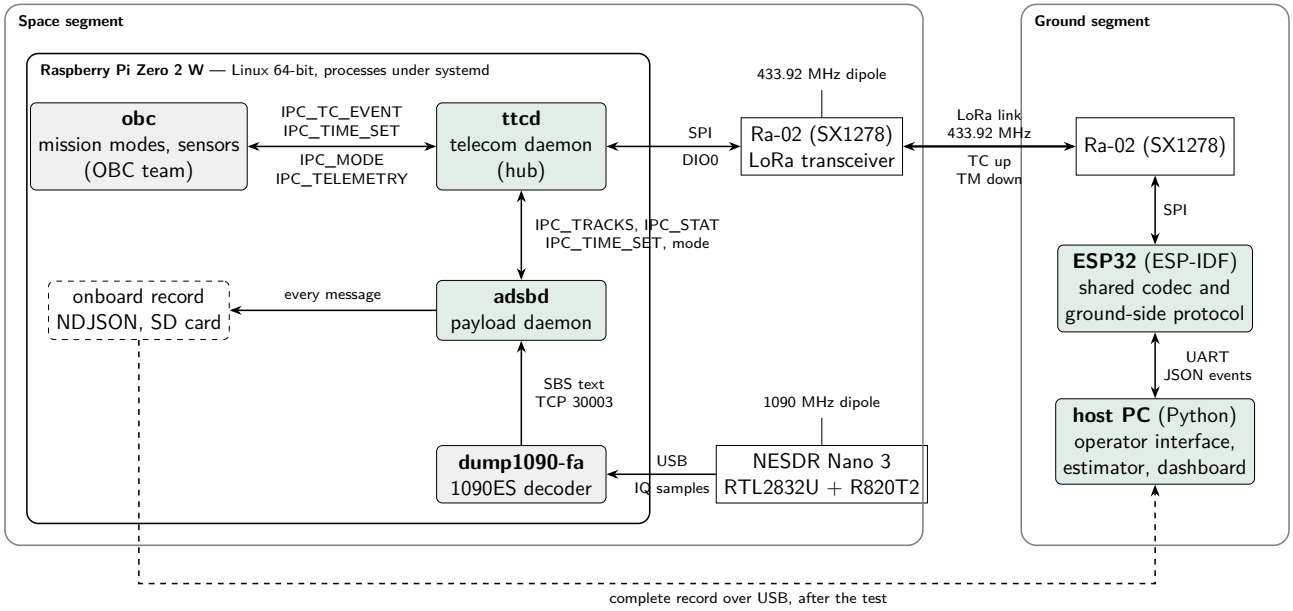


Figure 59: Onboard software architecture and ground segment. Green: software of the TT&C subsystem; grey: third-party decoder and the OBC software; dashed: the onboard record and its retrieval

Source: Authors (2026).

The telecom daemon is the hub: it listens on one local socket, and the payload daemon and the OBC software connect to it; they never talk to each other directly. The OBC is the single owner of the mission mode. The telecom daemon keeps no copy of the mode state machine: it translates a ground telecommand into an OBC event, reports the mode the OBC announces and forwards it to the payload daemon. Two state machines describing the same modes would drift apart, and the day they disagreed the satellite would report one mode and behave as another.

Both daemons are built as a *pure core* inside a thin *shell*. The core holds every decision — the link state machine, telecommand dispatch and telemetry scheduling in `ttcd`; the message parser, the track table, the record and the snapshots in `adsbd` — and makes no system call: it receives events stamped with a time (a frame arrived, a transmission ended, a timer expired) and acts through a small table of operations (transmit this frame, change profile, send this message). The shell owns the file descriptors, the clock and a single-threaded event loop built on `epoll` with timer, signal and event descriptors (Kerrisk, 2010); nothing in it blocks or sleeps. Because the clock is injected, the link protocol is exercised for hours of simulated time

in milliseconds, against the real ground-side code on a simulated half-duplex channel, and every run is repeatable (Section 11.4.2).

Table 45: Onboard processes

Process	Owns	Written by	Serves
`ttcd`	the LoRa radio, the ground protocol, telecommand dispatch, telemetry scheduling, the IPC hub and the wall-clock anchor	LoRa sub-team	COMM-01/02/03
`adsbd`	the ADS-B chain: starts and supervises the decoder, keeps the track table, writes the onboard record, builds the snapshots	ADS-B sub-team	ADS-01 to 05, 07
`dump1090-fa`	the SDR; demodulation and decoding of 1090ES messages, published as SBS text on a local TCP port	FlightAware, child of `adsbd`	ADS-01, ADS-02
`obc`	the mission modes, the platform sensors, the interface to the attitude controller, antenna deployment and the heaters	OBC team	SW-01, EPS-04

Source: Authors (2026), after ADR-0005 and ADR-0008 (Gama CubeDesign, 2026).

Inter-process communication

The processes exchange messages through a local socket of type `SOCK_SEQPACKET` (ADR-0006), which preserves message boundaries: a message arrives whole or not at all, so there is no stream to resynchronise, and the death of a peer is seen at once as the end of its connection (Kerrisk, 2010). Every message is one frame with exactly the header and checksum used on the radio, so the code that carries telecommands to the OBC is the code exercised by every radio test. The first frame of a connection identifies the peer and its interface version; a mismatch closes the connection. Every call is non-blocking: a message to a full queue is counted as dropped instead of waited on. Table 46 lists the messages. The OBC software needs one client of about a hundred lines, delivered by the TT&C team as a reference implementation that fills the radio polling function its main loop already calls.

Table 46: Messages between the onboard processes

Message	Direction	Payload	Content
`IPC_HELLO`	client → `ttcd`	2 B	role of the peer and interface version
`IPC_TC_EVENT`	`ttcd` → `obc`	1 B	mission event from a `SET_MODE` telecommand; end of the downlink phase
`IPC_TIME_SET`	`ttcd` → `obc`, `adsbd`	8 B	wall-clock anchor; also sent to a peer that connects later
`IPC_MODE`	`obc` → `ttcd` → `adsbd`	1 B	current mission mode, forwarded to the payload daemon†
`IPC_TELEMETRY`	`obc` → `ttcd`	14 B	battery voltage and current, battery and external temperatures, attitude
`IPC_TRACKS`	`adsbd` → `ttcd`	$6 + 20N$ B	snapshot of up to 24 aircraft, every 1 s
`IPC_STAT`	`adsbd` → `ttcd`	12 B	payload counters; decoder-alive flag†
callsign table	`adsbd` → `ttcd`	—	ICAO address to callsign, for the roster telemetry†

Source: Authors (2026), after the shared codec (Gama CubeDesign, 2026). † Part of protocol version 2, decided after the first integration tests.

Hardware access

The transceiver hangs off the SPI bus of the Raspberry Pi (chip select CE0) with two GPIO lines, the reset and the interrupt output DIO0, whose numbers are set in the configuration file. The radio is driven through the kernel's own interfaces, without any library (ADR-0012): SPI through `spidev`, and the GPIO lines through the character device (The Linux kernel development community, 2026b,a). DIO0 is requested with rising-edge detection and its descriptor goes directly into the event loop, so the end of every transmission and reception is one more event, ordered with all the others, with no thread and no callback. Membership of the `spi` and `gpio` groups replaces administrator privileges. The register logic sits behind an abstract bus, so the same driver serves the Raspberry Pi and the ESP32 of the ground station and is tested on a computer against an emulated chip. The SDR connects by USB; the decoder reads it at 2.4 MS/s (Nooelec, 2026; FlightAware, 2026). The decoder's own service is disabled and the kernel television driver that would claim the SDR is blocked, because the payload daemon starts the decoder itself (ADR-0008).

11.3.4 Physical Layer Configuration

The transceiver and its band are fixed by the procured hardware: the Ra-02 module, built on the SX1278, with a front end matched for 433 MHz (Shenzhen Ai-Thinker Technology, 2017). The spreading factor, bandwidth, coding rate and output power are not fixed: they are register values, and changing them costs one write (Semtech Corporation, 2020). The most robust common configuration, spreading factor 12 at 62.5 kHz with coding rate 4/8, would limit the whole mission to 2.6 KiB of downlink, three orders of magnitude below the payload output. The link is now configured as three fixed profiles at 125 kHz, selected from the ground by telecommand (ADR-0004, Table 47).

Table 47: LoRa rate profiles (125 kHz, explicit header, CRC on, preamble of 8 symbols)

Profile	SF	CR	ToA, 208 B	Goodput	600 s at 50 %	Sensitivity	Use
SF12, 62.5 kHz	12	4/8	23.35 s	71 bit/s	2.6 KiB	−140 dBm	reference only
`SAFE`	12	4/8	11.67 s	143 bit/s	5.2 KiB	−137 dBm	recovery, beacon
`NOMINAL`	9	4/5	1.046 s	1592 bit/s	58.3 KiB	−129 dBm	the mission
`FAST`	7	4/5	0.328 s	5074 bit/s	185.8 KiB	−123 dBm	spare capacity

Source: Authors (2026), Eq. (96) and datasheet sensitivities (Semtech Corporation, 2020).

The time on air of a frame of PL bytes follows the transceiver datasheet (Semtech Corporation, 2020, §4.1.1.7):

$$\begin{aligned} T_{\text{air}} &= (n_{\text{pre}} + 4.25)\, T_{\text{sym}} + n_{\text{pay}}\, T_{\text{sym}}, \qquad T_{\text{sym}} = \frac{2^{SF}}{BW}, \\ n_{\text{pay}} &= 8 + \max\!\left(\left\lceil \frac{8PL - 4SF + 28 + 16}{4\,(SF - 2DE)} \right\rceil (CR + 4),\ 0 \right), \end{aligned} \tag{96}$$

with $n_{\text{pre}} = 8$ preamble symbols, the explicit header and the payload CRC enabled, CR from 1 (4/5) to 4 (4/8), and the low data rate optimisation $DE = 1$ when a symbol lasts more than 16 ms. The same model, implemented once in the shared code, sets every timer of the protocol and every figure of the budgets; on the bench it matched the measured time on air within 1.9 % in the three profiles (Section 11.4.3).

The selection criterion is to take the lowest spreading factor whose margin remains positive after every pessimistic assumption that can be justified, and to spend the rest on data rate. At 100 m the nominal profile keeps 9.8 dB after the 70 dB allowance of the link budget

(Table 49); spreading factor 12 would add 8 dB at an eleven-fold cost in rate. The bandwidth of 125 kHz is the best characterised setting, the one for which the datasheet specifies sensitivity, and it leaves the channel well inside the band; 500 kHz would occupy almost a third of it. The coding rate 4/5 reflects that, with this margin, errors come from interference bursts rather than thermal noise, and those overwhelm 4/8 as readily as 4/5; the payload CRC turns them into detected losses, which the software counts (HLR-ADS-08). The safe profile keeps 4/8 because it must work when the assumptions are wrong.

Only these three combinations are reachable: the rate telecommand carries a profile index, never free values of spreading factor and bandwidth. Every change is acknowledged and reverts by itself if the ground is not heard at the new profile (Section 11.3.5). The failure being prevented is asymmetric and unrecoverable: a rate command that leaves the two ends on settings that cannot hear each other ends the mission, because the only channel for the correction is the one that just broke. Two alternatives were rejected. Automatic adaptation from the received signal strength would be a control loop with its own stability problems, needing link data that will not exist before the test, and a wrong adaptation is indistinguishable from a link failure. Moving to 915 MHz, the band of the ITU region where the test takes place, would avoid the crowded 433 MHz band, but the procured modules are 433 MHz parts.

The carrier is 433.92 MHz, at the centre of the 433.05 MHz to 434.79 MHz band for which the antennas and baluns were designed; the 125 kHz channel occupies 433.858 MHz to 433.983 MHz. The local conditions of use of the band at the venue are to be confirmed with the organisers. The flight transmit power is 14 dBm, set by configuration and changeable by telecommand from 2 dBm to 20 dBm; the reason is the protection of the ADS-B receiver (link budget, below). The bench tests run at 2 dBm, because at less than a metre a higher power would bring the receiver close to saturation. Because the radio is half-duplex, the budgets allocate at most half of the mission to transmission; the allocation uses 46 % (Table 43). The fast profile was first justified by a bulk download of the raw record over radio, which was dropped in favour of the USB retrieval; it stays available as spare capacity at short range.

Link budget

The received power and the margin over the receiver sensitivity S are

$$P_{\mathrm{rx}} = P_{\mathrm{tx}} + G_{\mathrm{tx}} + G_{\mathrm{rx}} - L - \mathrm{FSPL}(d, f), \qquad M = P_{\mathrm{rx}} - S, \tag{97}$$

with the free-space path loss $\mathrm{FSPL} = 20\log_{10} d_{\mathrm{km}} + 20\log_{10} f_{\mathrm{MHz}} + 32.44$ dB. Table 48 lists the parameters. The transmit power is 14 dBm, below the (18 ± 1) dBm the Ra-02 can deliver (Shenzhen Ai-Thinker Technology, 2017), because the same power reaches the ADS-B receiver through the coupling between the two antennas: with the simulated isolation of -23.2 dB, Eq. (92) gives -9.2 dBm at the SDR input, inside the 0 dBm protection criterion and close to the -10 dBm goal (Table 41).

Table 48: Link budget parameters

Parameter	Value	Source
Carrier frequency	433.92 MHz	centre of the band used for the antenna design
Transmit power, both ends	14 dBm	flight setting; commandable from 2 dBm to 20 dBm
Antenna gain, each end	2.15 dBi	half-wave dipole
Losses, both ends	2.3 dB	0.15 dB balun per end plus 2 dB of cables and connectors
Sensitivity	−123 / −129 / −137 dBm	spreading factor 7 / 9 / 12, datasheet typical (Semtech Corporation, 2020)
Propagation	free space	optimistic indoors; covered by the allowance in Table 50

Source: Authors (2026).

The rules do not fix the distance between the CubeSat and the ground station on site, so the margin is evaluated at 10 m, 100 m and 1 km (Table 49) and defended at 100 m. Free space is optimistic for an indoor test, so a profile is accepted only if its margin survives every degradation that can be justified, applied at once (Table 50). At 100 m the nominal profile keeps 9.8 dB after 70 dB of accumulated allowances and the safe profile 17.8 dB. The 20 dB allowance for antenna mismatch also covers operation with the LoRa antenna still stowed, before the deployment telecommand. The interference allowance is an estimate: 433 MHz is a crowded band and its noise floor at the venue has not been measured; the measurement is part of the on-site procedure.

Table 49: Link margin at 14 dBm and 433.92 MHz, by profile

Distance	FSPL	P_{rx}	Margin, SF7	Margin, SF9	Margin, SF12
10 m	45.2 dB	−29.2 dBm	93.8 dB	99.8 dB	107.8 dB
100 m	65.2 dB	−49.2 dBm	73.8 dB	79.8 dB	87.8 dB
1 km	85.2 dB	−69.2 dBm	53.8 dB	59.8 dB	67.8 dB
At 100 m, after the 70 dB allowance			3.8 dB	9.8 dB	17.8 dB

Source: Authors (2026), Eq. (97) and Table 48.

Table 50: Degradation allowance applied to the free-space margin

Source	Allowance
Indoor multipath and obstruction	20 dB
Antenna mismatch and polarisation loss inside the structure (including the stowed antenna)	20 dB
Interference in the 433 MHz band	10 dB
Antenna detuned by the CubeSat structure	10 dB
Unmodelled	10 dB
Total	**70 dB**

Source: Authors (2026), after the project link budget (Gama CubeDesign, 2026).

Coexistence with the ADS-B receiver

During a LoRa transmission, part of the transmitted power couples into the ADS-B receiver: −9.2 dBm at 14 dBm (Eq. (92)), far above the level of about −30 dBm at which the tuner stops being linear. The receiver is therefore expected to be blind while the radio transmits, which is up to 46 % of the mission. The software does not blank the receiver. A message corrupted during a transmission fails the decoder's parity check and is never recorded, and a message that survives the check is valid and is kept. The telecom daemon logs the start and the end of every transmission, so the blind time is known exactly and the reception loss it causes can be accounted for. The decoding target of HLR-ADS-08 is therefore evaluated over the messages broadcast while the LoRa radio is listening, and the fraction of the mission spent transmitting is reported with it. Track continuity is not affected: each aircraft broadcasts about two positions per second, while a snapshot needs only one position less than 10 s old. The actual effect on reception is to be measured on the bench (Section 11.4.7).

11.3.5 Link Protocol

The protocol (ADR-0007) defines what travels over the radio and when. Four constraints shape it: the radio is half-duplex; HLR-COMM-01 requires a response time defined by the team; HLR-ADS-08 requires loss and latency to be measured, so every frame must be countable and every track record must carry its own timing; and no frame may depend on another frame having arrived.

Frame

One frame format serves the radio and the IPC (Table 51). All multi-byte fields are little-endian and serialised field by field, never by copying a structure from memory; the exact bytes of every frame and record are pinned by reference vectors in the test suite, so any change of order, width, scale or byte order fails a test on purpose. Sequence numbers count per direction: the satellite numbers every frame it transmits, so the ground counts each lost downlink frame from the gaps; the ground numbers each telecommand once and reuses the number when it retransmits. The checksum is the 16-bit CRC with the CCITT polynomial over the header and the payload (Tanenbaum; Wetherall, 2011); the 248-byte payload limit comes from the 255-byte transceiver buffer.

Table 51: Frame format, on the radio and on the inter-process sockets

Field	Size	Content
`ver`	1 B	protocol version; a receiver rejects any version it does not know
`type`	1 B	message type, grouped by range: link control, telemetry, bulk transfer, internal messages, beacon
`seq`	2 B	per-direction counter; loss accounting (HLR-ADS-08) and retransmission identity
`len`	1 B	payload length, 0 to 248 bytes
payload	0–248 B	one telecommand, one acknowledgement or a whole number of fixed-size telemetry records
`crc16`	2 B	CRC-16 (CCITT polynomial, initial value `0xFFFF`) over header and payload

Source: Authors (2026), after the shared codec (Gama CubeDesign, 2026).

Telemetry and telecommands

Telemetry records have fixed sizes, so a receiver splits a payload into records by division, with no length prefix, and physical quantities travel as scaled integers with a resolution at or below that of the source (Table 52). Together the records cover HLR-COMM-02: system status (mission mode, link state, status flags, uptime), power parameters (battery voltage and

current), attitude (roll, pitch, yaw) and mission data (the ADS-B counters, the track snapshots and the statistics). A field without data reads as unknown, never as zero, so a missing sensor cannot pass for a real value. Table 53 lists the telecommands.

Table 52: Telemetry records

Record	Size	Content	Cadence
`TM_TRACKS`	20 B per aircraft, 12 per frame	ICAO address; latitude and longitude (about 1.2 m and 2.4 m); altitude (25 ft); ground speed (0.1 kt); track (0.01°); vertical rate (ft/min); age (100 ms); validity flags	5 s in the mission
`TM_HK`	25 B (27†)	mission mode, link state; battery voltage and current; external and processor temperatures, battery temperature†; roll, pitch, yaw; ADS-B message count; uptime; status flags	10 s (60 s in `SAFE`)
`TM_STAT`	26 B	messages received and decoded; aircraft tracked; frames sent; telecommands accepted and rejected; latency 95th percentile; decoder restarts	30 s in the mission
`TM_ROSTER`	11 B per aircraft	ICAO address to callsign	on request; downlink phase
`TC_ACK`	4 B	command, status, sequence number of the telecommand	every telecommand
`BEACON`	minimal	presence of the satellite	30 s in `SAFE`

Source: Authors (2026), after the shared codec (Gama CubeDesign, 2026). † Protocol version 2.

Table 53: Telecommands

Telecommand	Argument	Action
`PING`	—	link test
`SET_TIME`	seconds and nanoseconds	wall-clock anchor (Section 11.3.6); sent by the ground on every contact
`SET_MODE`	mission event (1 to 7)	delivers the event to the OBC; the acknowledgement means delivered, the mode is reported by the OBC
`SET_RATE`	profile (0 to 2)	changes the rate profile, with revert
`SET_TX_POWER`	2 dBm to 20 dBm	transmit power
`STREAM_START`, `STREAM_STOP`	period (3 s to 60 s)	starts or stops the snapshots by hand, overriding the mode
`REQ_HK`, `REQ_STAT`, `REQ_ROSTER`	—	sends that record now
`SHUTDOWN`	confirmation constant	stops the telecom daemon after its acknowledgement has left

Source: Authors (2026), after the shared codec (Gama CubeDesign, 2026). The acknowledgement status is one of: accepted, unknown command, bad arguments, rejected in the current state, failed.

Medium access

The satellite listens by default. After each of its frames it stays silent for a *reply gap* long enough for a reply from the ground to be detected: 20 ms of ground turnaround, one header time and two symbols, 112 ms at the nominal profile. Before every transmission it asks the transceiver whether a reception is in progress and defers if so (Listen Before Talk (LBT)). The ground transmits an operator telecommand as soon as its transceiver is not receiving,

and at once whenever a satellite frame ends; its keepalives go only in the reply gap, which is guaranteed free. Pure random access was rejected: while streaming, the satellite starts a frame about every 1.5 s, and every collision destroys a telecommand and a telemetry frame (Tanenbaum; Wetherall, 2011).

Acknowledgement and retransmission

Every telecommand receives exactly one acknowledgement, with a status, on every path: unknown command, wrong argument size, out-of-range argument, rejected in the current state, or failed while executing. The ground is stop-and-wait (Tanenbaum; Wetherall, 2011): the next telecommand waits for the acknowledgement of the current one, which times out after 328 ms at the nominal profile; the retransmission backoff starts at 250 ms and doubles up to 2 s, and a telecommand is abandoned after 60 s. Execution is at most once: a repeated telecommand with the same sequence number and payload within 60 s is answered by replaying the acknowledgement, so a lost acknowledgement never delivers the same mission event twice to the OBC. Two commands are exceptions. The rate change is always executed, because it is idempotent and because its retransmission is the only thing that can bring the two ends back together after the satellite has switched and reverted; the time telecommand is re-stamped at every attempt.

Profile change and contact loss

The rate change is acknowledged at the old profile and applied when that acknowledgement has left. The ground follows when it hears the acknowledgement, waits 50 ms for the satellite to reconfigure and confirms with a `PING` at the new profile. The satellite commits on any ground frame heard at the new profile and reverts after 20 s without one; the ground reverts after 30 s, so that it does not give up while the satellite is still waiting. After 120 s without hearing the ground, the satellite falls to the safe profile and transmits only beacons and housekeeping; there are no snapshots in it, because one of 20 aircraft would take 23 s at spreading factor 12. When the ground hears the satellite again it restores the profile it wanted, without an operator, and the snapshots resume by themselves. The ground sends a keepalive every 30 s so that a quiet operator does not trigger this.

Telemetry schedule

At each transmission opportunity the satellite sends, in priority order: acknowledgements, housekeeping (every 10 s), statistics (every 30 s while streaming), the track snapshot (every commanded period, 5 s by default) and the beacon. Only complete snapshots are sent, and a snapshot is never sent twice.

Track age and latency

The `age` field of a track record is the time from the aircraft's last update to the start of the transmission of the frame that carries it, in tenths of a second. The telecom daemon writes it immediately before transmitting, from the time the payload daemon stamps on each internal snapshot; both read the same monotonic clock. Each record is thus self-contained in time: the ground stamps the arrival of the frame with its own clock and recovers the time of the update as

$$t_{\mathrm{update}} = t_{\mathrm{arrival}} - T_{\mathrm{air}} - 0.1\,\mathrm{s} \times age, \tag{98}$$

in its own clock domain, with no synchronisation with the satellite. The age is also, record by record, the HLR-ADS-08 latency from reception to transmission. The satellite reports its 95th percentile in the statistics record, counting each update only at its first transmission, and the ground computes the maximum from the records themselves. The 95th percentile is bounded by the snapshot period at 5 s. The maximum is bounded by

$$L_{\max} \approx T_{\mathrm{period}} + T_{\mathrm{ipc}} + (T_{\mathrm{air},1} + T_{\mathrm{gap}}) + T_{\mathrm{preempt}} + T_{\mathrm{lbt}} = 5 + 1 + 1.34 + 0.97 + 0.17 \approx 8.5\,\mathrm{s}, \tag{99}$$

where T_{ipc} is the period of the internal snapshot, the third term is the delay of the second frame of a snapshot, T_{preempt} covers an acknowledgement, a housekeeping and a statistics frame due at the same moment, and T_{lbt} one ground telecommand on the air. The acceptance criteria are therefore a 95th percentile of at most 5 s to the start of transmission (7 s to the ground) and a maximum of 10 s to the start of transmission (12 s to the ground) at the nominal profile.

Telecommand response bound

HLR-COMM-01 requires a response time defined by the team. A telecommand is executed by the telecom daemon within a bound B_0, counted from its submission by the operator, if no frame is lost, and within B_1 if one frame is lost:

$$B_0 = T_{\mathrm{air}}(\text{longest satellite frame}) + T_{\mathrm{air}}(\text{telecommand}), \qquad B_1 = B_0 + T_{\mathrm{ack}} + T_{\mathrm{backoff}} + B_0, \quad (100)$$

where T_{ack} is the acknowledgement timeout and T_{backoff} the first backoff. The first term of B_0 is the wait for the satellite to finish its longest frame, a full snapshot frame while streaming. Table 54 gives the values; commands executed by the OBC add one cycle of its main loop, 1 s.

Table 54: Telecommand response bound, from submission to execution by the telecom daemon

Profile	No frame lost, B_0	One frame lost, B_1
`SAFE`	4.21 s	10.89 s
`NOMINAL`	**1.40 s**	**3.37 s**
`FAST`	0.44 s	1.30 s

Source: Authors (2026), Eq. (100), computed by `lora_budget.py` (Gama CubeDesign, 2026).

11.3.6 Time Reference

The Raspberry Pi Zero 2 W has no real-time clock: at boot its clock holds the time of the last shutdown, there is no network at the venue, and if a network ever appears the clock jumps. Time has three consumers. The live link needs the time of each track update at the ground, which the age of each record already provides with no synchronisation (Eq. (98)). The logs need ordering and a date. The onboard record needs a wall-clock time per message that is ordered, free of jumps and on the same timeline as everything downlinked, because the ground estimator orders observations by time and a jump in the middle of a mission splits or scrambles trajectories (HLR-ADS-07).

Every process stamps with the monotonic clock, which never jumps, and nothing sets the system clock (The Open Group, 2018). Wall-clock time is the monotonic time plus an offset that comes from the ground (ADR-0009). The ground station sends the time telecommand by itself every time it gains or regains contact; the telecom daemon anchors it at the ground's transmission start, subtracting the time on air of the frame, and forwards the anchor to the OBC and to the payload daemon, and to either of them that connects later, so a restarted process is re-anchored at once. The payload daemon stamps each message with its arrival time on the decoder socket, monotonic plus offset. Until the first anchor the offset is that of the system clock at start and the log says so; every new anchor is logged with the step it made, so the record can be re-timed afterwards. Sending the time on contact places the first, possibly large, step before the mission; after it, each anchor only corrects the drift of the crystal, 30 ms over a 10-minute mission at 50 ppm. The record keeps nanosecond resolution, and the link 100 ms, about 23 m at 450 kt. The timestamp is the arrival at the decoder output, not the radio-frequency reception; the decoder offers a 12 MHz receiver counter if that is ever needed.

11.3.7 Operation by Mission Mode

The mission modes belong to the OBC (Section 9.2.3); the TT&C software reacts to them. The telecom daemon forwards each mode the OBC announces to the payload daemon, and both adapt their behaviour (Table 55). The decoder runs only in `MISSION_ADSB`, and the snapshots start and stop with that mode, without a ground command; in the other modes the SDR stays powered and idle. The states of the link — idle, transmitting, streaming, safe — are not mission modes: they describe the radio, and they are reported separately in the housekeeping. The manual commands `STREAM_START` and `STREAM_STOP` remain as an override, and while the OBC is not connected the housekeeping reports its mode as unknown and the link keeps working.

Table 55: What the TT&C software does in each mission mode

Mission mode (OBC)	TT&C software
`PRE_TEST`	housekeeping every 10 s; telecommands answered; decoder stopped
`BASIC_INTERMEDIATE`	the mode is entered through `SET_MODE`, which triggers the antenna deployment; until then the link operates with the antenna stowed, which the link margin covers (Table 50)
`ADVANCED_AOCS`	housekeeping every 10 s carries the attitude reported by the OBC
`MISSION_ADSB`	the payload daemon starts the decoder, retrying every second until it opens the SDR; snapshots every 5 s and statistics every 30 s start by themselves
`MISSION_DOWNLINK`	the decoder stops; the roster of the aircraft observed and the final statistics are sent (260 B, 1.4 s on air); the telecom daemon then signals the end of the task to the OBC, which returns to `PRE_TEST`; the complete record is retrieved over USB
`ENV_SURVIVAL`	housekeeping, including the processor temperature; the onboard logs keep recording during the environmental tests; the mode is left by a `SET_MODE` telecommand

Source: Authors (2026). The behaviour in the downlink and survival modes is a TT&C proposal to be fixed in the interface control document with the OBC team.

11.3.8 Ground Segment

The ground station is an Ra-02 module on an ESP32 running ESP-IDF, connected by USB to a computer (Espressif Systems, 2025). The ESP32 firmware compiles the same shared code as the satellite: the frame codec, the telemetry and telecommand records, the ground side of the link protocol and the transceiver driver, which runs over the SPI and GPIO of ESP-IDF through the same abstract bus (ADR-0002). The reference byte vectors of the test suite are also built for the ESP32, which demonstrates that both ends of the link produce the same bytes instead of arguing that they should.

The ground side of the protocol is a pure state machine shared by the ESP32, the bench tools and the channel simulation. It sends one telecommand at a time and waits for its acknowledgement, retransmits with the same sequence number, talks only when the satellite leaves the channel free, follows profile changes and confirms them, sends keepalives, falls back to the safe profile when the satellite is not heard and restores the profile without an operator, and counts every lost downlink frame from the sequence gaps. It also sends the time telecommand by itself at every contact (Section 11.3.6).

The ESP32 sends the computer, over the serial port, one JSON object per event: each received frame with its raw bytes, received signal strength, signal-to-noise ratio and arrival time, and each acknowledgement, profile change and loss. The computer therefore needs no

second implementation of the codec. On the computer, a Python program lets the operator send telecommands, records the event log, and feeds an adapter that turns the received snapshots into the observation format of the ground estimator, recovering the time of each update by Eq. (98). The estimator reconstructs trajectories, estimates origin and destination and presents them on a dashboard (Section 12). The bench ground station — the same protocol code on a second Ra-02, driven from a command line — has already run against the flight software on the bench, so a Raspberry Pi with an Ra-02 is a working fallback ground station.

11.3.9 Data Products

Table 56 lists what the software records and why. Together the products answer HLR-SW-02. The data are *structured*: the telemetry is made of fixed binary records whose bytes are pinned by tests, and the logs are JSON objects, one per line (Bray, 2017; NDJSON Project, 2014). They are *documented*, in this section and in the interface control document of the link. They are *reproducible*: the ground keeps every frame as received, so any decoding can be repeated, and every figure of the budgets comes from a script. And they are *available for post-analysis*: the complete onboard record is retrieved over USB after the test, and the bench logs are kept in the repository.

The onboard record grows at about 12.5 KiB/s, 7.34 MiB per mission. The payload daemon caps it at a configured size (1 GiB by default) and stops appending instead of filling the file system, because a full root file system would also stop the radio process; the cap is two orders of magnitude above the mission volume and leaves room for the hours of the environmental tests.

Table 56: Data products of the TT&C and payload software

Product	Written by	Format	Purpose
Onboard record	`adsbd`	one JSON line per decoded message, 12.5 KiB/s, capped at 1 GiB	the complete mission data, retrieved over USB after the test
Event logs	`ttcd`, `adsbd`	one JSON line per event, with the monotonic time, capped at 64 MiB	every frame sent and received, every telecommand, profile change, transmission interval, time anchor and decoder restart; the OBC telemetry
Ground log	ground station	one JSON object per event, with the raw frame, received signal strength, signal-to-noise ratio and arrival time	repeating any decoding; counting the downlink loss from the sequence gaps
Estimator input	adapter	the onboard record format, rebuilt from the snapshots	trajectories, origin and destination
Bench logs	bench procedure	satellite and ground logs for each test step	test evidence, kept in the repository

Source: Authors (2026).

11.3.10 Implementation Practices

The software is written in C11 on both ends of the link (International Organization for Standardization, 2011) and follows a few practices that explain why it can be trusted before any measurement. Table 57 lists them with the reason for each. C gives no protection against memory errors in the parser, the component most exposed to malformed input; that cost is

paid with tooling: the build stays at zero warnings with conversion warnings enabled, because a truncated length or a lost sign on a latitude appears first as an implicit conversion, and the whole test suite also runs under AddressSanitizer and UndefinedBehaviorSanitizer (Serebryany *et al.*, 2012; Free Software Foundation, 2026).

Table 57: Implementation practices and the reasons for them

Practice	**Reason**
One codec, compiled from the same files on both ends	the wire format cannot diverge between satellite and ground
Every field serialised explicitly, little-endian	the format does not depend on the compiler
Reference byte vectors in the tests	any change of the format fails a test on purpose
Pure core in an event-loop shell; nothing blocks	the protocol is testable in virtual time and the radio always answers
Separate processes under systemd, without root	a payload fault cannot stop the radio
Local sockets that preserve message boundaries	no byte-by-byte reassembly; a dead peer is seen at once
Configuration file with command-line override; an unknown key is an error	no path compiled into the program
Monotonic clock with an anchor from the ground	there is no real-time clock
Every rejected frame counted	HLR-ADS-08 asks for the loss rate
Zero warnings with conversion warnings; tests under sanitizers	conversion and memory errors caught before the hardware
Requirement identifier cited in the code that meets it; figures produced by scripts checked in the tests	traceability, and documents that cannot drift from the code

Source: Authors (2026), from the decision records (Gama CubeDesign, 2026).

11.4 Software Verification and Validation

This subsection presents the evidence behind the software of Section 11.3.1 onwards: the definitions and methods used, the results of the software tests and of the hardware tests carried out so far, and what each test changed in the design. It ends with the derived requirements, the traceability matrix from each requirement to its acceptance criteria, verification and validation, and the validation that remains before the on-site test.

11.4.1 Strategy, Definitions and Methods

In this document, *verification* is the evidence, produced by analysis, inspection or software tests (unit tests, simulation, integration of the real programs), that the implementation meets an acceptance criterion. *Validation* is the evidence obtained on the flight hardware: the real radios, the real SDR and decoder, the onboard computer, the integrated chain and, finally, the on-site test. This usage is narrower than that of the systems engineering standards, where verification asks whether the product meets its requirements and validation whether it meets its intended use (National Aeronautics and Space Administration, 2016; European Cooperation for Space Standardization, 2018); it is adopted because, at this stage, the distinction that matters is between evidence produced in software and evidence produced on the hardware. A requirement can have several acceptance criteria, for instance one for each subsystem that

interprets it, and each criterion has its own verification and validation. Not every criterion has a validation: those verified by documentation review do not.

The rules assign a verification method to each requirement; Table 58 states what evidence each method produces in this project (HLR-VV-02). Software tests come before hardware tests on purpose. The properties of the link protocol — the telecommand response bound, the recovery from a lost acknowledgement, the fallback to the safe profile — depend on timing and on failures that cannot be produced at will on hardware, and that do not repeat when they happen. With the clock injected into the pure cores, the simulation produces them deterministically and in milliseconds. The bench then checks the model where the hardware can contradict it: the time on air, the detection of a reception in progress, the frame loss.

Table 58: Verification methods and the evidence they produce

Method	Evidence	Used for
Analysis	a calculation reproduced by a script in the repository; the test suite fails if a document quotes a value the script no longer gives	budgets, time on air, response and latency bounds
Documentation review	the document itself: decision records, budgets, this design review	requirements verified by documentation review
Test, in software	unit tests, channel simulation, integration tests of the real programs, the full chain without hardware	protocol rules, interfaces, fault handling
Test, on hardware	a written procedure with its acceptance criterion and the recorded logs	radio link, SDR, decoder, chain
Data inspection	raw data kept as received (frames, onboard record) and, for validation, a reference to compare with	telemetry content, loss, latency
Demonstration	the on-site test, run by the organisers	the competition tests

Source: Authors (2026).

Each level of evidence covers something the previous one does not (Table 59). Levels 1 to 7 are verification and are complete; levels 8 and 9 are validation and were carried out on 7 and 9 October 2026; levels 10 and 11 are planned.

Table 59: Levels of evidence: what each proves and what it does not

#	Level	Proves	Does not prove	Evidence
1	Analysis	capacity, time on air, margins, volumes and bounds close with the datasheet	that the real radio has these numbers	budget script, checked in the test suite
2	Format	the exact bytes of every frame and record; behaviour with out-of-range values	that the ESP32 produces the same bytes	reference vectors, codec tests
3	Rules	each rule of the protocol and of the payload, in isolation, in virtual time	behaviour with the other side or in real time	tests of the pure cores

Continued on the next page.

Table 59 — continued

#	Level	Proves	Does not prove	Evidence
4	Channel simulation	the whole protocol, both real sides, on a half-duplex channel with collisions and losses	how the transceiver detects a reception; real noise; driver timing	channel simulation
5	Emulated chip	the transceiver configuration, register by register	real SPI and GPIO; timing	driver test
6	Real programs, simulated radio	event loop, sockets, timers, signals, configuration, reconnections, decoder supervision	the hardware	integration tests
7	Chain, no hardware	nothing lost between processes; the downlink preserves the estimator's conclusions; the radio survives a dead payload	real radio, processor load, real traffic	chain test, 10-minute replay
8	Radios on the bench	time on air, frame loss, rate changes, lost acknowledgement and listen-before-talk on two Ra-02	the mission distance and the team's antennas	bench of 9 October
9	Onboard computer with SDR	decoder on the target, processor load, the payload daemon and the chain on hardware	20 aircraft; receiver gain	bench of 9 October
10	10-minute rehearsal	everything together, with the ADS-B mission metrics	the organisers' scenario	planned
11	On-site test	the organisers' scenario and the venue	—	planned

Source: Authors (2026).

11.4.2 Software Verification

The software builds with zero compiler warnings with all the usual warnings enabled, including the implicit conversion and sign conversion warnings (`-Wall`, `-Wextra`, `-Wpedantic`, `-Wconversion`, `-Wsign-conversion`). Its 23 test suites — 20 in C with 3427 checks, plus 49 tests of the ground estimator and 7 of the downlink adapter in Python — pass in the normal build and in a build instrumented with AddressSanitizer and UndefinedBehaviorSanitizer with warnings treated as errors. A script also checks that the 71 quotations of the rules in the requirements document match the official text verbatim. These results were obtained on 10 October 2026, at the repository version that the logs of the bench also record (Gama CubeDesign, 2026). Table 60 groups the suites by level.

Table 60: Test suites, grouped by level of evidence

Level	Suites (checks)	What they verify
1 Analysis	budget check	every figure quoted in the decision records is reproduced by the budget script
2 Format	telemetry records (1324), time on air (832), CRC (260), frame (58), reference vectors (50), telecommands (49), IPC payloads (55)	exact bytes of every record; quantisation and saturation; time on air for every spreading factor and size
3 Rules	telecom core (199), ground protocol (39), payload core (66), message parser (64), track table (51), configurations (45, 35)	every protocol and payload rule, including 20 aircraft for 10 minutes at 4 messages per second each
4 Simulation	channel simulation (63)	the scenarios of Table 61
5 Emulated chip	transceiver driver (95)	register sequences of the three profiles, including the low data rate optimisation and the modem status measured on the bench
6 Real programs	telecom daemon (29), payload daemon (46), IPC transport (36)	the real programs on the simulated radio; 10 000 frames each way between two processes without loss or blocking
7 Chain	ADS-B chain (31); estimator (49), adapter (7)	both daemons with the ground; a killed payload daemon restarted and re-anchored; the estimator and the adapter

Source: Authors (2026), test run of 10 October 2026 (Gama CubeDesign, 2026).

The channel simulation runs the real satellite core against the real ground-side code on a simulated half-duplex channel. Each frame occupies the air for its exact time on air, a collision destroys both frames, and a radio sees the other's transmission only after decoding its header and only if it was already listening, a pessimistic assumption. Time is virtual and runs are seeded, so the figures of Table 61 are reproduced exactly at every run.

Table 61: Channel simulation results (nominal profile, virtual time)

Scenario	**Result**
Idle link, 400 telecommands	median 0.12 s, maximum 0.83 s; 6 collisions, 6 retried, none failed
Streaming 20 aircraft, 401 telecommands	median 0.21 s, 95th percentile 1.78 s, maximum 2.40 s; maximum at the first attempt 1.27 s, below $B_0 = 1.40$ s; 69 collisions, 48 retried (12 %), none failed
The same, detecting a reception after 5 preamble symbols	12 collisions, 9 retried (2 %); 95th percentile 1.11 s
Streaming, 10 % frame loss each way	95th percentile 2.34 s, maximum 6.25 s, none failed; 191 of 1601 downlink frames lost, all counted by the ground
Streaming, keepalives only	56 keepalives acknowledged; no collision; no contact loss
Onboard latency, reception to transmission	95th percentile 3.2 s (target 5 s)
Rate change, acknowledgement lost	both ends at the fast profile after 23.4 s; on different profiles for 20.0 s; one satellite revert

Continued on the next page.

Table 61 — continued

Scenario	Result
Rate change, confirmation lost	both ends back at the nominal profile; desynchronised for 10.0 s
200 s uplink blackout while streaming	back at the nominal profile 58 s after the uplink returned; snapshots resumed
300 s blackout both ways	both ends fell to the safe profile; back at the nominal profile 32 s after the blackout
`SET_MODE`, 100 commands, one acknowledgement lost	exactly 100 events delivered to the OBC; worst case from submission to the OBC 2.42 s
`SET_TIME`, acknowledgement lost	clock anchor within 0 ms after 3 attempts

Source: Authors (2026), channel simulation of the test suite (Gama CubeDesign, 2026).

The full chain was also run without hardware: a simulated 10-minute mission with four flights and 1894 messages, replayed in real time as if it came from the decoder, went through the same programs that fly, with the simulated radio in place of the transceiver. The payload daemon recorded 5682 lines with none rejected; 231 downlink frames were sent and none lost; 386 track records reached the ground estimator; and the telecom daemon measured an onboard latency of 2.0 s at the 95th percentile. The estimator reached the same conclusions from the downlinked snapshots as from the complete stream, also with 10 % of the frames removed. The scenario is synthetic and the simulated radio loses nothing to noise, so these results verify the software, not the radio.

11.4.3 Hardware Validation

The hardware tests followed a written procedure, run on 7 and 9 October 2026 (Gama CubeDesign, 2026). The onboard computer is a Raspberry Pi Zero 2 W with the 64-bit operating system (Debian 13, compiler GCC 14.2). The two Ra-02 modules are soldered on a perforated board and wired to the same computer: one on the first SPI chip select, driven by the flight telecom daemon, and one on the second, driven by the bench ground station. They are 0.8 m apart and transmit at 2 dBm, which gives a received signal of about −28 dBm. The SDR uses a commercial 1090 MHz antenna, indoors. Each step of the radio bench starts a fresh telecom daemon, feeds a script of ground commands, records both logs and prints a summary; the logs are kept in the repository. The bench is shown in Figure 62.

On the target, the software compiles with no warning and passes the 23 test suites in the normal build and in a build with UndefinedBehaviorSanitizer; AddressSanitizer cannot run there, because the kernel of the board gives each process a 39-bit address space, so it runs on the development computer only. The sanitizer build on the target found one defect, in a test (Table 64). Table 62 gives the results of the radio bench and Table 63 those of the payload.

Table 62: Radio bench: two Ra-02 at 0.8 m and 2 dBm (9 October 2026)

Step	Acceptance criterion	Result	Status
1. Both chips answer; ping	version register reads `0x12`; 10 of 10 pings acknowledged	`0x12` on both modules; 10 of 10, signal −28 dBm, signal-to-noise ratio 9.4 dB	pass

Continued on the next page.

Table 62 — continued

Step	Acceptance criterion	Result	Status
2. Time on air	within 5 % of Eq. (96)	141 frames in the three profiles; worst 1.9 % (fast profile, 11 B); nominal 0.4 %; safe 0.1 %	pass
3. Frame loss	frame loss measured over 1000 frames per profile	nominal 1000 of 1000 (mean 276 ms), fast 1000 of 1000 (81 ms), safe 200 of 200 (2.70 s), all at the first attempt; 2220 downlink frames, none lost or corrupt; upper bound of the loss at 95 % confidence below 0.3 % (nominal, fast) and 1.5 % (safe)	partial
4. Rate change	every change between two profiles acknowledged and committed	7 of 7 at the first attempt; slowest when leaving the safe profile, 5.44 s	pass
5. Lost acknowledgement	both ends recover without operator	ground deaf for 3 s: satellite reverted at 20.000 s; both at the fast profile after 23.2 s (simulation: 23.4 s)	pass
6. Listen-before-talk	retries with header and with preamble detection	400 pings with housekeeping every second: 0 % retries in both; one deferral per ping	pass

Source: Authors (2026), bench logs (Gama CubeDesign, 2026).

Table 63: Payload bench: SDR, decoder and the chain on hardware (9 October 2026)

Stage	Acceptance criterion	Result	Status
SDR and decoder	the decoder opens the SDR at 2.4 MS/s without sample loss	dump1090-fa 11.1 on the 64-bit system; R820T tuner; 24 million samples in 10 s, none lost	pass
Processor load	decoder below 50 % of one core	21.9 % of one core; whole computer 5 % busy; 46 °C; no throttling; 283 MB free — with 0 to 3 aircraft in view	pass
Receiver gain	a gain that maximises accepted messages	six gains tested; traffic varied more between two minutes at the same gain than between gains; automatic gain kept	open
Payload daemon with the real decoder	starts, reads and stops the decoder; records every message	decoder connected in 1.0 s; 24 messages from 2 aircraft recorded, none rejected; clean stop, decoder terminated with it	pass
Chain: SDR, payload daemon, telecom daemon, LoRa, ground	data and time cross the whole chain	automatic time telecommand reached the payload daemon (anchor step −4 ms); 120 internal snapshots; 24 downlink frames, none lost; 6 telecommands acknowledged at the first attempt in 276 ms to 521 ms; **no track record**: the only aircraft in view sent no position	partial

Source: Authors (2026), bench logs (Gama CubeDesign, 2026).

These results have limits that are stated with them. The radio bench ran at 0.8 m with a signal about 100 dB above the sensitivity: a loss of zero says that the protocol, the driver and the timing work on the real chips, not what the loss will be at mission distance with the team's antennas. The payload tests saw at most three aircraft: the processor load is dominated by the sample rate, which is fixed, but the decoding share grows with traffic and must be measured with about 20 aircraft. These open points are in the remaining validation (Section 11.4.7).

11.4.4 How the Tests Shaped the Design

The tests did not only confirm the design; several changed it. Table 64 lists what each one found and what changed as a result.

Table 64: What the tests found and what changed

Finding	Level	Consequence
The first channel simulation had 170 collisions in 401 telecommands while streaming	4	reply gap after every satellite frame; the ground transmits at once when the gap opens; a waiting keepalive gives way to an operator command. Collisions fell to 69 and the 95th percentile from 2.44 s to 1.78 s (ADR-0007)
The simulation predicted that detecting a reception at the preamble would cut retries from 12 % to 2 %	4	a bench step was added to decide between header and preamble detection
On the bench, header and preamble detection both gave 0 % retries	8	header detection kept as default; the protocol record is ready for acceptance
On the bench, nothing went on air: the real SX1278 keeps its "reception on-going" status bit set during continuous reception, even on an empty channel	8	the driver no longer counts that bit as a busy channel; the emulated chip now uses the measured value
On the target, the sanitizer build failed a test that compared the padding bytes of a structure	2	the tests compare field by field; the codec was right and did not change; AddressSanitizer runs on the development computer only
Time on air within 1.9 % of the model in all profiles	8	the model behind the response bound, the protocol timers and the data budget holds
Lost acknowledgement recovered in 23.2 s, against 23.4 s in simulation	8	the simulation's timing model is confirmed for rate changes
Decoder load of 21.9 % of one core	9	the sample rate stays at 2.4 MS/s; the largest unquantified risk of the architecture is retired, pending a test with traffic
Receiver gain inconclusive with 0 to 3 aircraft	9	automatic gain kept; the sweep is repeated with traffic or a known source
Reading the decoder's source: the on-ground flag comes from the capability field of every message, and −1 means true; the prototype read it wrongly	1	parser corrected and pinned by tests
The record line computed from the real format is 160.4 B, not the 110 B first assumed	1	mission volume 7.34 MiB and ratio 129:1 instead of 88:1; errata added to the decision records; the case for onboard processing became stronger

Continued on the next page.

Table 64 — continued

Finding	Level	Consequence
Killing the payload daemon in the chain test: once restarted, it stamped with an unsynchronised clock until the next time telecommand	7	the telecom daemon re-sends the anchor to any peer that reconnects (ADR-0009)
In the chain on hardware the only aircraft sent no position, so no track record went down	9	the chain with tracks is repeated with traffic or a known source
The antenna design gives 2.15 dBi and the simulated isolation puts −5.2 dBm at the SDR at 18 dBm	1	flight power set to 14 dBm and carrier to 433.92 MHz; link budget recomputed (Section 4.3.5)

Source: Authors (2026). Levels as in Table 59.

11.4.5 Derived Requirements

HLR-GEN-04 asks each team to derive subsystem requirements from the system requirements. Table 65 lists the requirements of the TT&C and payload software, each traced to the high-level requirements it serves and to the decision that fixed its value.

Table 65: Derived requirements of the TT&C and payload software

ID	Requirement	Parent	Source
TTC-01	The link operates at 433.92 MHz, with its 125 kHz channel inside 433.05 MHz to 434.79 MHz.	COMM-03	this review
TTC-02	The link offers three rate profiles, selected by profile index: nominal (SF9, 4/5, the default), safe (SF12, 4/8) and fast (SF7, 4/5).	COMM-03	ADR-0004
TTC-03	The flight transmit power is 14 dBm, commandable from 2 dBm to 20 dBm.	COMM-03, ADS-01	this review
TTC-04	A telecommand is executed within 1.40 s of its submission with no frame lost and within 3.37 s with one frame lost, at the nominal profile; commands executed by the OBC add 1 s.	COMM-01	ADR-0007
TTC-05	Every telecommand receives exactly one acknowledgement with a status and is executed at most once.	COMM-01	ADR-0007
TTC-06	Housekeeping is transmitted every 10 s with the mission mode, link state, battery voltage, current and temperature, the other temperatures, the attitude and the ADS-B message count; a field without data reads as unknown.	COMM-02, EPS-04	ADR-0007
TTC-07	In `MISSION_ADSB`, a snapshot of up to 20 aircraft is transmitted every 5 s and the statistics every 30 s, started by the mode without a ground command.	ADS-05, ADS-08, GEN-03	ADR-0003, data budget
TTC-08	The radio transmits at most 50 % of the mission time.	COMM-01	data budget
TTC-09	The latency from the reception of a message to the start of the transmission carrying it is at most 5 s at the 95th percentile and 10 s at the maximum (7 s and 12 s to receipt on the ground).	ADS-08	ADR-0007, this review

Continued on the next page.

Table 65 — continued

ID	Requirement	Parent	Source
TTC-10	Every frame carries a per-direction sequence number, the ground counts the gaps, every rejected frame is counted, and the downlink loss is at most 5 %.	ADS-08	ADR-0007, data budget
TTC-11	Every ADS-B message is recorded on board with its arrival time on the ground's timeline; every track record carries its age with a 100 ms resolution.	ADS-07, SW-02	ADR-0003, ADR-0009
TTC-12	After 120 s without the ground the satellite falls to the safe profile, and the link recovers without an operator; an unconfirmed profile change reverts after 20 s.	GEN-03, COMM-03	ADR-0004, ADR-0007
TTC-13	A failure of the payload software does not interrupt the link; every decoder restart is counted.	GEN-03, ADS-05	ADR-0005, ADR-0008
TTC-14	The onboard record is capped and retrieved over USB; the ground keeps every frame as received, with signal strength, signal-to-noise ratio and arrival time.	SW-02	ADR-0003
TTC-15	The decoder uses less than half of one processor core at 2.4 MS/s.	ADS-01, ADS-05	ADR-0008
TTC-16	Both ends of the link use the same codec, and its reference vectors give the same bytes on the onboard computer and on the ESP32.	SW-02, SYS-01	ADR-0002

Source: Authors (2026). Parent requirements as in Section 3; decision records in Table 42.

11.4.6 Traceability Matrix

Table 66 traces each high-level requirement served by the software to its acceptance criteria. Each criterion has an identifier made of the requirement and a sequence number, its verification (software, analysis or documents) and its validation (hardware), and a status: **A** verified by analysis, **D** verified by documentation review, **S** verified in software, **H** validated on hardware, **P** planned. A requirement interpreted by other subsystems has their criteria too; they are referred to where they are defined: the antenna requirements R1 to R6 (Tables 36 and 41) and the payload metrics (Tables 73 and 74).

Table 66: Traceability of the requirements served by the TT&C and payload software

ID	Acceptance criterion	Verification	Validation	Status
HLR-GEN-03 — autonomous operation with telecommands and telemetry				
GEN-03.1	The link recovers from a contact loss and from a lost acknowledgement without operator (TTC-12)	simulation: 200 s and 300 s blackouts, lost acknowledgement and lost confirmation (Table 61)	bench step 5 (23.2 s); ground switched off for 2 min in the rehearsal	S, H, P
GEN-03.2	A payload failure does not stop the link (TTC-13)	chain test: payload daemon killed and restarted while the radio kept transmitting	on the target under systemd, decoder killed in the rehearsal	S, P
GEN-03.3	The snapshots start and stop with the mission mode (TTC-07)	— (protocol version 2)	integration with the OBC	P
HLR-GEN-04 — derived subsystem requirements				

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Table 66 — continued

ID	Acceptance criterion	Verification	Validation	Status
GEN-04.1	Software requirements derived, documented and traced	Tables 65 and 66	—	D
HLR-COMM-01 — telecommands executed within a bounded response time				
COMM-01.1	Executed within $B_0 = 1.40$ s at the nominal profile, no frame lost (TTC-04)	Eq. (100); simulation: first-attempt maximum 1.27 s while streaming	bench step 3: 1000 pings, maximum 281 ms; fast 82 ms, safe 2.70 s ($B_0 = 4.21$ s)	A, S, H
COMM-01.2	Executed within $B_1 = 3.37$ s with one frame lost	simulation: maximum 2.40 s while streaming	bench step 1: first ping, retransmitted once, in 977 ms	S, H
COMM-01.3	Exactly one acknowledgement with status; execution at most once (TTC-05)	core tests; simulation: 100 `SET_MODE` with a lost acknowledgement, exactly 100 events	bench: 2211 telecommands, each acknowledged	S, H
COMM-01.4	Commands executed by the OBC within the bound plus 1 s	—	OBC log of reception and transition, after integration	P
COMM-01.5	RF interpretation: LoRa antenna matching (R1)	Table 41	Table 41	see R1
HLR-COMM-02 — structured telemetry: status, power, attitude, mission data				
COMM-02.1	Housekeeping with status, power, attitude and mission data every 10 s (TTC-06)	reference vectors; integration test of the real daemon	bench chain: 17 housekeeping frames in 150 s, decoded on the ground; power and attitude unknown until the OBC client is integrated	S, H, P
COMM-02.2	Snapshots and statistics carry the mission data (TTC-07)	chain tests	bench chain: statistics with the payload counters; no snapshot without a position in view	S, H, P
HLR-COMM-03 — communication architecture defined, justified and consistent				
COMM-03.1	Physical layer, protocol and rates defined and justified	ADR-0004, ADR-0007; budgets (Sections 4.3.4 and 4.3.5)	—	D, A
COMM-03.2	Time on air within 5 % of Eq. (96)	time-on-air test (832 checks)	bench step 2: worst 1.9 %	A, S, H
COMM-03.3	Frame loss measured over 1000 frames per profile	simulation	bench step 3: none lost at 0.8 m; mission distance with the team's antennas	S, H, P
COMM-03.4	Rate change acknowledged and reverted when unconfirmed (TTC-12)	simulation: lost acknowledgement, lost confirmation	bench steps 4 and 5	S, H
COMM-03.5	Carrier at 433.92 MHz, channel inside the band (TTC-01)	driver test of the frequency registers, updated to the new carrier	register read-back on the bench	P
HLR-ADS-01 — reception of ADS-B at 1090 MHz during the mission				
ADS-01.1	The onboard chain receives and records 1090ES messages	parser tests on recorded traffic; decoder supervision with a substitute	bench: decoder on the target, 24 messages from 2 aircraft recorded (commercial antenna)	S, H

Continued on the next page.

Table 66 — continued

ID	Acceptance criterion	Verification	Validation	Status
ADS-01.2	Decoder below half of one core (TTC-15)	—	bench: 21.9 % with 0 to 3 aircraft; about 20 aircraft	H, P
ADS-01.3	RF interpretation: ADS-B antenna matching and receiver protection (R2, R3)	Table 41	Table 41	see R2, R3
HLR-ADS-02 — ICAO address, position, altitude and velocity extracted				
ADS-02.1	The fields are extracted into the track record (speed, track and vertical rate for velocity)	parser and track table tests; reference vectors	bench: ICAO and altitude of real aircraft; functional test of the payload team (Section 12); known scenario	S, H, P
HLR-ADS-03 — several aircraft at the same time				
ADS-03.1	Table of 128 aircraft; snapshots of up to 24	track table test: 20 aircraft for 10 min, 48 000 updates	bench: 2 aircraft; 20 in the rehearsal	S, H, P
HLR-ADS-04 — processing architecture implemented and justified				
ADS-04.1	Hybrid split defined, justified and implemented (Section 11.3.2)	data budget; ADR-0003; downlink emulation	bench chain on hardware	A, D, S, H
HLR-ADS-05 — continuous 10-minute mission with up to 20 aircraft				
ADS-05.1	600 s without unexplained discontinuity, 20 aircraft	track table test; 10-minute chain replay in real time, no frame lost	10-minute rehearsal	S, P
ADS-05.2	Decoder supervised; every restart counted (TTC-13)	integration test: substitute decoder killed, restarted and counted	bench: real decoder started and stopped with the daemon; killed in the rehearsal	S, H, P
HLR-ADS-07 — timestamp for every message				
ADS-07.1	Every message stamped on arrival, on the ground's timeline (TTC-11)	payload core tests; chain: restarted daemon re-anchored	bench chain: anchor reached the payload daemon, step −4 ms	S, H
ADS-07.2	Track age at 100 ms resolution; the ground recovers the time by Eq. (98)	codec, simulation and adapter tests	chain with traffic	S, P
ADS-07.3	Anchor error measured on the target	simulation: 0 ms after a retry	ground time against a known event	S, P
HLR-ADS-08 — quantitative mission success criteria				
ADS-08.1	At least 95 % of the messages broadcast while the LoRa radio listens are decoded; the fraction of time spent transmitting is reported	— (needs a reference scenario)	rehearsal with a known scenario; bench test with the LoRa transmitting	P
ADS-08.2	Downlink loss at most 5 %, counted from sequence gaps (TTC-10)	simulation: every lost frame counted at 10 % loss	bench: 2220 frames and chain: 24 frames, none lost; mission distance	S, H, P
ADS-08.3	Latency at most 5 s to transmission and 7 s to the ground at the 95th percentile (TTC-09)	simulation 3.2 s; chain replay 2.0 s	rehearsal with traffic	S, P

Continued on the next page.

Table 66 — continued

ID	Acceptance criterion	Verification	Validation	Status
ADS-08.4	Latency at most 10 s to transmission and 12 s to the ground at the maximum (TTC-09)	Eq. (99): 8.5 s	rehearsal: maximum of the received ages	A, P
ADS-08.5	Payload and RF interpretations: payload metrics; receiver protection (R3)	Table 74; Table 41	idem	see tables
HLR-SW-02 — structured, documented and reproducible data				
SW-02.1	Structured: fixed binary records pinned by reference vectors; JSON-line logs	reference vectors	bench logs decoded	S, H
SW-02.2	Documented: formats described in Section 11.3.5 and in the link interface control document	this review	—	D
SW-02.3	Reproducible: every frame kept as received, with signal and time (TTC-14)	ground log of the bench tool	bench logs kept in the repository	S, H
SW-02.4	Available for post-analysis: onboard record capped and retrieved over USB	payload daemon tests	bench: record retrieved from the target	S, H
HLR-EPS-04, **HLR-ENV-02** — TT&C share				
EPS-04.1	Battery voltage, current and temperature in the housekeeping	codec tests (voltage, current); temperature in protocol version 2	with the OBC client	S, P
ENV-02.1	Telemetry and onboard logs before, during and after the environmental tests; processor temperature in the housekeeping	codec tests	during the environmental tests	S, P
HLR-VV-01, **HLR-VV-02**, **HLR-COST-02**, **HLR-SYS-01** — documentation				
VV-01.1	Verification and validation strategy for the software requirements	this subsection	—	D
VV-02.1	Methods defined and justified	Table 58	—	D
COST-02.1	Decisions justified by performance, cost and complexity	Table 42; capacity 22 times that of spreading factor 12 at 62.5 kHz, at no hardware cost	—	D
SYS-01.1	One figure, one place: budgets reproduced by a script in the test suite	budget check	—	A, D

Source: Authors (2026).

11.4.7 Remaining Validation

Table 67 lists the validation still to be done before the on-site test, each with its procedure and acceptance criterion. The tests that can change a design decision come first.

Table 67: Remaining validation of the TT&C and payload software

Test	Procedure	Acceptance criterion	Criteria
Link at mission distance	bench steps 2 and 3 with the team's antennas at the on-site distance, 14 dBm, antenna deployed and stowed	frame loss over 1000 frames per profile; at most 5 %	COMM-03.3, ADS-08.2
ADS-B with the LoRa transmitting	decoder message rate with the LoRa at 14 dBm switched on and off; power at the SDR input with a calibrated attenuator	reception loss during transmission quantified; at most 0 dBm at the SDR	ADS-08.1, R3
SDR with traffic	gain sweep and processor load with about 20 aircraft or a known signal source	load below half of one core; gain chosen by accepted messages	ADS-01.2
Chain with track records	the hardware chain with aircraft positions in view	track records decoded on the ground; update times recovered by Eq. (98)	COMM-02.2, ADS-07.2
10-minute rehearsal	known scenario, 20 aircraft, 600 s; decoder killed once; ground switched off for 2 min	ADS-08.1 to ADS-08.4; recovery without operator; restarts counted	ADS-03 to ADS-08, GEN-03
OBC integration	reference client in the OBC software; mode events, telemetry and mode reporting	commands executed within the bound plus 1 s; power, attitude and battery temperature in the housekeeping	COMM-01.4, COMM-02.1, EPS-04.1, GEN-03.3
Reference vectors on the ESP32	the vector test built for the ESP32	identical bytes on both ends	TTC-16
Anchor error	ground time against the anchored time of a known event	error recorded	ADS-07.3
Carrier frequency	433.92 MHz configured; frequency register read back	channel inside the band	COMM-03.5
Noise at the venue	spectrum around 433.92 MHz before the test	link margin re-evaluated with the measured noise	COMM-03

Source: Authors (2026).

12 ADS-B Payload

The ADS-B payload of the Cubetschek mission is responsible for acquiring and initially processing aeronautical surveillance signals, providing observations that enable aircraft tracking and the estimation of their origin and destination airports. To fulfil this function, the project adopts a hybrid architecture: reception at 1090 MHz, decoding, time association, and message structuring are performed onboard, while trajectory reconstruction, inference algorithms, and result presentation are performed in the ground segment. A complete record of the decoded messages is also maintained onboard, enabling subsequent analysis independently of the data reduction applied to telemetry.

This division was established considering link capacity and the different needs of each stage. Functions directly related to acquisition remain close to the receiver, allowing observations to be recorded independently of the execution of ground analysis algorithms. These algorithms, in turn, require a time history, access to geographical databases, and parameter adjustments during development. Their execution at the ground station provides greater flexi-

bility for processing and updates, while also enabling a web interface. This choice corresponds to the definition of a hybrid processing architecture specified in requirement HLR-ADS-04 (CubeDesign, 2026).

12.0.1 Theoretical Background

Aeronautical surveillance comprises the resources used to determine and monitor aircraft states. Unlike primary radar, which uses reflected electromagnetic energy, cooperative surveillance depends on information transmitted by the aircraft itself. In this context, *Automatic Dependent Surveillance–Broadcast* (ADS-B) enables the automatic transmission of data produced by onboard navigation systems and sensors. Its name reflects these characteristics: transmissions are automatic, depend on information obtained onboard, and are broadcast to receivers capable of interpreting them (Strohmeier *et al.*, 2014; Sun, 2021).

In the 1090ES mode, known as *1090 MHz Extended Squitter*, this information is transmitted over the Mode S link at 1090 MHz. In a DF17 ADS-B message, the 112 bits are organised into fields for downlink format, transponder capability, ICAO address, ADS-B content, and parity. This structure can be represented as

$$M = [DF \mid CA \mid ICAO \mid ME \mid PI], \tag{101}$$

where the ICAO address occupies 24 bits and the message field ME occupies 56 bits. The first five bits of ME form the *Type Code* (TC), which identifies the nature of the transmitted information (Sun, 2021). Table 68 summarises the types relevant to interpreting the observations used in the project.

Table 68: Main ADS-B Type Codes related to aircraft information

Type Code	Main information
1–4	Aircraft identification
5–8	Surface position
9–18	Airborne position with barometric altitude
19	Airborne velocity
20–22	Airborne position with GNSS altitude
28	Aircraft status
31	Operational status

Source: Adapted from Sun (2021).

Since identification, position, and velocity are distributed across different message types, a single reception does not necessarily provide the complete aircraft state. The absence of a field therefore does not correspond to a physically measured zero value, nor does it independently demonstrate a communication loss. This characteristic supports the individual preservation of observations and their subsequent association in the ground segment. Position also requires interpreting the *Compact Position Reporting* (CPR) encoding, whose reconstruction may use message pairs or a known spatial reference. Thus, not every received message immediately produces a new valid position (Sun, 2021).

Reception using *Software Defined Radio* (SDR) allows some signal-processing functions to be performed in software. In this project, receiver acquisition is followed by demodulation and decoding performed by *dump1090-fa*; the program developed by the team consumes the resulting messages without reimplementing this chain or CPR decoding. The use of ADS-B receivers and processing in research systems has precedents such as the OpenSky network,

described by Schäfer *et al.* (2014), which demonstrates the organisation of observations collected by a sensor network for air traffic analysis.

After decoding, observations can be associated by ICAO address and ordered in time. The trajectory is then represented by the sequence $\mathcal{T} = (\mathbf{x}_1, \ldots, \mathbf{x}_n)$, where

$$\mathbf{x}_i = [t_i, \phi_i, \lambda_i, h_i, v_i, \psi_i, \dot{h}_i], \tag{102}$$

with t_i denoting the time associated with the observation, ϕ_i and λ_i the latitude and longitude, h_i the altitude, v_i the horizontal speed, ψ_i the direction of motion over the ground, and $\dot{h}_i$ the vertical rate. Components of this vector may be absent from individual observations. This representation, defined to organise the project's data, provides the basis for analysing flight evolution rather than assigning meaning to an isolated position. Flight extraction and phase identification from ADS-B time series are discussed by Sun, Ellerbroek, and Hoekstra (2017).

One relevant quantity for this analysis is the vertical trend. When estimated from altitude, it may be affected by outlying samples or short intervals between measurements. The prototype uses a median of slopes between sample pairs, related to the estimator of Sen (1968). For the selected pairs, the trend is expressed as

$$\widehat{\beta} = \operatorname{median}_{(i,j)\in\mathcal{P}} \left(\frac{h_j - h_i}{t_j - t_i} \right), \tag{103}$$

where $\mathcal{P}$ represents the set of pairs admitted by the algorithm. In the implementation, the number of samples and the temporal separation of the pairs are limited, making the routine a computational adaptation of this approach. For altitude in feet and time in seconds, converting $\widehat{\beta}$ to feet per minute requires multiplication by 60. The resulting trend and the received vertical rate provide evidence for classifying climb, level flight, and descent; the classification rule adopted in the project is not a complete reproduction of the phase-identification method presented in the literature.

Vertical analysis is complemented by the geometry between the aircraft and airports. The prototype uses a spherical approximation of the Earth and calculates great-circle distance using the Haversine formula (Sinnott, 1984):

$$a = \sin^2\left(\frac{\Delta\phi}{2}\right) + \cos\phi_1 \cos\phi_2 \sin^2\left(\frac{\Delta\lambda}{2}\right), \tag{104}$$

$$d = 2R \arcsin\sqrt{a}, \tag{105}$$

with angles in radians and $R = 6371.0088$ km in the implementation. Although more rigorous ellipsoidal geodesic methods exist, such as those discussed by Karney (2013), distance is used in this prototype as one of the pieces of evidence for ranking candidates. It is analysed together with its evolution over time, altitude, and flight phase, since proximity or instantaneous direction alone does not determine the origin or destination airport.

12.0.2 Modeling

The functional architecture shown in Figure 61 separates observation recording from the interpretation of its history. The planned reception chain comprises the 1090 MHz dipole antenna, the balun, the SDR receiver, and the USB connection to the onboard computer. The balun interfaces the balanced dipole feed with the receiver's unbalanced connection. The construction and evaluation of the dipole–balun assembly must be distinguished from the reception tests already performed with a commercial antenna.

The selected receiver is the Nooelec NESDR Nano 3, comprising the RTL2832U circuit, responsible for the USB interface and digitisation, and the R820T2 tuner. The manufacturer's

specified tuning range, approximately 25 to 1750 MHz, covers the 1090 MHz used by ADS-B. The module has a temperature-compensated oscillator with a specified stability better than 0.5 ppm, an aluminium enclosure, and an external heat sink to assist heat transfer. The module's antenna input is MCX; connection to SMA cables requires a suitable adapter. Its small dimensions facilitate integration with the onboard computer but do not eliminate the need for thermal and mechanical verification of the final assembly (Nooelec, 2026).

Figure 60: Nooelec NESDR Nano 3 SDR receiver used in the ADS-B payload

Source: Amazon Brazil product listing (Amazon Brazil, n.d.).

The onboard segment uses a Raspberry Pi Zero 2 W with a 1 GHz quad-core processor and 512 MB of RAM (Raspberry Pi Ltd, 2026). The acquisition service starts and supervises the decoder, receives its messages over a local connection, and organises the information by ICAO address. Classification considers the message type and the associated aircraft; the latest state is made available to the communication service, while decoded observations are recorded onboard. Acquisition supervision and communication run in separate processes, allowing a receiver interruption to be handled without necessarily interrupting the link. This functional distribution integrates the payload into the OBC computing platform without assigning direct SDR operation to the satellite's overall management software.

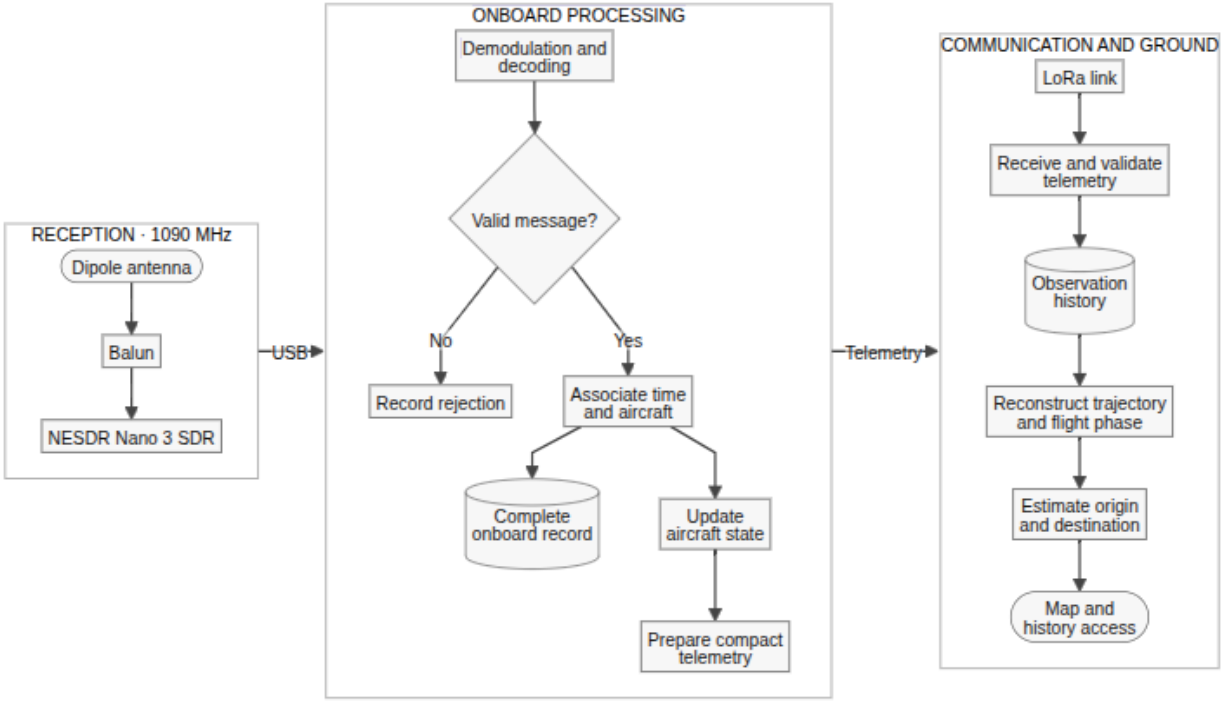
The processing distribution was sized according to the expected observation volume and transmission capacity. For 20 aircraft tracked over 600 s, assuming four messages per second per aircraft and the average record size adopted in the project, an estimated volume of 7.34 MiB is obtained. The calculated capacity of the nominal link, considering the time reserved for receiving telecommands, is approximately 58.3 KiB over the same interval. Since the volume of textual records exceeds this capacity by approximately 129 times, transmitting all messages in full is incompatible with the communication budget. The adopted solution is to retain complete observations onboard and transmit a compact representation of each aircraft's state. The difference between these volumes is a sizing estimate, not a measurement of integrated-system performance.

The volume follows from the record format and the broadcast rates. A tracked aircraft broadcasts airborne position and airborne velocity at about 2 Hz each and its identification about every 5 s, roughly four messages per second (International Civil Aviation Organization, 2026; Sun, 2021). The size of one line of the onboard record was computed from the record format for a representative aircraft and weighted by this broadcast mix (Table 69). For N_a aircraft broadcasting r messages per second over a mission of duration T, the volume is

$$V = N_a \, r \, T \, \bar{L}, \qquad \bar{L} = \frac{\sum_i f_i L_i}{\sum_i f_i}, \tag{106}$$

where f_i and L_i are the rate and the line size of each message type. With $N_a = 20$, $r = 4$ msg/s and $T = 600$ s (HLR-ADS-05), the mission produces 48 000 messages and 7.34 MiB (Table 70); the link capacity of Eq. (95) is 58.3 KiB.

Figure 61: ADS-B data reception, processing, and analysis flow



Source: Authors (2026).

Table 69: Size of one line of the onboard record, by message type

Message type	Rate per aircraft	Line size
Airborne position	2 Hz	156 B
Airborne velocity	2 Hz	170 B
Identification	0.2 Hz	109 B
Weighted mean, $\bar{L}$	4.2 Hz	160.4 B

Source: Authors (2026), computed by `lora_budget.py` (Gama CubeDesign, 2026).

Table 70: ADS-B volume over the mission against the link capacity

Messages per aircraft	Messages in 600 s	Onboard record	Ratio to capacity
2 msg/s	24 000	3.67 MiB	65:1
4 msg/s	**48 000**	**7.34 MiB**	**129:1**
6 msg/s	72 000	11.02 MiB	194:1

Source: Authors (2026), Eqs. (106) and (95).

States are transmitted every 5 s, resulting in approximately 48.5 KiB over the planned mission. This interval reduces the transmitted volume and allows link time to be reserved for statistics, operational information, and telecommand acknowledgements. The interval was chosen considering that the radio cannot transmit and receive simultaneously; therefore, the payload data stream must share the available time with control communications. The ground station uses the received states to reconstruct trajectories and perform inference, while the onboard record enables subsequent analysis at a higher temporal resolution.

Each event receives a time reference associated with the application's reception of the decoded message. The current implementation uses a monotonic time base to measure intervals

and a wall-clock reference provided by the ground segment to relate them to mission time. Updates to this reference are recorded, allowing synchronisation changes to be identified. Before this reference is obtained, the system may produce records without synchronised absolute time. When expressed in nanoseconds, the representation uses $t_{ns} = 10^9 t_s + t_n$, where t_s and t_n are the seconds and nanoseconds components; the storage unit does not imply nanosecond accuracy. The recorded instant also does not identify the exact time of transmission by the aircraft or acquisition of a radiofrequency sample. The state age is also included in telemetry, enabling the ground segment to account for observation delay and ageing. Timing accuracy relative to an independent reference remains to be characterised.

Events are represented in JSON according to RFC 8259, a technical document in the *Request for Comments* series that defines the syntax and interoperability rules of this data-interchange format. Its adoption allows records to be interpreted consistently by different programs; the document does not define the ADS-B protocol or the accuracy of recorded values. NDJSON framing, specified separately, places one JSON object on each line and enables incremental writing and sequential reading without waiting for a structure containing the entire session to be closed (Bray, 2017; NDJSON Project, 2014). This organisation preserves the information actually received. Table 71 presents the quantities used by the ground software; their availability depends on the content of the decoded messages.

Table 71: Main event information used in ADS-B processing

Information	Unit/format	Description
ICAO address	Hexadecimal	Aircraft's 24-bit address.
Time reference	ns	Time associated with the application's reception of the event.
Callsign	Text	Textual identification, when received.
Latitude and longitude	Degrees	Valid geographical coordinates.
Altitude	ft	Received altitude, according to message type.
Horizontal speed	kt	Speed over the ground.
Direction of motion	Degrees	Ground-track direction, distinct from aircraft heading.
Vertical rate	ft/min	Received vertical speed.
Flight condition	Indicator	On ground, airborne, or unknown.

Source: Adapted from Sun (2021) for the ADS-B information; the application reception time reference and event organisation were defined by the authors.

In the ground segment, the reading and validation stage checks message structure and numerical field limits before incorporating observations into the history. Valid records are stored in a SQLite database in the ground segment, together with the capture session, ICAO address, and event time. This database retains previous observations and the estimates produced, allowing past sessions to be consulted and each aircraft's history to be reconstructed. Removing an aircraft from the live view does not delete these records; it only indicates that the observations no longer meet the freshness criterion. Missing fields remain absent from the original record, while the derived state may temporarily reuse earlier values, retaining their age and provenance. In the ground visualisation and inference prototype, the position freshness limit was set to 60 s, or one minute. This interval is a heuristic tolerance for brief interruptions, chosen to avoid retaining old positions indefinitely; it is neither an ADS-B protocol rule nor an experimentally calibrated parameter. In the current onboard processing, position has a stricter limit of 10 s, while the remaining fields and the expiry of a state without updates use 60 s in the reference configuration. These limits belong to different stages and must be considered together when interpreting telemetry. Expiry may remove an aircraft from the live view, but it is not evidence of landing and does not erase its history.

The sequence of valid positions provides the observed trajectory and allows data quality to be assessed before inference. For flight-phase classification, the software combines the vertical trend with received information and applies persistence rules. Entry into climb and descent states uses thresholds of approximately +300 and −300 ft/min; once the phase has been established, retention thresholds of +150 and −150 ft/min introduce hysteresis. Changes based on the vertical trend require 10 s of persistence. The ground condition can be identified directly from the received indicator, while approach classification requires descent and persistent geometric compatibility with a runway. In the absence of sufficient evidence or when inconsistencies occur, the classification remains undetermined. These thresholds are project parameters, not universal definitions of flight phases.

Origin estimation uses the initial tracking window and evidence of climb. To prevent movement away from the departure point from favouring intermediate airports, the manager preserves a takeoff reference corresponding to the last position explicitly received with an indication that the aircraft was on the ground. This reference remains available after the transition to climb. When present, an airport's proximity to the ground position receives greater weight than its proximity to the current position; when absent, no artificial takeoff point is created. The hypothesis remains subject to the decision's quality and persistence criteria.

For the destination, aircraft geometry and its evolution over time are evaluated relative to airports and runways in the local database, derived from OurAirports data (OurAirports, 2026). Evidence includes distance, direction, convergence, vertical behaviour, and, where applicable, runway alignment and deceleration. The approach history allows useful information to be retained during small course changes or level segments in a terminal context. Inference does not depend on querying the aircraft's identity through online services or associating it with a registered commercial route.

The available evidence is combined into a normalised score,

$$C(a,t) = \frac{\sum_{i\in\mathcal{E}} w_i s_i(a,t)}{\sum_{i\in\mathcal{E}} w_i}, \tag{107}$$

where $s_i(a,t)$ represents a compatibility component for airport a, w_i its weight, and $\mathcal{E}$ the set of available components. The weighted combination is a decision rule defined within the project. The weights vary according to the observed context and are heuristic in nature. In level flight, the weight of the approach trend was reduced to prevent movement towards an airport alone from providing dominant destination evidence. Under this condition, outside a previously identified terminal context, the score may be penalised by a simplified descent-feasibility assessment calculated from relative height, direct distance, and horizontal speed. This assessment is a heuristic, not a certified aircraft performance model.

The resulting score expresses compatibility between observations and each candidate, rather than a calibrated probability. Thus, a value of 0.82 does not imply an 82% probability of being correct. To reduce dependence on a single observation, compatibility scores are integrated through an exponential memory based on the principle of exponential smoothing, which assigns progressively smaller weights to earlier observations (National Institute of Standards and Technology, 2026):

$$M_t = (1-\alpha)M_{t-1} + \alpha C_t, \qquad \alpha = 1 - \exp\left(-\frac{\min(\Delta t, 120\text{ s})}{\tau}\right), \tag{108}$$

with $\tau = 30$ s in the implementation. The choice of this constant, the treatment of irregular intervals, and the time limits are project adaptations. Memory is updated from new positions, with a minimum interval of 10 s, and reset after gaps exceeding 120 s between updates. Each candidate's first value initialises its memory. The purpose of this stage is to integrate evidence over time; it is not a Bayesian probability update.

The final decision distinguishes an accepted hypothesis, a tentative hypothesis, and a situation with insufficient evidence. An accepted hypothesis requires at least three persistent updates over 20 s, a minimum quality of 0.35, minimum evidence coverage of 0.55, accumulated compatibility of at least 0.65, and a margin of 0.08 over the second candidate, in addition to trend and relative-height criteria. For the destination, tentative classification allows a nearby hypothesis to be presented when persistent evidence exists but not all acceptance criteria have been met. The minimum compatibility in this condition is 0.65 during descent or approach, or 0.55 when a terminal context exists. Therefore, a tentative hypothesis may result either from insufficient separation between candidates or from compatibility below the final acceptance threshold. When the other evidence filters are not satisfied, the result remains undetermined.

This distinction preserves the option to abstain when the origin or destination is not observable within the received segment. Decisions and their evidence are stored and made available on the dashboard, which enables active aircraft to be monitored and historical sessions to be consulted. Presentation of an airport hypothesis remains separate from physical confirmation of takeoff or landing, which cannot be inferred solely from the estimator's acceptance state.

12.0.3 Verification

The verification presented comprises functional acquisition tests, software tests, data-transport simulation, and bench tests of the link. This separation distinguishes reception-chain operation from inference correctness: correctly processing a file does not demonstrate radiofrequency performance, just as decoding messages does not establish correct airport identification. Integrated validation, in turn, assesses the behaviour of the physical assembly in the intended configuration and operating conditions. The following results bring together tests performed in different configurations and at different integration levels, making the scope of each piece of evidence explicit.

Separate tests were performed to assess decoding and acquisition continuity. Functional reception-chain tests demonstrated extraction of ICAO address, position, altitude, and velocity. In another test, focused on continuous acquisition operation, 24 messages from two aircraft were accepted over five minutes, with no recorded rejections, internal discards, or restarts. This sample contained altitude and velocity information but no new positions. The tests provide complementary evidence: the former demonstrates field extraction, while the latter records continuous operation under the observed traffic conditions. Quantifying the correct-decoding rate requires comparing the obtained messages and fields against a known set of transmissions, since accepting the messages made available by the decoder does not independently measure the proportion of all transmissions received.

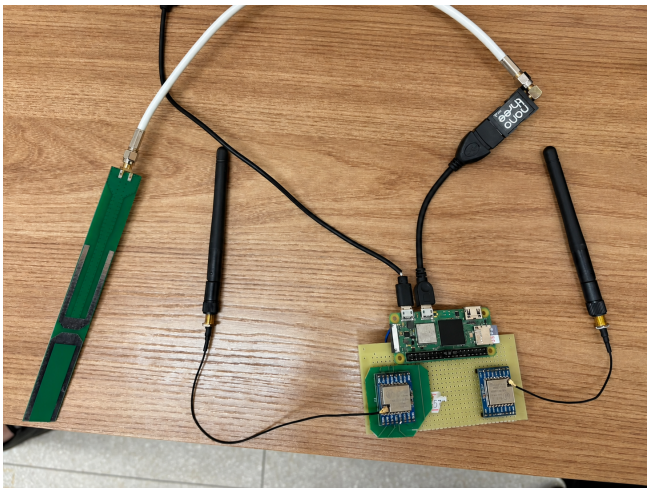
Computational evaluation with the SDR operating at 2.4 million samples per second indicated average decoder utilisation of approximately 21.9% of one core, with no recorded sample loss, under low-traffic conditions with up to three observed aircraft. These results support the functional feasibility of acquisition on the platform used, but they do not constitute a maximum-load measurement with 20 aircraft. Operation under that load still needs to be assessed in the integrated test.

Software tests include a scenario with 20 simultaneous aircraft over 600 s of simulated time, with 48,000 updates and checks on the preservation of individual states. This test extends coverage beyond the seven aircraft used to evaluate the estimator, although it does not represent ten minutes of physical operation of the entire chain. Integration tests also cover acquisition, state organisation, and communication over simulated transport. In the link simulation with 20 aircraft, the 95th percentile of latency until transmission start was 3.2 s, below the 5 s target under that scenario's conditions; this result must remain separate from actual radio

measurements.

The communication test used the configuration Raspberry Pi → LoRa → LoRa → the same Raspberry Pi, with both radio modules connected to the same computing platform. The transmitted message could therefore be compared with the received message to assess link errors and losses on the bench. The radios were approximately 80 cm apart, with transmit power set to 2 dBm. Request-and-acknowledgement tests completed 1,000 operations in the nominal profile, 1,000 in the fast profile, and 200 in the more robust profile. The ground-log summary counted 2,220 received frames, with no sequence gaps or invalid frames detected in that set. This result characterises loss and integrity of the observed frames, not a bit-error rate or ADS-B decoding accuracy.

Figure 62: Test bench for the ADS-B payload and its communication link



Source: Authors (2026).

In the joint acquisition and communication test performed on the same bench, operation of the inter-process connection, statistics exchange, and time synchronisation were observed. However, the low traffic activity did not produce enough position states to demonstrate transmission of trajectories for 20 aircraft. Likewise, round-trip times measured through radio requests are not equivalent to the latency between ADS-B reception and delivery of an observation to the ground segment. These tests demonstrate stages of operation while retaining the need for a complete run with a known scenario.

The estimator evaluation used seven synthetic aircraft with predefined expected answers. The set includes four complete flights, two partial observations, and one cruise case, allowing origin, destination, and abstention to be analysed separately. In total, 3,637 events were accepted by the reading and validation stage, with no rejections. This result concerns ingestion of the supplied file and does not measure the correct-decoding rate of ADS-B signals. Table 72 presents the hypotheses retained in the final summary of each scenario.

In the final summary, the five observable origins and five observable destinations matched the expected airports. Four of these destination hypotheses remained tentative, and the four deliberately unobservable endpoints remained undetermined. Agreement with the expected answer must therefore be distinguished from the degree of acceptance assigned by the algorithm. These results characterise the synthetic set used and do not represent an estimate of operational accuracy in real traffic.

The temporal analysis also identified a limitation that is not apparent in the final table: cases F10002, F10004, and F10005 produced incorrect intermediate destination hypotheses before converging to the expected airport. In F10005, some of these hypotheses reached accepted status. Therefore, agreement in the final results does not demonstrate an absence of

Table 72: Final results of synthetic origin and destination estimation scenarios

Case	Expected origin	Obtained origin	Expected destination	Obtained destination
F10001	SBGO	SBGO (E)	SBBR	SBBR (T)
F10002	SBBR	SBBR (E)	SBGO	SBGO (T)
F10003	SBKP	SBKP (E)	SBGR	SBGR (T)
F10004	SBGR	SBGR (E)	SBKP	SBKP (T)
F10005	Not observable	Undetermined	SBBR	SBBR (E)
F10006	SBBR	SBBR (E)	Not observable	Undetermined
F10007	Not observable	Undetermined	Not observable	Undetermined

E: accepted hypothesis; T: tentative hypothesis.

Source: Authors (2026).

errors throughout tracking. In addition to the airport recorded at the end, estimator verification should consider the time to the first hypothesis, decision stability, and the occurrence of incorrect candidates along the trajectory.

Interpretation of the results also depends on flight observability. When reception begins after takeoff, the history may not contain sufficient evidence for the origin; similarly, a capture ending before approach may not allow the destination to be inferred. These situations justify assessing the two endpoints separately and retaining an undetermined result as a valid output. The weights and thresholds remain heuristic, and their statistical evaluation would require independent and representative data beyond the scenarios used during development.

Regarding the mission, the described architecture meets the need to define and justify the processing distribution, while time association, organisation by aircraft, and position reconstruction provide the mechanisms used for tracking and estimating origin and destination (CubeDesign, 2026). The presence of these mechanisms in the software and the bench results are related to requirements in Table 73. The criteria presented make the requirements operational for the project; they do not replace the wording of the regulations. The secondary mission concept is identified in the matrix to maintain traceability, but its development is outside the scope of this ADS-B payload revision.

Table 73: Traceability of ADS-B requirements, acceptance criteria, and available evidence

Requirement	Acceptance criterion	Verification	Integrated validation	Current status
HLR-ADS-01	Receive valid ADS-B messages at 1090 MHz in the test scenario.	Reception using an SDR and commercial antenna; inspection of acquisition records.	Repeat reception with the dipole and balun integrated into the final platform.	Reception demonstrated on the bench. The mission antenna and balun assembly remains to be built and verified.
HLR-ADS-02	Correctly extract ICAO address, position, altitude, and velocity from applicable messages.	Complete decoding confirmed in the functional test reported by the team; data-interpretation tests.	Compare fields from the known scenario with the physical chain's output and record their correspondence.	Functionally tested and validated on the bench, according to the test reported by the team.

Continued on the next page.

Table 73 — continued

Requirement	Acceptance criterion	Verification	Integrated validation	Current status
HLR-ADS-03	Maintain 20 simultaneous aircraft with separate identification and states.	Simulation with 20 aircraft and inspection of update association; estimator evaluation with seven scenarios.	Inject traffic from 20 aircraft and check identities and states on the ground.	Capacity verified in software; equivalent load has not yet been demonstrated in the complete physical chain.
HLR-ADS-04	Define, implement, and justify the processing distribution.	Review of the architecture, data budget, and onboard and ground service tests.	Exercise the complete chain in the final configuration as evidence complementing the documentation review.	Hybrid architecture implemented and justified; stages tested on the bench. Final integration remains to be demonstrated.
HLR-ADS-05	Operate continuously for 600 s with up to 20 aircraft, without unhandled failure.	Test over 600 s of simulated time with 20 aircraft and 48,000 updates.	Run for 600 s of real time with 20 aircraft, recording continuity, losses, and states.	Verified in the software scenario; the examined records do not demonstrate the complete physical mission at this load.
HLR-ADS-06	Present a secondary mission concept compatible with the primary mission.	Documentation review of the concept and its compatibility; implementation is not mandatory.	Not required for a conceptual-proposal requirement.	Must be traced in the secondary mission section; this ADS-B analysis does not declare it fulfilled.
HLR-ADS-07	Associate time with observations, enabling ordering and identification of synchronisation adjustments.	Tests with known times and inspection of recorded references and ages.	Measure effective resolution and error relative to an independent time reference.	Time association and synchronisation implemented and exercised. Timing accuracy has not yet been characterised.
HLR-ADS-08	Apply quantitative decoding, loss, and latency targets according to Table 74.	Calculate metrics in known scenarios and inspect records containing counters and times.	Measure the same indicators in the integrated chain with 20 aircraft over 600 s.	Targets defined, including maximum latencies of 10 s until transmission start and 12 s until ground reception. Partial simulation and bench evidence is available, compliance with these limits remains to be demonstrated in the integrated test..

Source: Authors, based on the mission requirements (CubeDesign, 2026) and the tests described in this section.

Requirement HLR-ADS-08 establishes how mission success will be measured quantitatively. It does not add a new decoding function: it requires criteria that distinguish a merely

functional system from one that achieves the defined performance. The targets in Table 74 constitute the quantitative criteria adopted for the project. The decoding rate must consider known messages emitted by the test generator and the correctness of their fields. Link loss must be calculated separately from expected and received frames, excluding duplicates and accounting for session restarts. Thus, zero loss in the LoRa test does not by itself demonstrate a 100% ADS-B decoding rate.

Table 74: Quantitative targets adopted to evaluate the payload and its data transport

Indicator	Target	Assessment method
Correct decoding	$\geq 95\%$	Compare decoded messages and fields with the injected set and its reference answers, over the messages broadcast while the LoRa radio is not transmitting; report the transmit fraction separately.
Frame loss on the link	$\leq 5\%$	Compare expected and received sequences in the ground segment.
Latency until transmission start	$p_{95} \leq 5$ s	Measure the interval between reception by the onboard application and the start of the corresponding transmission.
Latency until ground reception	$p_{95} \leq 7$ s	Measure the interval until the observation is received on the ground, accounting for the time references.
Maximum latency	≤ 10 s; ≤ 12 s to the ground	Largest track age received on the ground (Eq. (99)).
Load and duration	20 aircraft; 600 s	Check continuity and correct state association throughout the test.

Source: Authors (2026).

The 95th percentile, denoted by p_{95}, is the value below which 95% of the observed latencies fall. It does not establish an absolute upper bound for all messages. Since the requirement also calls for a maximum acceptable latency, an upper bound of 10 s from reception to the start of transmission is adopted, above the analytical worst case of about 8.5 s (Eq. (99)); the largest observed delay is recorded alongside the percentile, from the age carried by every track record. Integrated validation must bring these metrics together in the same test, with a known scenario, documented configuration, and input and output records. This distinction preserves the results already demonstrated and precisely identifies the evidence still needed to declare full compliance with the requirements.

13 Extra Payload

This section summarizes the plasma physics concepts that underlie the acceleration principle of the thruster and the performance estimates presented below.

A plasma is a medium in which an appreciable fraction of a gas has been ionized, so that free electrons, ions, and neutral species coexist (Bittencourt, 2004; Chen, 2016).

13.1 Plasma physics

Unlike a neutral gas, which is governed by short-range collisions, a plasma is dominated by the long-range Coulomb force, which gives the medium a collective response. An ionized gas is treated as a plasma when it simultaneously meets three conditions (Chen, 2016): a characteristic dimension much larger than the Debye length ($\lambda_D \ll L$), which ensures quasi-

neutrality, $n_e \approx n_i$; a large number of electrons within the Debye sphere ($N_D = \frac{4}{3}\pi n_e \lambda_D^3 \gg 1$); and electrostatic oscillations faster than collisions with neutrals ($\omega_{pe}\,\tau_n > 1$).

13.1.1 Characteristic Scales

A potential perturbation in the plasma is compensated by the redistribution of the surrounding charge carriers, and the field decays exponentially over the length scale

$$\lambda_D = \left(\frac{\varepsilon_0\, k_B T_e}{n_e\, e^2} \right)^{1/2} . \tag{109}$$

For $T_e \approx 5$ eV and $n_e \approx 5 \times 10^{18}$ m^{-3}, values representative of propulsion sources, $\lambda_D \approx 7\ \mu$m, three orders of magnitude below the dimensions of the ionization chamber. The thruster volume can therefore be treated as quasi-neutral.

The separation between electrons and ions also produces a restoring field that makes the electrons oscillate at the plasma frequency

$$\omega_{pe} = \left(\frac{n_e\, e^2}{\varepsilon_0\, m_e} \right)^{1/2} , \tag{110}$$

of approximately 20 GHz for the density above. Since waves with frequencies below ω_{pe} do not propagate in an unmagnetized plasma, high-density sources rely on modes that propagate in the presence of a magnetic field, such as helicon waves.

13.1.2 Electron Temperature and Boltzmann Relation

The electrons are assumed to follow a Maxwellian distribution at temperature T_e, with mean energy $\frac{3}{2}k_B T_e$ (Lieberman; Lichtenberg, 2005). The energy delivered to the propellant is absorbed mostly by the electrons and, since energy exchange with the ions is limited by the mass ratio m_e/m_i, it follows that $T_i \ll T_e$. This disparity is the starting condition for ambipolar propulsion.

Neglecting electron inertia, the balance between the electric and pressure forces acting on the electrons leads to the Boltzmann relation (Chen, 2016):

$$n_e = n_0 \exp\left[\frac{e\,(\phi - \phi_0)}{k_B T_e} \right] . \tag{111}$$

Density and potential are thus coupled: a tenfold reduction in density implies a potential drop of about $2.3\, k_B T_e / e$.

13.1.3 Sheath and Bohm Criterion

Because they are faster than the ions, the electrons reach the surfaces in contact with the plasma first, and these surfaces become negatively charged. A non-neutral layer a few λ_D thick then forms, the sheath, in which the potential difference between the plasma and the wall is concentrated (Hutchinson, 2002). For this layer to be stable, the ions must reach its edge with a minimum velocity

$$u_B = \left(\frac{k_B T_e}{m_i} \right)^{1/2} , \tag{112}$$

known as the Bohm velocity. The resulting ion flux to the walls represents a loss of particles and energy, and the associated bombardment must be taken into account in the selection of the chamber materials.

13.1.4 Ambipolar Diffusion and Magnetization

If electrons and ions diffused freely, the electrons would leave the plasma first. However, the incipient charge separation generates an electric field, the ambipolar field, which slows down the electrons and accelerates the ions until both fluxes are equal, $\Gamma_e = \Gamma_i$ (Lieberman; Lichtenberg, 2005). In the differential form of Eq. (111),

$$\mathbf{E}_{\text{amb}} = -\frac{k_B T_e}{e}\,\frac{\nabla n}{n}, \tag{113}$$

the field points toward the lower-density regions and transfers part of the electron thermal energy to the ions.

In the presence of a magnetic field, the particles describe helical trajectories around the field lines, with cyclotron frequency and Larmor radius

$$\omega_c = \frac{|q|\,B}{m}, \qquad r_L = \frac{m\,v_\perp}{|q|\,B}. \tag{114}$$

For $B = 0.05$ T, a 5 eV electron has $r_L \approx 0.1$ mm, whereas an argon ion accelerated to 11 km/s has $r_L \approx 9$ cm. The electrons therefore remain magnetized, while the ions do not. The magnetic field guides the electrons and, through the ambipolar field, indirectly guides the ions.

13.2 Ambipolar thruster

The ambipolar thruster is an electric thruster in which thrust results from the conversion of the electron thermal energy into directed ion kinetic energy, by means of the ambipolar electric field that forms during plasma expansion (Charles, 2009; Takahashi, 2019). Unlike gridded ion thrusters and Hall-effect thrusters (Goebel; Katz, 2008), there are no accelerating electrodes in contact with the plasma, and the emitted beam is quasi-neutral and carries no net current. For this reason, the thruster does not require a neutralizer cathode.

Its operation can be divided into three stages. In the first, the gaseous propellant is injected into an ionization chamber and converted into plasma by electromagnetic energy supplied without internal electrodes, usually by radiofrequency power coupled through an external antenna (helicon-type source) or by microwaves under electron cyclotron resonance (ECR) conditions (Takahashi, 2019; Cannat *et al.*, 2015). Under this condition, the wave frequency matches the electron cyclotron frequency [Eq. (114)], and the resonance field is

$$B_{\text{ECR}} = \frac{2\pi f\, m_e}{e}, \tag{115}$$

which corresponds to about 0.0875 T for $f = 2.45$ GHz. In both cases, the energy is absorbed predominantly by the electrons, which reach temperatures ranging from a few to tens of electron-volts, while the ions remain relatively cold.

In the second stage, the plasma is guided out of the chamber by a divergent magnetic field, produced by coils or permanent magnets, known as a magnetic nozzle (Ahedo; Merino, 2010). The magnetized electrons follow the field lines and expand with them. As the cross-section of the flux tube increases, the density decreases and, according to Eq. (111), the electric potential also decreases along the nozzle.

In the third stage, the weakly magnetized or unmagnetized ions are accelerated by this potential drop. Neglecting collisions and ion thermal energy, energy conservation along a streamline gives

$$\tfrac{1}{2} m_i u_i^2 + e\,\phi = \tfrac{1}{2} m_i u_{i0}^2 + e\,\phi_0, \tag{116}$$

so that, for a total potential drop $\Delta\phi$, the ion exhaust velocity is

$$u_e \approx \sqrt{\frac{2\,e\,\Delta\phi}{m_i}}. \tag{117}$$

The ions enter the expansion region at the Bohm velocity [Eq. (112)] and exit supersonically. Since the potential drop is sustained by the electron temperature, $\Delta\phi$ is typically a few times $k_B T_e/e$. As an illustration, for argon and $\Delta\phi = 25$ V, Eq. (117) yields $u_e \approx 11$ km/s. In some configurations, the potential drop is concentrated in a narrow, current-free layer called a double layer (Charles; Boswell, 2003); the underlying physical mechanism remains the charge separation imposed by the difference in mobility between electrons and ions.

To a first approximation, the thrust is given by the product of the mass flow rate and the exhaust velocity,

$$F = \dot{m}\,u_e, \qquad I_{sp} = \frac{u_e}{g_0}, \tag{118}$$

where g_0 is the standard gravitational acceleration. In terms of the forces acting on the spacecraft, the thrust has two contributions (Ahedo; Merino, 2010; Takahashi, 2019). The first is the plasma pressure on the walls of the ionization chamber. The second is the magnetic force: the plasma expansion in the nozzle induces azimuthal diamagnetic currents, which interact with the radial component of the applied field. The reaction to this force acts on the coils or magnets and is transmitted to the satellite structure.

For the thrust to be effective, the plasma must detach from the magnetic field lines, which close back toward the thruster. Detachment occurs as the field weakens downstream and the electrons become demagnetized. The efficiency of this process, together with particle and energy losses to the chamber walls, still limits the performance of these thrusters, especially at small scales (Takahashi, 2019).

13.2.1 Advantages and Limitations for CubeSats

The absence of electrodes in contact with the plasma eliminates grid and cathode erosion, which is one of the main wear mechanisms of conventional electric thrusters. The architecture also accepts different propellants, such as xenon, krypton, and argon, and does not require a neutralizer, which reduces the number of components. When the nozzle is formed by permanent magnets, the magnetic field consumes no electrical power, a relevant feature for the power budget of a CubeSat.

The limitations include the still modest efficiency at low power, the mass of the magnets or coils, and the need for a radiofrequency or microwave power unit. For integration with the satellite, the magnetic field of the thruster must also be regarded as a source of disturbance: the dipole moment of the magnets interacts with the geomagnetic field and produces a residual torque on the spacecraft, and it may also affect the readings of the magnetometers of the attitude control subsystem. These effects are addressed in the integration section.

13.3 Thruster developed by LPF

This section covers only the prototype developed at the Plasma Physics Laboratory of the University of Brasília (LPF-UnB) (Costa *et al.*, 2024; Miranda *et al.*, 2024; Silvano *et al.*, 2025), based on the thruster described by Sheehan et al (Sheehan *et al.*, 2014). The operating principle is the one described in the previous section. Figure 63 shows a prototype and Table 75 summarizes its parameters.

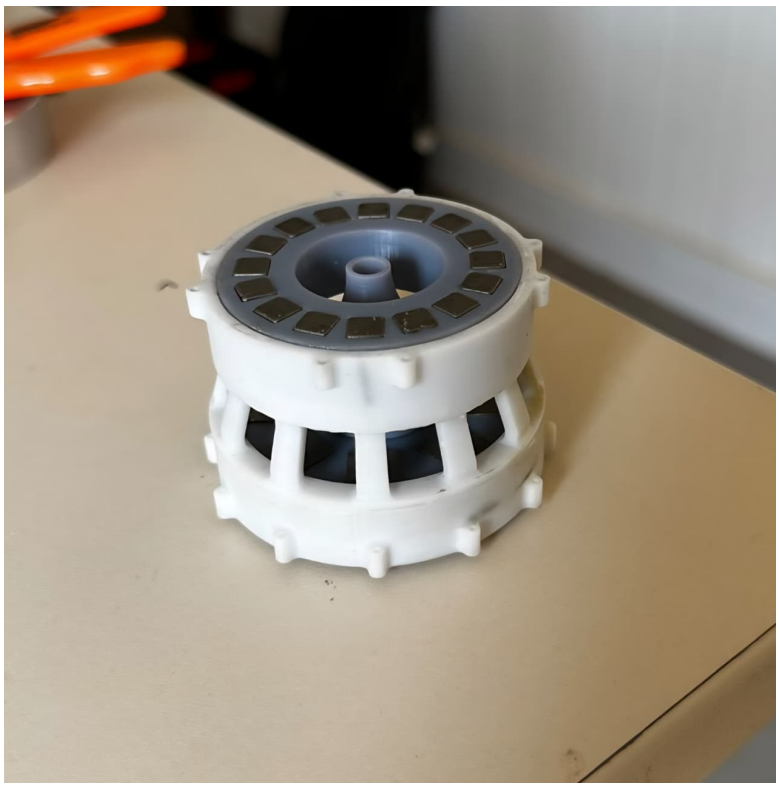


Figure 63: Prototype of the ambipolar thruster developed at LPF-UnB, showing the permanent magnet ring and the central gas inlet. Source: LPF-UnB.

Table 75: Parameters of the LPF-UnB thruster model used in the simulations (Costa *et al.*, 2024; Silvano *et al.*, 2025).

Parameter	Value
Propellant	Argon
Flow rate	12 sccm
Injector plate	1 mm thick, 25 holes of 0.25 mm radius
Magnetic field	Permanent magnets
Antenna position	$\approx$ 10 mm from the plate

The group focuses on the neutral gas distribution in the discharge chamber. Molecular flow simulations in COMSOL Multiphysics compare normal injection, with holes perpendicular to the plate, and vortex injection, with inclined holes that add a radial component to the flow (Ding *et al.*, 2017). The vortex mode raised the neutral density by approximately 43% according to the authors (Costa *et al.*, 2024; Silvano *et al.*, 2025), and the profiles were fitted by fourth-degree polynomials for particle-in-cell simulations (PIC-MCC) (Silvano *et al.*, 2025).

These results come from simulations without plasma feedback; no thrust or specific impulse measurements are reported in these works.

13.4 Integration with the satellite

The satellite is compared in two configurations. In the 2U configuration, the vehicle contains only the service module, which accommodates the power, communications, onboard computer, and attitude-control subsystems. The envelope is approximately 100 mm $\times$ 100 mm $\times$ 227 mm (The CubeSat Program, Cal Poly SLO, 2022a), and integration is limited to this module: there is no thruster, propellant tank, or power electronics. The dispenser, the rail

length, and the center-of-mass window are those of a 2U CubeSat (The CubeSat Program, Cal Poly SLO, 2022a).

In the 3U configuration, the propulsion system occupies the lower unit, which extends approximately 113.5 mm along the Z axis. This unit is part of the continuous structure of the satellite, built to the standard of a 3U CubeSat, and is not a module attached at a later stage. The integrated envelope is approximately 100 mm × 100 mm × 340.5 mm, in accordance with the CubeSat Design Specification (The CubeSat Program, Cal Poly SLO, 2022a). The four rails run the full length of the structure. An internal panel separates the service module from the propulsion volume, reserves that volume for the thruster, the propellant tank, and the power unit, and provides the mounting support for the engine.

The propulsion assembly is installed in the lower unit, with the nozzle axis aligned with the satellite's longitudinal axis and oriented in the negative Z direction. The exhaust is directed outward, away from the service module, in order to reduce the interaction of the exhaust plume with the solar panels, the antennas, and the sensors.

The electrical interface between the propulsion unit and the service module passes through the internal separation panel. Conductor routing includes securing and strain relief of the harness, as well as physical segregation of the high-voltage lines from the data lines, in order to limit electromagnetic interference and the risk of electrical discharge.

13.4.1 Mass comparison and attitude performance

In the 3U configuration, the structure of the lower unit, the thruster, the propellant tank, the propellant, the power electronics, and the design margin are included. Under the usual reference of 1.33 kg per unit, the launch mass limit increases from about 2 kg for the 2U to about 4 kg for the 3U. The design mass remains below these limits. The propellant is accounted for in both the full and the empty condition.

With the tank full, the center of mass shifts toward the lower unit. As the propellant is consumed, the center of mass moves back toward the service module. The moments of inertia increase, particularly for rotations about the transverse axes, because the additional mass is located at the end of a longer body. The attitude control of the 3U must provide torque and slew time compatible with this inertia. The thrust axis must pass through the center of mass; misalignment produces residual torque during firing and greater use of the actuators. In the 2U there is no thrust, and the center of mass remains that of the service platform.

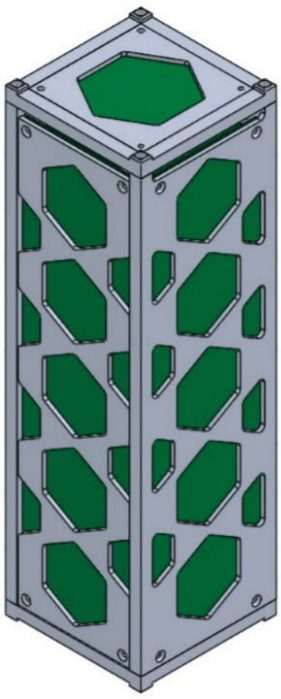


Figure 64: External structure of the 3U configuration, with side closures and top plate cut with hexagonal openings
Source: Authors(2026).

13.4.2 Electrical Power System

The electrical power generation of the 3U satellite remains that of the photovoltaic array already adopted for the 2U. The solar cells are mounted on printed circuit boards fixed to the walls of the service module, in the same position and with the same collecting area as in the previous configuration. The lower unit does not carry solar cells. The increase in body length therefore does not change the generated power, and the external electrical design of the service module is retained.

Within the power subsystem, these boards serve as the substrate of the solar panels. Their attachment to the walls follows the lateral faces of the service module, which provide the surface available for collection. Each board carries the cells, the interconnection traces, and the connection points to the power system, and it closes the corresponding face of the structure. The same type of board, without cells, may be used to close the walls of the lower unit, solely to preserve the continuity of the external skin and the mechanical protection of the propulsion volume.

Closing the faces reduces the direct exposure of the internal components to the external environment. These boards, however, do not constitute the thermal insulation of the satellite: temperature control of the lower unit is treated separately, by means of surface finishes, heat rejection from the thruster and, where required, insulation and heating of the propellant tank.

Retention of this arrangement depends on the power budget of the 3U. The margin verified for the 2U must also cover the thruster during operation, taking into account the peak power during firing, the average consumption over the orbit, and the battery depth of discharge, including eclipse. If this margin is not confirmed, the generating area shall be revised before the arrangement is finalized.

13.4.3 Thermal Comparison

In the 2U, internal heat originates from the dissipation of the service electronics. Passive control conducts this heat from the component to the board on which it is mounted, from the board to the structure, and from the structure to the external faces. The boards fixed to the walls take part in this path: they support the circuits, close the faces, and limit the direct exposure of the interior to the environment. The surface finish determines the fraction rejected to space. Where heat exchange is to be reduced, multilayer insulation is applied.

The battery remains within the temperature range already qualified for the 2U. Its control combines conductive coupling to the structure, temperature sensors, and heaters, which are activated when the temperature approaches the lower limit, particularly during eclipse. Heater command is assigned to the onboard computer, with a protective thermostat.

In the 3U, the service module retains this arrangement. The lower unit adds two conditions. During firing, the thruster and the power unit concentrate the dissipation in that volume. Heat is conducted to one face of the lower unit, left with a high-emissivity finish and without insulation, for rejection to space. This rejection forms part of the thermal control during firing and does not replace it.

Outside firing, in eclipse or with the engine off, the lower unit loses heat. The propellant tank is provided with a sensor, a heater, and a thermostat, so that it remains within the operating range of the propellant. The internal separation panel receives thermal insulation, so that conduction along the rails does not raise the temperature of the battery and the computer above the limits already accepted for the service module.

The boards without cells that close the lower unit repeat the closing function used on the 2U. The face assigned to thruster heat rejection remains free of insulating cover, with the high-emissivity finish defined for firing.

13.4.4 Structural comparison

Both configurations use 6061-T6 aluminum for the primary structure and thin closeout panels on the faces. These parts are joined with 10M3×8 screws and nuts. The closeout panels are plates with hexagonal cutouts: the cutout removes mass uniformly while keeping the plate connected to the rails, and it also allows the internal air to escape during launch. The external structure of the 3U is that of a conventional 3U CubeSat. Internally, the lower unit is reserved for the thruster and is separated from the service module by the internal panel.

The estimated mass of the 2U, including the external aluminum structure and all internal components, is approximately 2373.5 g. For the 3U, 102.10 g of additional external aluminum structure and 93.80 g of payload are added. A margin of 30% is applied to the 93.80 g, corresponding to 28.14 g, because the mass of the payload external structure and the mass of the propellant with the tank full were not found. The estimated mass of the 3U is therefore 2597.54 g. For construction uncertainties, a variation of ±10% about this total is allowed, corresponding to the range from 2337.79 g to 2857.29 g.

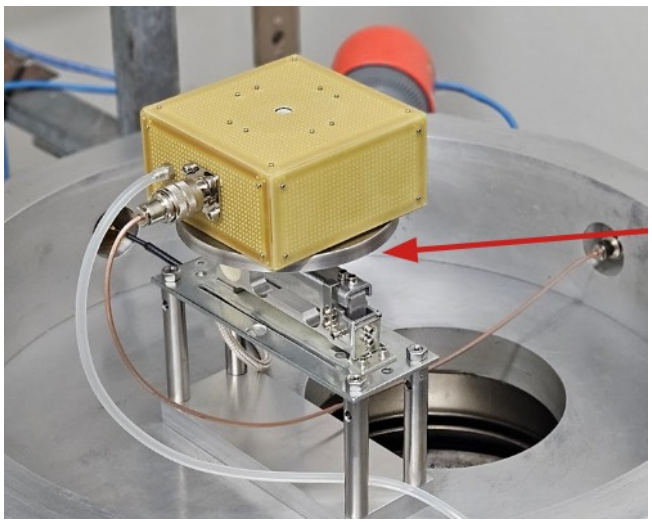


Figure 65: External structure of the ambipolar thruster prototype
Source: LPF-UnB.

14 Integration Plan

14.1 Assembly Manual

This section aims to briefly present how the assembly of the Cubetschek project is performed. Below is the step-by-step procedure in chronological order to facilitate the understanding of the structural parts and the organization of the subsystems. The structure is in its final machined version, while the PCBs are 3D-printed in Polylactic Acid (PLA), which are used to test the internal space and ensure the proper placement of all boards.

The structure begins with the four beams and the two walls that were machined together, alongside the bottom cover. The third wall, screwed to the base, can be kept in place or removed to improve internal access. Further explanations regarding the structural parts are provided in Section 5.2.

Figure 66: Open structure
Source: Authors (2026).

Next, the rods are positioned, passing through the bottom cover, and are screwed in place. The PCBs and spacers are then installed.

Figure 67: Structure with rods and PCBs
Source: Authors (2026).

Once the interior is completed, the final wall, the top cover, and the closure frames can be screwed on. Furthermore, the antennas and solar panels will be added to complete the project's structure, which can be seen in Figure 4.

Figure 68: Closed structure
Source: Authors (2026).

15 Conclusion

The development of Cubetschek represented a comprehensive integration across multidisciplinary engineering domains, combining expertise in electronics, structures, and control systems to accomplish the mission proposed for the 5th CubeDesign. Each module was conceived, modeled, and evaluated to ensure the autonomous and reliable operation of the nanosatellite, fulfilling all technical and operational requirements established by the competition.

The outcomes obtained thus far through simulation and experimental phases validate the satellite's performance, demonstrating its resilience against simulated environmental conditions. However, at the time of writing, full end-to-end integration across all hardware units has not yet been conducted. Leveraging accessible technologies alongside tailored solutions exemplifies

the team's ability to translate theoretical principles into practical space systems engineering.

Upcoming milestones in the overall project lifecycle will concentrate on full subsystem integration, as well as comprehensive testing to verify multi-module interoperability. Certain stages presented notable risk management hurdles, particularly internal layout allocation and tasks associated with the ADS-B mission. The team views these challenges as invaluable learning experiences for future endeavors.

Ultimately, this technical report consolidates the expertise gained by the Gama CubeDesign team. It serves not only as a record of Cubetschek's design and validation milestones, but also as a foundation for refining future space missions and fostering next-generation CubeSat initiatives for aerospace research and innovation in Brazil.

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A Project Management and Schedule

Table 76: Master Schedule for the Cubetschek Mission

ID	Activity	Start	End	Duration (Days)
1	Project Outline	08/04/26	29/04/26	16
2	Subsystem Definition	08/04/26	29/04/26	16
3	Literature Review	09/04/26	16/05/26	28
4	CAD Modeling and Design	01/05/26	14/09/26	93
5	Engineering Simulations	01/06/26	09/10/26	93
6	Component Procurement	05/06/26	21/09/26	75
7	Manufacturing and Assembly	01/08/26	10/11/26	72
8	Testing and Verification	08/09/26	20/11/26	53
9	CDR Documentation and Reporting	15/09/26	09/10/26	19

Source: Authors (2026).

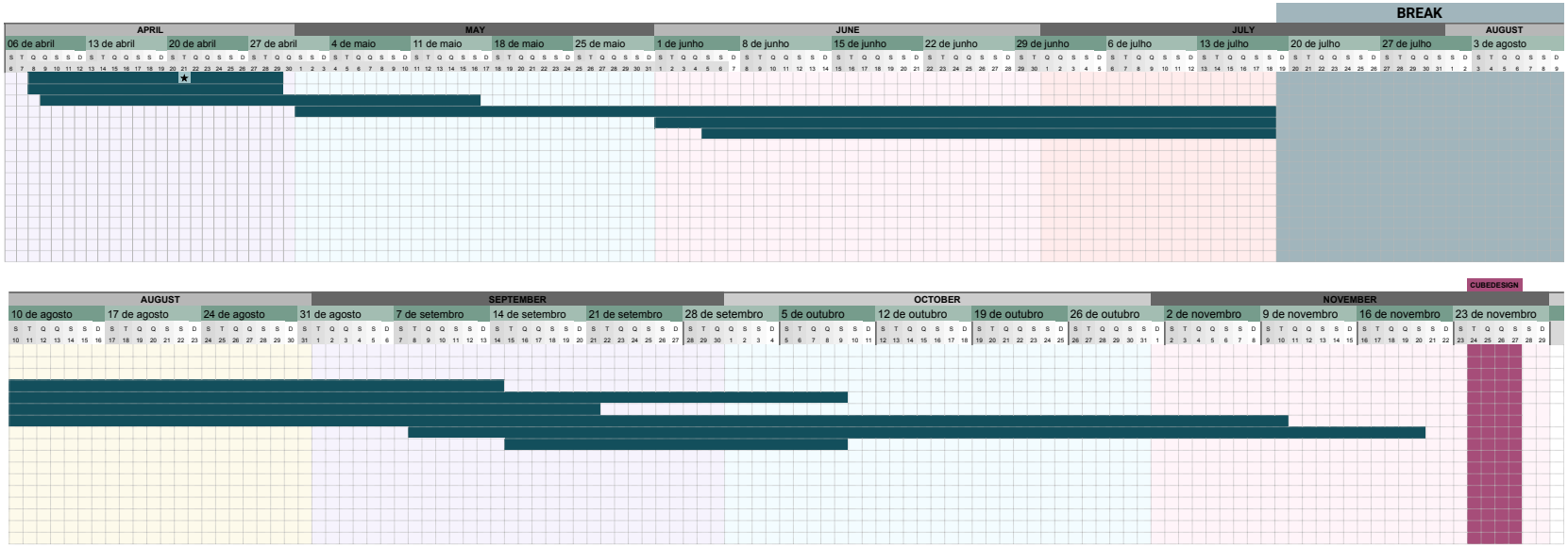


Figure 69: Gantt chart of the overall schedule for the Cubetschek mission
Source: Authors (2026).

B Structures

B.1 Technical drawing

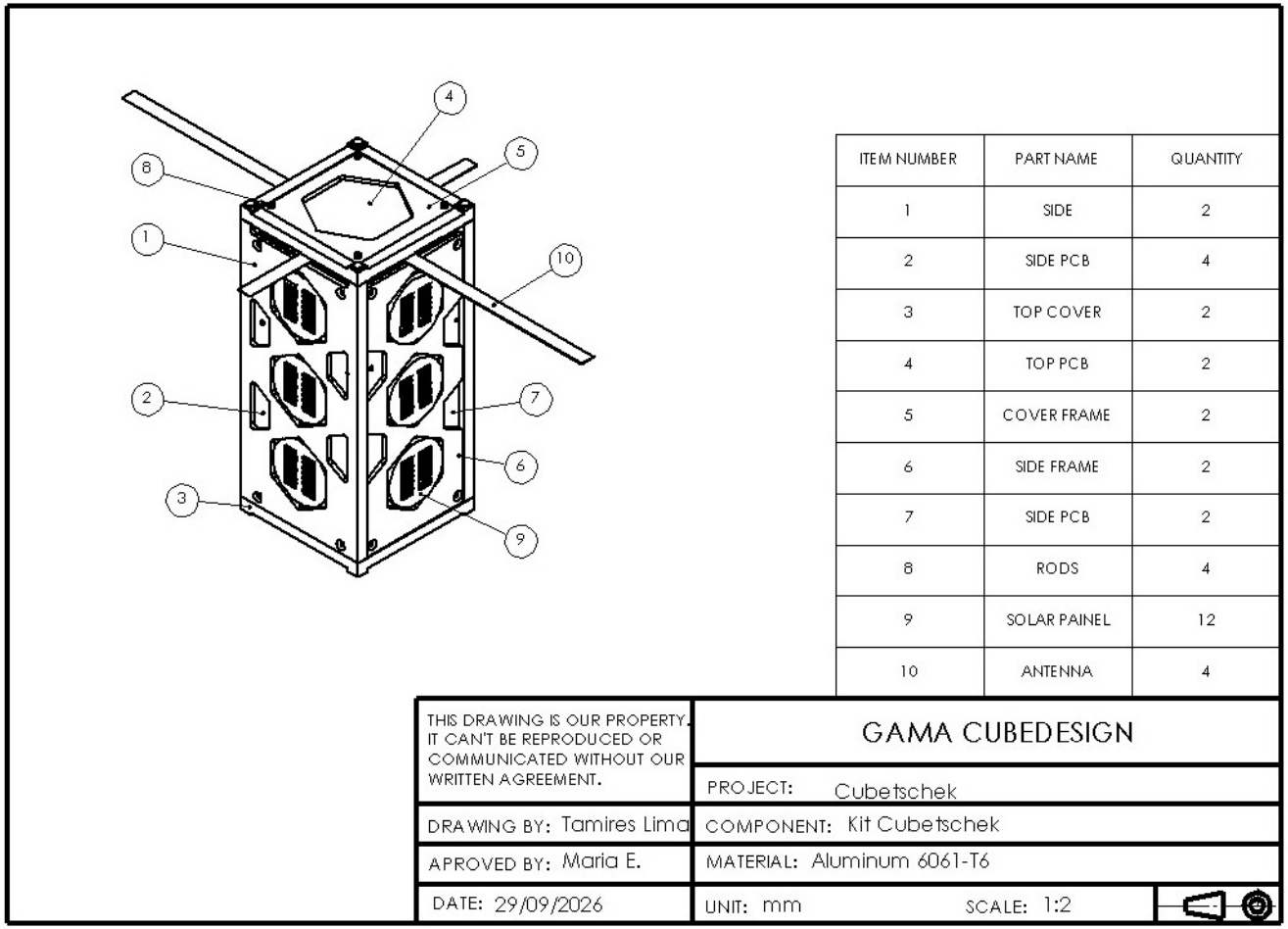


Figure 70: Closed Cubetschek
Source: Authors (2026).

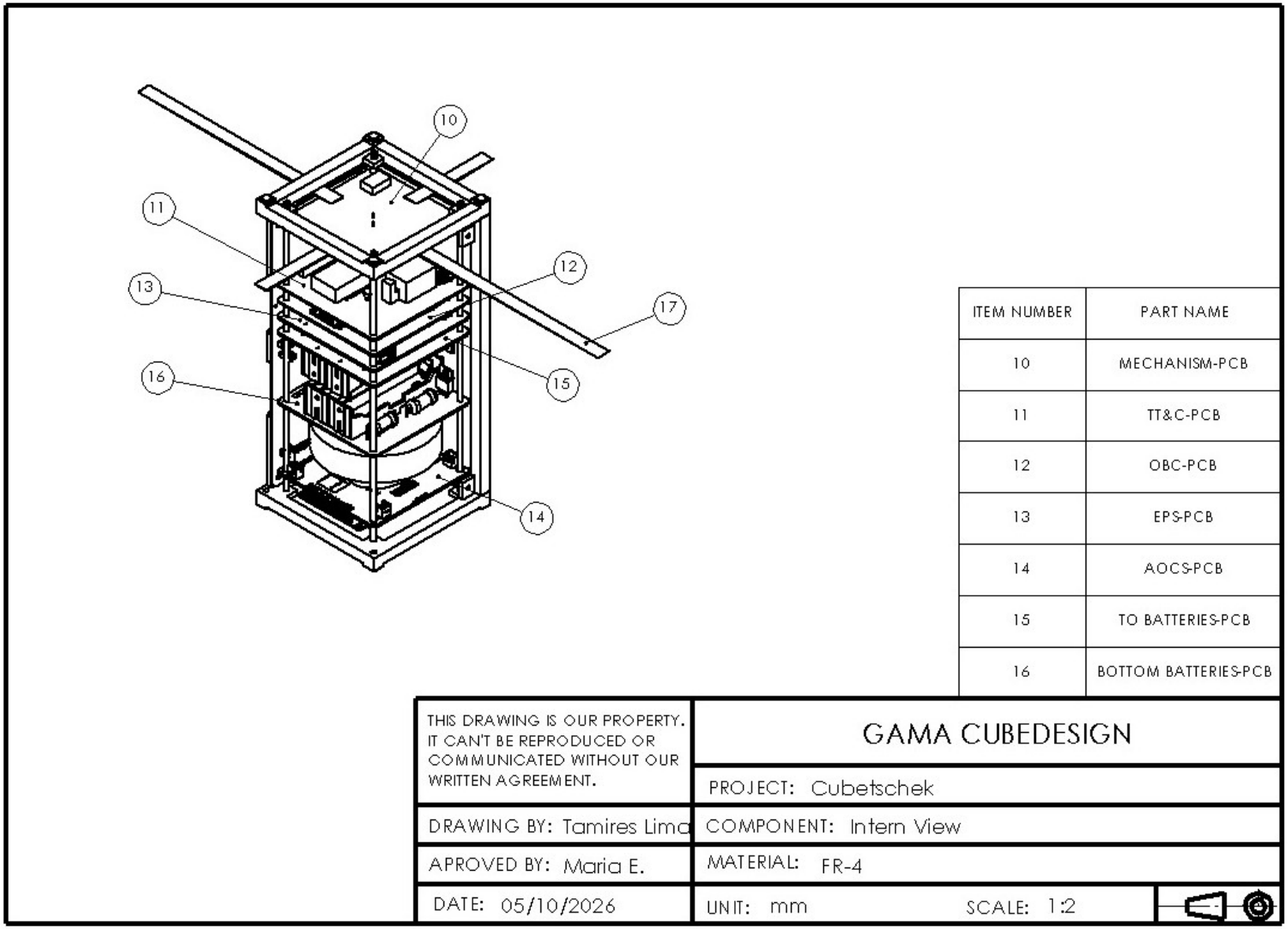


Figure 71: Open Cubetschek
Source: Authors (2026).

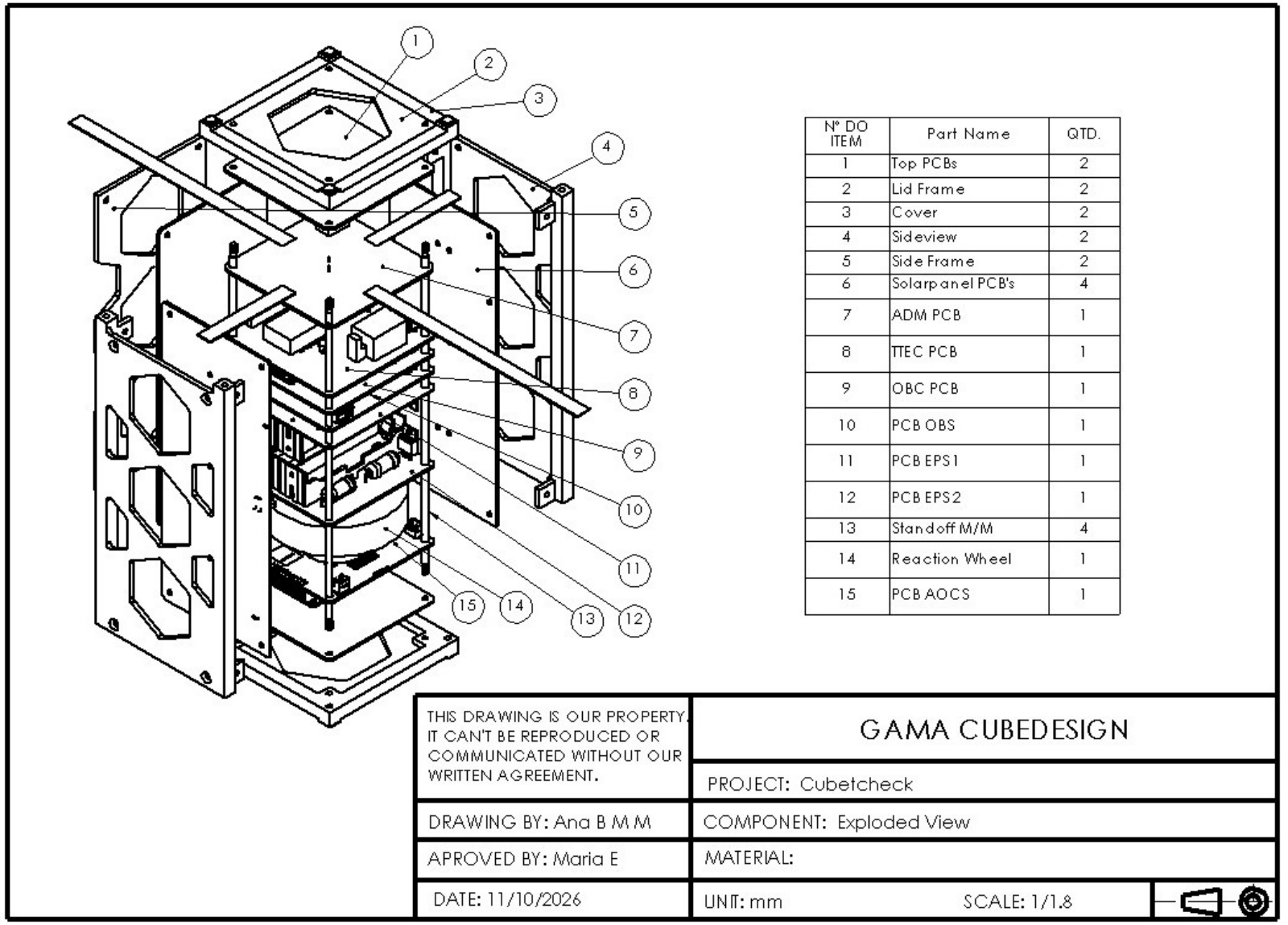


Figure 72: Exploded View Cubetschek
Source: Authors (2026).

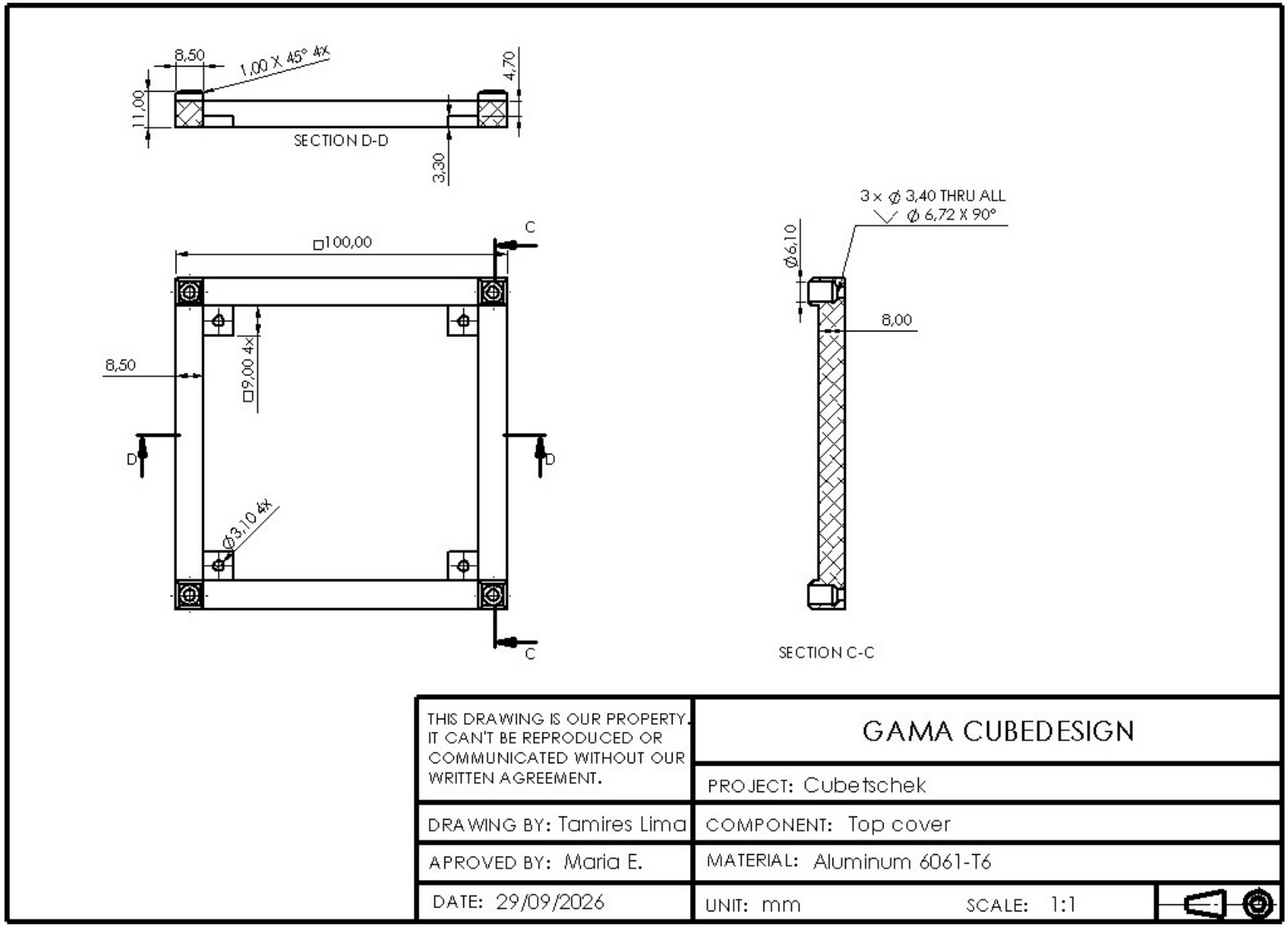


Figure 73: Top/Bottom Cover
Source: Authors (2026).

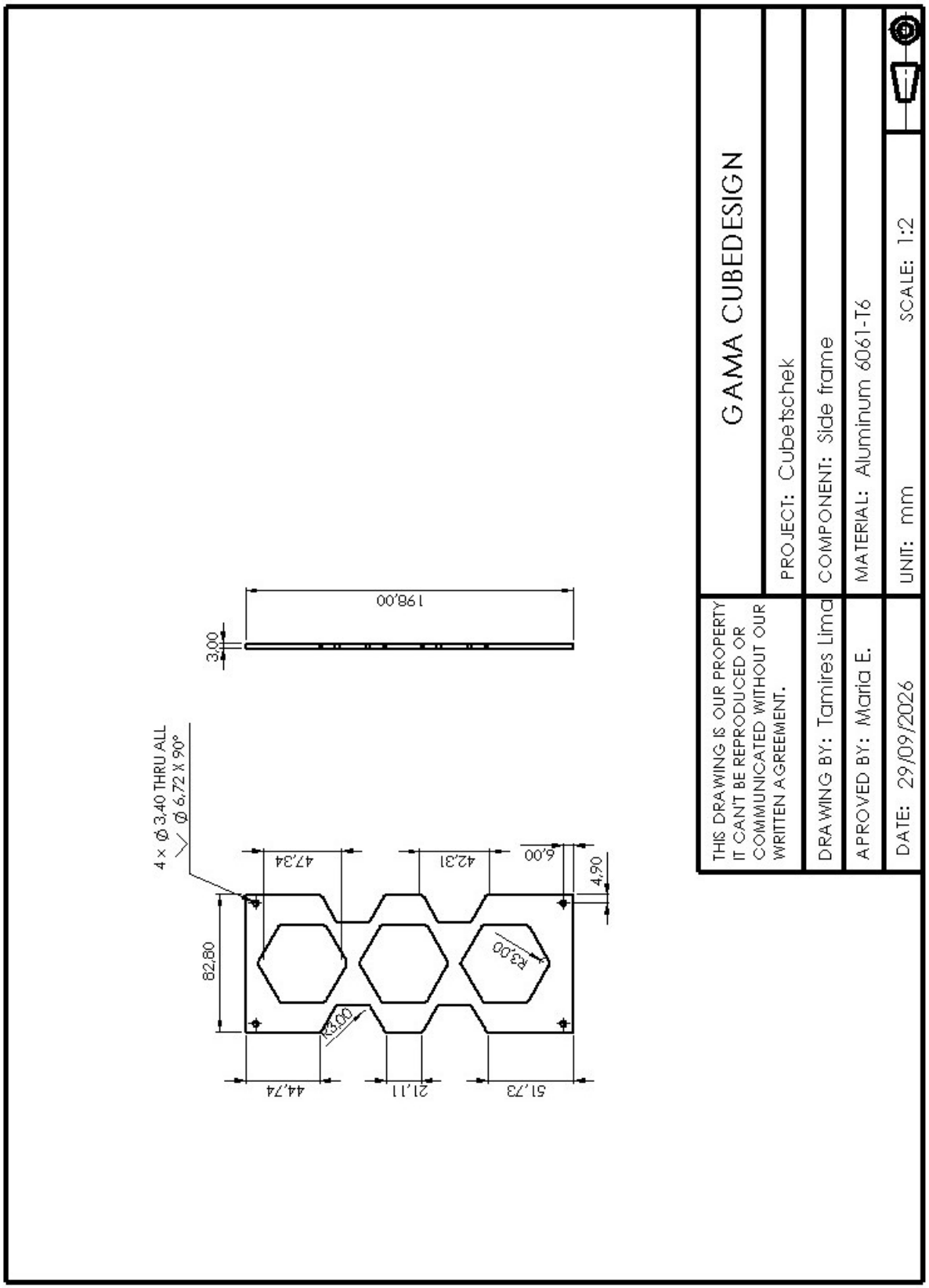


Figure 74: Side frame
Source: Authors (2026).

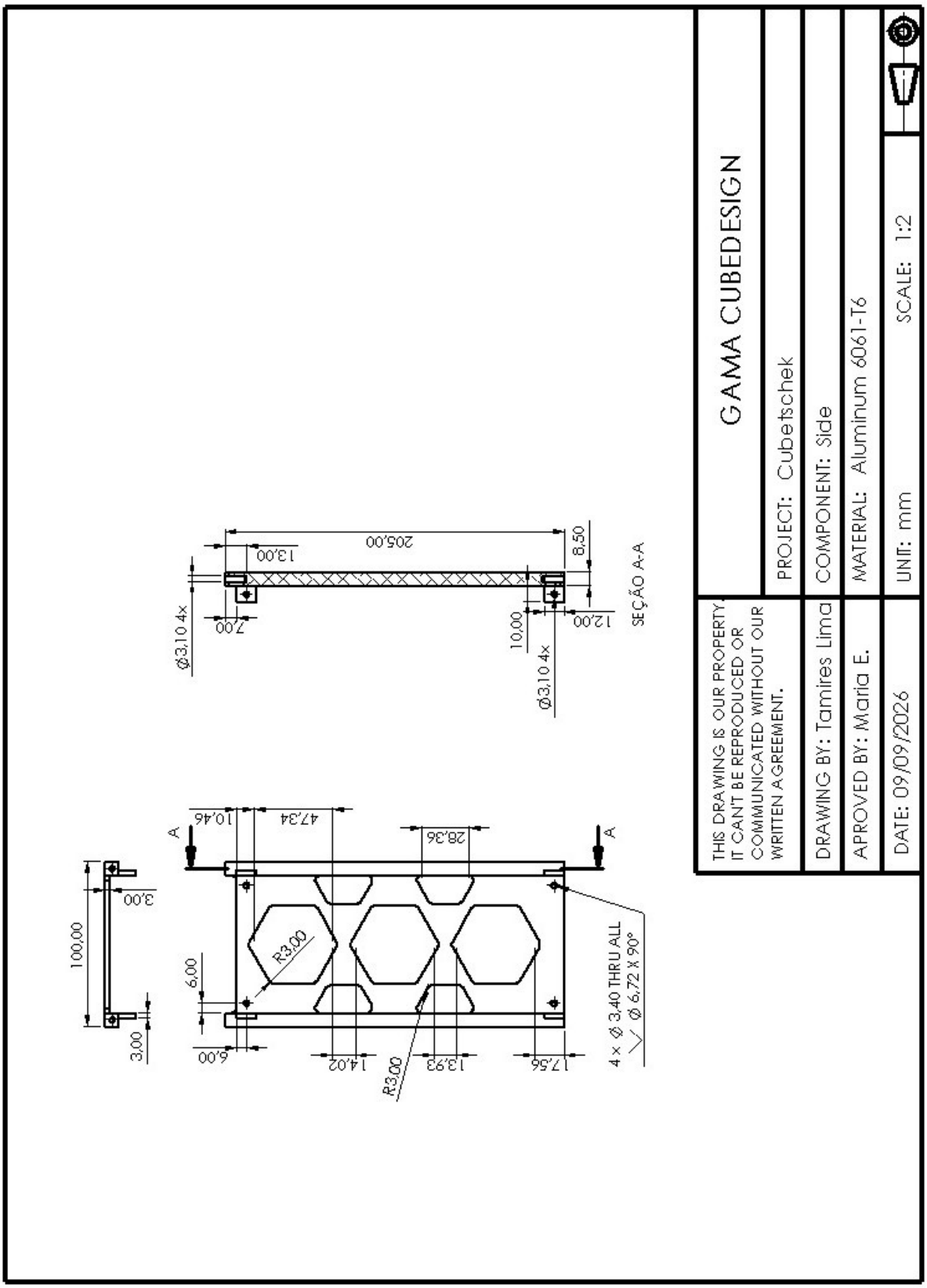


Figure 75: Side
Source: Authors (2026).

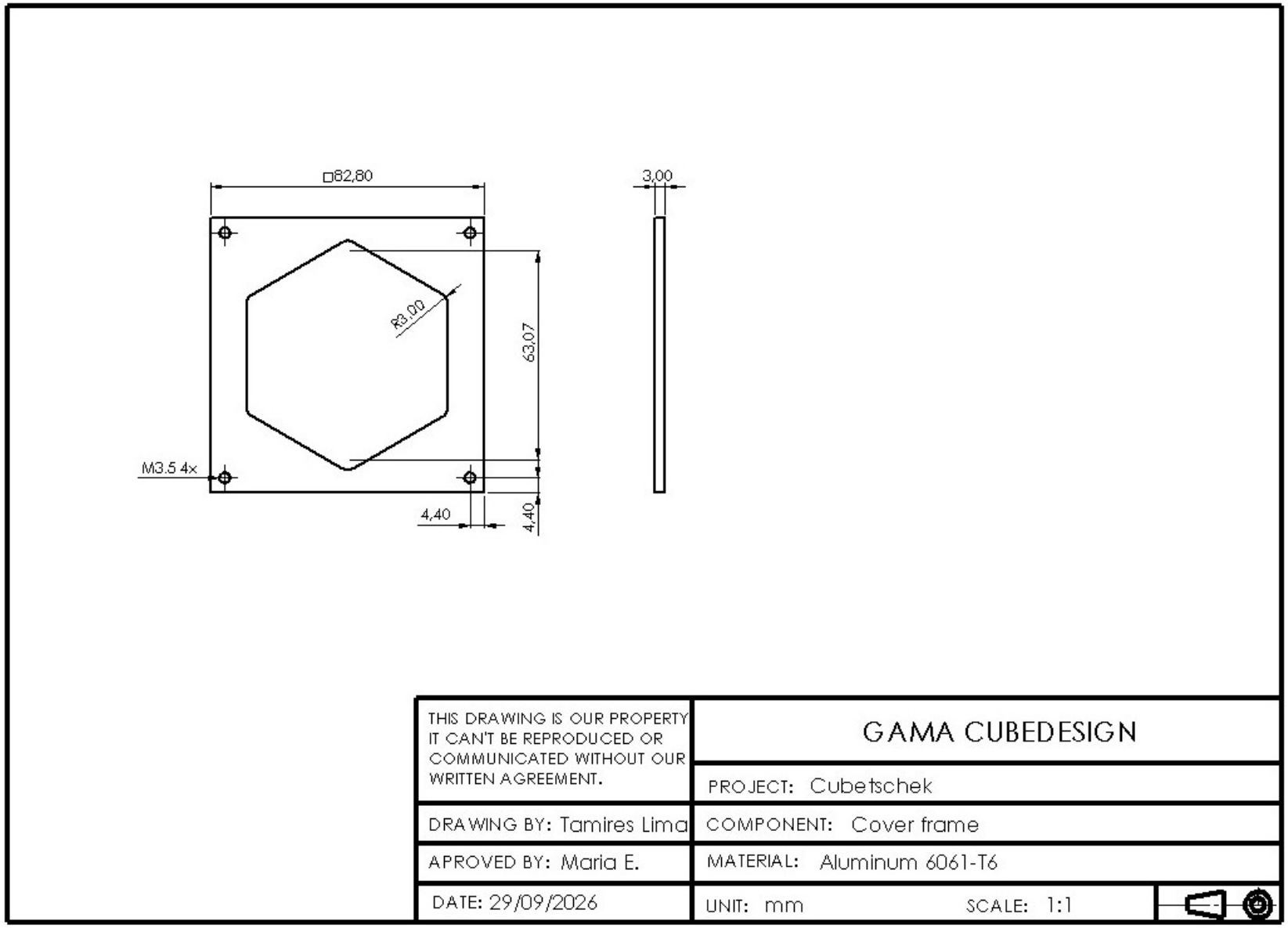


Figure 76: Cover frame
Source: Authors (2026).

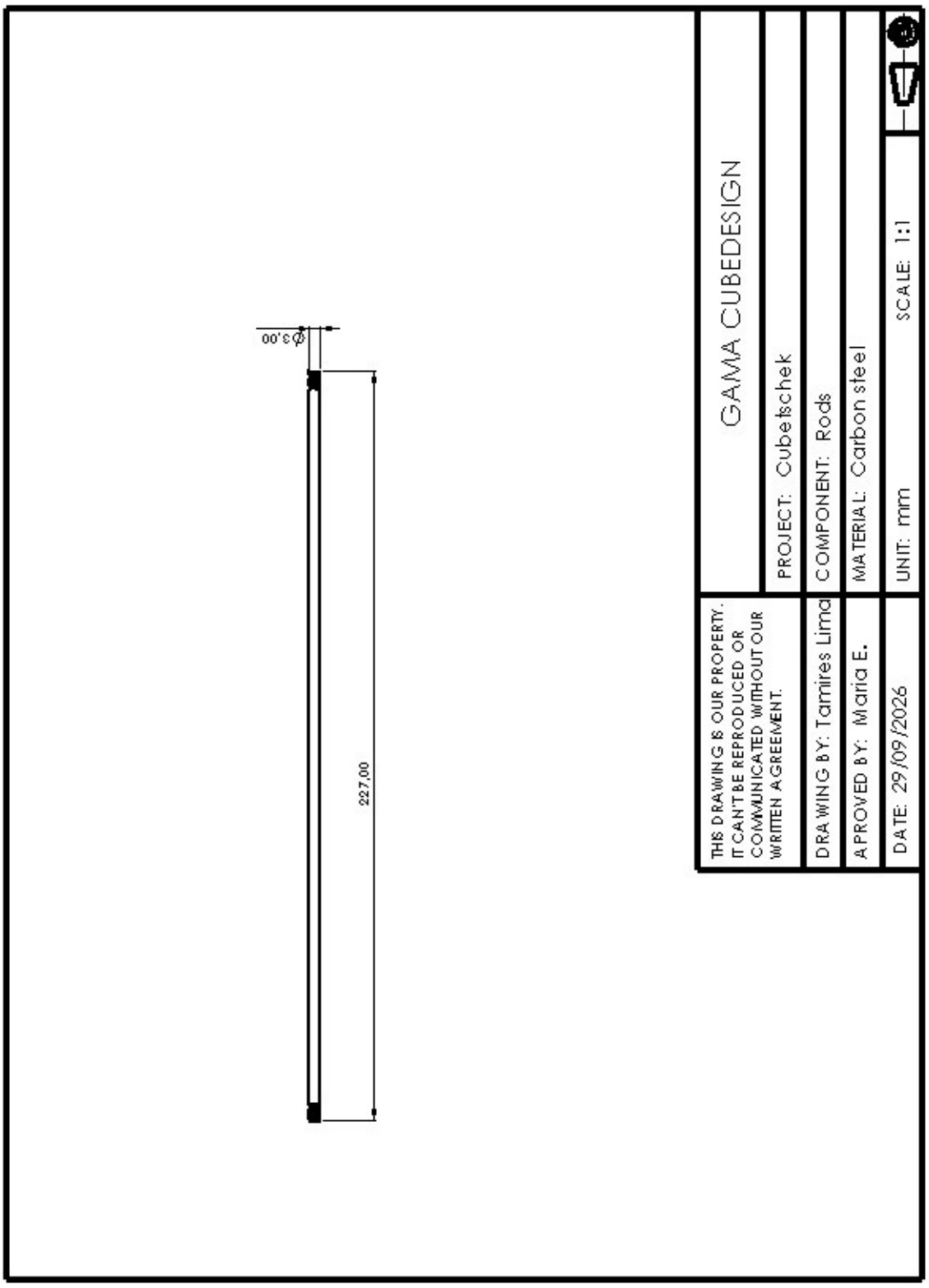


Figure 77: Rods
Source: Authors (2026).

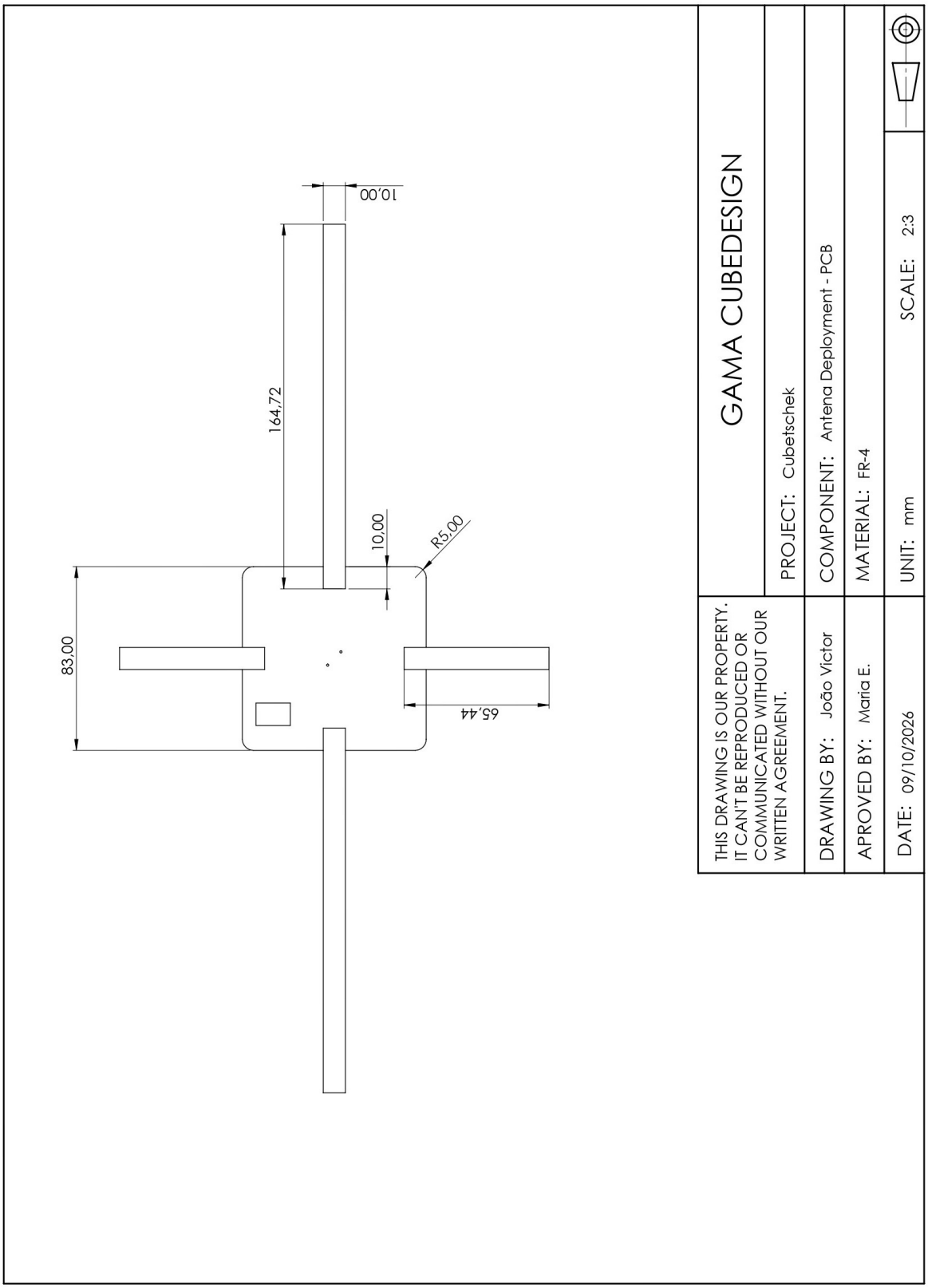


Figure 78: Mechanism-PCB
Source: Authors (2026).

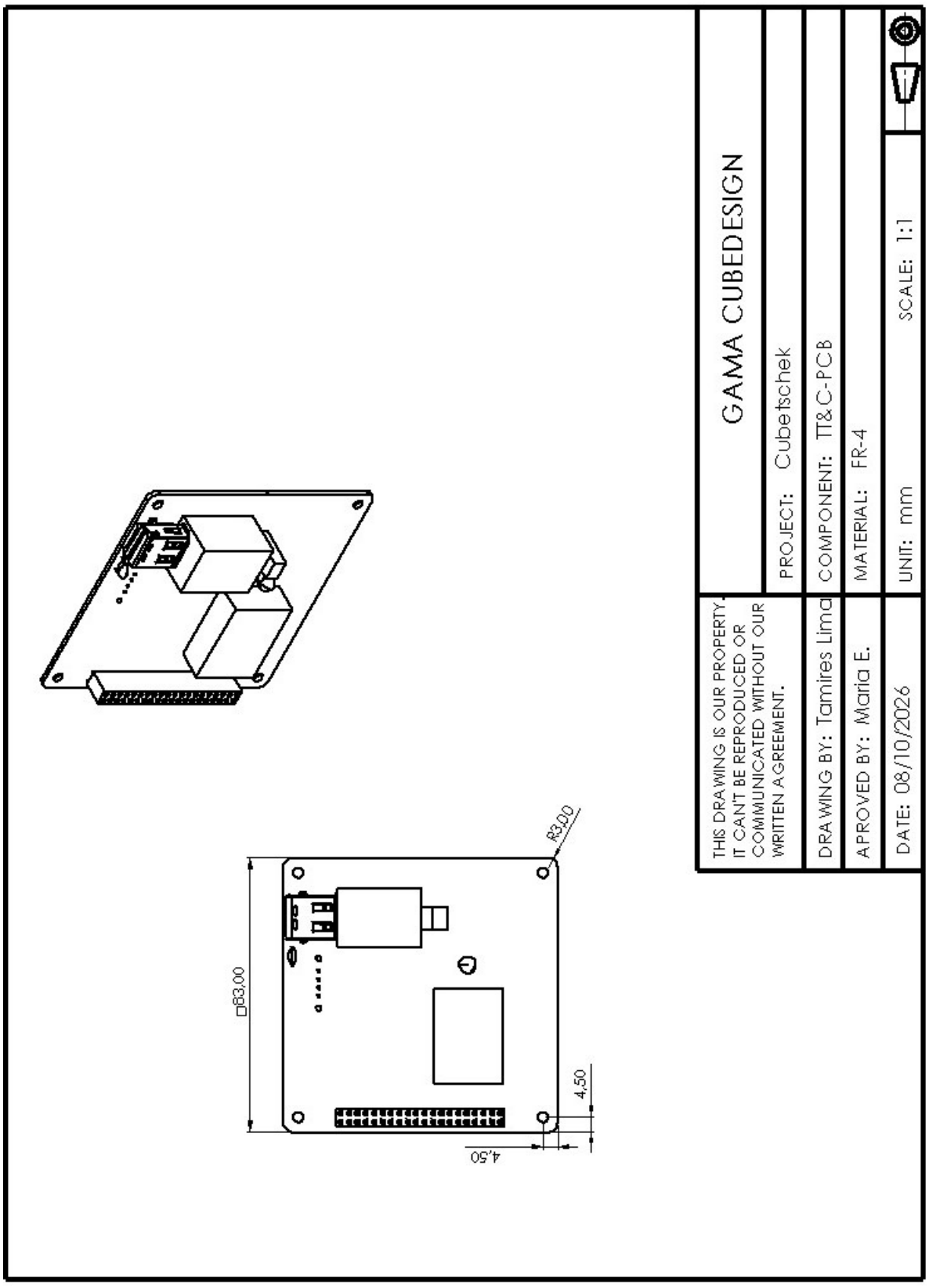


Figure 79: TT&C-PCB
Source: Authors (2026).

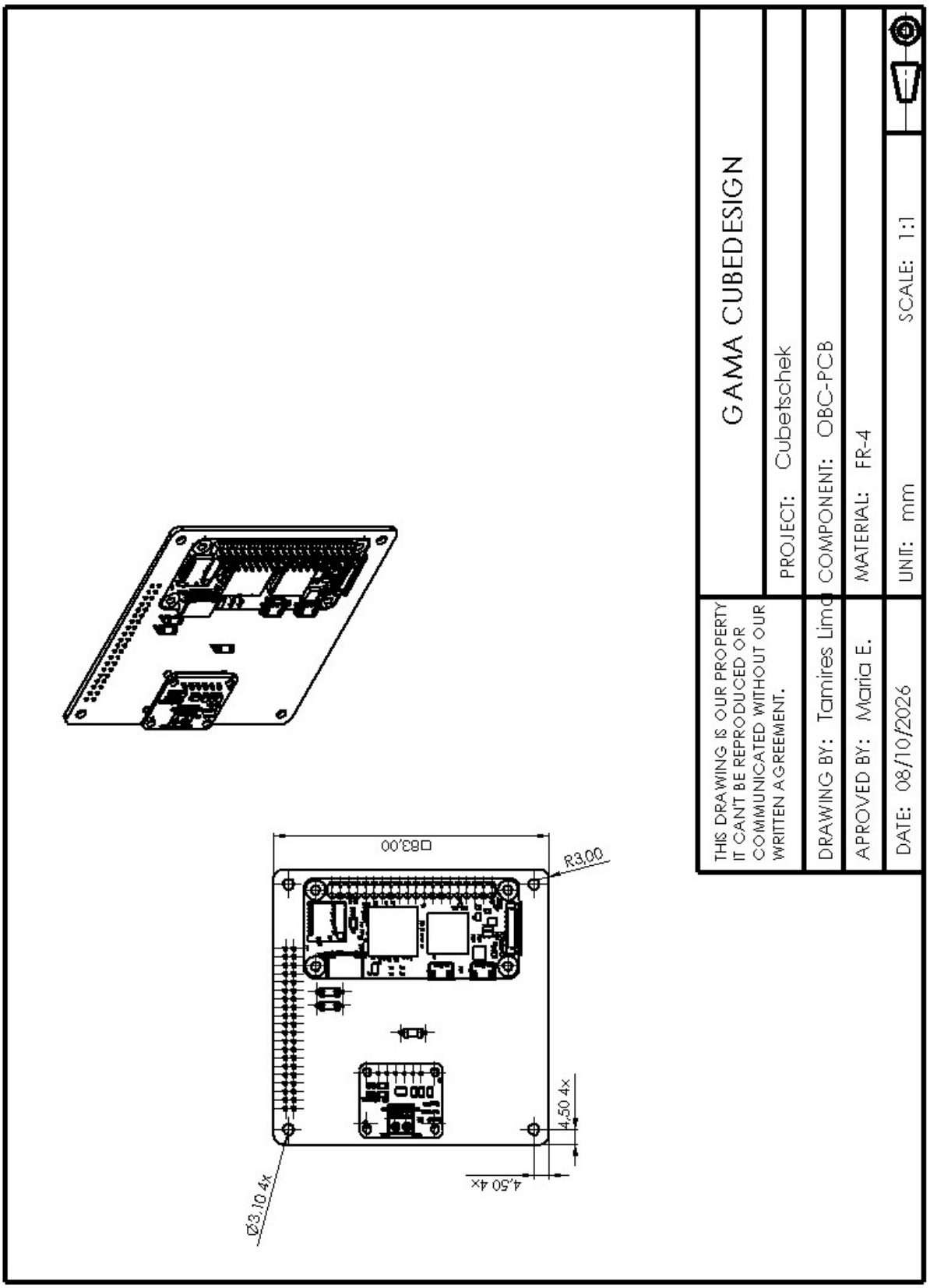


Figure 80: OBC-PCB
Source: Authors (2026).

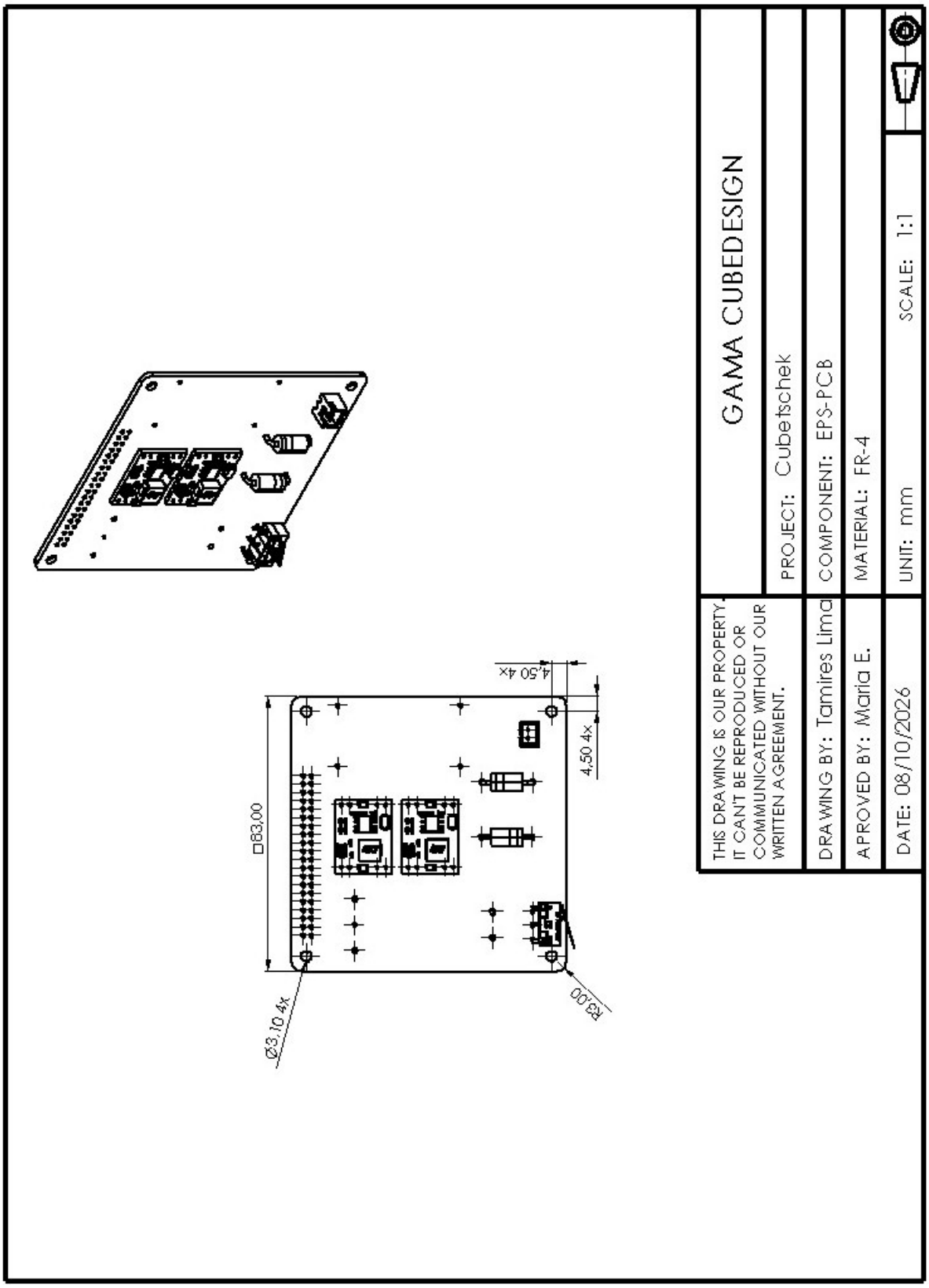


Figure 81: EPS-PCB
Source: Authors (2026).

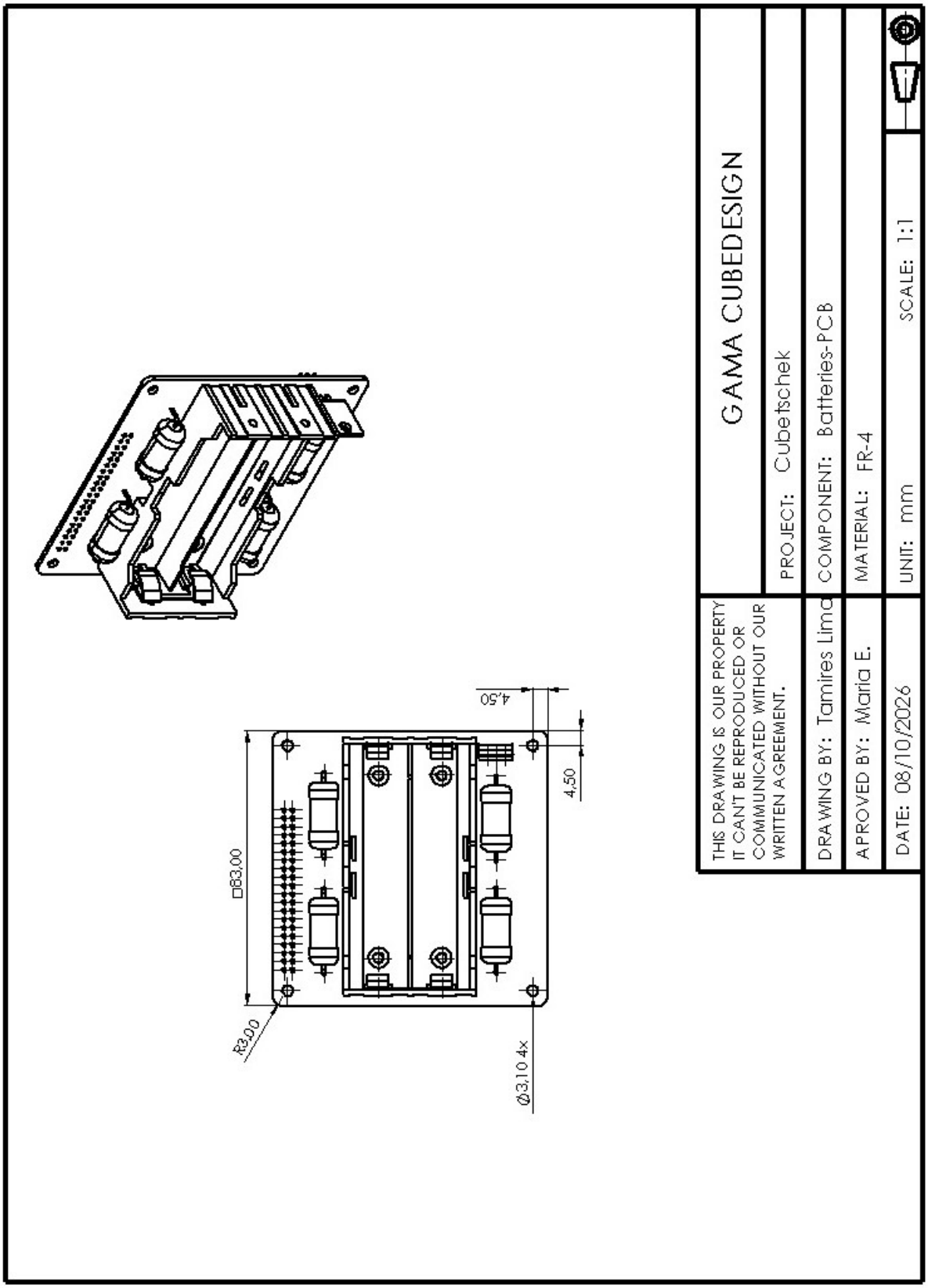


Figure 82: Batteries-PCB
Source: Authors (2026).

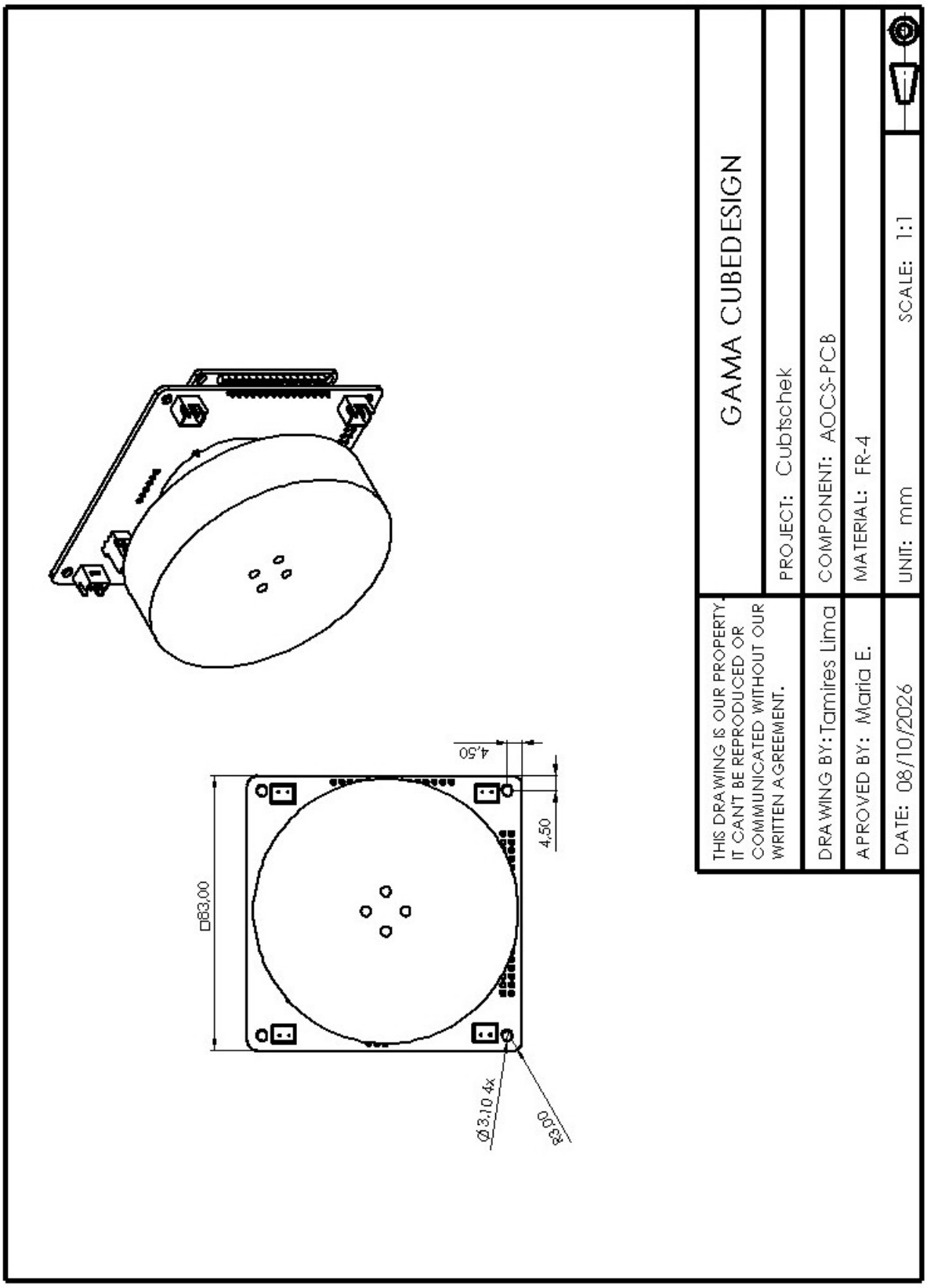


Figure 83: AOCS-PCB
Source: Authors (2026).

B.2 Thermal Control

```
%% ========================================================================
%  THERMAL SIMULATION AND POWER ACTIVATION FOR CUBESAT
% ========================================================================

import numpy as np
from scipy.optimize import fsolve
from scipy.integrate import solve_ivp
import matplotlib.pyplot as plt

# ========================================================================
# CONSTANTS & PROPERTIES
# ========================================================================

# General Constants
sigma = 5.67e-8  # Stefan-Boltzmann constant (W/m^2*K^4)
h = 10           # Convective heat transfer coefficient (placeholder)

# CubeSat Properties
generated_power_value = 20.0  # W -- active control power
area_rad = 0.01               # m^2
alpha = 0.2                   # Solar absorptivity
epsilon = 0.7                 # Emissivity
mass = 2.0                    # kg
cp = 1700                     # Specific heat capacity (J/kg*K)

# Wall Conduction
k_aluminum = 152
thickness_face = 0.0045
area_face = 0.01373734

# Aluminum Cover Conduction
thickness_cover_alum = 0.004
area_cover_alum = 0.00646791

# PCB Cover Conduction
thickness_cover_pcb = 0.002
area_cover_pcb = 0.01

# EPDM Conduction
k_epdm = 0.26
thickness_epdm = 0.002
area_epdm = 0.01

# PCB Conduction
k_pcb = 0.25
thickness_pcb = 0.002
area_pcb = 0.02


# ========================================================================
# THERMAL PROFILE PARAMETERS
# ========================================================================

n_cycles = 5
temp_initial = 25.0              # °C (start)
temp_high = 50.0                 # °C (high plateau)
temp_low = -10.0                 # °C (low plateau)
heating_rate_min = 1.5           # °C per minute
hold_min = 30                    # minutes at each plateau
temp_activation_C = 15

# Conversions and Durations (in seconds)
heating_rate_s = heating_rate_min / 60.0                        # °C/s -> 0.025
t_rise_s = int(round((temp_high - temp_initial) / heating_rate_s))       # 1000 s
t_desc_s = int(round((temp_high - temp_low) / heating_rate_s))           # 2400 s
t_rise_2_s = int(round((temp_initial - temp_low) / heating_rate_s))
t_hold_s = hold_min * 60                                         # 1800 s

# Build ONE cycle (without duplicating points between segments)
rise_segment = np.linspace(temp_initial, temp_high, t_rise_s, endpoint=False)
plateau_high = np.full(t_hold_s, temp_high)
desc_segment = np.linspace(temp_high, temp_low, t_desc_s, endpoint=False)
plateau_low = np.full(t_hold_s, temp_low)
rise_segment_2 = np.linspace(temp_low, temp_initial, t_rise_2_s, endpoint=False)

cycle = np.concatenate([rise_segment, plateau_high, desc_segment, plateau_low, rise_segment_2])
```

```

# Repeat cycles for the full profile
t_oven_profile = np.tile(cycle, n_cycles)

# Time vector in seconds (step = 1 s) matched to the profile
time_s = np.arange(t_oven_profile.size, dtype=int)

# Time vector in minutes (optional)
time_min = time_s / 60.0


def t_oven_interp(t):
    return np.interp(t, time_s, t_oven_profile)


# Activation log tracker
activation_log = np.zeros((len(time_s), 1))


# ============================================================================
# DIFFERENTIAL EQUATION FUNCTION
# ============================================================================

def dTdt(t, T_K):
    T_C = T_K - 273.15
    t_oven_K = t_oven_interp(t) + 273.15

    # Thermal Resistances
    r_cond_alum = thickness_face / (k_aluminum * area_face)
    r_cond_cover_alum = thickness_cover_alum / (k_aluminum * area_cover_alum)
    r_cond_cover_pcb = thickness_cover_pcb / (k_pcb * area_cover_pcb)
    r_cond_cover_epdm = thickness_epdm / (k_epdm * area_epdm)
    r_cond_pcb = thickness_pcb / (k_pcb * area_pcb)
    r_rad = 1 / (epsilon * sigma * area_rad * ((t_oven_K)**2 + (T_K)**2) * (t_oven_K + T_K))

    r_lateral_face = (r_cond_alum + r_cond_pcb + 8 * r_cond_cover_epdm) + r_rad
    r_superior_face = (r_cond_cover_alum + r_cond_cover_pcb + r_cond_cover_epdm) + r_rad
    r_inferior_face = r_cond_cover_alum + r_cond_cover_pcb + r_cond_cover_epdm

    r_total = 1 / (1 / r_superior_face + 4 / r_lateral_face + 1 / r_inferior_face)
    q_total = (t_oven_K - T_K) / r_total

    if int(t) == 100:
        print("r_lateral_face =", r_lateral_face)
        print("r_superior_face =", r_superior_face)
        print("r_total =", r_total)

    # Internal heating power activated only if T_C <= temp_activation_C
    if T_K <= (temp_activation_C + 273.15):
        generated_power = generated_power_value
        activation_log[tempo] = 1
    else:
        generated_power = 0

    q_total = q_total + generated_power
    return q_total / (mass * cp)


# ============================================================================
# NUMERICAL INTEGRATION (EULER METHOD)
# ============================================================================

t0_K = 25 + 273.15
t_sat_K = np.ones((len(time_s), 1))
t_sat_K[0] = t0_K

for tempo in range(1, len(time_s)):
    t_sat_K[tempo] = dTdt(tempo, t_sat_K[tempo - 1]) * (time_s[tempo] - time_s[tempo - 1]) + t_sat_K[tempo - 1]

# Convert temperatures to Celsius
t_sat_C = t_sat_K - 273.15

# Satellite Maximum Temperature
temp_max = np.max(t_sat_C)
print(f"Satellite Maximum Temperature: {temp_max:.2f} °C")

# Satellite Minimum Temperature
temp_min = np.min(t_sat_C)
```

```python
print(f"Satellite Minimum Temperature: {temp_min:.2f} °C")


# ============================================================================
# PLOTTING RESULTS
# ============================================================================

plt.figure(figsize=(12, 6))
plt.plot(time_min, t_oven_profile, "--", label="Temperature of the Oven (°C)", linewidth=1.5)
plt.plot(time_min, t_sat_C, label="Temperature of the Satellite (°C)", linewidth=2)
plt.axhline(temp_activation_C, color="r", linestyle=":", label="Activation Temperature")

# Add activation background shading
activation_array = np.array(activation_log)
for i in range(1, len(activation_array)):
    if activation_array[i] == 1:
        plt.axvspan(time_min[i - 1], time_min[i], color='orange', alpha=0.15)

plt.xlabel("Time (min)")
plt.ylabel("Temperature (°C)")
plt.title("Thermal Simulation with Internal Power Activation")
plt.grid(True)
plt.legend()
plt.tight_layout()
plt.show()
```

C AOCS

C.1 Electrical Schematic

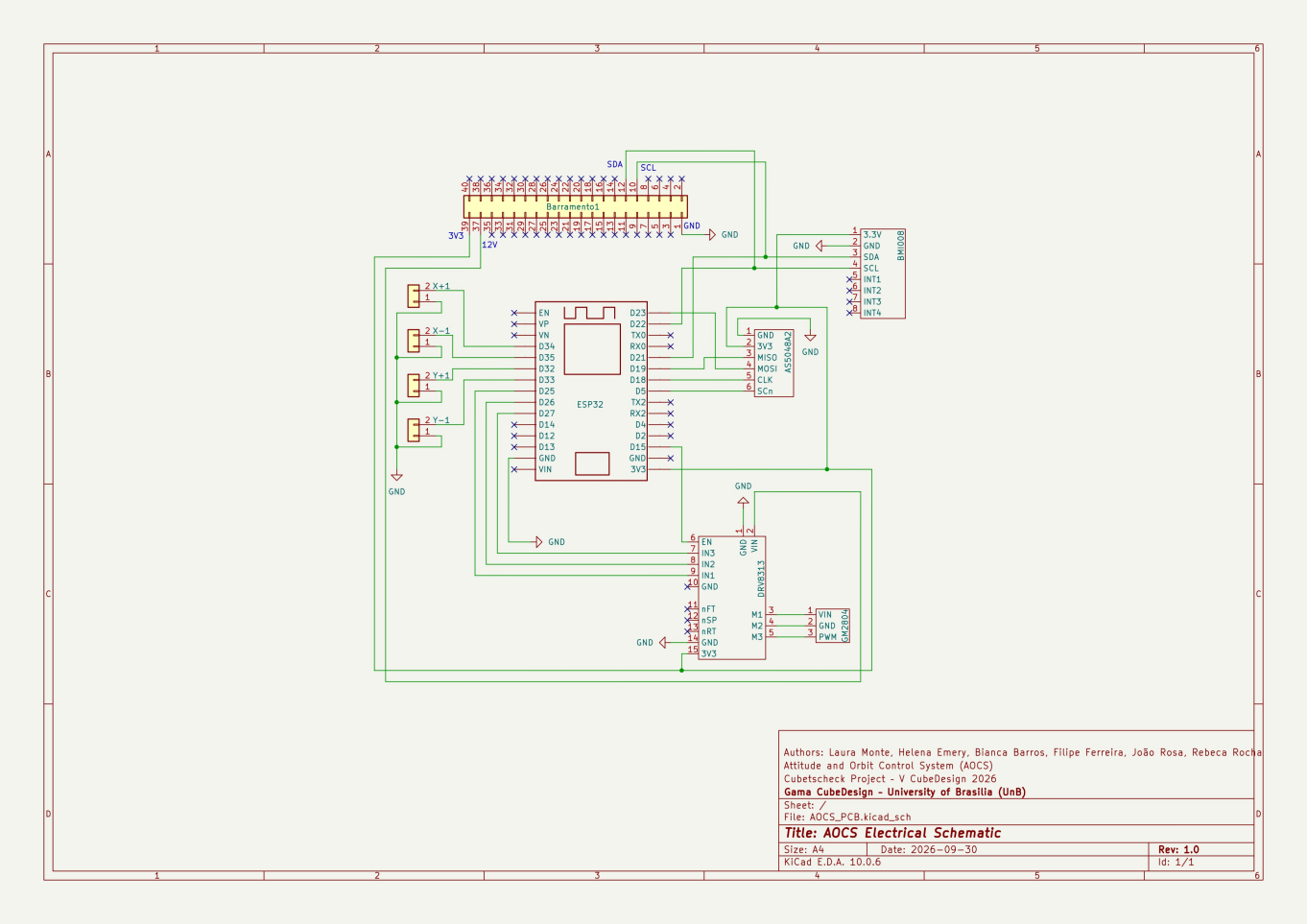
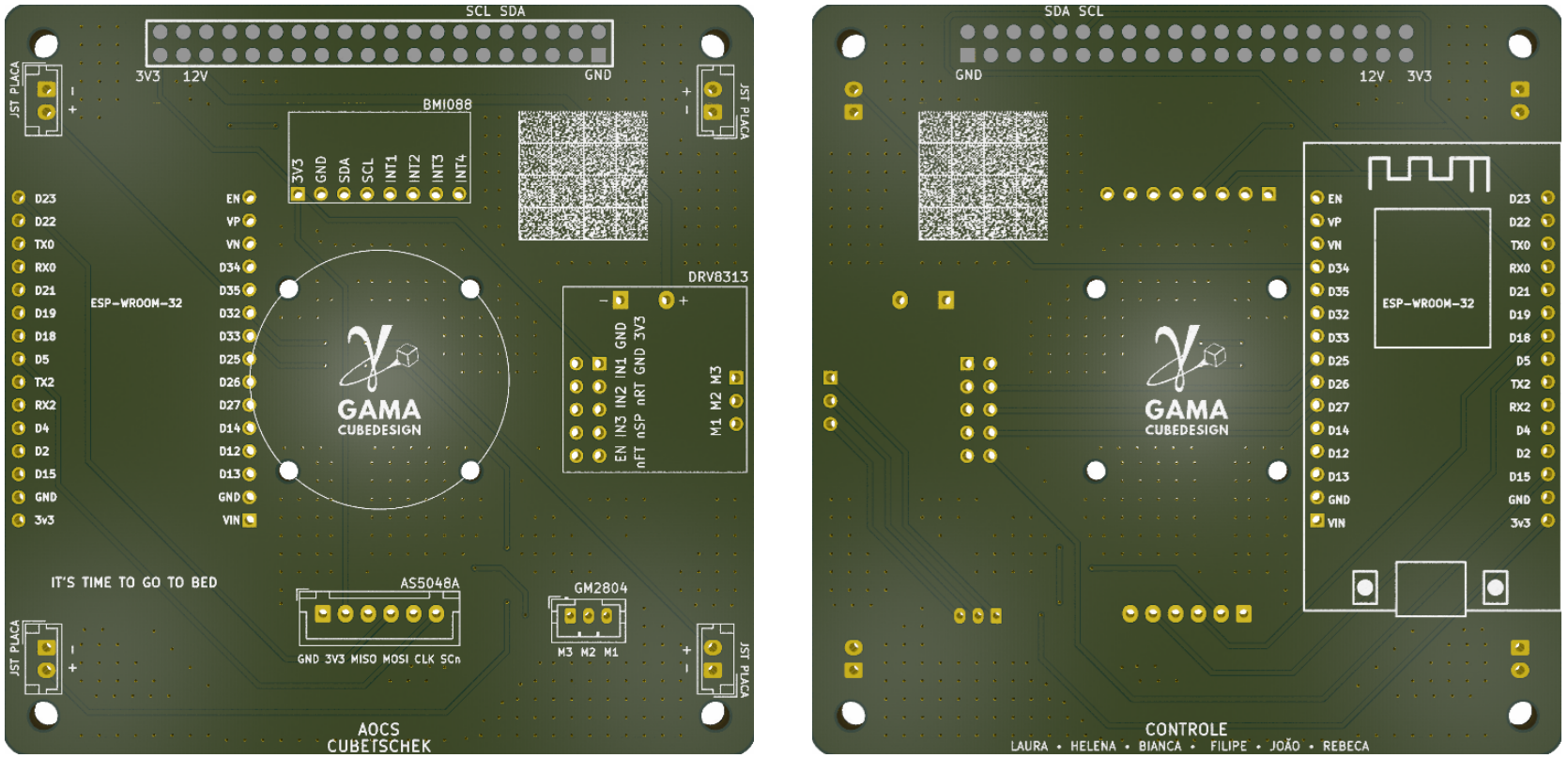


Figure 84: Electrical schematic of the attitude control subsystem
Source: Authors (2026).

C.2 Printed Circuit Board



(a) Front

(b) Back

Figure 85: 3D model of the AOCS PCB

Source: Authors (2026).

C.3 Reaction Wheel Technical Drawing

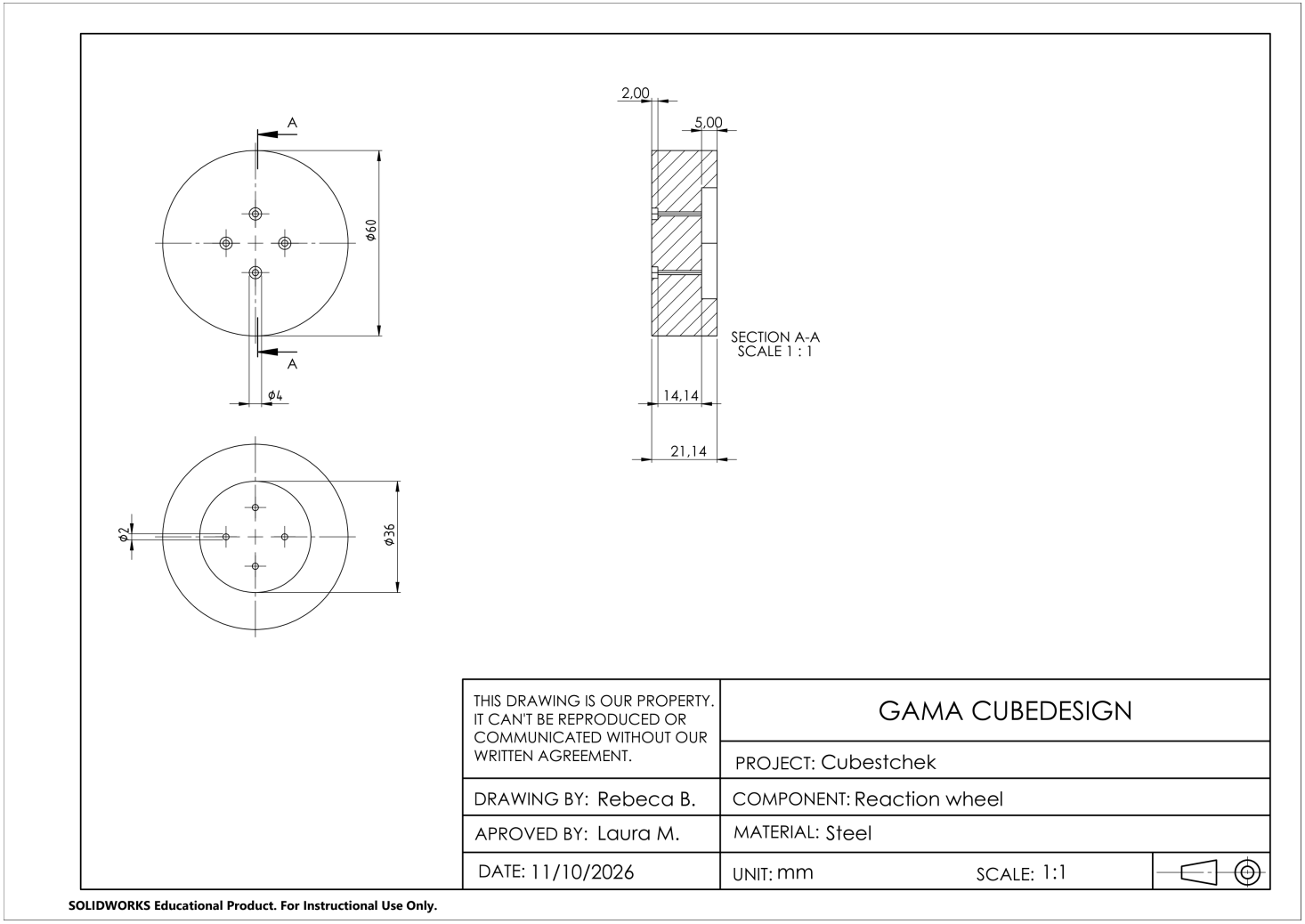


Figure 86: Technical drawing of the reaction wheel
Source: Authors (2026).

C.4 Reaction Wheel Sizing

```
%% ========================================================================
%  REACTION WHEEL SIZING FOR CUBESAT DETUMBLING
%  ========================================================================
% Description:
%     This script performs a preliminary sizing of a reaction wheel
%     intended to detumble a 2U CubeSat for the CubeDesign 2026.
%
%     The input mass is the satellite mass WITHOUT the reaction wheel.
%     The wheel is sized so that, starting from the initial tumbling
%     rate with the wheel at rest relative to the body, the body reaches
%     the final angular velocity while the wheel reaches its operating
%     speed (relative to the body).
%
%     Wheel geometry: solid cylinder of radius R and thickness h, with a
%     coaxial cylindrical recess (radius r_c, depth h_c) on one face to
%     couple the wheel to the motor. The unknown is the thickness h,
%     computed for each material.
%
% Main calculations:
%     1. Satellite (without wheel) moment of inertia
%     2. Required reaction wheel spin inertia (coupled solution)
%     3. Wheel thickness and mass for each material
%     4. Total satellite mass, inertia and angular momentum
%
% Assumptions:
%     - Rotation about the Z axis only (single-axis problem)
%     - Satellite modeled as a homogeneous rectangular prism
%     - Wheel spin axis parallel to Z, at a distance "wheel_axis_offset"
%       from the satellite center of mass (0 = centered on the Z axis)
%     - Recess coaxial with the wheel
%     - Motor rotor, hub and fasteners inertia neglected
%     - No external torques (angular momentum is conserved)
%     - Wheel starts at rest relative to the body
%
% Units:
%     Mass                    -> kg
%     Length                  -> m
%     Time                    -> s
%     Angular velocity        -> rad/s
%     Moment of inertia       -> kg.m^2
%     Torque                  -> N.m
%     Angular momentum        -> N.m.s
%
% Company:     Gama CubeDesign
% Subsystem:   Attitude and Orbit Control Systems
% Project:     Cubetscheck
% Event:       V CubeDesign - Buenos Aires - November 24, 2026
% Authors:     Laura Monte, Helena Emery, Bianca Barros, Filipe Ferreira,
%              João Rosa, Rebeca Rocha
% Institution: University of Brasilia (UnB)

clear; clc; close all;

%% ========================================================================
%  1. DESIGN PARAMETERS
%  ========================================================================
% Reaction wheel outer radius
wheel_radius = 0.03;                % [m]

% Motor coupling recess (coaxial cylindrical cavity on one face)
recess_diameter = 0.036;            % [m]
recess_depth = 0.005;               % [m]

% Reaction wheel operating speed (relative to the body, end of detumbling)
wheel_speed_rpm = 1000;             % [rpm]

% Distance between the wheel spin axis and the satellite Z axis
wheel_axis_offset = 0;              % [m]

% Materials
materials(1).name = 'Aluminum';
materials(1).density = 2700;        % [kg/m^3]

materials(2).name = 'Steel';
materials(2).density = 7850;        % [kg/m^3]

```

```
77
78  %% ======================================================================
79  %  2. CUBESAT PARAMETERS
80  %  ======================================================================
81  % Satellite mass WITHOUT the reaction wheel
82  satellite_mass_without_wheel = 1.9235;    % [kg]
83
84  % Satellite dimension in X
85  satellite_side_x = 0.10;              % [m]
86
87  % Satellite dimension in Y
88  satellite_side_y = 0.10;              % [m]
89
90
91  %% ======================================================================
92  %  3. DETUMBLING REQUIREMENTS
93  %  ======================================================================
94  % Initial satellite angular velocity
95  initial_satellite_speed_rpm = 60;     % [rpm]
96
97  % Final satellite angular velocity
98  final_satellite_speed_rpm = 0;        % [rpm]
99
100
101 %% ======================================================================
102 %  4. MOTOR PARAMETERS
103 %  ======================================================================
104 % Motor maximum torque
105 motor_max_torque_gf_cm = mean([250 350]); % [gf.cm]
106
107
108 %% ======================================================================
109 %  5. UNIT CONVERSIONS
110 %  ======================================================================
111 initial_satellite_speed = ...
112     initial_satellite_speed_rpm * (2*pi/60);   % [rad/s]
113
114 final_satellite_speed = ...
115     final_satellite_speed_rpm * (2*pi/60);     % [rad/s]
116
117 wheel_speed = ...
118     wheel_speed_rpm * (2*pi/60);               % [rad/s]
119
120 motor_max_torque = ...
121     motor_max_torque_gf_cm * 9.81e-5;          % [N.m]
122
123 recess_radius = recess_diameter / 2;           % [m]
124
125 if recess_radius >= wheel_radius
126     error('Recess diameter must be smaller than the wheel diameter.');
127 end
128
129
130 %% ======================================================================
131 %  6. SATELLITE MOMENT OF INERTIA (WITHOUT WHEEL)
132 %  ======================================================================
133 % For rotation about the Z axis:
134 %     J_sat = (1/12) * m_sat * (a^2 + b^2)
135 satellite_inertia_without_wheel = ...
136     (satellite_mass_without_wheel / 12) * ...
137     (satellite_side_x^2 + satellite_side_y^2);
138
139
140 %% ======================================================================
141 %  7. DETUMBLING COUPLING TERMS
142 %  ======================================================================
143 % Required change in satellite angular velocity:
144 %     delta_w = w0 - wf
145 delta_speed = ...
146     initial_satellite_speed - final_satellite_speed;
147
148 % Wheel axis offset factor (parallel axis theorem):
149 %     k = 2 * d^2 / R^2
150 k_offset = ...
151     2 * wheel_axis_offset^2 / wheel_radius^2;
152
153 % Denominator of the coupled solution (must be positive):
154 %     W - (1 + k) * delta_w
```

```
denominator = ...
    wheel_speed - (1 + k_offset) * delta_speed;

if denominator <= 0
    error(['Wheel operating speed is too low to detumble the satellite. ' ...
           'Increase wheel_speed_rpm or reduce wheel_axis_offset.']);
end


%% =========================================================================
%  8. REQUIRED REACTION WHEEL SPIN INERTIA
%  =========================================================================
% Material densities (one value per material)
material_densities = [materials.density];

% Recess mass adjustment:
%     m_adj = rho * pi * r_c^2 * h_c * (1 - r_c^2 / R^2)
recess_adjustment = recess_mass_adjustment( ...
    material_densities, wheel_radius, recess_radius, recess_depth);

% Conservation of angular momentum:
%     J_s = (J_b - m_adj * d^2) * delta_w / (W - (1 + k) * delta_w)
wheel_inertia = ...
    (satellite_inertia_without_wheel - ...
    recess_adjustment * wheel_axis_offset^2) * ...
    delta_speed / denominator;


%% =========================================================================
%  9. REACTION WHEEL THICKNESS AND MASS
%  =========================================================================
% Wheel thickness for the required spin inertia:
%     h = (2 * J_s / (rho * pi) + r_c^4 * h_c) / R^4
wheel_thickness = wheel_thickness_from_inertia( ...
    wheel_inertia, material_densities, ...
    wheel_radius, recess_radius, recess_depth);

% Wheel mass:
%     m_w = rho * pi * (R^2 * h - r_c^2 * h_c)
wheel_mass = wheel_mass_from_thickness( ...
    wheel_thickness, material_densities, ...
    wheel_radius, recess_radius, recess_depth);

% Consistency check:
%     J_s = (1/2) * rho * pi * (R^4 * h - r_c^4 * h_c)
wheel_inertia_check = wheel_inertia_from_thickness( ...
    wheel_thickness, material_densities, ...
    wheel_radius, recess_radius, recess_depth);

if any(abs(wheel_inertia_check - wheel_inertia) > 1e-12)
    error('Inconsistent wheel geometry.');
end

% Recess must fit inside the wheel thickness
recess_fits = wheel_thickness > recess_depth;


%% =========================================================================
%  10. TOTAL SATELLITE PROPERTIES (WITH WHEEL)
%  =========================================================================
% Total mass:
%     m_total = m_sat + m_w
total_mass = ...
    satellite_mass_without_wheel + wheel_mass;

% Wheel mass fraction:
%     f_w = m_w / m_total
wheel_mass_fraction = ...
    wheel_mass ./ total_mass;

% Total moment of inertia (wheel at rest relative to the body):
%     J_total = J_sat + J_s + m_w * d^2
total_inertia = ...
    satellite_inertia_without_wheel + wheel_inertia + ...
    wheel_mass * wheel_axis_offset^2;

% Initial angular momentum:
%     H0 = J_total * w0
```

```
initial_momentum = ...
    total_inertia * initial_satellite_speed;

% Final angular momentum (must be equal to H0):
%     Hf = J_total * wf + J_s * W
final_momentum = ...
    total_inertia * final_satellite_speed + ...
    wheel_inertia * wheel_speed;


%% ========================================================================
%  11. DISPLAY RESULTS
%  ========================================================================
fprintf('\n');
fprintf('============================================================\n');
fprintf('              REACTION WHEEL SIZING RESULTS                 \n');
fprintf('============================================================\n');

fprintf('\n----- Satellite Parameters (without wheel) -----\n');
fprintf('Satellite mass                : %10.4f kg\n', ...
    satellite_mass_without_wheel);
fprintf('Satellite side X              : %10.4f m\n', satellite_side_x);
fprintf('Satellite side Y              : %10.4f m\n', satellite_side_y);
fprintf('Satellite moment of inertia   : %10.6e kg.m^2\n', ...
    satellite_inertia_without_wheel);

fprintf('\n----- Detumbling Requirements -----\n');
fprintf('Initial angular velocity      : %10.4f rad/s (%g rpm)\n', ...
    initial_satellite_speed, initial_satellite_speed_rpm);
fprintf('Final angular velocity        : %10.4f rad/s (%g rpm)\n', ...
    final_satellite_speed, final_satellite_speed_rpm);

fprintf('\n----- Reaction Wheel Geometry -----\n');
fprintf('Wheel outer radius            : %10.4f m\n', wheel_radius);
fprintf('Recess diameter               : %10.4f m\n', recess_diameter);
fprintf('Recess depth                  : %10.4f m\n', recess_depth);
fprintf('Wheel axis offset             : %10.4f m\n', wheel_axis_offset);
fprintf('Wheel operating speed         : %10.0f rpm\n', wheel_speed_rpm);
fprintf('Wheel angular velocity        : %10.4f rad/s\n', wheel_speed);

fprintf('\n----- Motor Parameters -----\n');
fprintf('Maximum motor torque          : %10.6e N.m\n', motor_max_torque);

fprintf('\n----- Material Comparison -----\n');
for i = 1:numel(materials)
    fprintf('\n%s:\n', materials(i).name);
    fprintf('Density                       : %10.2f kg/m^3\n', ...
        material_densities(i));
    fprintf('Required wheel spin inertia   : %10.6e kg.m^2\n', ...
        wheel_inertia(i));
    fprintf('Wheel thickness               : %10.3f mm\n', ...
        wheel_thickness(i) * 1000);
    fprintf('Wheel mass                    : %10.6f kg\n', ...
        wheel_mass(i));
    fprintf('Total satellite mass          : %10.4f kg\n', ...
        total_mass(i));
    fprintf('Wheel mass fraction           : %10.2f %%\n', ...
        wheel_mass_fraction(i) * 100);
    fprintf('Total moment of inertia       : %10.6e kg.m^2\n', ...
        total_inertia(i));
    fprintf('Initial angular momentum      : %10.6e N.m.s\n', ...
        initial_momentum(i));
    fprintf('Final angular momentum (check) : %10.6e N.m.s\n', ...
        final_momentum(i));
    if ~recess_fits(i)
        fprintf('WARNING: wheel thickness is smaller than the recess depth!\n');
    end
end

fprintf('\n============================================================\n');
fprintf('                    END OF ANALYSIS                         \n');
fprintf('============================================================\n\n');


%% ========================================================================
%  LOCAL FUNCTIONS
%  ========================================================================
function m_w = wheel_mass_from_thickness(h, rho, R, r_c, h_c)
```

```
    % m_w = rho * pi * (R^2 * h - r_c^2 * h_c)
    m_w = rho .* pi .* (R^2 .* h - r_c^2 * h_c);
end

function J_s = wheel_inertia_from_thickness(h, rho, R, r_c, h_c)
    % J_s = (1/2) * rho * pi * (R^4 * h - r_c^4 * h_c)
    J_s = 0.5 .* rho .* pi .* (R^4 .* h - r_c^4 * h_c);
end

function h = wheel_thickness_from_inertia(J_s, rho, R, r_c, h_c)
    % h = (2 * J_s / (rho * pi) + r_c^4 * h_c) / R^4
    h = (2 .* J_s ./ (rho .* pi) + r_c^4 * h_c) ./ R^4;
end

function m_adj = recess_mass_adjustment(rho, R, r_c, h_c)
    % m_adj = rho * pi * r_c^2 * h_c * (1 - r_c^2 / R^2)
    m_adj = rho .* pi .* r_c^2 .* h_c .* (1 - r_c^2 / R^2);
end
```

C.5 Detumbling Controller

```
%% ========================================================================
%  REACTION WHEEL DETUMBLING CONTROLLER
%  ========================================================================
% Description:
%     This script simulates the detumbling of a 2U CubeSat about its Z axis
%     using a reaction wheel driven by a BLDC motor, for the CubeDesign 2026.
%
%     The controller uses velocity feedback to reduce the satellite angular
%     velocity. The motor + wheel assembly is modeled as a DC motor with
%     back-EMF: the wheel speed is a state, and the motor torque follows
%     from the armature current, T = Kt*(V - Kt*omega_w)/R.
%
%     Optional back-EMF compensation (feed-forward Kt*omega_w in the voltage
%     command) represents a driver that controls torque/current, such as
%     SimpleFOC with phase_resistance and KV_rating configured.
%
%     The reaction wheel is a steel disk with a coaxial cylindrical recess
%     to couple it to the motor, sized for an operating speed of 1000 rpm.
%
% Main calculations:
%     1. Satellite moment of inertia
%     2. Reaction wheel inertia from geometry
%     3. Angular momentum to be absorbed during detumbling
%     4. Motor constants from the datasheet
%     5. Controller gain from the desired closed-loop pole
%     6. Motor + wheel time constant (reference only)
%     7. Closed-loop simulation with back-EMF
%     8. Simulation results and Simulink parameters
%
% Units:
%     Mass                    -> kg
%     Length                  -> m
%     Time                    -> s
%     Angular velocity        -> rad/s
%     Moment of inertia       -> kg.m^2
%     Torque                  -> N.m
%     Angular momentum        -> N.m.s
%     Voltage                 -> V
%     Current                 -> A
%     Resistance              -> ohm
%
% Company:     Gama CubeDesign
% Subsystem:   Attitude and Orbit Control Systems
% Project:     Cubetscheck
% Event:       V CubeDesign - Buenos Aires - November 24, 2026
% Authors:     Laura Monte, Helena Emery, Bianca Barros, Filipe Ferreira,
%              João Rosa, Rebeca Rocha
% Institution: University of Brasilia (UnB)

clear; clc; close all;


%% ========================================================================
%  1. REACTION WHEEL PARAMETERS
% ========================================================================
% Values from the reaction wheel sizing script (steel, 1000 rpm).
wheel_radius = 0.030;                % [m]
wheel_speed_rpm = 1000;              % [rpm]
material_density = 7850;             % [kg/m^3] steel
wheel_mass = 0.429158;               % [kg]
wheel_thickness = 0.021136;          % [m]

% Motor coupling recess (coaxial cylindrical cavity on one face).
recess_radius = 0.036 / 2;           % [m]
recess_depth = 0.005;                % [m]


%% ========================================================================
%  2. CUBESAT PARAMETERS
% ========================================================================
% Satellite mass without the reaction wheel.
satellite_mass = 1.9235;             % [kg]

% Satellite x and y sides.
satellite_side_x = 0.10;             % [m]
satellite_side_y = 0.10;             % [m]
```

```
77
78
79  %% ========================================================================
80  %  3. DETUMBLING REQUIREMENTS
81  % ========================================================================
82  % Initial satellite angular velocity.
83  initial_satellite_speed_rpm = 60;     % [rpm]
84
85  % Target angular velocity used only as a reference.
86  final_satellite_speed_rpm = 0;        % [rpm]
87
88  % Criterion used to determine the detumbling time.
89  speed_tolerance_rpm = 5;              % [rpm]
90
91
92  %% ========================================================================
93  %  4. MOTOR PARAMETERS (DATASHEET)
94  % ========================================================================
95  % Armature resistance.
96  motor_resistance = 9;                 % [ohm]
97
98  % Supply voltage used to derive Kt from the datasheet no-load speed.
99  motor_voltage = 12;                   % [V]
100
101 % Maximum voltage the driver can actually apply to the motor.
102 driver_voltage_limit = 12;            % [V]
103
104 % No-load current.
105 motor_no_load_current = 0.09;         % [A]
106
107 % No-load speed (mean of the datasheet range).
108 motor_no_load_speed_rpm = mean([1602 1771]);  % [rpm]
109
110 % Motor maximum torque (mean of the datasheet range).
111 motor_max_torque_gf_cm = mean([250 350]);     % [gf.cm]
112
113
114 %% ========================================================================
115 %  5. CONTROLLER PARAMETERS
116 % ========================================================================
117 % For K1 = 0 and an ideal torque actuator:
118 %      J*omega_dot = -K2*omega
119 %      K2 = p*J
120 closed_loop_pole = 0.5;               % [1/s]
121
122 % K1 = 0 for pure detumbling by velocity feedback.
123 K1 = 0;                               % [N.m/rad]
124
125 % Back-EMF compensation in the voltage command:
126 %   true  -> V_c = -u_c/f + Kt*omega_w  (driver controls torque/current)
127 %   false -> V_c = -u_c/f               (plain voltage command)
128 backemf_compensation = true;
129
130
131 %% ========================================================================
132 %  6. SIMULATION PARAMETERS
133 % ========================================================================
134 time_step = 0.005;                    % [s]
135 final_time = 30;                      % [s]
136 time = 0:time_step:final_time;        % [s]
137
138
139 %% ========================================================================
140 %  7. UNIT CONVERSIONS
141 % ========================================================================
142 % Initial satellite angular velocity.
143 initial_satellite_speed = ...
144     initial_satellite_speed_rpm * (2*pi/60);   % [rad/s]
145
146 % Final satellite angular velocity.
147 final_satellite_speed = ...
148     final_satellite_speed_rpm * (2*pi/60);     % [rad/s]
149
150 % Reaction wheel reference speed.
151 wheel_speed = ...
152     wheel_speed_rpm * (2*pi/60);               % [rad/s]
153
154 % Motor no-load speed.
```

```
155 motor_no_load_speed = ...
156     motor_no_load_speed_rpm * (2*pi/60);       % [rad/s]
157
158 % Motor maximum torque.
159 motor_max_torque = ...
160     motor_max_torque_gf_cm * 9.81e-5;          % [N.m]
161
162
163 %% ========================================================================
164 %  8. SATELLITE MOMENT OF INERTIA
165 % ========================================================================
166 % Rotation about the Z axis:
167 %       J = (1/12)*m*(a^2 + b^2)
168 % The reaction wheel is not included in this value: in the dynamics, the
169 % wheel inertia is accounted for separately in J_wheel, since the wheel
170 % angular momentum is J_wheel*(omega + omega_w).
171 satellite_inertia = ...
172     (satellite_mass / 12) * ...
173     (satellite_side_x^2 + satellite_side_y^2);
174
175
176 %% ========================================================================
177 %  9. REACTION WHEEL INERTIA
178 % ========================================================================
179 % Full disk minus the coaxial recess disk:
180 %       J_wheel = (1/2)*rho*pi*(R^4*h - r_c^4*h_c)
181 wheel_inertia = ...
182     0.5 * material_density * pi * ...
183     (wheel_radius^4 * wheel_thickness - recess_radius^4 * recess_depth);
184
185 % Geometric volume of the wheel (full cylinder minus recess).
186 wheel_volume = ...
187     pi * (wheel_radius^2 * wheel_thickness - ...
188     recess_radius^2 * recess_depth);
189
190 % Mass calculated from geometry and density.
191 wheel_mass_from_geometry = ...
192     material_density * wheel_volume;
193
194
195 %% ========================================================================
196 %  10. ANGULAR MOMENTUM TO BE ABSORBED
197 % ========================================================================
198 % Initially the wheel rotates together with the satellite, so the wheel
199 % must absorb the angular momentum of both:
200 %       H_0 = (J + J_wheel)*omega_0
201 initial_satellite_momentum = ...
202     (satellite_inertia + wheel_inertia) * initial_satellite_speed;
203
204
205 %% ========================================================================
206 %  11. MOTOR CONSTANTS
207 % ========================================================================
208 % At no load:
209 %       V - IO*R = Ke*omega_nl
210 % In the SI approximation used here:
211 %       Kt = Ke
212 % Therefore:
213 %       Kt = (V - IO*R)/omega_nl
214 %       f  = Kt/R
215 % where f is the torque-per-voltage factor at zero wheel speed (stall).
216 motor_torque_constant = ...
217     (motor_voltage - motor_no_load_current * motor_resistance) / ...
218     motor_no_load_speed;
219
220 motor_torque_per_volt = ...
221     motor_torque_constant / ...
222     motor_resistance;
223
224
225 %% ========================================================================
226 %  12. CONTROLLER GAIN
227 % ========================================================================
228 % Pure velocity feedback:
229 %       u_c = -K2*omega   (desired torque on the satellite)
230 % For an ideal torque actuator:
231 %       J*omega_dot = -K2*omega
232 % and therefore:
```

```
%      K2 = p*J
K2 = ...
    closed_loop_pole * satellite_inertia;


%% ========================================================================
%  13. MOTOR + WHEEL TIME CONSTANT (REFERENCE ONLY)
% ========================================================================
% Mechanical time constant of the wheel SPEED for a voltage input:
%      tau = J_wheel*R/Kt^2
% It is no longer used as a torque lag: the back-EMF dynamics are now
% represented explicitly by the wheel speed state.
time_constant = ...
    wheel_inertia * motor_resistance / ...
    motor_torque_constant^2;


%% ========================================================================
%  14. ACTUATOR LIMITS
% ========================================================================
% Voltage saturation in the Simulink model.
Vmax = driver_voltage_limit;          % [V]

% Torque saturation in the Simulink model.
Tmax = motor_max_torque;              % [N.m]

% Back-EMF needed to reach the wheel reference speed.
backemf_at_reference_speed = ...
    motor_torque_constant * wheel_speed;       % [V]


%% ========================================================================
%  15. REACTION WHEEL ANGULAR MOMENTUM
% ========================================================================
% Angular momentum stored at the specified wheel reference speed:
%      H_wheel = J_wheel*omega_wheel
wheel_angular_momentum = ...
    wheel_inertia * wheel_speed;


%% ========================================================================
%  16. DISPLAY PARAMETERS
% ========================================================================
fprintf('\n');
fprintf('============================================================\n');
fprintf('          DETUMBLING CONTROLLER - PARAMETERS                \n');
fprintf('============================================================\n');

fprintf('\n----- Satellite Parameters -----\n');
fprintf('Satellite mass                 : %10.4f kg\n', satellite_mass);
fprintf('Satellite moment of inertia    : %10.6e kg.m^2\n', satellite_inertia);
fprintf('Inverse of inertia (1/J)       : %10.4f 1/(kg.m^2)\n', ...
    1/satellite_inertia);
fprintf('Initial angular velocity       : %10.4f rad/s\n', ...
    initial_satellite_speed);
fprintf('Final angular velocity         : %10.4f rad/s\n', ...
    final_satellite_speed);
fprintf('Momentum to absorb (60->0 rpm) : %10.6e N.m.s\n', ...
    initial_satellite_momentum);

fprintf('\n----- Reaction Wheel Parameters -----\n');
fprintf('Wheel material                 : Steel\n');
fprintf('Wheel density                  : %10.2f kg/m^3\n', ...
    material_density);
fprintf('Wheel radius                   : %10.2f mm\n', ...
    wheel_radius*1e3);
fprintf('Wheel diameter                 : %10.2f mm\n', ...
    2*wheel_radius*1e3);
fprintf('Recess diameter                : %10.2f mm\n', ...
    2*recess_radius*1e3);
fprintf('Recess depth                   : %10.2f mm\n', ...
    recess_depth*1e3);
fprintf('Wheel mass                     : %10.6f kg\n', ...
    wheel_mass);
fprintf('Wheel thickness                : %10.3f mm\n', ...
    wheel_thickness*1e3);
fprintf('Wheel reference speed          : %10.2f rpm\n', ...
    wheel_speed_rpm);
```

```
fprintf('Wheel inertia                 : %10.6e kg.m^2\n', ...
    wheel_inertia);
fprintf('Wheel angular momentum        : %10.6e N.m.s\n', ...
    wheel_angular_momentum);
fprintf('Mass from geometry            : %10.6f kg\n', ...
    wheel_mass_from_geometry);

fprintf('\n----- Motor Parameters -----\n');
fprintf('Supply voltage (Kt derivation) : %10.4f V\n', ...
    motor_voltage);
fprintf('Driver voltage limit          : %10.4f V\n', ...
    Vmax);
fprintf('Torque constant Kt (= Ke)     : %10.6f N.m/A\n', ...
    motor_torque_constant);
fprintf('Torque per volt f = Kt/R      : %10.6f N.m/V\n', ...
    motor_torque_per_volt);
fprintf('Maximum motor torque          : %10.6e N.m\n', ...
    motor_max_torque);
fprintf('Back-EMF at reference speed   : %10.4f V\n', ...
    backemf_at_reference_speed);

fprintf('\n----- Controller Parameters -----\n');
fprintf('Desired pole parameter p      : %10.4f 1/s\n', ...
    closed_loop_pole);
fprintf('Velocity gain K2              : %10.6e N.m/(rad/s)\n', ...
    K2);
fprintf('Position gain K1              : %10.4f N.m/rad\n', ...
    K1);
fprintf('Back-EMF compensation         : %10d\n', ...
    backemf_compensation);
fprintf('Wheel time constant tau (ref.) : %10.6f s\n', ...
    time_constant);
fprintf('Voltage saturation            : +/- %8.4f V\n', ...
    Vmax);
fprintf('Torque saturation             : +/- %8.6e N.m\n', ...
    Tmax);


%% ========================================================================
%  17. CLOSED-LOOP LINEAR ANALYSIS (NO SATURATION)
% ========================================================================
% This section is only used to calculate the poles of the unsaturated
% closed-loop model. It is not the nonlinear simulation used for the final
% results.
%
% States (K1 = 0): z = [omega; omega_w]
%   Motor torque on the wheel:  T_m = f*(V - Kt*omega_w)
%   Satellite:                  J*omega_dot = -T_m
%   Wheel (relative speed):     omega_w_dot = T_m/J_wheel - omega_dot
%
% With compensation:    V = K2*omega/f + Kt*omega_w  ->  T_m = K2*omega
% Without compensation: V = K2*omega/f              ->  T_m = K2*omega - f*Kt*omega_w
J = satellite_inertia;
Jw = wheel_inertia;
Kt = motor_torque_constant;
f = motor_torque_per_volt;

if backemf_compensation
    Tm_row = [K2, 0];
else
    Tm_row = [K2, -f*Kt];
end

A_cl = [-Tm_row/J;
         Tm_row*(1/Jw + 1/J)];

closed_loop_poles = eig(A_cl);

fprintf('\n----- Unsaturated Closed-Loop Poles -----\n');
fprintf('Pole 1                        : %10.6f 1/s\n', ...
    closed_loop_poles(1));
fprintf('Pole 2                        : %10.6f 1/s\n', ...
    closed_loop_poles(2));

% Both cases have one pole at 0 (a line of equilibria where T_m = 0).
% With compensation, T_m = 0 only when omega = 0, so omega -> 0 with pole -p.
% Without compensation, T_m = 0 whenever K2*omega = f*Kt*omega_w, so the
% satellite stops at a nonzero speed once the wheel back-EMF builds up.
```

```
389
390 fprintf('(pole at 0: equilibrium where the motor torque vanishes)\n');
391
392
393 %% ========================================================================
394 %  18. NONLINEAR SIMULATION
395 % ========================================================================
396 % Simulink block-by-block correspondence:
397 %   K2 feedback + K1 feedback
398 %   u_c = -(K1*theta + K2*omega)            desired torque on the satellite
399 %   V_c = -u_c/f (+ Kt*omega_w if compensated)
400 %   voltage saturation
401 %   T_m = f*(V - Kt*omega_w)                motor torque on the wheel
402 %   torque saturation
403 %   satellite: omega_dot = -T_m/J
404 %   wheel:     omega_w_dot = T_m/J_wheel - omega_dot
405 %
406 % The state vector is:
407 %      x(1) = theta
408 %      x(2) = omega     (satellite)
409 %      x(3) = omega_w   (wheel, relative to the satellite)
410
411 % Initial condition: satellite tumbling, wheel at rest relative to it.
412 initial_state = ...
413     [0;
414      initial_satellite_speed;
415      0];
416
417 % ODE options.
418 options = odeset('RelTol',1e-8,'AbsTol',1e-10);
419
420 % Numerical integration.
421 [t_sim, x] = ode45( ...
422     @(t,x) closed_loop_dynamics( ...
423         t, x, J, Jw, Kt, f, K1, K2, Vmax, Tmax, backemf_compensation), ...
424     time, ...
425     initial_state, ...
426     options);
427
428
429 %% ========================================================================
430 %  19. SIMULINK-COMPATIBLE SIGNALS
431 % ========================================================================
432 % State variables.
433 theta_satellite = x(:,1);
434 satellite_speed = x(:,2);
435 wheel_relative_speed = x(:,3);
436
437 % Controller output (desired torque on the satellite).
438 controller_output = ...
439     -(K1*theta_satellite + K2*satellite_speed);
440
441 % Voltage command before the voltage saturation block.
442 voltage_unsaturated = -controller_output / f;
443 if backemf_compensation
444     voltage_unsaturated = voltage_unsaturated + Kt*wheel_relative_speed;
445 end
446
447 % Voltage applied after saturation.
448 voltage = saturation( ...
449     voltage_unsaturated, Vmax);
450
451 % Motor torque on the wheel (with back-EMF) and torque on the satellite.
452 motor_torque = saturation( ...
453     f * (voltage - Kt*wheel_relative_speed), Tmax);
454 delivered_torque = -motor_torque;
455
456 % Motor current.
457 motor_current = motor_torque / Kt;
458
459 % Satellite angular velocity in rpm.
460 satellite_speed_rpm = ...
461     satellite_speed * 60/(2*pi);
462
463 % Satellite angular acceleration.
464 satellite_acceleration = ...
465     delivered_torque / J;
466
```

```
% Wheel speed (relative to the satellite) in rpm.
wheel_speed_rpm_sim = ...
    wheel_relative_speed * 60/(2*pi);

% Wheel angular momentum (inertial).
wheel_momentum = ...
    Jw * (wheel_relative_speed + satellite_speed);


%% =======================================================================
%  20. DETUMBLING TIME AND MAXIMUM VALUES
% =======================================================================
settling_index = find( ...
    abs(satellite_speed_rpm) < speed_tolerance_rpm, ...
    1, 'first');

if isempty(settling_index)
    settling_time = NaN;
else
    settling_time = t_sim(settling_index);
end

max_controller_output = ...
    max(abs(controller_output));

max_voltage_unsaturated = ...
    max(abs(voltage_unsaturated));

max_voltage = ...
    max(abs(voltage));

max_delivered_torque = ...
    max(abs(delivered_torque));

max_motor_current = ...
    max(abs(motor_current));

max_wheel_momentum = ...
    max(abs(wheel_momentum));

max_wheel_speed_rpm = ...
    max(abs(wheel_speed_rpm_sim));


%% =======================================================================
%  21. SIMULATION RESULTS
% =======================================================================
fprintf('\n============================================================\n');
fprintf('              CLOSED-LOOP SIMULATION RESULTS              \n');
fprintf('============================================================\n');

fprintf('\n----- Reaction Wheel Simulation -----\n');
fprintf('Wheel inertia                  : %10.6e kg.m^2\n', ...
    wheel_inertia);
fprintf('Maximum controller output      : %10.4f mN.m\n', ...
    max_controller_output*1e3);
fprintf('Maximum voltage before sat.    : %10.4f V\n', ...
    max_voltage_unsaturated);
fprintf('Maximum voltage applied        : %10.4f V\n', ...
    max_voltage);
fprintf('Maximum delivered torque       : %10.4f mN.m\n', ...
    max_delivered_torque*1e3);
fprintf('Motor torque limit             : %10.4f mN.m\n', ...
    Tmax*1e3);
fprintf('Maximum motor current          : %10.4f A\n', ...
    max_motor_current);
fprintf('Maximum wheel angular momentum : %10.6e N.m.s\n', ...
    max_wheel_momentum);
fprintf('Maximum wheel speed (relative) : %10.4f rpm\n', ...
    max_wheel_speed_rpm);
fprintf('Final satellite speed          : %10.4f rpm\n', ...
    satellite_speed_rpm(end));
fprintf('Minimum satellite speed        : %10.4f rpm\n', ...
    min(satellite_speed_rpm));
fprintf('Time to |omega| < %.1f rpm     : %10.4f s\n', ...
    speed_tolerance_rpm, settling_time);


```

```
%% ========================================================================
%  22. VALIDATION CHECKS
% ========================================================================
fprintf('\n----- Validation Checks -----\n');

if isnan(settling_time)
    fprintf('Detumbling requirement           : FAIL (not reached)\n');
else
    fprintf('Detumbling requirement           : PASS\n');
end

if max_voltage_unsaturated <= Vmax + 1e-12
    fprintf('Voltage within driver limit      : PASS\n');
else
    fprintf('Voltage within driver limit      : FAIL (saturated)\n');
end

if max(abs(f * (voltage - Kt*wheel_relative_speed))) <= Tmax + 1e-12
    fprintf('Motor torque limit               : PASS\n');
else
    fprintf('Motor torque limit               : FAIL (saturated)\n');
end

% The wheel was sized to end at the reference speed, so only a small
% margin is allowed.
if max_wheel_speed_rpm <= wheel_speed_rpm*1.02
    fprintf('Wheel reference speed limit      : PASS\n');
else
    fprintf('Wheel reference speed limit      : CHECK REQUIRED\n');
end


%% ========================================================================
%  23. PLOTS
% ========================================================================
figure('Name', ...
    'CubeSat 2U Detumbling - Simulink Equivalent');

subplot(3,1,1);
plot(t_sim, satellite_speed_rpm, ...
    'LineWidth', 1.5);
grid on;
hold on;
yline(0, '--k', 'HandleVisibility', 'off');
yline(speed_tolerance_rpm, '--', 'HandleVisibility', 'off');
yline(-speed_tolerance_rpm, '--', 'HandleVisibility', 'off');
ylabel('\omega_{sat} [rpm]');
title('Satellite angular velocity');

subplot(3,1,2);
plot(t_sim, voltage_unsaturated, ...
    'LineWidth', 1.5);
grid on;
hold on;
yline(Vmax, '--', 'HandleVisibility', 'off');
yline(-Vmax, '--', 'HandleVisibility', 'off');
ylabel('Voltage [V]');
title('Motor voltage command');

subplot(3,1,3);
plot(t_sim, delivered_torque*1e3, ...
    'LineWidth', 1.5);
grid on;
hold on;
yline(Tmax*1e3, '--', 'HandleVisibility', 'off');
yline(-Tmax*1e3, '--', 'HandleVisibility', 'off');
ylabel('Torque [mN.m]');
xlabel('Time [s]');
title('Torque applied to satellite');

figure('Name', ...
    'CubeSat 2U Detumbling - Control Signals');

subplot(3,1,1);
plot(t_sim, controller_output*1e3, ...
    'LineWidth', 1.5);
grid on;
ylabel('u_c [mN.m]');
```

```
623 title('Controller output before voltage conversion');
624
625 subplot(3,1,2);
626 plot(t_sim, voltage, ...
627     'LineWidth', 1.5);
628 grid on;
629 ylabel('V [V]');
630 title('Applied motor voltage');
631
632 subplot(3,1,3);
633 plot(t_sim, wheel_speed_rpm_sim, ...
634     'LineWidth', 1.5);
635 grid on;
636 ylabel('\omega_{wheel} [rpm]');
637 xlabel('Time [s]');
638 title('Reaction-wheel speed (relative to satellite)');
639
640
641 %% =======================================================================
642 %  24. SIMULINK PARAMETERS
643 % =======================================================================
644 fprintf('\n============================================================\n');
645 fprintf('                  SIMULINK PARAMETERS                       \n');
646 fprintf('============================================================\n');
647 fprintf('1/J                          : %10.6f\n', ...
648     1/satellite_inertia);
649 fprintf('1/J_wheel                    : %10.6f\n', ...
650     1/wheel_inertia);
651 fprintf('K1                           : %10.6f\n', ...
652     K1);
653 fprintf('K2                           : %10.9f\n', ...
654     K2);
655 fprintf('Kt                           : %10.9f N.m/A\n', ...
656     motor_torque_constant);
657 fprintf('f = Kt/R                     : %10.9f N.m/V\n', ...
658     motor_torque_per_volt);
659 fprintf('Vmax                         : %10.6f V\n', ...
660     Vmax);
661 fprintf('Tmax                         : %10.9f N.m\n', ...
662     Tmax);
663 fprintf('Initial omega                : %10.6f rad/s\n', ...
664     initial_satellite_speed);
665 fprintf('Simulation step              : %10.6f s\n', ...
666     time_step);
667
668 fprintf('\n============================================================\n');
669 fprintf('                    END OF ANALYSIS                         \n');
670 fprintf('============================================================\n\n');
671
672
673 %% =======================================================================
674 %  LOCAL FUNCTIONS
675 % =======================================================================
676 % CLOSED_LOOP_DYNAMICS
677 % Satellite + reaction wheel + DC motor with back-EMF.
678 %
679 % State:
680 %   x(1) = theta
681 %   x(2) = omega     (satellite)
682 %   x(3) = omega_w   (wheel, relative to the satellite)
683 %
684 % Controller:
685 %   u_c = -(K1*theta + K2*omega)          desired torque on the satellite
686 %
687 % Voltage command:
688 %   V_c = -u_c/f  (+ Kt*omega_w if backemf_comp)
689 %   V   = saturation(V_c, +/-Vmax)
690 %
691 % Motor (torque on the wheel):
692 %   T_m = saturation(f*(V - Kt*omega_w), +/-Tmax)
693 %
694 % Satellite and wheel:
695 %   omega_dot   = -T_m/J
696 %   omega_w_dot =  T_m/Jw - omega_dot
697 %   theta_dot   =  omega
698
699 function dx = closed_loop_dynamics(~, x, J, Jw, Kt, f, K1, K2, Vmax, Tmax, backemf_comp)
700     % Current values of the system states.
```

```
    theta = x(1);
    omega = x(2);
    omega_w = x(3);

    % Controller output (desired torque on the satellite).
    u_c = -(K1*theta + K2*omega);

    % Voltage command: the wheel must receive the opposite torque.
    V_command = -u_c / f;
    if backemf_comp
        V_command = V_command + Kt*omega_w;
    end
    V_applied = saturation(V_command, Vmax);

    % Motor torque on the wheel, including back-EMF.
    T_motor = saturation(f * (V_applied - Kt*omega_w), Tmax);

    % Satellite and wheel dynamics.
    omega_dot = -T_motor / J;
    omega_w_dot = T_motor / Jw - omega_dot;
    theta_dot = omega;

    dx = [theta_dot;
          omega_dot;
          omega_w_dot];
end

% SATURATION
% Symmetric saturation equivalent to a Simulink Saturation block.
function y = saturation(u, limit)
    y = min(max(u, -limit), limit);
end
```

C.6 Simulink Model

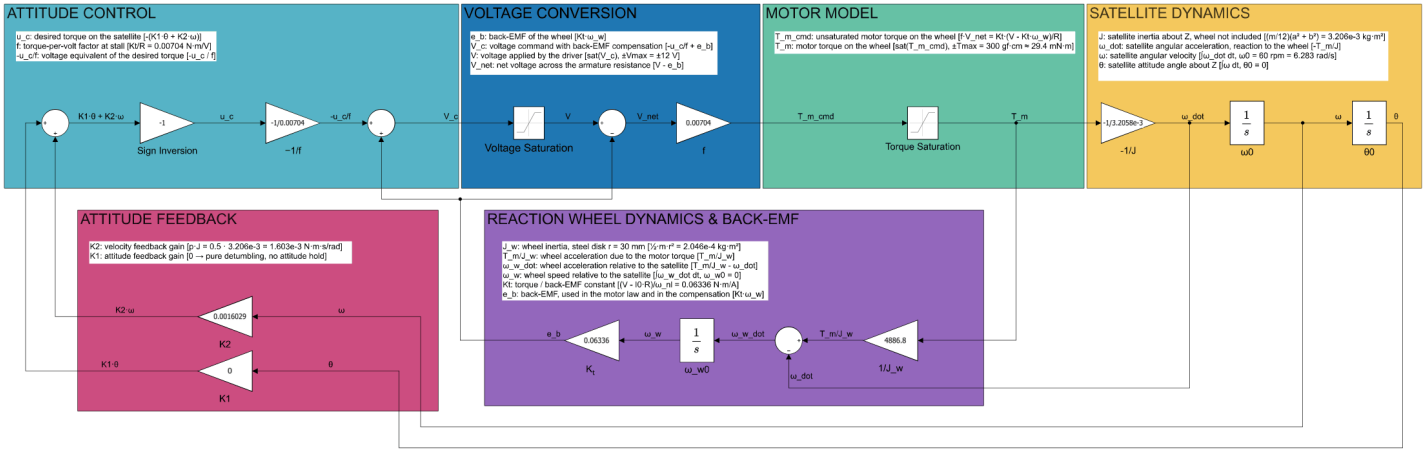


Figure 87: Block diagram of the Simulink model
Source: Authors (2026).

D Electronics

This appendix gives the electrical schematic and the 3D model of each printed circuit board of the electronics: TT&C, EPS, solar panels and OBC. The schematics were drawn and the boards laid out in KiCad.

D.1 TT&C

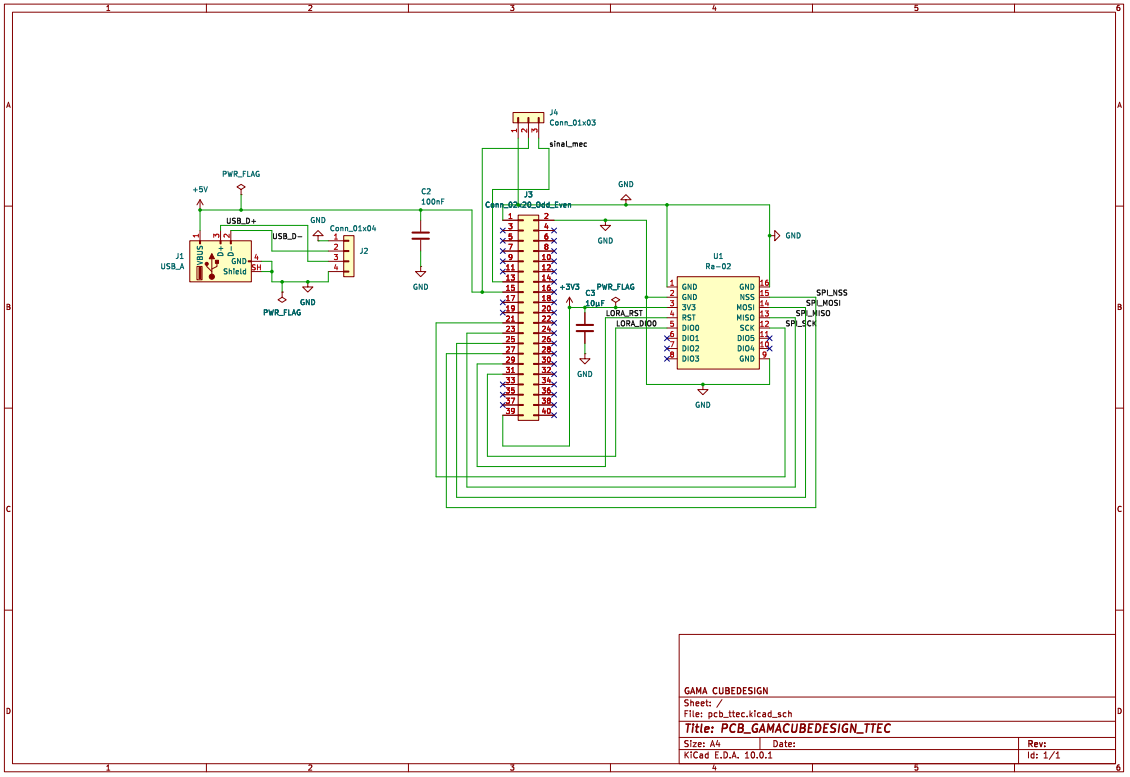
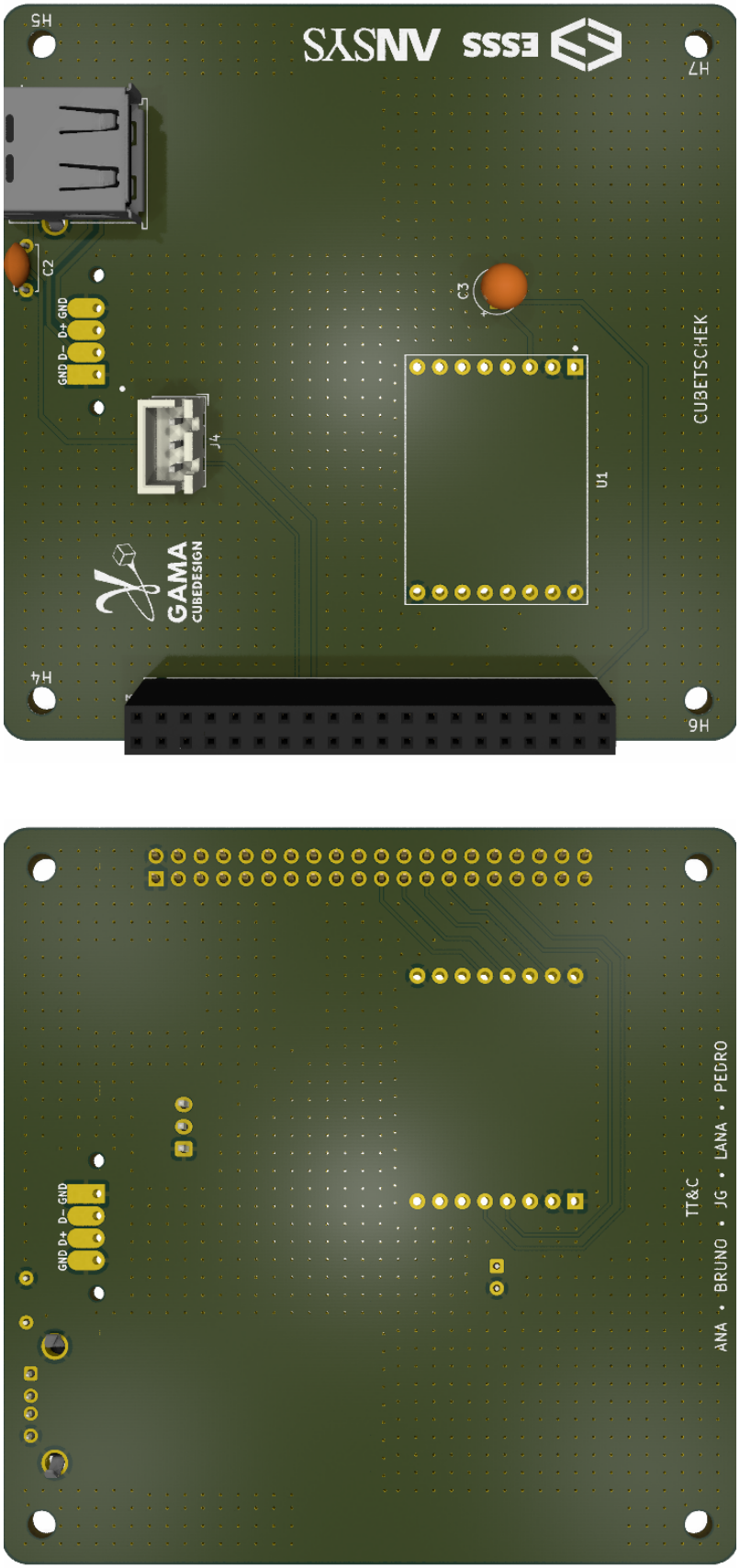


Figure 88: Electrical schematic of the TT&C board
Source: Authors (2026).



(a) Front (b) Back

Figure 89: 3D model of the TT&C board
Source: Authors (2026).

D.2 EPS

The EPS uses three boards: the power board, and the upper and lower battery boards, which also carry the active thermal control.

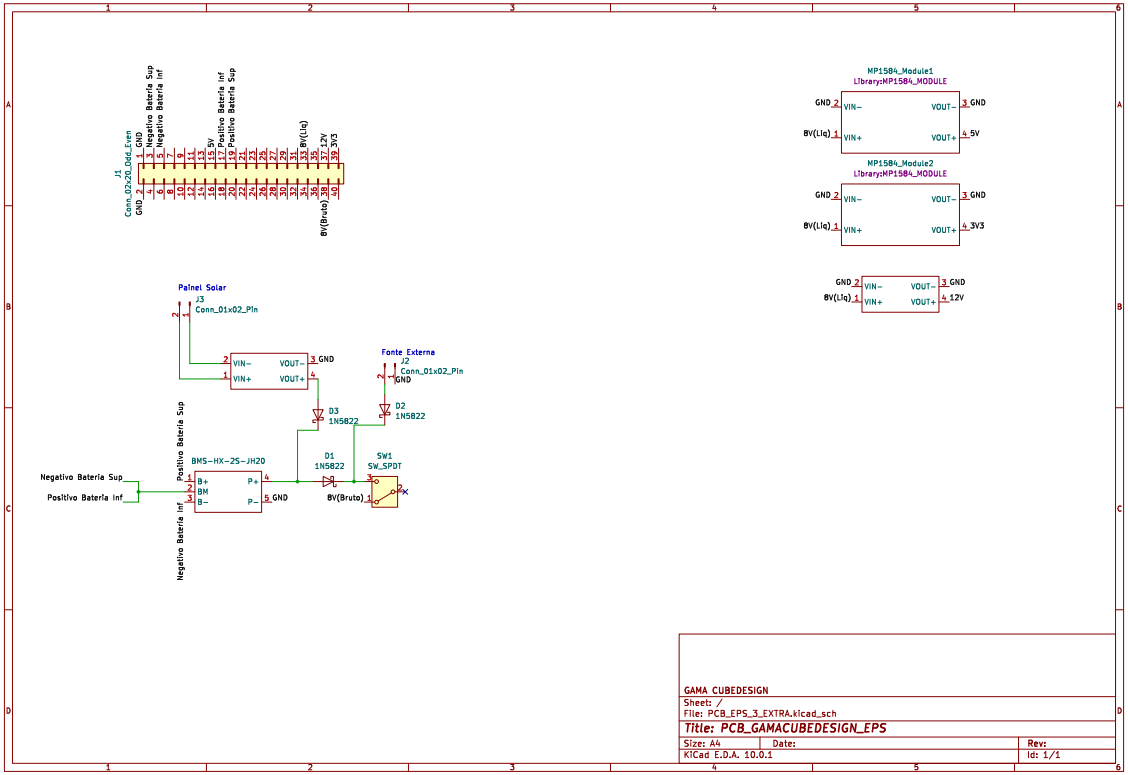
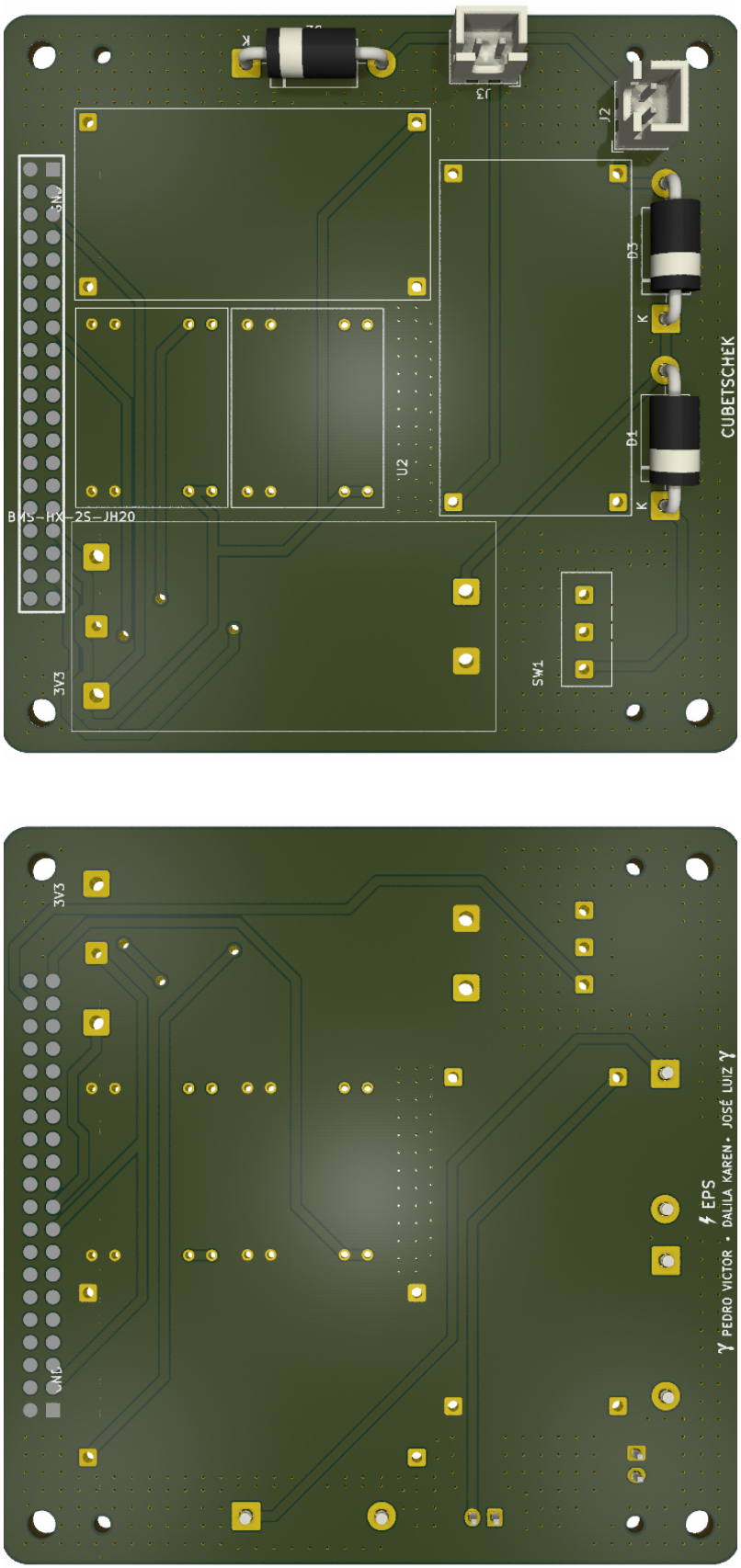


Figure 90: Electrical schematic of the EPS power board
Source: Authors (2026).



(a) Front (b) Back

Figure 91: 3D model of the EPS power board
Source: Authors (2026).

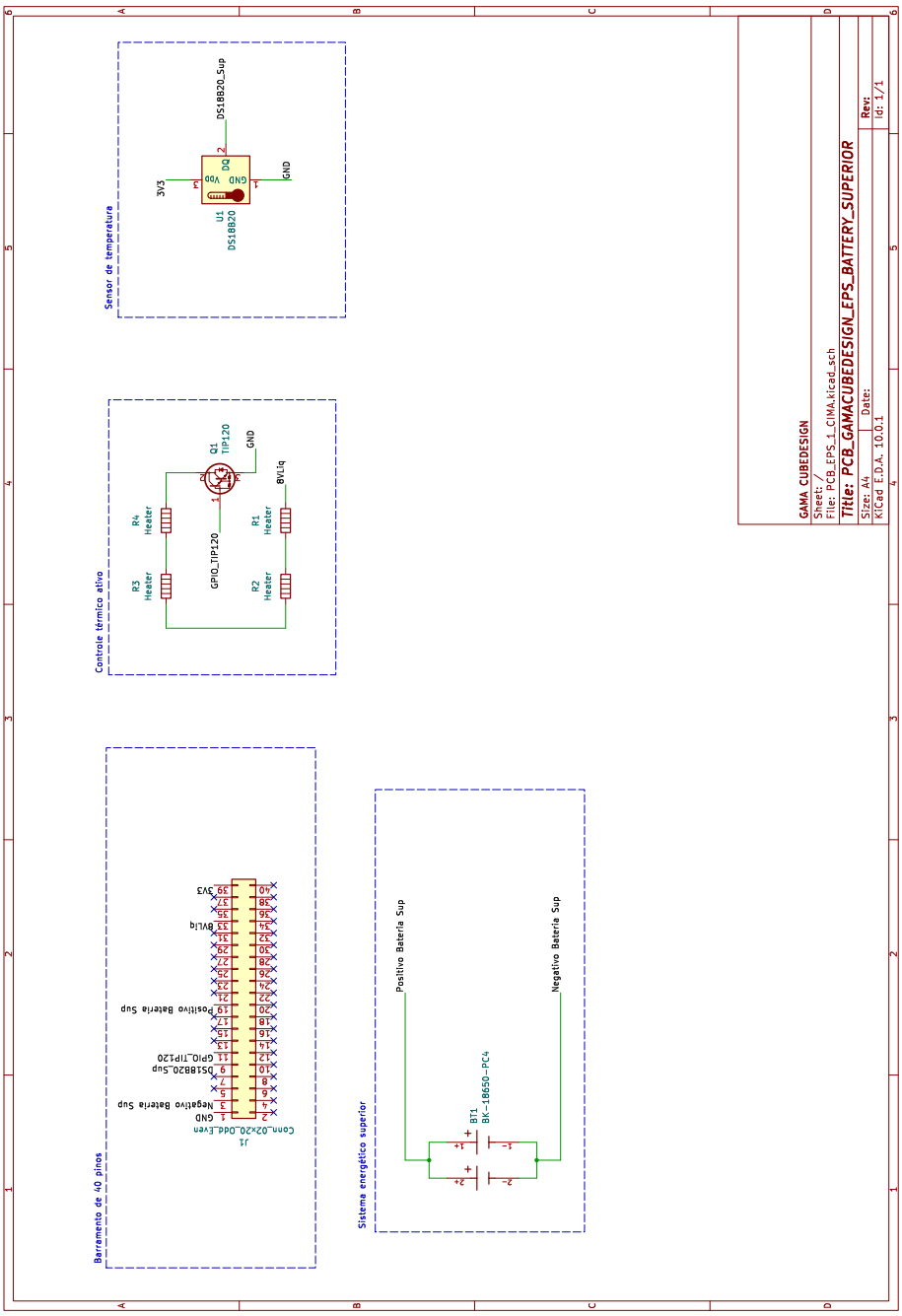
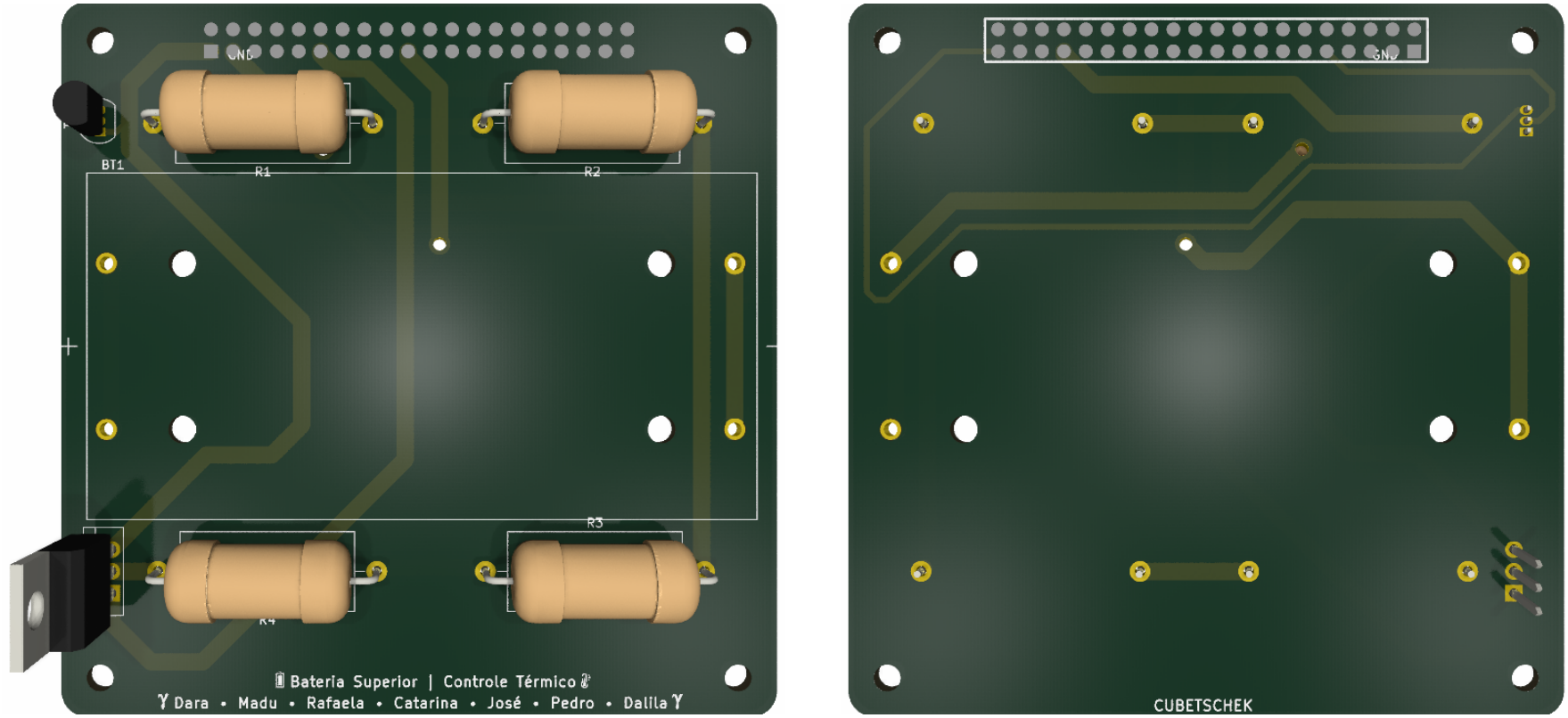


Figure 92: Electrical schematic of the upper battery and thermal control board
Source: Authors (2026).



(a) Front

(b) Back

Figure 93: 3D model of the upper battery and thermal control board

Source: Authors (2026).

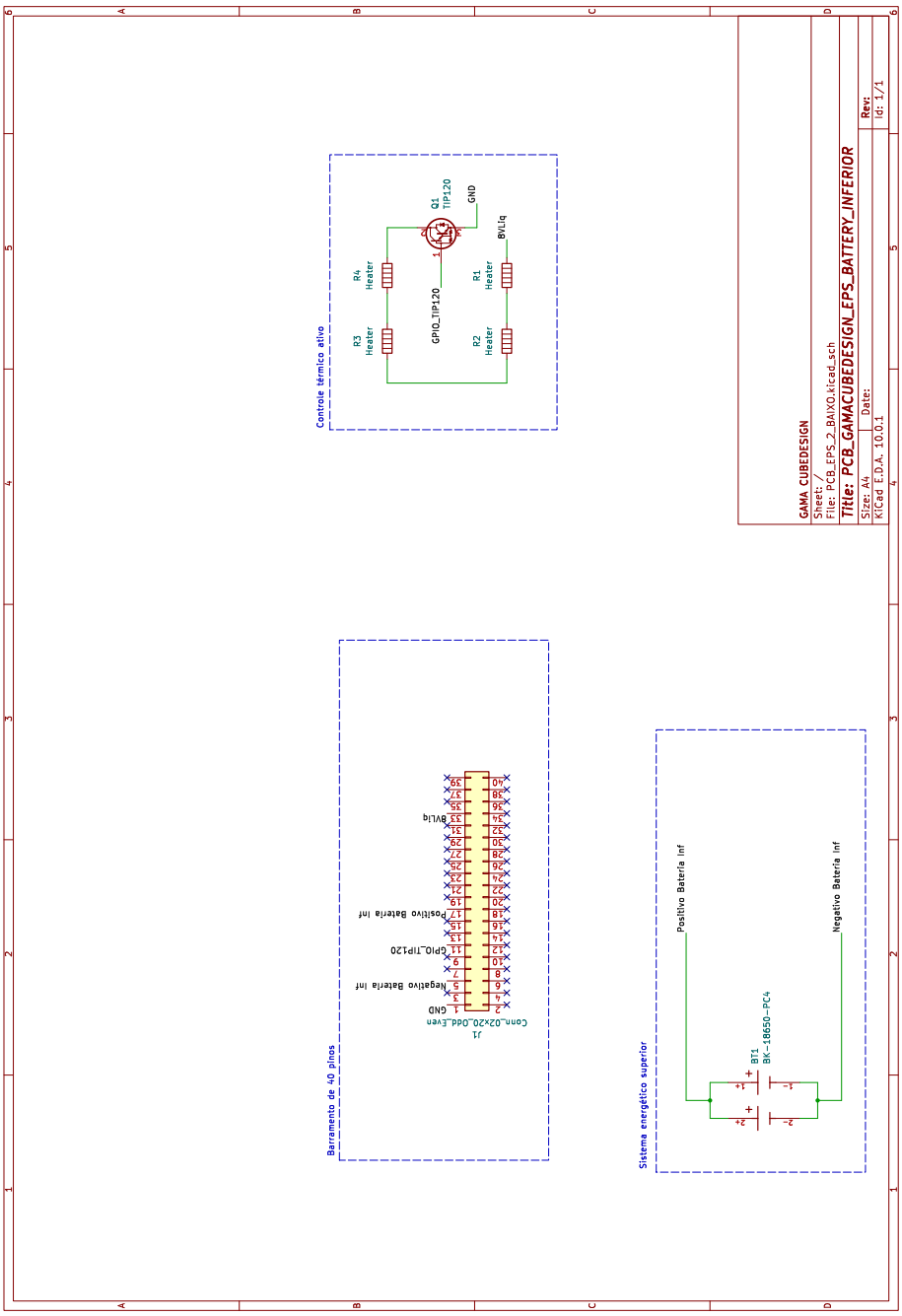
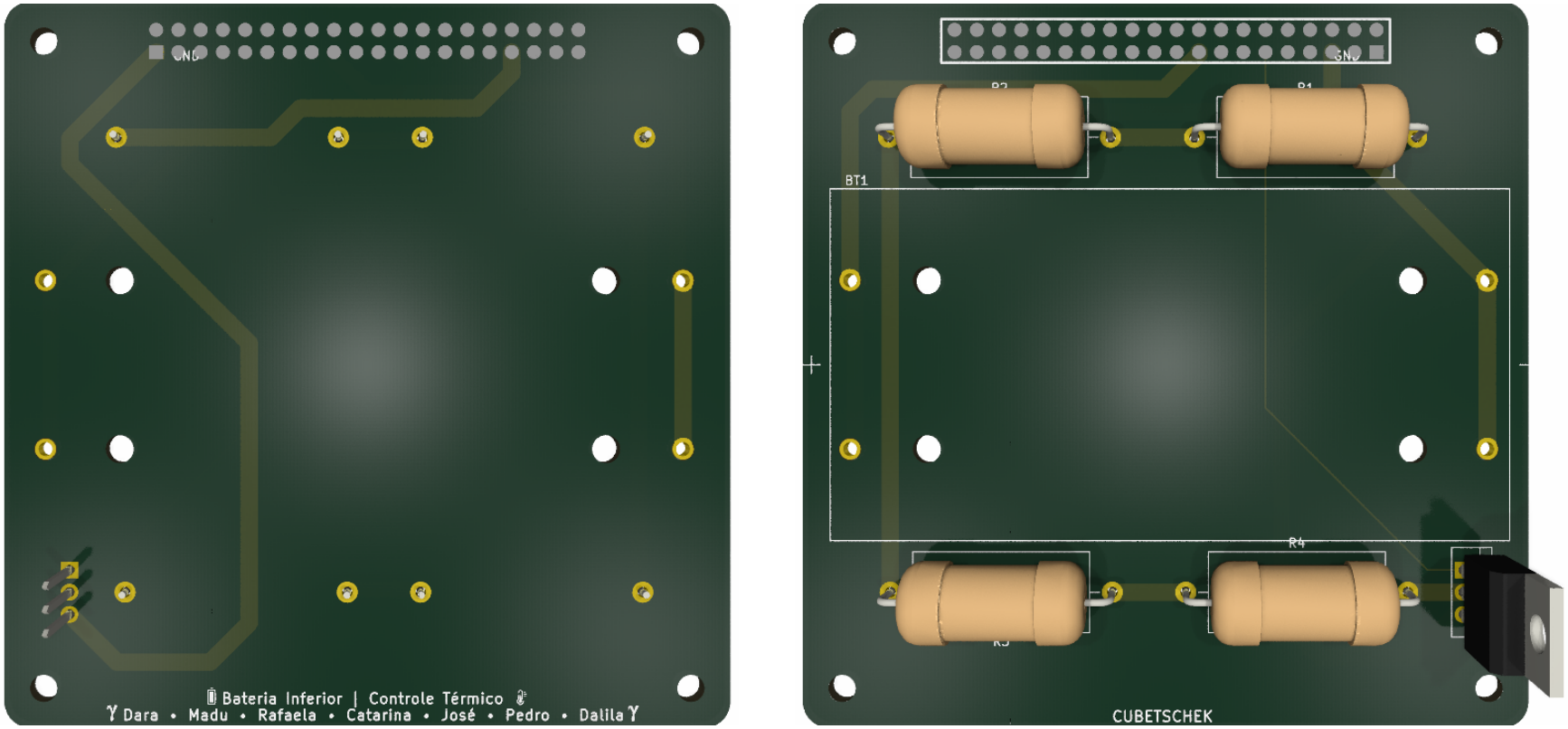


Figure 94: Electrical schematic of the lower battery and thermal control board
Source: Authors (2026).



(a) Front

(b) Back

Figure 95: 3D model of the lower battery and thermal control board

Source: Authors (2026).

D.3 Solar Panel

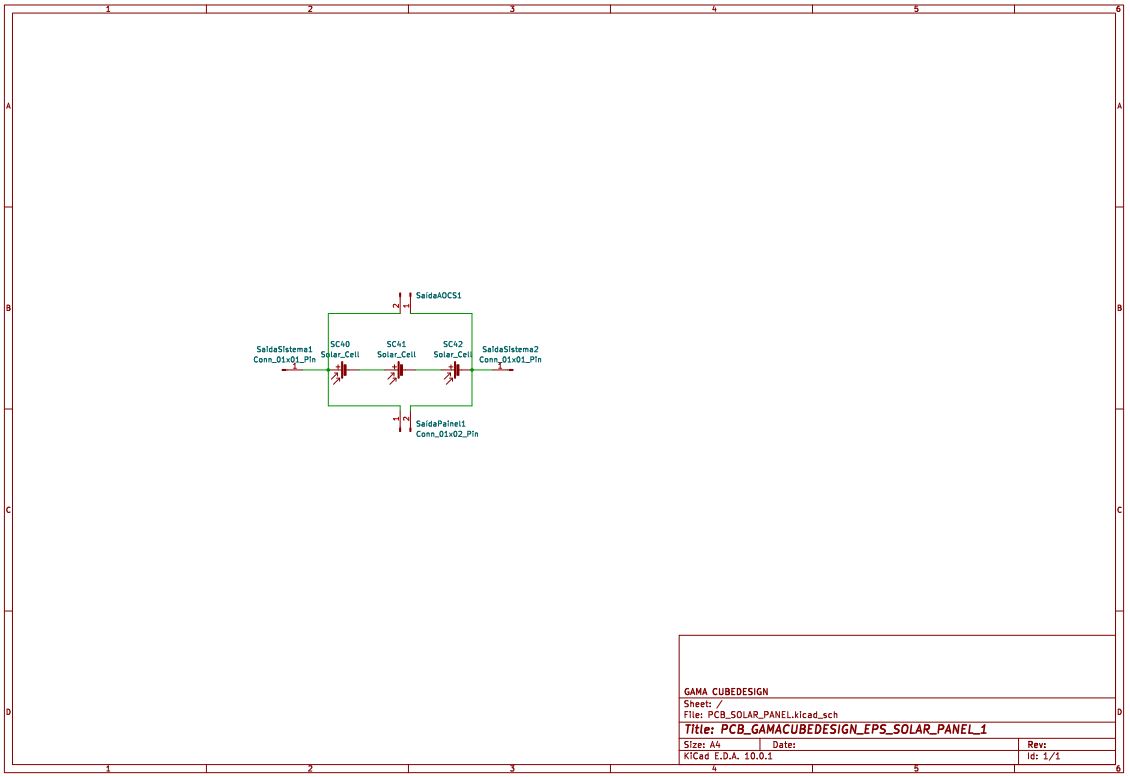


Figure 96: Electrical schematic of the solar panel board (sheet 1)
Source: Authors (2026).

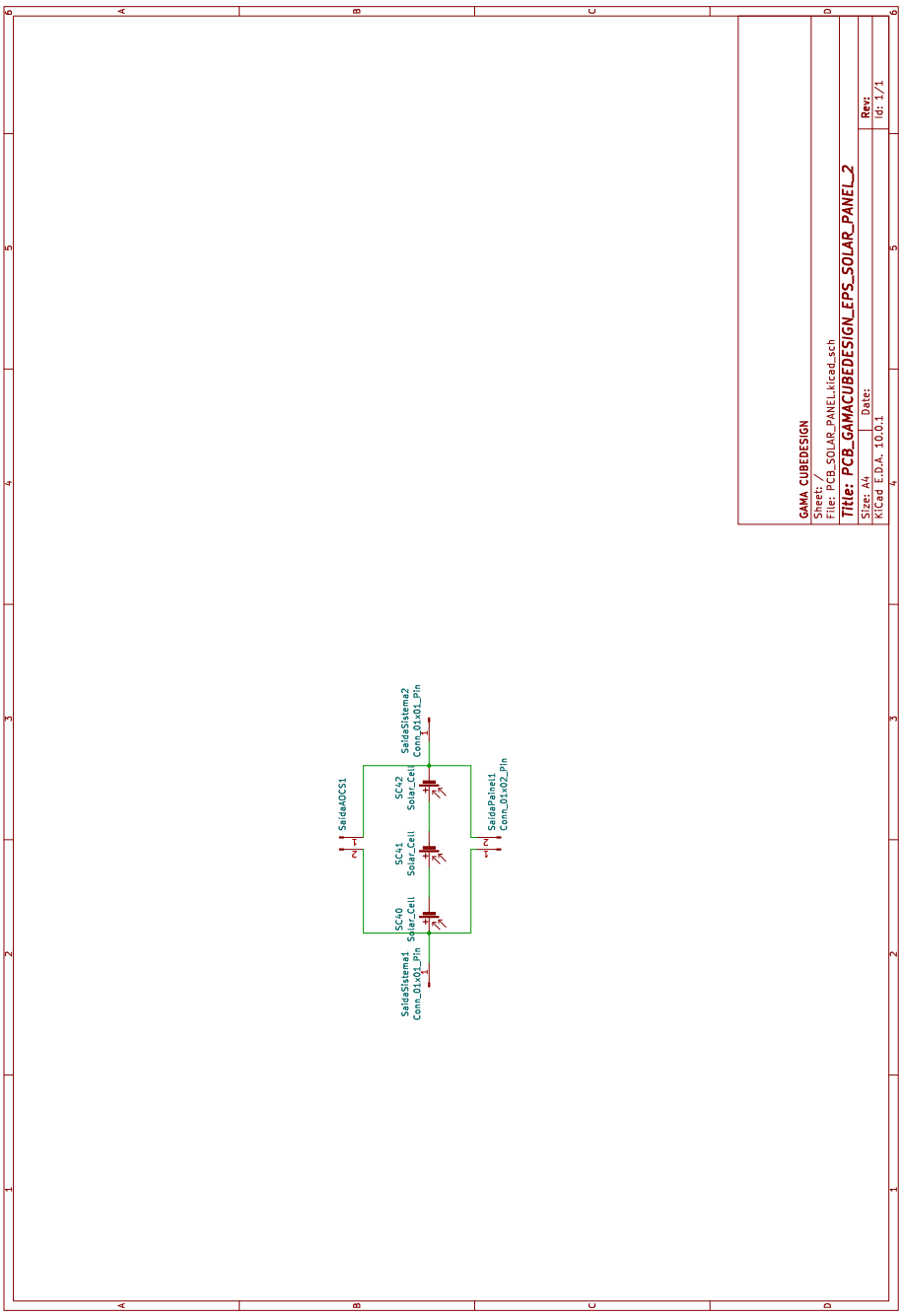


Figure 97: Electrical schematic of the solar panel board (sheet 2)
Source: Authors (2026).

(a) Front (b) Back

Figure 98: 3D model of the solar panel board
Source: Authors (2026).

D.4 OBC

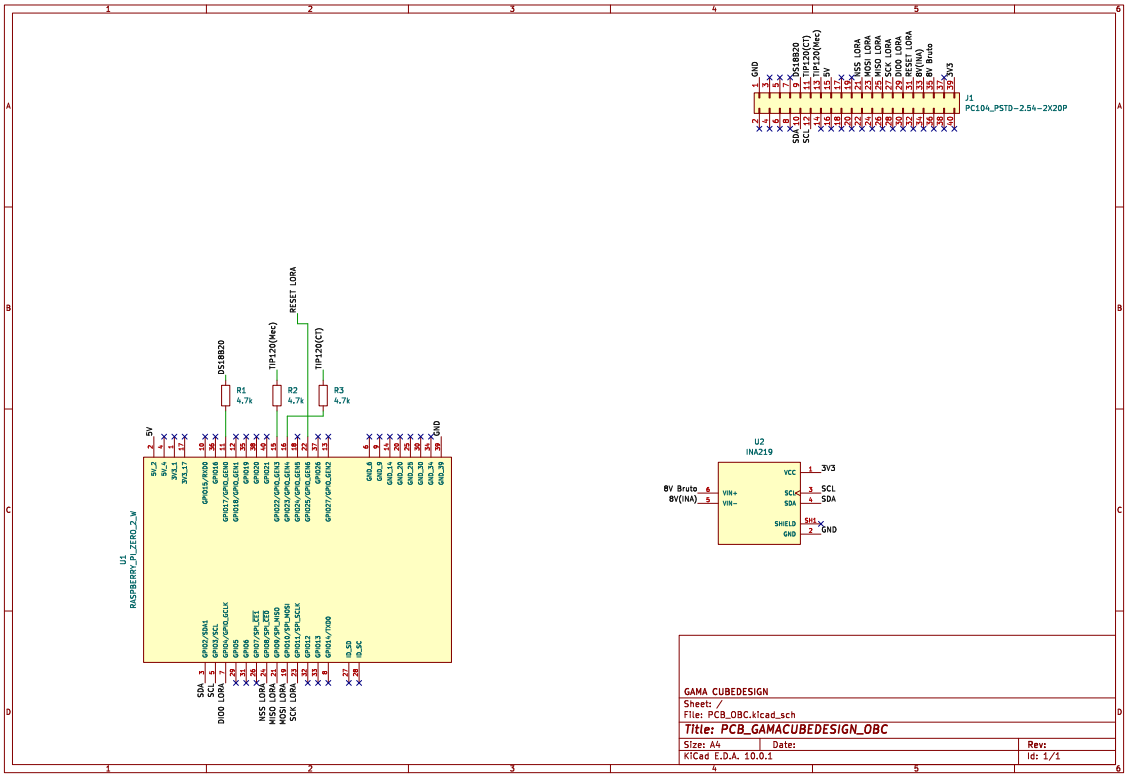


Figure 99: Electrical schematic of the OBC board

Source: Authors (2026).

(a) Front

(b) Back

Figure 100: 3D model of the OBC board

Source: Authors (2026).